COURSE SLIDES · WSQ

# AWS Certified SysOps Administrator Associate Training



WSQ Course Code: TGS-2024051413

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v6.0 · 23 August 2026

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ADMI
N

COURSE ADMINISTRATION

# Welcome & Course Orientation

Three days · 24 hours · 12 labs · 2 assessment hours


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# Digital Attendance (Mandatory)

1

## AM, PM and Assessment

Complete every required SSG digital attendance and wait for confirmation.

2

## Trainer displays the QR

Scan only the current session's QR code using your mobile camera.

3

## 75% minimum

Attendance and assessment requirements determine course-completion eligibility.

4

## Evidence

Notify the trainer immediately if a submission does not confirm.


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# About the Trainer




## Your Trainer

General trainer profile template

**NAME**

**TITLE / DESIGNATION**

**QUALIFICATIONS**

**AWS AND CLOUD EXPERTISE**

**INDUSTRY EXPERIENCE**

**CONTACT**

personal information is optional


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YOUR TRAINER

# About the Trainer




## Dr. Alfred Ang

Founder & Chief Training Officer
Tertiary Infotech Academy Pte Ltd

**EXPERIENCE**

More than 20 years across software engineering, AI, data analytics and cloud computing.

**QUALIFICATIONS**

PhD (NUS), DACE, ACTA and GCWLP among professional credentials.

**COURSE FOCUS**

AWS observability, automation, resilience, security and evidence-led operations.

**TEACHING APPROACH**

Connects architecture choices to measurable service outcomes and controlled change.


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ICE-BREAKER

# Let's Know Each Other

1

## Introduce

Share your name, organisation and role.

2

## Experience

Name one cloud service you have operated or supported.

3

## Aim

Choose one outcome: stronger monitoring, recovery, automation, security or troubleshooting.


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SAFE CLOUD LEARNING

# Ground Rules

1. **Participate**
   Ask, challenge and contribute evidence.

2. **Respect**
   Critique the system or idea, not the person.

3. **Protect access**
   Never share AWS passwords, access keys or account details.

4. **Bound risk**
   Use learner-owned resources and the trainer-approved Region.

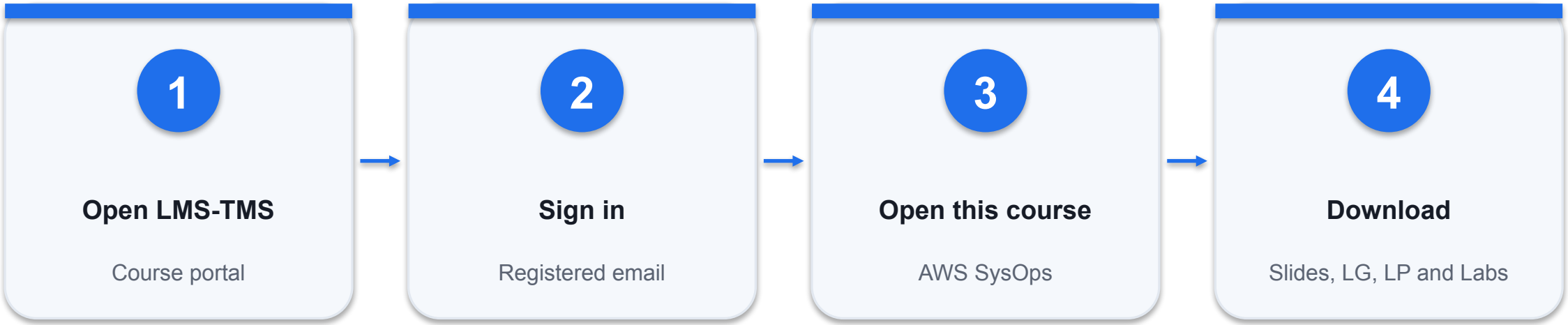
5. **Be punctual**
   Return on time and complete attendance.

6. **Clean up**
   Verify deletion of chargeable resources after every lab.


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VISUAL DOWNLOAD FLOW · LMS-TMS.TERTIARYINFOTECH.COM

# Download Your Course Material




**PROCEDURE BOUNDARY**

Detailed lab procedures are in the Learner Guide and the individual lab folders.


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22 HOURS DELIVERY · 2 HOURS ASSESSMENT · 4 LABS/DAY

# Lesson Plan — Three Days

1

**Day 1**

Foundations, observability, EC2 and fleet operations · Labs 01–04

2

**Day 2**

Reliability, continuity, deployment automation and security · Labs 05–08

3

**Day 3**

Networking, data operations, integrated troubleshooting and assessment · Labs 09–12

4

**Assessment**

WA-SAQ: 60 min · Case Study: 60 min


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WHAT YOU WILL BE ABLE TO DO

# Learning Outcomes

1

**LO1**

Maintain business continuity of AWS cloud operations to ensure stable service delivery.

2

**LO2**

Implement secure AWS cloud infrastructure setup such as data loading and performance monitoring.

3

**LO3**

Apply service quality standards to ensure security and compliance of AWS cloud operations.

4

**LO4**

Analyse and resolve processing issues to ensure continuity and disaster recovery in AWS cloud applications.

5

**LO5**

Troubleshoot networking performance issues at scale for optimal AWS infrastructure operation.


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# Skills Framework Alignment — Knowledge

1. **K1**
   Cloud computing infrastructure components and technologies
2. **K2**
   Installation, configuration and troubleshooting of cloud computing infrastructures
3. **K3**
   Installation, configuration and troubleshooting of software packages supporting cloud computing infrastructures
4. **K4**
   Design, configuration, maintenance and troubleshooting of cloud computing applications
5. **K5**
   Virtualisation technologies


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# Skills Framework Alignment — Abilities

1

**A1**
Implement data loading, security, performance monitoring and tuning, capacity planning, backup and recovery

2

**A2**
Implement performance diagnostics and troubleshooting at scale in a live environment

3

**A3**
Maintain availability and performance of cloud computing operations and ensure secure services with no unplanned downtime

4

**A4**
Analyse processing issues and data anomalies and identify solutions

5

**A5**
Implement and release cloud computing operations changes

6

**A6**
Implement IT risk-management measures for business continuity and disaster recovery

7

**A7**
Implement service-quality standards


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CURRENT CLOUDOPS BLUEPRINT · 10 TOPICS · 12 LABS

# Course Outline

1 **Topic 01 · Cloud Operations Foundations**
Shared responsibility · regions and zones · operational excellence · SOA-C03 transition

2 **Topic 02 · Monitoring, Logging and Incident Response**
CloudWatch · CloudTrail · EventBridge · alarms · anomaly detection · remediation

3 **Topic 03 · Compute, Virtualisation and Systems Management**
EC2 lifecycle · AMIs · placement · Systems Manager · patching · Image Builder

4 **Topic 04 · Reliability, Elasticity and High Availability**
Elastic Load Balancing · Auto Scaling · health checks · containers · graceful degradation

5 **Topic 05 · Backup, Restore and Business Continuity**
RTO/RPO · AWS Backup · snapshots · replication · pilot light · warm standby

6 **Topic 06 · Deployment, Provisioning and Automation**
CloudFormation · StackSets · change sets · drift · Systems Manager Automation · Git and IaC

7 **Topic 07 · Security, Identity and Compliance**
IAM · Organizations · KMS · Config · Security Hub · GuardDuty · Inspector · Macie

8 **Topic 08 · Networking, DNS and Content Delivery**
VPC · routes · security groups · NACLs · endpoints · Route 53 · CloudFront

9 **Topic 09 · Storage, Data and Database Operations**
S3 · EBS · EFS · RDS · DynamoDB · lifecycle · performance · data protection

10 **Topic 10 · Performance, Cost and Integrated Troubleshooting**
Bottleneck analysis · Cost Explorer · Trusted Advisor · runbooks · game days · capstone


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HANDS-ON CLOUD OPERATIONS

# AWS Lab Environment

1

## Learner account

Use only the account and Region authorised by the trainer.

2

## AWS-native assets

Each folder includes editable YAML, JSON, CLI, shell or Python files.

3

## Visual reference

Every lab includes an EXPECTED-ARCHITECTURE.jpg image.

4

## Evidence

Retain outputs, decisions, verification and cleanup records.

5

## Cost control

Use small resources, bounded tests and explicit cleanup.

6

## Schedule

Exactly four labs are completed on each training day.


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WA-SAQ + CASE STUDY

# Assessment Briefing

1

## Individual work

No discussion, recording or sharing of assessment scripts.

2

## Open book

Use slides, Learner Guide and approved materials only.

3

## Written Assessment

Five open-ended questions · K1–K5 · 60 minutes.

4

## Case Study

Five connected tasks · A1–A7 · 60 minutes.

5

## Submission

Complete and upload the candidate papers through LMS-TMS.

6

## Outcome

Competent or Not Yet Competent; reassessment and appeal rights apply.


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ON ASSESSMENT DAY

# Assessment Flow

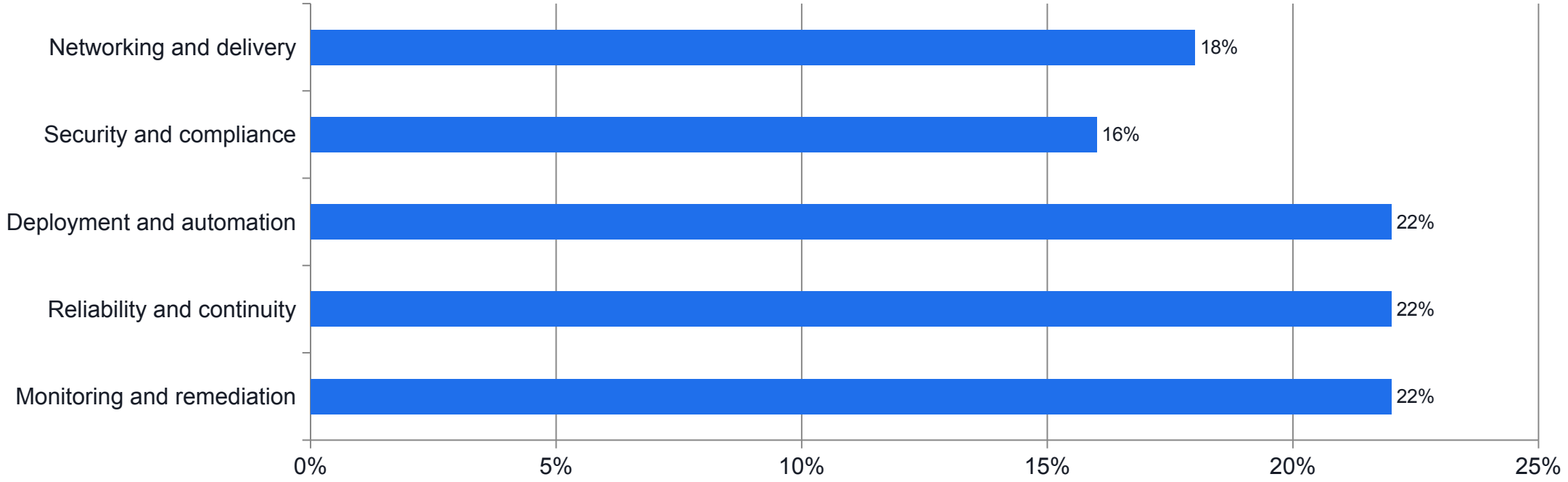
1. **TRAQOM**
   Course feedback
2. **Attendance**
   Assessment QR
3. **WA-SAQ**
   5 questions · 60 min
4. **Case Study**
   5 tasks · 60 min
5. **Submit**
   LMS-TMS and sign-off

**CONFIDENTIALITY**

Candidate papers are learner-facing; answer keys remain trainer-only.


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# Current SOA-C03 Exam Domain Weighting




**WHAT THE DATA SHOWS**

Three domains carry equal 22% weight; the course integrates all five through observability, controlled change, recovery and troubleshooting.


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CURRENT AWS AUTHORITY · HTTPS://EXAMS.TERTIARYINFOTECH.COM

# Certification Readiness

1

## Current exam

AWS Certified CloudOps Engineer — Associate (SOA-C03).

2

## Exam format

65 questions · 130 minutes · scaled passing score 720.

3

## Role transition

The WSQ title remains SysOps; exam-readiness content follows the current CloudOps guide.

4

## Practice Exam

Open exams.tertiaryinfotech.com and select the matching AWS CloudOps / SysOps practice exam; never use exam dumps.

5

## Readiness loop

Explain the service choice, apply the labs, interpret evidence and justify recovery.

6

## Scope

Current guide version 1.1, published 1 June 2026.


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VERSION v6.0 · 23 August 2026

# Source and Version Authority

1. **WSQ authority**
   Live Tertiary course page and Assessment Plan v5.0 — Course code, 24-hour duration, five LO, K1–K5, A1–A7 and WA+CS design.

2. **Legacy content**
   Reference v3/v4 PowerPoints and Drive Activities dataset — Preserved useful EC2, SSM, CloudFormation, storage, monitoring, security and networking coverage; procedures moved out of the deck.

3. **Certification authority**
   AWS Certified CloudOps Engineer - Associate SOA-C03 Exam Guide v1.1 — Current five-domain blueprint and scope, including containers, multi-account operations, automation and IaC.

4. **Architecture guidance**
   AWS Well-Architected Framework and official AWS service documentation — Operational excellence, security, reliability, performance, cost and sustainability practices.

5. **Assessment authority**
   Drive-held WA/CS v2 papers and Assessment Plan v5.0 — Instrument type, five-item counts, 60-minute timings and controlled K/A coverage.


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SOURCE INTEGRATION · TRACEABLE LEGACY SLIDE NUMBERS

# How the Original 927-Slide PPT Is Blended

1

### Coverage authority

The original v4 deck supplies its broad EC2, SSM, scaling, CloudFormation, storage, database, monitoring, account, security, DNS and VPC coverage.

2

### Modern visual treatment

Retained technical slides are rebuilt as layered cards, process visuals and diagram-plus-analysis layouts—not pasted as legacy bullet walls.

3

### Current AWS authority

SOA-C03 and current AWS documentation govern terminology, operating practices and features; historical limits are explicitly treated as items to verify.

4

### Procedure boundary

Console click paths, activities, quizzes and assessment pages are excluded. Detailed steps remain in the Learner Guide and the 12 lab folders.


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01

TOPIC 01 · LO1 · LO2 · K1 · K5 · A1

# Cloud Operations Foundations

Shared responsibility · regions and zones · operational excellence · SOA-C03 transition


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# Key Ideas — Cloud Operations Foundations

1

## Operate for outcomes

Availability, security, performance and cost are managed as measurable service outcomes.

2

## Know the responsibility boundary

AWS secures the cloud; customers secure configurations, identities, data and workloads in the cloud.

3

## Design for failure domains

Regions, Availability Zones and edge locations create distinct resilience and latency choices.

4

## Use evidence before change

Metrics, logs, configuration history and events establish the operational baseline.


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IMAGEGEN VISUAL · TOPIC 01

# Operate Across Accounts and Regions



READ THE SYSTEM

Read the scene as a governed operating model: isolated workload accounts, shared observability, policy, automation, regional resilience and human authority.


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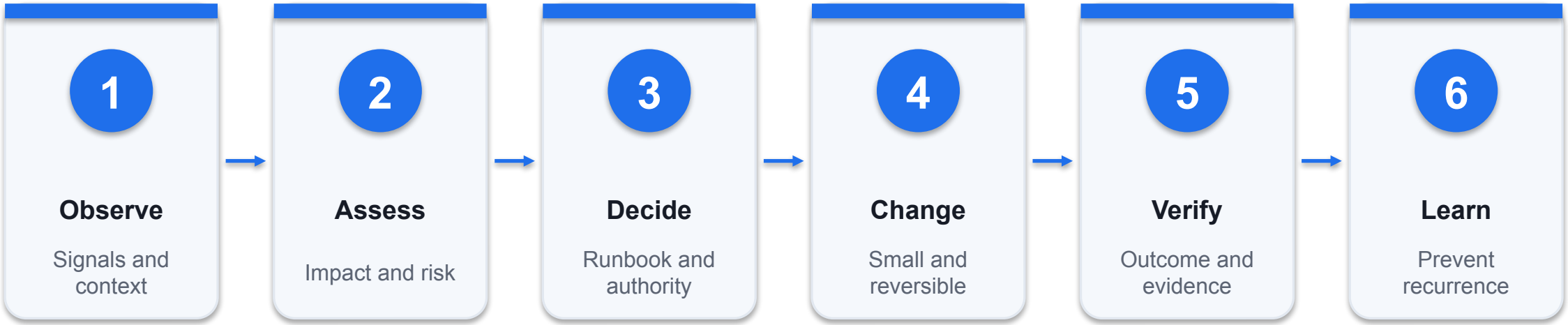
# SysOps to CloudOps — What Changed

| Area | Legacy SOA-C02 emphasis | Current SOA-C03 emphasis |
|---|---|---|
| **Role name** | SysOps Administrator | CloudOps Engineer |
| **Workloads** | EC2-heavy operations | Compute, containers, serverless and hybrid operations |
| **Scope** | Six broad domains | Five task-oriented domains |
| **Automation** | AWS-native operations | IaC, Git and third-party automation are explicit |

**WHEN IT MATTERS** The WSQ course title and code remain unchanged; exam-readiness content follows the current SOA-C03 guide.


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# The Cloud Operations Control Loop




**PROFESSIONAL STANDARD**

Every operational change must have a reason, owner, verification signal and rollback path.


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# AWS Global Infrastructure

1

## Region

Geographic boundary for sovereignty, latency and disaster-recovery decisions.

2

## Availability Zone

Independent failure domain connected with low-latency networking.

3

## Edge

CloudFront and Route 53 move delivery and routing closer to users.

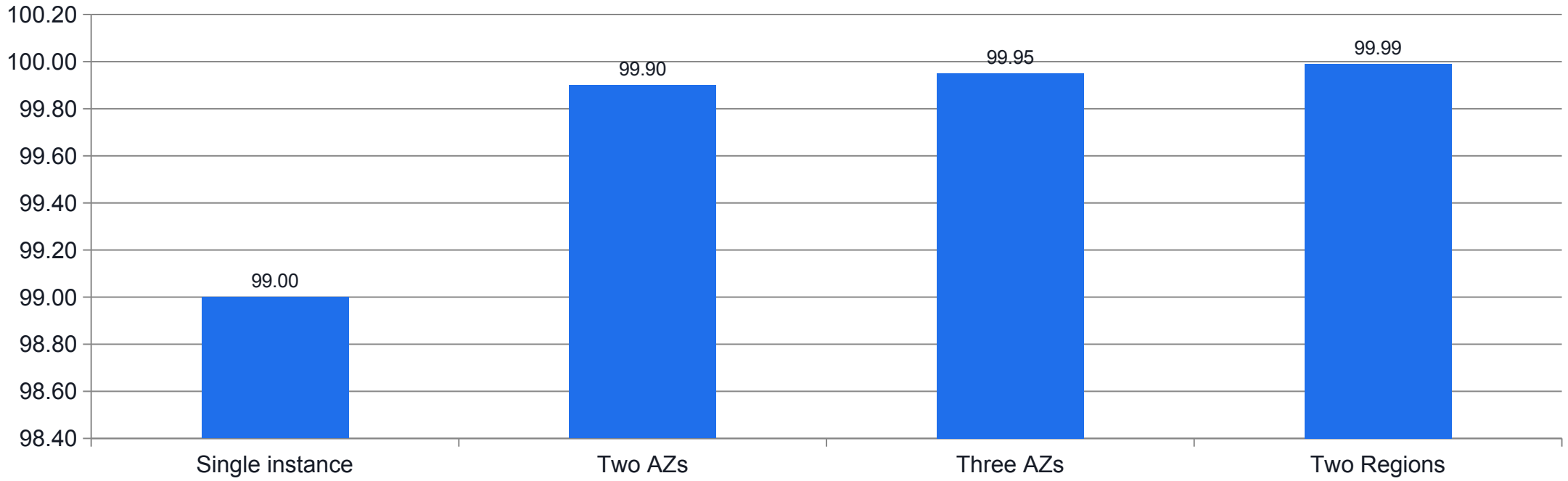
4

## Local and hybrid

Outposts, Local Zones and connected environments extend the operating boundary.


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# Availability Improves with Independent Failure Domains




**WHAT THE DATA SHOWS**

More failure domains increase potential resilience only when dependencies, health checks and failover are also designed correctly.


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# Shared Responsibility by Layer

| Layer | AWS responsibility | Customer responsibility |
| --- | --- | --- |
| **Facilities and hardware** | Physical security, hosts and core network | Choose Regions and resilience strategy |
| **Managed service** | Service infrastructure and platform | Identity, configuration, data and use |
| **EC2 workload** | Underlying facilities and hypervisor | Guest OS, patches, application and traffic controls |
| **Data** | Durable service mechanisms | Classification, encryption, retention and access |


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# Well-Architected Operational Questions

## Prepare

- What must be observable?
- What runbooks and permissions exist?

## Operate

- What is normal?
- Who responds to deviation?

## Evolve

- What did evidence teach?
- Which control prevents recurrence?


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Operations Mindset

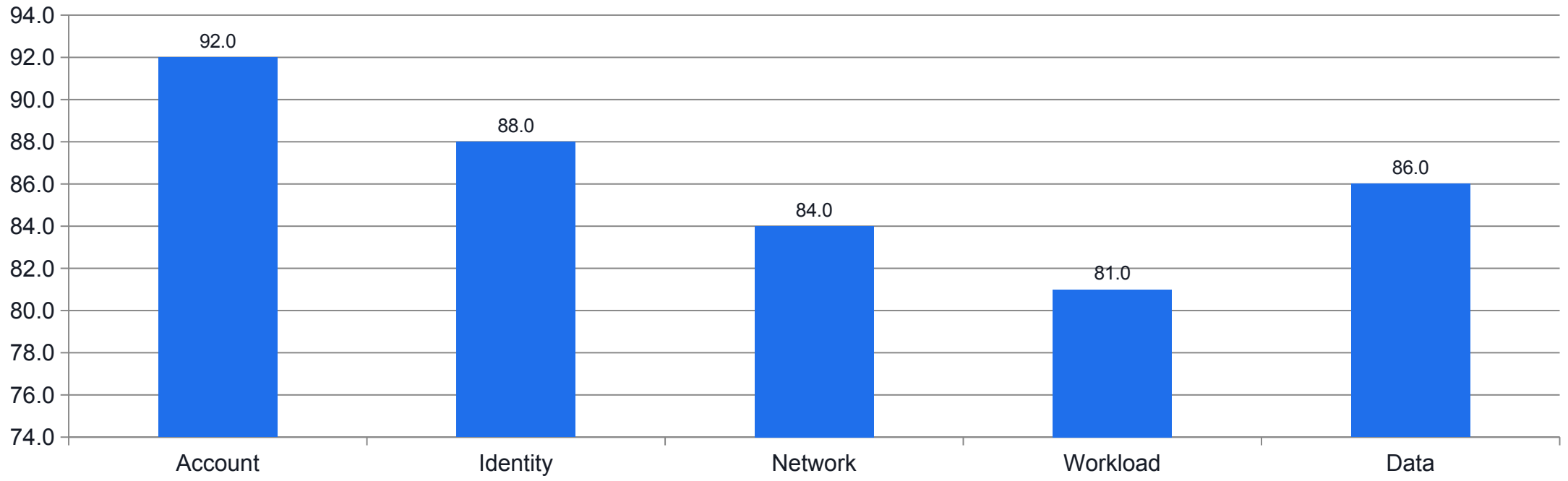
# The console is an interface—not the source of truth.

Treat configuration, telemetry, change history and verification evidence as the durable operational record.


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TOPIC 01 · NATIVE EDITABLE TECHNICAL CHART

# Operational Coverage by Control Layer

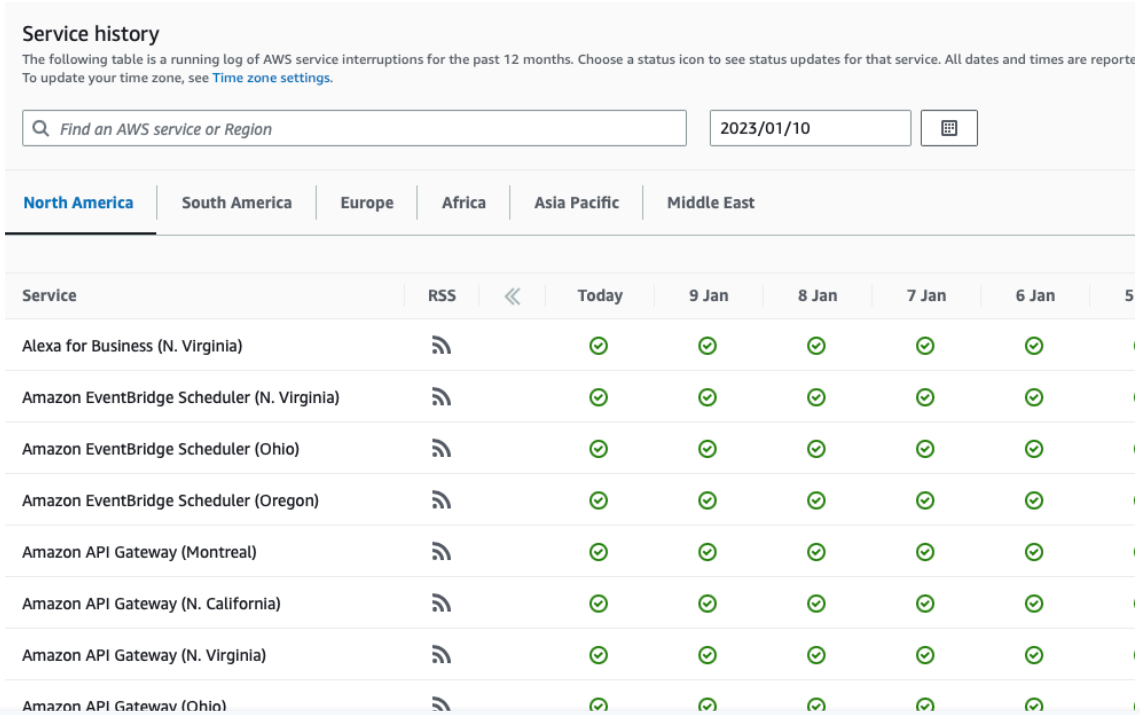



**WHAT THE DATA SHOWS**

Controls are only effective when ownership and evidence span every layer; the lowest-covered layer defines residual operational risk.


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# AWS Health Dashboard - Service History




**1** **Shows all regions, all services**

Shows all regions, all services health
Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and…

**2** **Failure mode**

Failure mode: an SCP sets a maximum permission boundary but grants nothing; troubleshooting must still evaluate…

**3** **Evidence: organization trails…**

Evidence: organization trails, Config aggregation, Health events, tag-policy compliance and cost-allocation…

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# AWS Health Dashboard - Your Account

### 1 Previously called AWS Personal…

Previously called AWS Personal Health Dashboard (PHD)

the performance and availability of the AWS services underlying your AWS resources.

### 2 AWS Account Health Dashboard…

AWS Account Health Dashboard provides alerts and remediation guidance when AWS is experiencing events that may impact you.

The dashboard displays relevant and timely information to help you

### 3 While the Service Health Dashboard

While the Service Health Dashboard displays the general status of AWS

manage events in progress and provides proactive notification to help you plan for scheduled activities.

### 4 services, Account Health…

services, Account Health Dashboard gives you a personalized view into.

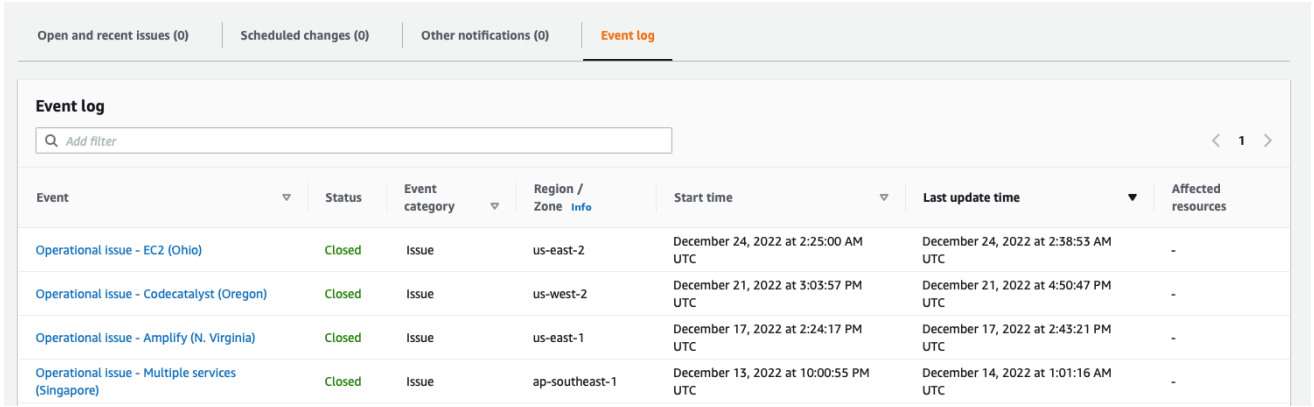
Can aggregate data from an entire AWS Organization

**CURRENT OPERATIONS LENS**

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# AWS Health Dashboard - Your Account




### 1 Global service

Global service
Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and…

### 2 Failure mode

Failure mode: an SCP sets a maximum permission boundary but grants nothing; troubleshooting must still evaluate…

### 3 Evidence: organization trails…

Evidence: organization trails, Config aggregation, Health events, tag-policy compliance and cost-allocation…

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# Health Event Notifications

### 1 Use EventBridge to react to

Use EventBridge to react to changes for AWS Health events in your AWS account

AWS Health Dashboard

### 2 Example: receive email…

Example: receive email notifications when EC2 instances in your AWS account are scheduled for updates

EventBridge

### 3 This is possible for Account

This is possible for Account events (resources that are affected in your account) and Public Events (Regional availability of a service)

Lambda SNS

### 4 Use cases

Use cases: send notifications, capture event information, take corrective action...

Kinesis Data Streams

**CURRENT OPERATIONS LENS**

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# AWS Health Dashboard - Examples

### 1 AWS Health Dashboard

AWS Health Dashboard

Remediating Exposed IAM Access Keys

Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and observability administration to…

### 2 EventBridge

EventBridge

Instance Scheduled for Retirement

Failure mode: an SCP sets a maximum permission boundary but grants nothing; troubleshooting must still evaluate identity, resource, boundary and…

### 3 Exposed Access Keys

Exposed Access Keys

Restart Instance

Evidence: organization trails, Config aggregation, Health events, tag-policy compliance and cost-allocation coverage prove whether governance is…

### 4 Delete Access Keys

Delete Access Keys

Restarting Instances That are Scheduled for Retirement

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# AWS Organizations

### 1 Global service

Global service

Member accounts can only be part of one organization

API is available to automate AWS account creation

### 2 Allows to manage multiple AWS

Allows to manage multiple AWS accounts

Consolidated Billing across all accounts - single payment method

### 3 The main account is the

The main account is the management account

Pricing benefits from aggregated usage (volume discount for EC2, S3…)

### 4 Other accounts are member accounts

Other accounts are member accounts

Shared reserved instances and Savings Plans discounts across accounts

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# AWS Organizations

### 1 Root Organizational Unit (OU)

Root Organizational Unit (OU)

Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and observability administration to…

### 2 Management Account

Management Account

Failure mode: an SCP sets a maximum permission boundary but grants nothing; troubleshooting must still evaluate identity, resource, boundary and…

### 3 Member Accounts

Member Accounts

Evidence: organization trails, Config aggregation, Health events, tag-policy compliance and cost-allocation coverage prove whether governance is…

### 4 OU (Finance)

OU (Finance)

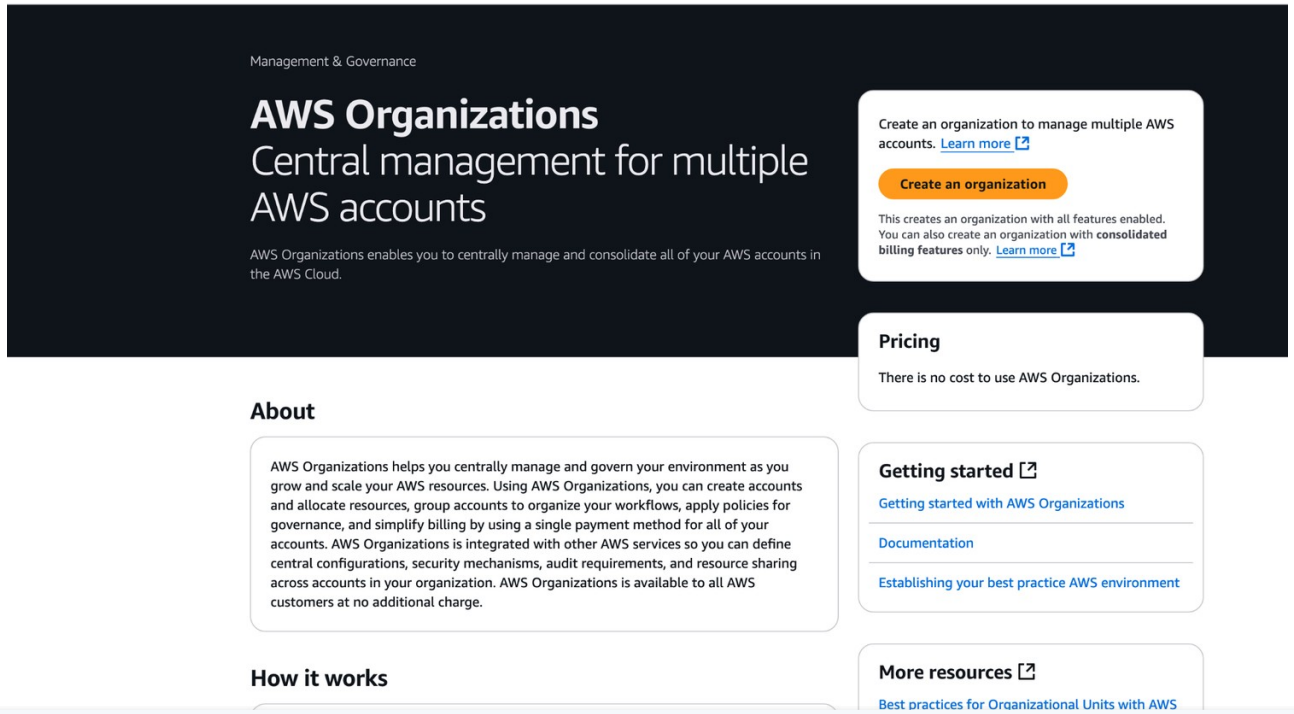
Decision point: define which controls are preventive, detective or responsive and which team owns exceptions, expiry and escalation.

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# AWS Organizations




**1 Operating model**

Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and…

**2 Failure mode**

Failure mode: an SCP sets a maximum permission boundary but grants nothing; troubleshooting must still evaluate…

**3 Evidence: organization trails…**

Evidence: organization trails, Config aggregation, Health events, tag-policy compliance and cost-allocation…

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# Organizational Units (OU) - Examples

### 1 Business Unit Environmental…

Business Unit Environmental Lifecycle

Sales Account 2

Finance Account 2

### 2 Management Account

Management Account

Retail Account 1

Prod Account 1

Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and observability administration to…

### 3 Finance OU

Finance OU

Retail Account 2

Prod Account 2

Failure mode: an SCP sets a maximum permission boundary but grants nothing; troubleshooting must still evaluate identity, resource, boundary and…

### 4 Sales Account 1

Sales Account 1

Finance Account 1

Dev Account 1

Evidence: organization trails, Config aggregation, Health events, tag-policy compliance and cost-allocation coverage prove whether governance is…

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# AWS Organizations

## 1 Advantages

Advantages

Send CloudWatch Logs to central logging account

They do not apply to the management account (full admin power)

## 2 Multi Account vs One Account

Multi Account vs One Account Multi VPC

Establish Cross Account Roles for Admin purposes

Must have an explicit allow from the root through each OU in the direct path to the target account (does not allow anything by default – like IAM)

## 3 Use tagging standards for billing

Use tagging standards for billing purposes

Security: Service Control Policies (SCP)

## 4 Enable CloudTrail on all accounts,

Enable CloudTrail on all accounts, send logs to central S3 account

IAM policies applied to OU or Accounts to restrict Users and Roles

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# SCP Hierarchy

### 1 Management Account

Management Account

Can do anything

Can access EC2

Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and observability administration to…

### 2 OU (Sandbox)

OU (Sandbox)

EXCEPT S3 (explicit Deny from Sandbox OU)

Prod OU & Account E & F

### 3 FullAWSAccess

FullAWSAccess

EXCEPT EC2 (explicit Deny)

OU (Workloads)

Failure mode: an SCP sets a maximum permission boundary but grants nothing; troubleshooting must still evaluate identity, resource, boundary and…

### 4 Can do anything (no SCP

Can do anything (no SCP apply)

Account B & C

Deny Athena FullAWSAccess

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# SCP Examples



```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "ec2:*",
                "cloudwatch:*"
            ],
            "Resource": "*"
        }
    ]
}
```

1 **Blocklist and Allowlist strategies**

Blocklist and Allowlist strategies
Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and…

2 **Failure mode**

Failure mode: an SCP sets a maximum permission boundary but grants nothing; troubleshooting must still evaluate…

3 **Evidence: organization trails…**

Evidence: organization trails, Config aggregation, Health events, tag-policy compliance and cost-allocation…

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# AWS Organizations - Reserved Instances

### 1 For billing purposes, the…

For billing purposes, the consolidated billing feature of AWS Organizations treats all the accounts in the organization as one account.

This means that RIs and Savings Plans discounts aren't shared between any accounts that have sharing turned off.

### 2 This means that all accounts

This means that all accounts in the organization can receive the hourly cost benefit of Reserved Instances that are purchased by any other account.

To share an RI or Savings Plans discount with an account, both accounts must have sharing turned on.

### 3 The payer account (management…

The payer account (management account) of an organization can turn off

### 4 Reserved Instance (RI) discount…

Reserved Instance (RI) discount and Savings Plans discount sharing for any accounts in that organization, including the payer account

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# AWS Organizations - IAM Policies

### 1 Use AWS

Use AWS:PrincipalOrgID condition key in your resource-based policies to restrict access to IAM principals from accounts in an AWS Organization

IAM User (stephane)

Use the aws:PrincipalOrgID condition

### 2 AWS Organization (o-exampleorgid)

AWS Organization (o-exampleorgid)

Member Account (222222222222)

Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and observability administration to…

### 3 Management Account (111111111111)

Management Account (111111111111)

IAM User (Olivia)

Failure mode: an SCP sets a maximum permission boundary but grants nothing; troubleshooting must still evaluate identity, resource, boundary and…

### 4 Member Account (123456789012)

Member Account (123456789012)

(financial-data-2021)

Evidence: organization trails, Config aggregation, Health events, tag-policy compliance and cost-allocation coverage prove whether governance is…

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# AWS Organizations - Tag Policies

```
{
    "tags": {
        "costcenter": {
            "tag_key": {
                "@@assign": "CostCenter"
            },
            "tag_value": {
                "@@assign": ["100", "200"]
            },
            "enforced_for": {
                "@@assign": ["secretsmanager:*"]
            }
        }
    }
}
```

1

### Helps you standardize tags across

Helps you standardize tags across resources in an AWS Organization

2

### Operating model

Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and…

3

### Failure mode

Failure mode: an SCP sets a maximum permission boundary but grants nothing; troubleshooting must still evaluate…

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# AWS Control Tower

### 1 Easy way to set up

Easy way to set up and govern a secure and compliant mutli-account AWS environment based on best practices

Monitor compliance through an interactive dashboard

### 2 Automate the set up of

Automate the set up of your environment in a few clicks

AWS Control Tower runs on top of AWS Organizations

### 3 Automate ongoing policy…

Automate ongoing policy management using guardrails

It automatically sets up AWS Organizations to organize accounts and implement SCPs (Service Control Policies)

### 4 Detect policy violations and…

Detect policy violations and remediate them

Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and observability administration to…

**CURRENT OPERATIONS LENS**

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# AWS Service Catalog

### 1 Users that are new to

Users that are new to AWS have too many options, and may create stacks that are not compliant / in line with the rest of the organization

Enter AWS Service Catalog!

### 2 Some users just want a

Some users just want a quick self-service portal to launch a set of

### 3 authorized products pre-defined…

authorized products pre-defined by admins

Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and observability administration to…

### 4 Includes: virtual machines…

Includes: virtual machines, databases, storage options, etc…

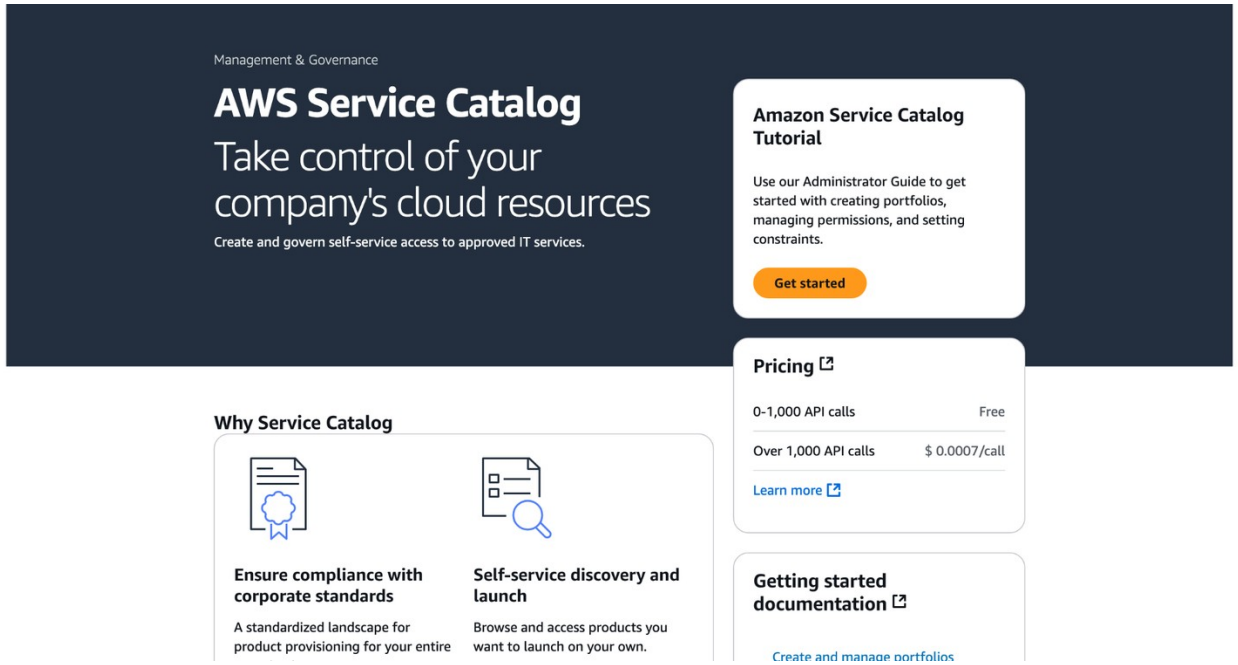
Failure mode: an SCP sets a maximum permission boundary but grants nothing; troubleshooting must still evaluate identity, resource, boundary and…

**CURRENT OPERATIONS LENS**

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# AWS Service Catalog




**1** **Operating model**

Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and…

**2** **Failure mode**

Failure mode: an SCP sets a maximum permission boundary but grants nothing; troubleshooting must still evaluate…

**3** **Evidence: organization trails…**

Evidence: organization trails, Config aggregation, Health events, tag-policy compliance and cost-allocation…

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# Service Catalog diagram

### 1 Product Portfolio

Product Portfolio

IAM Permissions to Access Portfolios

Provisioned Products

Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and observability administration to…

### 2 CloudFormation Templates

CloudFormation Templates

Product List

Ready to use Properly Configured Properly Tagged

### 3 Collection of Products

Collection of Products

USER TASKS

Failure mode: an SCP sets a maximum permission boundary but grants nothing; troubleshooting must still evaluate identity, resource, boundary and…

### 4 ADMIN TASKS

ADMIN TASKS

Authorized by IAM

Evidence: organization trails, Config aggregation, Health events, tag-policy compliance and cost-allocation coverage prove whether governance is…

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# AWS Service Catalog - Sharing Catalogs

### 1 Share portfolios with individual…

Share portfolios with individual AWS accounts or AWS Organizations

Ability to add products from the

share (in-sync)

### 2 Sharing options

Sharing options

imported portfolio to the local portfolio

(re-deploy updates)

### 3 Share a reference of the

Share a reference of the portfolio, then import the shared portfolio in the recipient account (stays in-sync with the original portfolio)

Portfolio A (port-xxxxxxxxxx)

import portfolio & launch products

### 4 Deploy a copy of the

Deploy a copy of the portfolio into the recipient account (must re-deploy any updates)

Product 1 Product 2 Product 3

Portfolio A - Copy (port-yyyyyyyyyy)

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# AWS Service Catalog - TagOptions Library

### 1 Easily manage tags on provisioned

Easily manage tags on provisioned products

Use cases: proper resources tagging, defined allowed tags, …

### 2 Key-value pair managed in AWS

Key-value pair managed in AWS Service Catalog

Can be shared with other AWS accounts and AWS Organizations

### 3 Used to create an AWS

Used to create an AWS Tag

Portfolio A (port-xxxxxxxxxx)

Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and observability administration to…

### 4 Can be associated with Portfolios

Can be associated with Portfolios and Products

Product 1 Product 2 Product 3

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# AWS Billing Alarms

### 1 Billing data metric is stored

Billing data metric is stored in CloudWatch us-east-1

Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and observability administration to…

### 2 Billing data are for overall

Billing data are for overall worldwide

Failure mode: an SCP sets a maximum permission boundary but grants nothing; troubleshooting must still evaluate identity, resource, boundary and…

### 3 It’s for actual cost, project

It’s for actual cost, project costs

Evidence: organization trails, Config aggregation, Health events, tag-policy compliance and cost-allocation coverage prove whether governance is…

### 4 Let’s create a billing alarm

Let’s create a billing alarm together!

Decision point: define which controls are preventive, detective or responsive and which team owns exceptions, expiry and escalation.

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# Cost Explorer

### 1 Visualize, understand, and manage…

Visualize, understand, and manage your AWS costs and usage over time

Choose an optimal Savings Plan(to lower prices on your bill)

### 2 Create custom reports that analyze

Create custom reports that analyze cost and usage data.

Forecast usage up to 12 months based on previous usage

### 3 Analyze your data at a high level

Analyze your data at a high level: total costs and usage across all accounts

### 4 Or Monthly, hourly, resource level

Or Monthly, hourly, resource level granularity

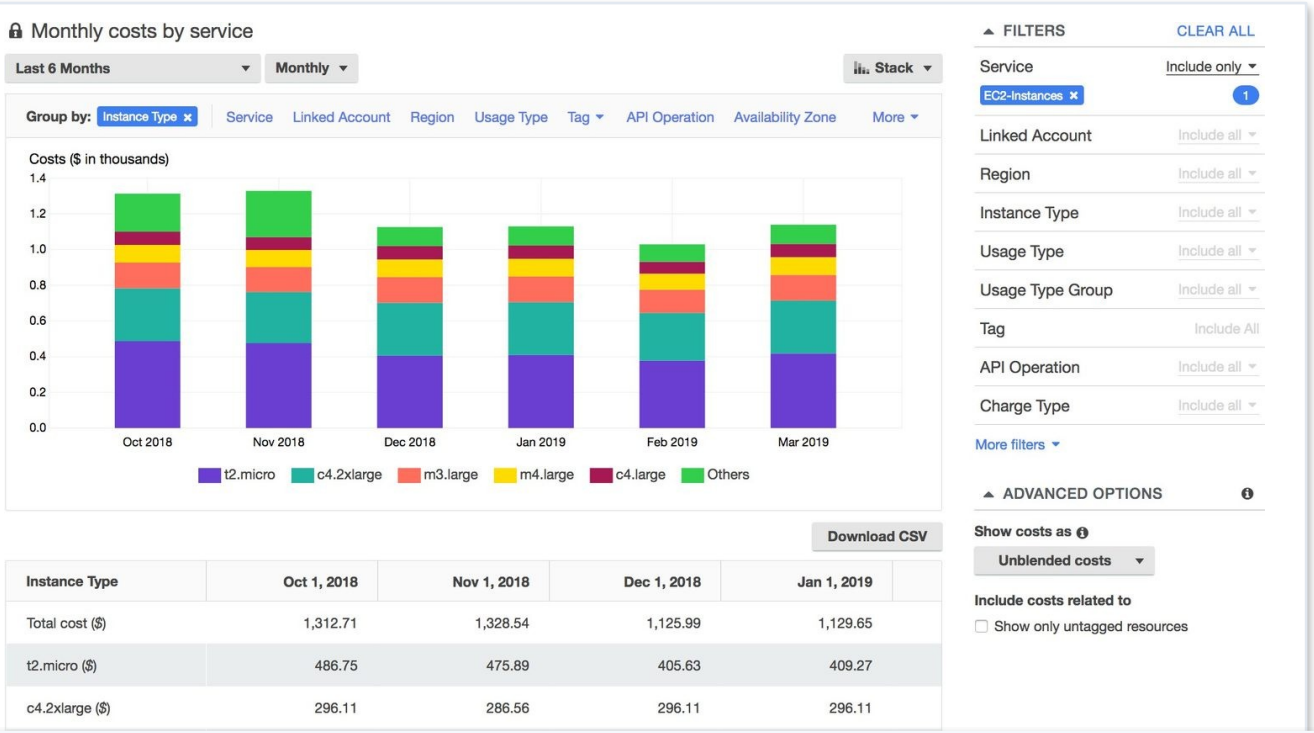
Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and observability administration to…

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# Cost Explorer - Monthly Cost by AWS Service




**1** **Operating model**

Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and…

**2** **Failure mode**

Failure mode: an SCP sets a maximum permission boundary but grants nothing; troubleshooting must still evaluate…

**3** **Evidence: organization trails…**

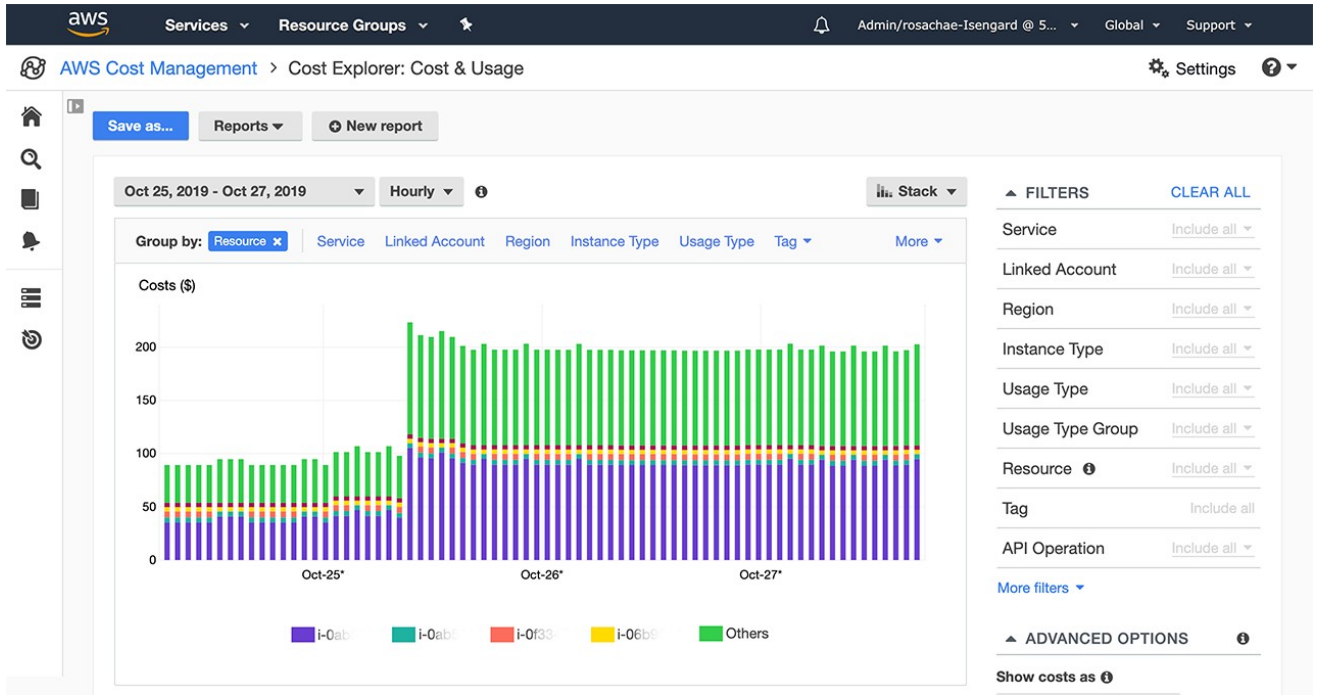
Evidence: organization trails, Config aggregation, Health events, tag-policy compliance and cost-allocation…

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# Cost Explorer - Hourly & Resource Level




**1** **Operating model**

Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and…

**2** **Failure mode**

Failure mode: an SCP sets a maximum permission boundary but grants nothing; troubleshooting must still evaluate…

**3** **Evidence: organization trails…**

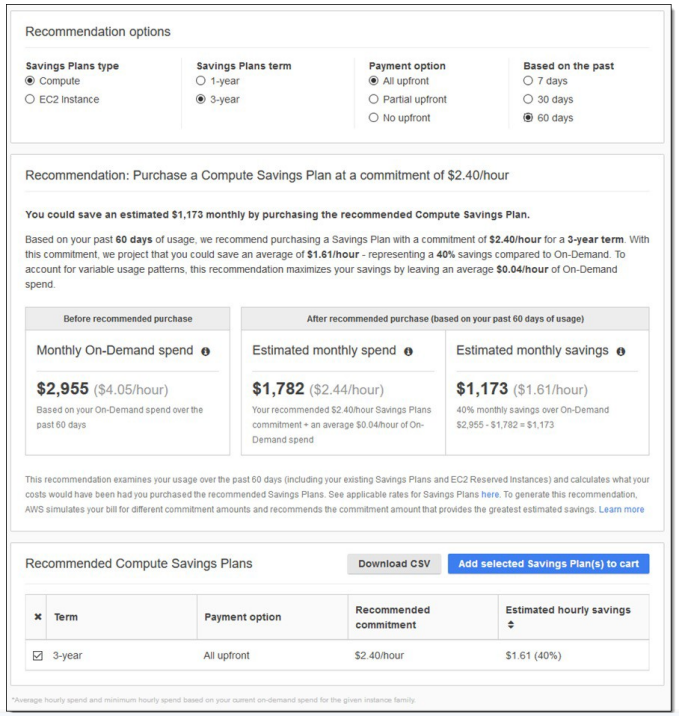
Evidence: organization trails, Config aggregation, Health events, tag-policy compliance and cost-allocation…

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# Cost Explorer - Savings Plan Alternative to Reserved Instances




**1** **Operating model**

Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and…

**2** **Failure mode**

Failure mode: an SCP sets a maximum permission boundary but grants nothing; troubleshooting must still evaluate…

**3** **Evidence: organization trails…**

Evidence: organization trails, Config aggregation, Health events, tag-policy compliance and cost-allocation…

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# Cost Explorer - Forecast Usage




**1** **Operating model**

Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and…

**2** **Failure mode**

Failure mode: an SCP sets a maximum permission boundary but grants nothing; troubleshooting must still evaluate…

**3** **Evidence: organization trails…**

Evidence: organization trails, Config aggregation, Health events, tag-policy compliance and cost-allocation…

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# AWS Budgets

### 1 Create budget and send alarms

Create budget and send alarms when costs exceeds the budget

Supports EC2, ElastiCache, RDS, Redshift

2 budgets are free, then $0.02/day/budget

### 2 4 types of budgets

4 types of budgets: Usage, Cost, Reservation, Savings Plans

Up to 5 SNS notifications per budget

### 3 For Reserved Instances (RI)

For Reserved Instances (RI)

Can filter by: Service, Linked Account,Tag, Purchase Option, Instance Type, Region, Availability Zone, API Operation, etc…

### 4 Track utilization

Track utilization

Same options as AWS Cost Explorer!

Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and observability administration to…

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# Cost Allocation Tags

| Total Cost | user:Owner | user:Stack | user:Cost Center | user:Application |
|---|---|---|---|---|
| 0.95 | DbAdmin | Test | 80432 | Widget2 |
| 0.01 | DbAdmin | Test | 80432 | Widget2 |
| 3.84 | DbAdmin | Prod | 80432 | Widget2 |
| 6.00 | DbAdmin | Test | 78925 | Widget1 |
| 234.63 | SysEng | Prod | 78925 | Widget1 |
| 0.73 | DbAdmin | Test | 78925 | Widget1 |
| 0.00 | DbAdmin | Prod | 80432 | Portal |
| 2.47 | DbAdmin | Prod | 78925 | Portal |

1

### Use cost allocation tags to

Use cost allocation tags to track your AWS costs on a detailed level

2

### Operating model

Operating model: keep the Organizations management account low-touch; delegate security, backup, Config and…

3

### Failure mode

Failure mode: an SCP sets a maximum permission boundary but grants nothing; troubleshooting must still evaluate…

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# Cost and Usage Reports

1

### Dive deeper into your AWS

Dive deeper into your AWS costs and usage

service category used by an account

Can be integrated with Athena, Redshift or QuickSight

2

### The AWS Cost & Usage

The AWS Cost & Usage Report contains the most comprehensive set of AWS cost and usage data available

in hourly or daily line items

AWS Cost & Usage Report

3

### Includes additional metadata…

Includes additional metadata about AWS services, pricing, and reservations (e.g. Amazon EC2 Reserved Instances (RIs))

any tags that you have activated for cost allocation purposes

4

### The AWS Cost & Usage

The AWS Cost & Usage Report lists AWS usage for each

Can be configured for daily exports to S3

**CURRENT OPERATIONS LENS**

Turn the architecture choice into an operating model: ownership boundary, account scope, guardrail, evidence and escalation. Validate service limits and regional availability against current AWS documentation.


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# Original PPT Coverage Retained — Topic 01

1

### Modernised slides

30 technical slides extracted from the original v4 PowerPoint and rebuilt in the current visual system.

2

### Traceability

Every blended slide names its original v4 source slide number in the kicker.

3

### Current validation

Terminology is normalised; service limits and Region availability remain explicit current-documentation checks.

4

### Procedure boundary

Click-by-click sequences, activities and quizzes are kept out of the deck and remain in the Learner Guide or lab package.


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# Recap — Cloud Operations Foundations

1

**Explain**

Availability, security, performance and cost are managed as measurable service outcomes.

2

**Distinguish**

AWS secures the cloud; customers secure configurations, identities, data and workloads in the cloud.

3

**Apply**

Use the control loop and responsibility boundary to frame an operational decision.

4

**Evidence**

Identify the configuration, telemetry, change and verification record required.


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02

TOPIC 02 · LO2 · LO4 · K1 · K3 · K4 · A1 · A4

# Monitoring, Logging and Incident Response

CloudWatch · CloudTrail · EventBridge · alarms · anomaly detection · remediation


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# Key Ideas — Monitoring, Logging and Incident Response

1

## Metrics show behaviour

Dimensions, periods and statistics turn raw measurements into service signals.

2

## Logs explain events

Structured records provide detail for search, correlation, retention and audit.

3

## Traces connect dependencies

A request path exposes latency and failure across distributed components.

4

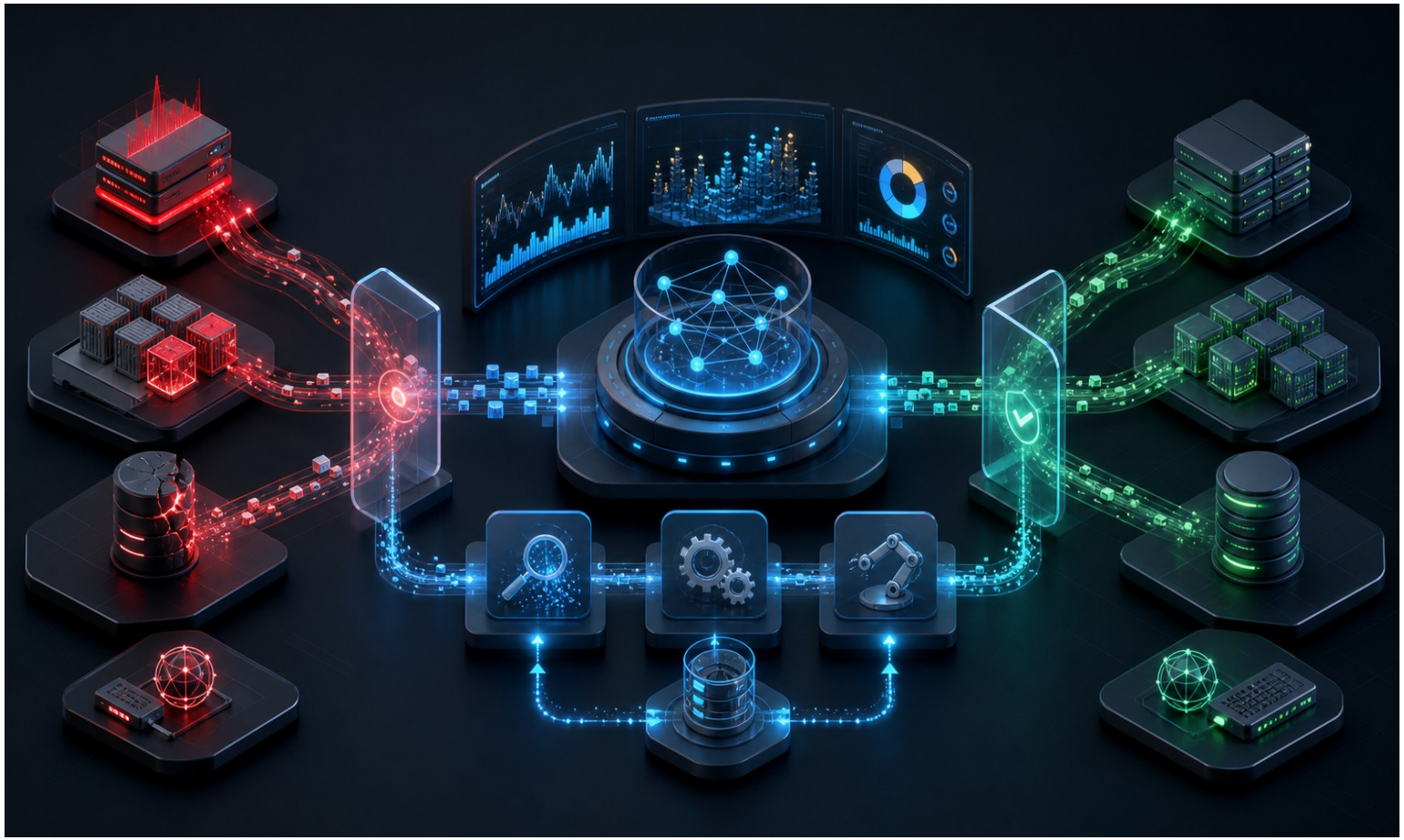
## Incidents require controlled response

Detection, triage, containment, recovery and learning form one evidence chain.


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IMAGEGEN VISUAL · TOPIC 02

# Observe, Decide, Remediate



READ THE SYSTEM

Read the scene as an operational control loop: signal, context, safe action, verification and escalation.


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# Telemetry Pipeline




**DESIGN TEST**

A signal is useful only when it identifies impact, owner and next safe action.

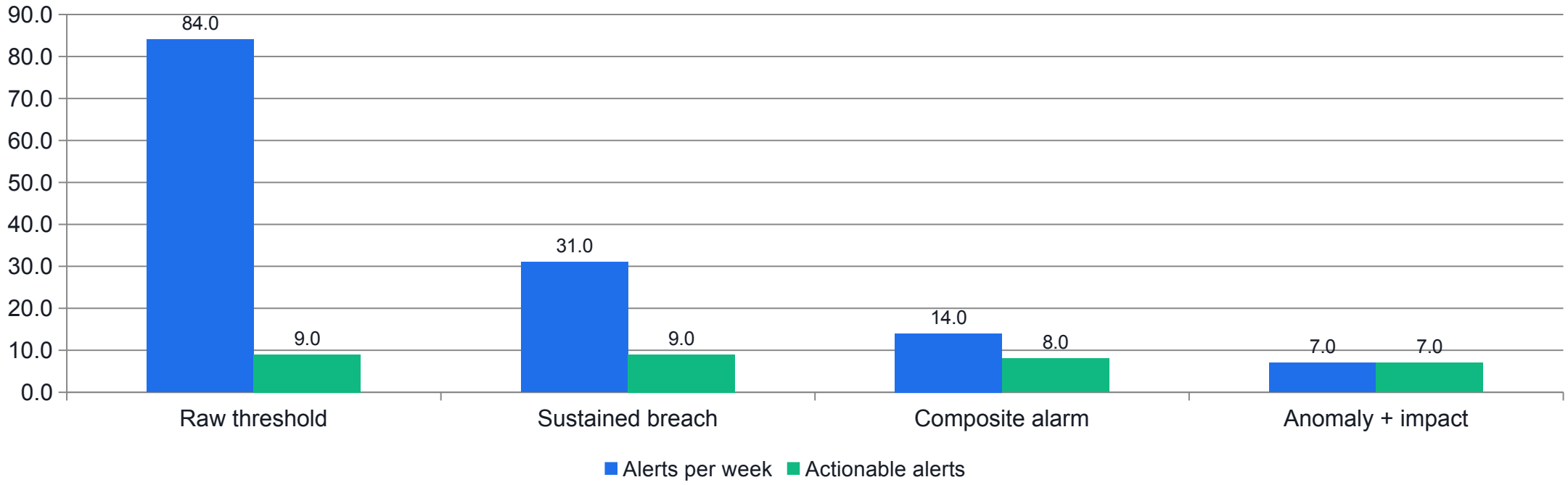

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# CloudWatch, CloudTrail and Config

| Service | Primary question | Typical evidence |
| --- | --- | --- |
| **CloudWatch** | How is the workload behaving? | Metrics, logs, alarms and dashboards |
| **CloudTrail** | Who called which AWS API, when and from where? | Management and data events |
| **AWS Config** | What configuration existed and was it compliant? | Configuration history and rule evaluation |


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# Composite Signals Reduce Alert Noise




**WHAT THE DATA SHOWS**

Filtering transient and context-free signals can preserve actionable detection while reducing noise.


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# Alarm Design

1

## Statistic

Average can hide peaks; percentiles expose tail latency.

2

## Period

Short periods detect quickly but may amplify noise.

3

## Missing data

Choose breaching, not breaching, ignore or missing deliberately.

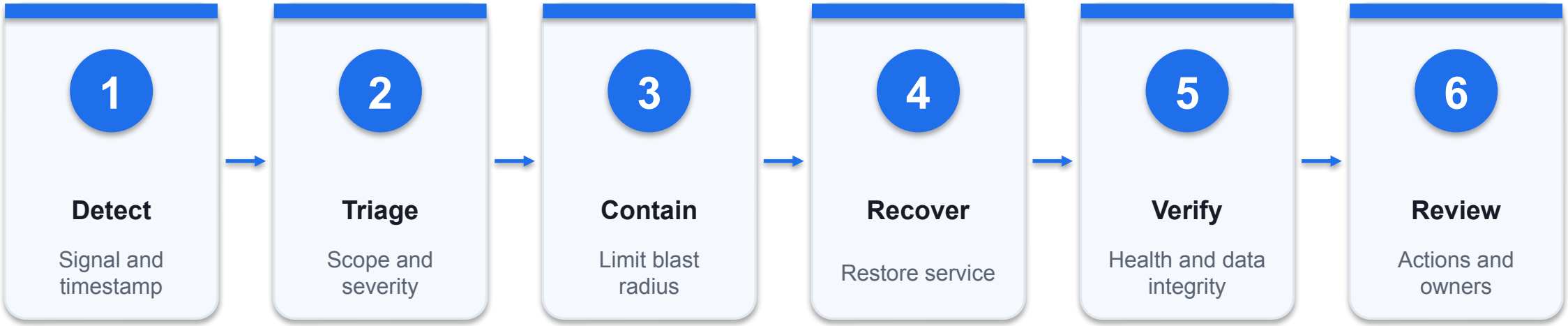
4

## Action

Notify, scale or remediate only with tested permissions and rollback.


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# Incident Lifecycle




**A4 CONNECTION**

Separate observed evidence, hypothesis, decision and result throughout the incident record.


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# Threshold vs Anomaly Detection

| Choice | Strength | Use when |
| --- | --- | --- |
| **Static threshold** | Transparent and predictable | A meaningful limit is known |
| **Anomaly detection** | Adapts to time-varying baseline | Seasonality makes one limit noisy |
| **Composite alarm** | Combines symptom and impact | One signal alone is insufficient |
| **Metric math** | Derives rates and ratios | Raw counts need context |


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# Automation Guardrails

## Trigger

- Use sustained evidence
- Prevent recursive events

## Authority

- Least-privilege execution role
- Separate approval for high-risk action

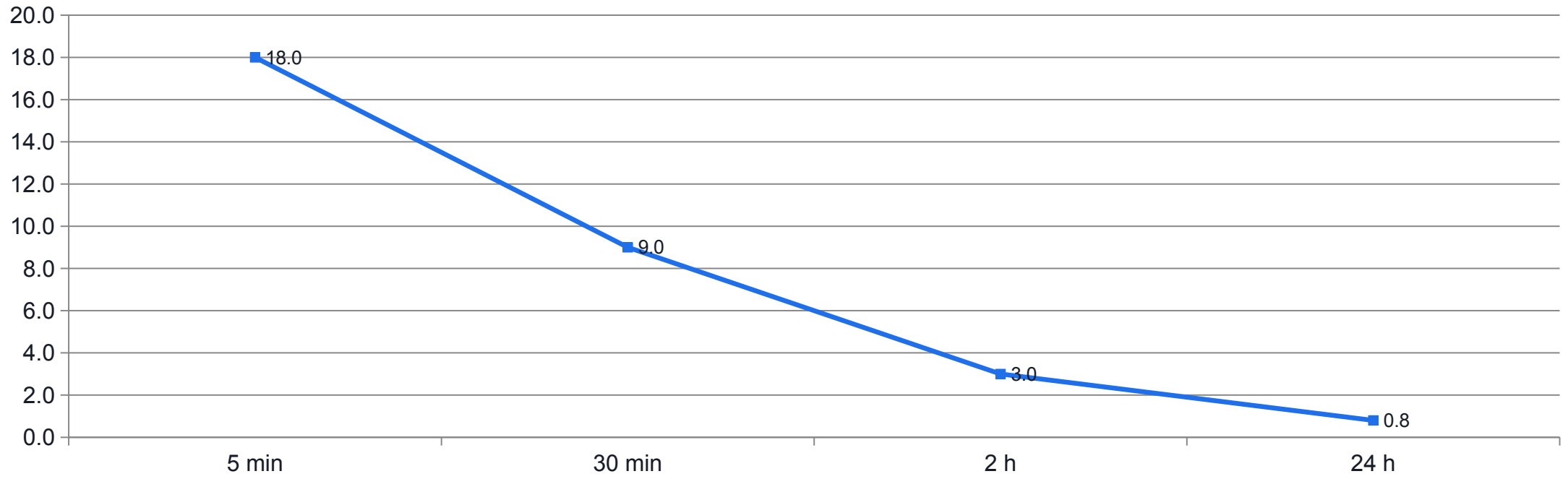
## Verification

- Check intended state
- Escalate when remediation fails


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TOPIC 02 · NATIVE EDITABLE TECHNICAL CHART

# SLO Error-Budget Burn by Time Window

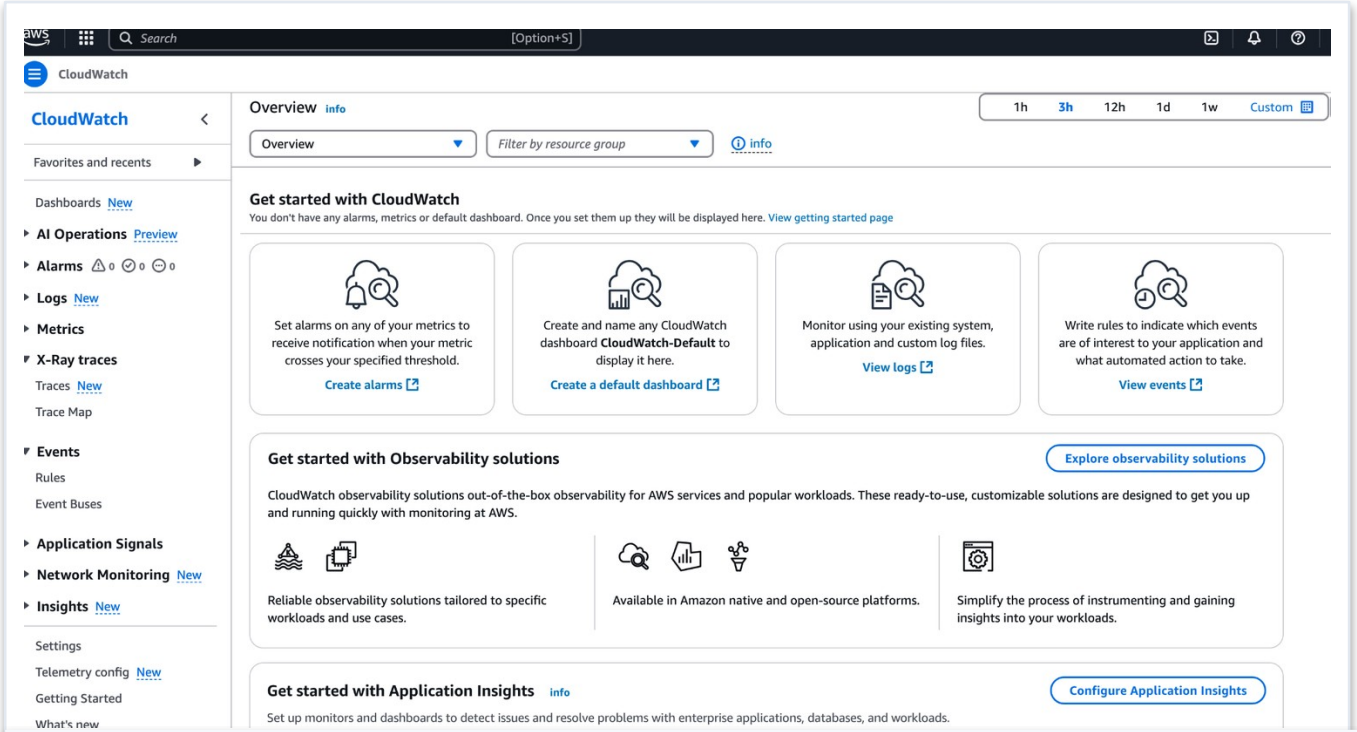



**WHAT THE DATA SHOWS**

Fast windows detect acute impact; slow windows protect the long-term objective. Multi-window alarms reduce both late detection and transient noise.


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# AWS CloudWatch




**1 Mechanism: metrics expose…**

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration…

**2 Failure mode**

Failure mode: average statistics, broad dimensions or short retention can hide tail latency, tenant impact and the…

**3 Evidence: alarm state history…**

Evidence: alarm state history, Logs Insights queries, CloudTrail event identity, trace correlation and SLO burn…

**CURRENT OPERATIONS LENS**

Connect every signal to service impact, an accountable owner, a safe response and a verification condition. Validate service limits and regional availability against current AWS documentation.


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# AWS CloudWatch Metrics

### 1 CloudWatch provides metrics for…

CloudWatch provides metrics for every services in AWS

Up to 30 dimensions per metric

### 2 Metric is a variable to

Metric is a variable to monitor (CPUUtilization, NetworkIn…)

Metrics have timestamps

### 3 Metrics belong to namespaces

Metrics belong to namespaces

Can create CloudWatch dashboards of metrics

### 4 Dimension is an attribute of

Dimension is an attribute of a metric (instance id, environment, etc…).

**CURRENT OPERATIONS LENS**

Connect every signal to service impact, an accountable owner, a safe response and a verification condition. Validate service limits and regional availability against current AWS documentation.


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# EC2 Detailed monitoring

### 1 EC2 instance metrics have metrics

EC2 instance metrics have metrics “every 5 minutes”

Note: EC2 Memory usage is by default not pushed (must be pushed from inside the instance as a custom metric)

### 2 With detailed monitoring (for a

With detailed monitoring (for a cost), you get data “every 1 minute”

### 3 Use detailed monitoring if you

Use detailed monitoring if you want to scale faster for your ASG!

### 4 The AWS Free Tier allows

The AWS Free Tier allows us to have 10 detailed monitoring metrics

**CURRENT OPERATIONS LENS**

Connect every signal to service impact, an accountable owner, a safe response and a verification condition. Validate service limits and regional availability against current AWS documentation.


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# CloudWatch Custom Metrics

### 1 Possibility to define and send

Possibility to define and send your own custom metrics to CloudWatch

Instance.id

High Resolution: 1/5/10/30 second(s) - Higher cost

### 2 Example: memory (RAM) usage, disk

Example: memory (RAM) usage, disk space, number of logged in users …

Environment.name

Important: Accepts metric data points two weeks in the past and two hours in the future (make sure to configure your EC2 instance time correctly)

### 3 Use API call PutMetricData

Use API call PutMetricData

Metric resolution (StorageResolution API parameter - two possible value)

### 4 Ability to use dimensions…

Ability to use dimensions (attributes) to segment metrics

Standard: 1 minute (60 seconds)

**CURRENT OPERATIONS LENS**

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# CloudWatch Dashboards

### 1 Great way to setup custom

Great way to setup custom dashboards for quick access to key metrics and alarms

You can setup automatic refresh (10s, 1m, 2m, 5m, 15m)

### 2 Dashboards are global

Dashboards are global

Dashboards can be shared with people who don't have an AWS account (public, email address, 3rd party SSO provider through Amazon Cognito)

### 3 Dashboards can include graphs from

Dashboards can include graphs from different AWS accounts and regions

3 dashboards (up to 50 metrics) for free

### 4 You can change the time

You can change the time zone & time range of the dashboards

$3/dashboard/month afterwards

**CURRENT OPERATIONS LENS**

Connect every signal to service impact, an accountable owner, a safe response and a verification condition. Validate service limits and regional availability against current AWS documentation.


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# CloudWatch Logs

### 1 Logs group

Logs group: arbitrary name, usually representing an application

Amazon S3 (exports)

OpenSearch

### 2 Log stream

Log stream: instances within application / log files / containers

Kinesis Data Streams

Logs are encrypted by default

### 3 Can define log expiration policies

Can define log expiration policies (never expire, 1 day to 10 years…)

Kinesis Data Firehose

Can setup KMS-based encryption with your own keys

### 4 CloudWatch Logs can send logs

CloudWatch Logs can send logs to

AWS Lambda

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration history explains what changed.

**CURRENT OPERATIONS LENS**

Connect every signal to service impact, an accountable owner, a safe response and a verification condition. Validate service limits and regional availability against current AWS documentation.


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# CloudWatch Logs - Sources

### 1 SDK, CloudWatch Logs Agent…

SDK, CloudWatch Logs Agent, CloudWatch Unified Agent

VPC Flow Logs:VPC specific logs

### 2 Elastic Beanstalk

Elastic Beanstalk: collection of logs from application

API Gateway

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration history explains what changed.

### 3 ECS: collection from containers

ECS: collection from containers

CloudTrail based on filter

Failure mode: average statistics, broad dimensions or short retention can hide tail latency, tenant impact and the evidence needed for a…

### 4 AWS Lambda

AWS Lambda: collection from function logs

Route53: Log DNS queries

**CURRENT OPERATIONS LENS**

Connect every signal to service impact, an accountable owner, a safe response and a verification condition. Validate service limits and regional availability against current AWS documentation.


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# CloudWatch Logs Insights




**1 Mechanism: metrics expose…**

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration…

**2 Failure mode**

Failure mode: average statistics, broad dimensions or short retention can hide tail latency, tenant impact and the…

**3 Evidence: alarm state history…**

Evidence: alarm state history, Logs Insights queries, CloudTrail event identity, trace correlation and SLO burn…

**CURRENT OPERATIONS LENS**

Connect every signal to service impact, an accountable owner, a safe response and a verification condition. Validate service limits and regional availability against current AWS documentation.


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# CloudWatch Logs Insights

### 1 Search and analyze log data

Search and analyze log data stored in CloudWatch Logs

Fetch desired event fields, filter based on conditions, calculate aggregate statistics, sort events, limit number of events…

### 2 Example: find a specific IP

Example: find a specific IP inside a log, count occurrences of “ERROR” in your logs…

Can save queries and add them to CloudWatch Dashboards

### 3 Provides a purpose-built query…

Provides a purpose-built query language

Can query multiple Log Groups in different AWS accounts

### 4 Automatically discovers fields…

Automatically discovers fields from AWS services and JSON log events

It’s a query engine, not a real-time engine

**CURRENT OPERATIONS LENS**

Connect every signal to service impact, an accountable owner, a safe response and a verification condition. Validate service limits and regional availability against current AWS documentation.


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# CloudWatch Logs - S3 Export

### 1 Log data can take up

Log data can take up to 12 hours to become available for export

### 2 The API call is CreateExportTask No…

The API call is CreateExportTaskNot near-real time or real-time… use Logs Subscriptions instead

### 3 CloudWatch Logs

CloudWatch Logs

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration history explains what changed.

### 4 Failure mode

Failure mode: average statistics, broad dimensions or short retention can hide tail latency, tenant impact and the evidence needed for a…

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# CloudWatch Logs Subscriptions

### 1 Get a real-time log events

Get a real-time log events from CloudWatch Logs for processing and analysis

Subscription Filter

near real-time

### 2 Send to Kinesis Data Streams,

Send to Kinesis Data Streams, Kinesis Data Firehose, or Lambda

Kinesis Data Firehose

EC2 Lambda

### 3 Subscription filter

Subscription filter - filter which logs are events delivered to your destination

Kinesis Data Streams

### 4 CloudWatch Logs

CloudWatch Logs

OpenSearch Service

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration history explains what changed.

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# CloudWatch Logs Aggregation Multi-Account & Multi Region

### 1 Kinesis Data Streams

Kinesis Data Streams

CloudWatch Logs Subscription Filter

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration history explains what changed.

### 2 Kinesis Data Firehose

Kinesis Data Firehose

ACCOUNT B REGION 2

Failure mode: average statistics, broad dimensions or short retention can hide tail latency, tenant impact and the evidence needed for a…

### 3 Near Real Time

Near Real Time

ACCOUNT B REGION 3

Evidence: alarm state history, Logs Insights queries, CloudTrail event identity, trace correlation and SLO burn rate should tell one consistent story.

### 4 ACCOUNT A REGION 1

ACCOUNT A REGION 1

Decision point: alert only when a signal identifies impact, urgency, owner and a safe next action; everything else belongs in a dashboard or…

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# CloudWatch Logs Subscriptions

### 1 Cross-Account Subscription

Cross-Account Subscription - send log events to resources in a different AWS account (KDS, KDF)

Subscription

(RecipientStream)

### 2 Account - Sender

Account - Sender

Account - Recipient

Subscription Destination

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration history explains what changed.

### 3 (111111111111)

(111111111111)

(999999999999)

Destination Access Policy

Failure mode: average statistics, broad dimensions or short retention can hide tail latency, tenant impact and the evidence needed for a…

### 4 CloudWatch Logs

CloudWatch Logs

Kinesis Data Streams

IAM Role (Cross-Account)

Evidence: alarm state history, Logs Insights queries, CloudTrail event identity, trace correlation and SLO burn rate should tell one consistent story.

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# CloudWatch Alarms

### 1 Alarms are used to trigger

Alarms are used to trigger notifications for any metric

Length of time in seconds to evaluate the metric

### 2 Various options (sampling, %, max,

Various options (sampling, %, max, min, etc…)

High resolution custom metrics: 10 sec, 30 sec or multiples of 60 sec

### 3 Alarm States

Alarm States

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration history explains what changed.

### 4 INSUFFICIENT_DATA

INSUFFICIENT_DATA

Failure mode: average statistics, broad dimensions or short retention can hide tail latency, tenant impact and the evidence needed for a…

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# CloudWatch Alarm Targets

### 1 Stop,Terminate, Reboot, or…

Stop,Terminate, Reboot, or Recover an EC2 Instance

EC2 Auto Scaling

### 2 Trigger Auto Scaling Action

Trigger Auto Scaling Action

Amazon SNS

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration history explains what changed.

### 3 Send notification to SNS (from

Send notification to SNS (from which you can do pretty much anything)

### 4 Amazon EC2

Amazon EC2

Failure mode: average statistics, broad dimensions or short retention can hide tail latency, tenant impact and the evidence needed for a…

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# CloudWatch Alarms - Composite Alarms

### 1 CloudWatch Alarms are on a

CloudWatch Alarms are on a single metric

EC2 Instance

Composite Alarm

### 2 Composite Alarms are monitoring…

Composite Alarms are monitoring the states of multiple other alarms

monitor CPU

CW Alarm - A

### 3 AND and OR conditions

AND and OR conditions

monitor IOPS

CW Alarm - B

### 4 Helpful to reduce “alarm noise”

Helpful to reduce “alarm noise” by creating complex composite alarms

Amazon SNS

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# EC2 Instance Recovery

### 1 CloudWatch Alarm

CloudWatch Alarm

System status = check the underlying hardware

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration history explains what changed.

### 2 StatusCheckFailed_System

StatusCheckFailed_System

Attached EBS status = check attached EBS volumes

Failure mode: average statistics, broad dimensions or short retention can hide tail latency, tenant impact and the evidence needed for a…

### 3 Status Check

Status Check

EC2 Instance

Evidence: alarm state history, Logs Insights queries, CloudTrail event identity, trace correlation and SLO burn rate should tell one consistent story.

### 4 Instance status = check the

Instance status = check the EC2 VM

Recovery: same Private, Public, Elastic IP, metadata, placement group

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# CloudWatch Alarm: good to know

### 1 Alarms can be created based

Alarms can be created based on CloudWatch Logs Metrics Filters

CloudWatch

### 2 Amazon SNS

Amazon SNS

Metric Filter

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration history explains what changed.

### 3 To test alarms and notifications,

To test alarms and notifications, set the alarm state to Alarm using CLI

### 4 aws cloudwatch set-alarm-state…

aws cloudwatch set-alarm-state --alarm-name "myalarm" --state-value ALARM --state-reason "testing purposes"

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# CloudWatch Synthetics Canary

### 1 Configurable script that monitor…

Configurable script that monitor your APIs, URLs, Websites, …

Integration with CloudWatch Alarms

CloudWatch Alarm

### 2 Reproduce what your customers do

Reproduce what your customers do programmatically to find issues before customers are impacted

Scripts written in Node.js or Python

Lambda Function

### 3 Checks the availability and…

Checks the availability and latency of your

Programmatic access to a headless Google Chrome browser

update DNS record

### 4 endpoints and can store load

endpoints and can store load time data and screenshots of the UI

Can run once or on a regular schedule

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# CloudWatch Synthetics Canary Blueprints

### 1 Heartbeat monitor

Heartbeat monitor - load URL, store screenshot and an HTTP archive file

Canary recorder - used with CloudWatch Synthetics Recorder (record your actions on a website and automatically generates a script for that)

### 2 API canary

API canary - test basic read and write functions of REST APIs

GUI flow builder - verifies that actions can be taken on your webpage (e.g., test a webpage with a login form)

### 3 Broken link checker

Broken link checker - check all links inside the URL that you are testing

### 4 Visual monitoring

Visual monitoring - compare a screenshot taken during a canary run with a baseline screenshot

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# Amazon EventBridge (formerly Amazon EventBridge)

### 1 Schedule: Cron jobs (scheduled…

Schedule: Cron jobs (scheduled scripts)

SNS Topic with Email Notification

### 2 Event Pattern

Event Pattern: Event rules to react to a service doing something

Schedule Every hour

### 3 Trigger Lambda functions, send…

Trigger Lambda functions, send SQS/SNS messages…

Trigger script on Lambda function

### 4 IAM Root User Sign in

IAM Root User Sign in Event

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration history explains what changed.

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# Amazon EventBridge Rules

### 1 Example Source

Example Source

(ex: upload object)

(any API call)

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration history explains what changed.

### 2 EC2 Instance

EC2 Instance

Trusted Advisor

Schedule or Cron

Failure mode: average statistics, broad dimensions or short retention can hide tail latency, tenant impact and the evidence needed for a…

### 3 (ex: Start Instance)

(ex: Start Instance)

(ex: new Finding)

(ex: every 4 hours)

### 4 (ex: failed build)

(ex: failed build)

CloudTrail

Amazon EventBridge

Evidence: alarm state history, Logs Insights queries, CloudTrail event identity, trace correlation and SLO burn rate should tell one consistent story.

**CURRENT OPERATIONS LENS**

Connect every signal to service impact, an accountable owner, a safe response and a verification condition. Validate service limits and regional availability against current AWS documentation.


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# Amazon EventBridge

### 1 Event buses can be accessed

Event buses can be accessed by other AWS accounts using Resource-based Policies

Default Event Bus

### 2 You can archive events…

You can archive events (all/filter) sent to an event bus (indefinitely or set period)

Partner Event Bus

### 3 Ability to replay archived events

Ability to replay archived events

Custom Event Bus

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration history explains what changed.

### 4 AWS Services

AWS Services

Custom Apps

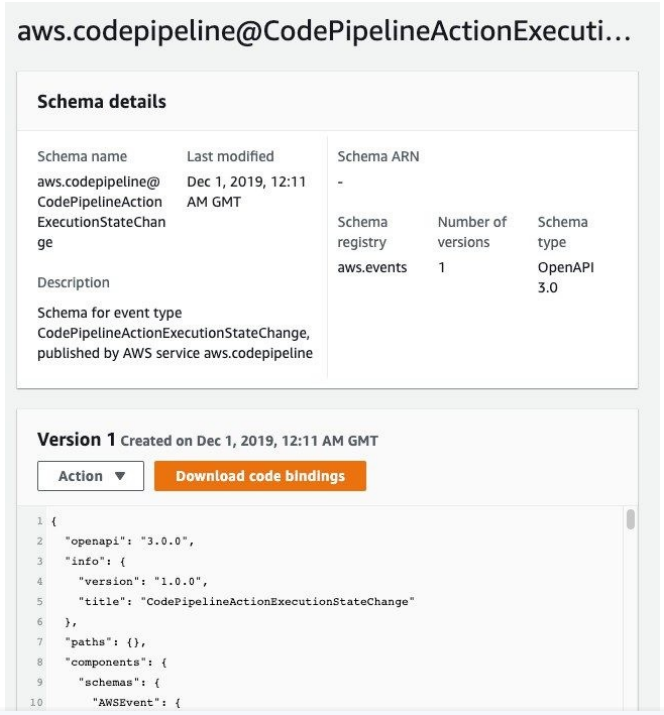
Failure mode: average statistics, broad dimensions or short retention can hide tail latency, tenant impact and the evidence needed for a…

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# Amazon EventBridge - Schema Registry




**1 EventBridge can analyze the events**

EventBridge can analyze the events in your bus and infer the schema

**2 Mechanism: metrics expose…**

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration…

**3 Failure mode**

Failure mode: average statistics, broad dimensions or short retention can hide tail latency, tenant impact and the…

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# Amazon EventBridge - Resource-based Policy

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": "events:PutEvents",
            "Principal": { "AWS": "111122223333" },
            "Resource": "arn:aws:events:us-east-1:123456789012:
event-bus/central-event-bus"
        }
    ]
}
```

1

### Manage permissions for a specific

Manage permissions for a specific Event Bus
Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration…

2

### Failure mode

Failure mode: average statistics, broad dimensions or short retention can hide tail latency, tenant impact and the…

3

### Evidence: alarm state history…

Evidence: alarm state history, Logs Insights queries, CloudTrail event identity, trace correlation and SLO burn…

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# Service Quotas CloudWatch Alarms

### 1 AWS Service Quotas

AWS Service Quotas

Helps you know if you need to request a quota increase or shutdown resources before limit is reached

### 2 Notify you when you’re close

Notify you when you’re close to a service quota value threshold

AWS Lambda Quota

### 3 Create CloudWatch Alarms on the

Create CloudWatch Alarms on the Service Quotas console

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration history explains what changed.

### 4 Example: Lambda concurrent…

Example: Lambda concurrent executions

Failure mode: average statistics, broad dimensions or short retention can hide tail latency, tenant impact and the evidence needed for a…

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# Alternative:Trusted Advisor + CW Alarms

### 1 Limited number of Service Limits

Limited number of Service Limits checks in Trusted Advisor (~50)

CloudWatch

### 2 Trusted Advisor publishes its…

Trusted Advisor publishes its check results to CloudWatch

Service Limits

### 3 You can create CloudWatch Alarms

You can create CloudWatch Alarms on service quota usage (Service Limits)

Lambda Concurrent Executions: 1000

### 4 AWS Trusted Advisor

AWS Trusted Advisor

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration history explains what changed.

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# AWS CloudTrail

### 1 Provides governance, compliance…

Provides governance, compliance and audit for your AWS Account

Can put logs from CloudTrail into CloudWatch Logs or S3

### 2 CloudTrail is enabled by default!

CloudTrail is enabled by default!

A trail can be applied to All Regions (default) or a single Region

### 3 Get a history of events

Get a history of events / API calls made within your AWS Account by

If a resource is deleted in AWS, investigate CloudTrail first!

### 4 AWS Services

AWS Services

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration history explains what changed.

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# CloudTrail Diagram

### 1 CloudTrail Console

CloudTrail Console

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration history explains what changed.

### 2 Inspect & Audit

Inspect & Audit

Failure mode: average statistics, broad dimensions or short retention can hide tail latency, tenant impact and the evidence needed for a…

### 3 CloudWatch Logs

CloudWatch Logs

Evidence: alarm state history, Logs Insights queries, CloudTrail event identity, trace correlation and SLO burn rate should tell one consistent story.

### 4 IAM Users & IAM Roles

IAM Users & IAM Roles

Decision point: alert only when a signal identifies impact, urgency, owner and a safe next action; everything else belongs in a dashboard or…

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# CloudTrail Events

### 1 Management Events

Management Events

Setting up logging (AWS CloudTrail CreateTrail)

By default, data events are not logged (because high volume operations)

### 2 Operations that are performed on

Operations that are performed on resources in your AWS account

By default, trails are configured to log management events.

Amazon S3 object-level activity (ex: GetObject, DeleteObject, PutObject): can separate Read and Write Events

### 3 Configuring security (IAM…

Configuring security (IAM AttachRolePolicy)

Can separate Read Events (that don't modify resources) from Write Events (that may modify resources)

AWS Lambda function execution activity (the Invoke API)

### 4 Configuring rules for routing data

Configuring rules for routing data (Amazon EC2 CreateSubnet)

Data Events

CloudTrail Insights Events

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# CloudTrail Insights

### 1 Enable CloudTrail Insights to…

Enable CloudTrail Insights to detect unusual activity in your account

Gaps in periodic maintenance activity

Event is sent to Amazon S3

### 2 inaccurate resource provisioning

inaccurate resource provisioning

CloudTrail Insights analyzes normal management events to create a baseline

An EventBridge event is generated (for automation needs)

### 3 hitting service limits

hitting service limits

And then continuously analyzes write events to detect unusual patterns

Management Events

### 4 Bursts of AWS IAM actions

Bursts of AWS IAM actions

Anomalies appear in the CloudTrail console

Continuous analysis

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# CloudTrail Events Retention

### 1 Events are stored for 90

Events are stored for 90 days in CloudTrail

Data Events

### 2 To keep events beyond this

To keep events beyond this period, log them to S3 and use Athena

Management Events

### 3 CloudTrail

CloudTrail

Insights Events

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration history explains what changed.

### 4 S3 Bucket Long-term retention

S3 Bucket Long-term retention

90 days retention

Failure mode: average statistics, broad dimensions or short retention can hide tail latency, tenant impact and the evidence needed for a…

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# Amazon EventBridge - Intercept API Calls

### 1 CloudTrail

CloudTrail

DeleteTable API Call

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration history explains what changed.

### 2 (any API call)

(any API call)

Failure mode: average statistics, broad dimensions or short retention can hide tail latency, tenant impact and the evidence needed for a…

### 3 Amazon EventBridge

Amazon EventBridge

Evidence: alarm state history, Logs Insights queries, CloudTrail event identity, trace correlation and SLO burn rate should tell one consistent story.

### 4 Log API call

Log API call

Decision point: alert only when a signal identifies impact, urgency, owner and a safe next action; everything else belongs in a dashboard or…

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# Amazon EventBridge + CloudTrail

## 1 CloudTrail

CloudTrail

AuthorizeSecurityGroupIngress

Mechanism: metrics expose behaviour, logs provide event detail, traces connect dependencies and configuration history explains what changed.

## 2 EventBridge

EventBridge

Security Group

Failure mode: average statistics, broad dimensions or short retention can hide tail latency, tenant impact and the evidence needed for a…

## 3 AssumeRole

AssumeRole

edit SG Inbound Rules

Evidence: alarm state history, Logs Insights queries, CloudTrail event identity, trace correlation and SLO burn rate should tell one consistent story.

## 4 API Call logs

API Call logs

Decision point: alert only when a signal identifies impact, urgency, owner and a safe next action; everything else belongs in a dashboard or…

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# CloudTrail - Log File Integrity Validation

### 1 Digest files

Digest files

Hashing using SHA-256, Digital Signing using SHA- 256 with RSA

contains a hash of each log file

### 2 References the log files for

References the log files for the last hour and contains a hash of each

Protect the S3 bucket using bucket policy, versioning, MFA Delete protection, encryption, object lock

Digest File

### 3 Stored in the same S3

Stored in the same S3 bucket as log files (different folder)

Protect CloudTrail using IAM

Amazon EC2

### 4 Helps you determine whether a

Helps you determine whether a log file was modified/deleted after CloudTrail delivered it

Log Files (1 hour)

start instance

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# Original PPT Coverage Retained — Topic 02

1

### Modernised slides

35 technical slides extracted from the original v4 PowerPoint and rebuilt in the current visual system.

2

### Traceability

Every blended slide names its original v4 source slide number in the kicker.

3

### Current validation

Terminology is normalised; service limits and Region availability remain explicit current-documentation checks.

4

### Procedure boundary

Click-by-click sequences, activities and quizzes are kept out of the deck and remain in the Learner Guide or lab package.


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TOPIC 02 · CONCEPT-LED LAB BRIEF

# Build an Observability and Incident Detection Baseline

**LAB 01**

Apply K1, K3, K4, A1 and A4 by creating a dashboard, alarm, CloudTrail evidence query and EventBridge detection rule for one tagged workload.

Create measurable service signals, route one actionable event and record an evidence-led incident hypothesis.

### YOU'LL BUILD

A CloudWatch dashboard, alarm definition, EventBridge pattern, evidence log and validation screenshots.

### TOOLCHAIN

CloudWatch · CloudTrail · EventBridge · SNS · AWS CLI · CloudShell

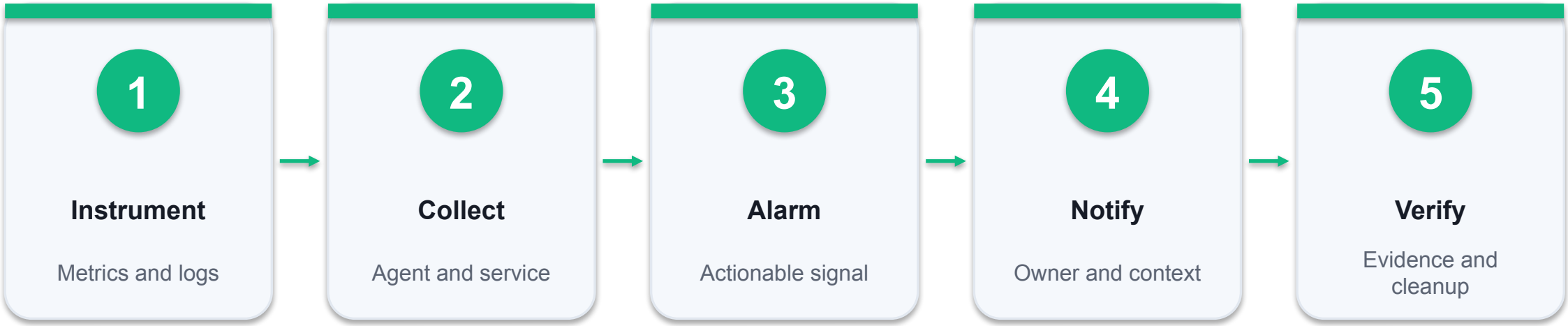
### DONE WHEN

The dashboard and alarm reference the assigned workload; CloudTrail evidence names the actor and event; the EventBridge pattern is valid; the incident record distinguishes observation from hypothesis; validation emits LAB 01 PASS.


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LAB 01 · CONCEPTUAL WORKFLOW

# Lab 01 Evidence Model




**PROCEDURE BOUNDARY**

Detailed commands and numbered steps are in the Learner Guide and lab folder—not on these slides.


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LAB 01 · ACCEPTANCE EVIDENCE

# Build an Observability and Incident Detection Baseline

## ✅ Expected result

The dashboard and alarm reference the assigned workload; CloudTrail evidence names the actor and event; the EventBridge pattern is valid; the incident record distinguishes observation from hypothesis; validation emits LAB 01 PASS.

IF IT DOESN'T WORK

**Input mismatch**

Reconcile identifiers, required fields and units with the supplied data dictionary.

**Validation stop**

Inspect the named rule and source row; preserve the check rather than bypassing it.

**Evidence differs**

Check the denominator, time window and assumptions before rerunning from clean inputs.


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TOPIC 02 · CONCEPT-LED LAB BRIEF

# Automate Event-Driven Incident Triage

**LAB 02**

Apply K3, K4, A1 and A4 by routing a controlled AWS event through EventBridge, invoking a least-privilege remediation function and preserving an auditable triage record.

Build an event pattern and Lambda triage function that records evidence and recommends a bounded next action.

### YOU'LL BUILD

An EventBridge rule, Lambda triage function, test event, CloudWatch Logs evidence and incident decision record.

### TOOLCHAIN

EventBridge · Lambda · CloudTrail · CloudWatch Logs · IAM

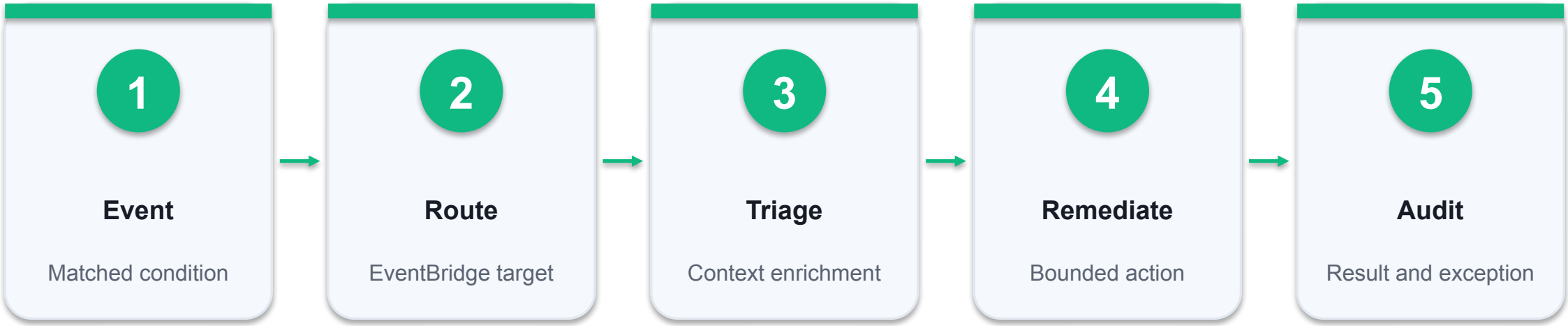
### DONE WHEN

The test and real events reach the Lambda target, logs contain the event ID and actor context, no automatic network modification occurs, the decision record names authority and verification, and validation emits LAB 02 PASS.


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LAB 02 · CONCEPTUAL WORKFLOW

# Lab 02 Evidence Model




**PROCEDURE BOUNDARY**

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LAB 02 · ACCEPTANCE EVIDENCE

# Automate Event-Driven Incident Triage

## ✅ Expected result

The test and real events reach the Lambda target, logs contain the event ID and actor context, no automatic network modification occurs, the decision record names authority and verification, and validation emits LAB 02 PASS.

IF IT DOESN'T WORK

### Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

### Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

### Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.


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# Recap — Monitoring, Logging and Incident Response

1

**Explain**

Dimensions, periods and statistics turn raw measurements into service signals.

2

**Distinguish**

Structured records provide detail for search, correlation, retention and audit.

3

**Apply**

K1, K3, K4, A1 and A4 by creating a dashboard, alarm, CloudTrail evidence query and EventBridge detection rule for one tagged workload.

4

**Evidence**

The dashboard and alarm reference the assigned workload; CloudTrail evidence names the actor and event; the EventBridge pattern is valid; the incident record…


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03

TOPIC 03 · LO1 · LO2 · K3 · K5 · A3 · A5

# Compute, Virtualisation and Systems Management

EC2 lifecycle · AMIs · placement · Systems Manager · patching · Image Builder


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# Key Ideas — Compute, Virtualisation and Systems Management

1

### Treat instances as managed nodes

Inventory, patch, access and automate consistently across a fleet.

2

### Images are controlled supply-chain inputs

Build, scan, test and version AMIs instead of hand-configuring servers.

3

### Placement affects failure and performance

Spread, partition and cluster strategies solve different operational needs.

4

### Lifecycle actions need coordination

Launch, stop, terminate and scale transitions require hooks and evidence.

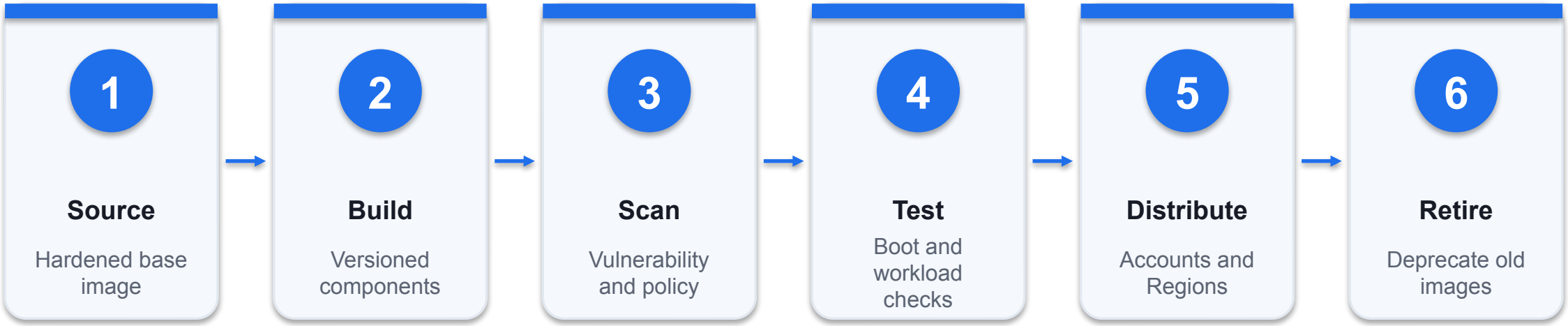

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# EC2 Purchase and Capacity Models

| Model | Operational value | Risk / constraint |
| --- | --- | --- |
| **On-Demand** | Flexible baseline and unpredictable workloads | Higher unit cost |
| **Savings Plans** | Commitment discount across eligible usage | Forecast and commitment risk |
| **Reserved capacity** | Capacity assurance in an AZ | Scope and utilisation management |
| **Spot** | Large discount for interruptible work | Two-minute interruption and capacity variability |


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# Golden Image Pipeline

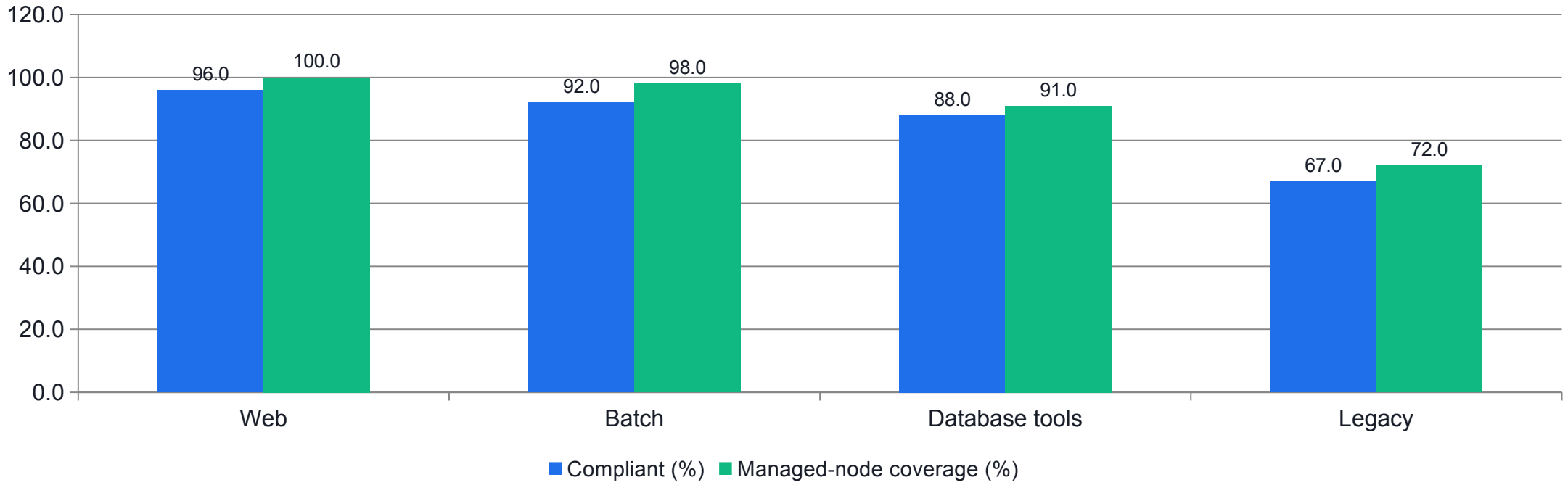



**A5 CONTROL**

An image release is an operational change with evidence, version and rollback.


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# Patch Compliance by Fleet Segment




**WHAT THE DATA SHOWS**

Patch compliance cannot exceed managed-node visibility; onboard and inventory the fleet before enforcing targets.


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# Systems Manager Capabilities

1

## Session Manager

Auditable shell access without inbound SSH or bastion exposure.

2

## Run Command

Execute controlled commands across tagged targets.

3

## Patch Manager

Baseline, maintenance window, compliance and remediation.

4

## Automation

Runbooks coordinate multi-step operational changes.

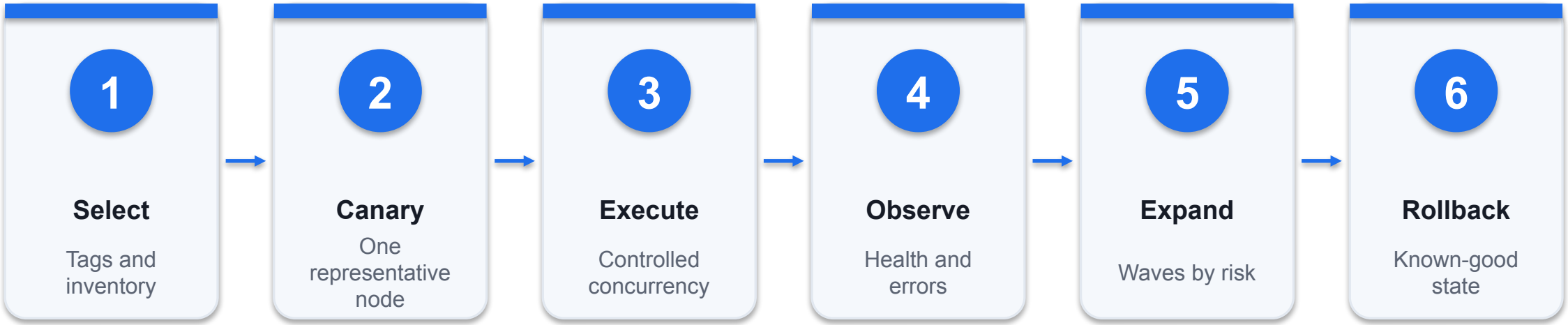

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# Placement Strategies

| Strategy | Optimises | Failure characteristic |
| --- | --- | --- |
| **Cluster** | Low-latency high-throughput east-west traffic | Single-AZ concentration |
| **Spread** | Small number of critical instances | Distinct underlying hardware |
| **Partition** | Large distributed workloads | Partition-aware failure isolation |


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# Safe Fleet Change




**OPERATIONS RULE**

Scale the change only after the canary evidence meets the acceptance threshold.


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Fleet Principle

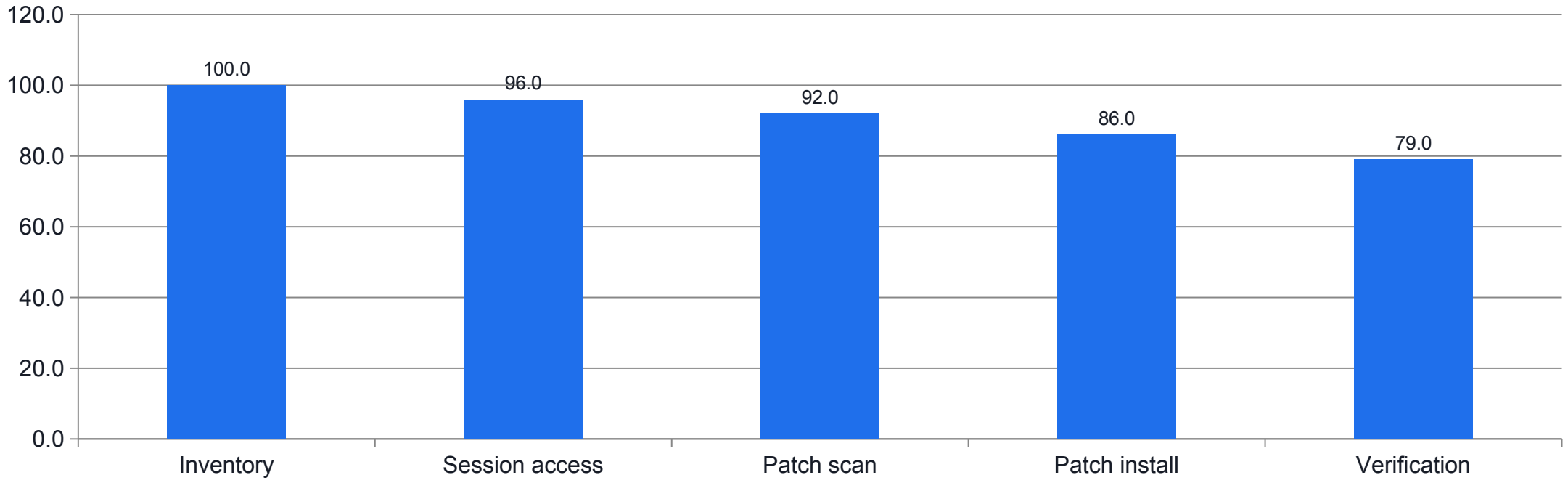
# If a server cannot be identified, accessed, patched and replaced consistently, it is not operationally managed.

Use tags, roles, inventory, image pipelines and Systems Manager to turn instances into a governed fleet.


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TOPIC 03 · NATIVE EDITABLE TECHNICAL CHART

# Fleet Management Coverage

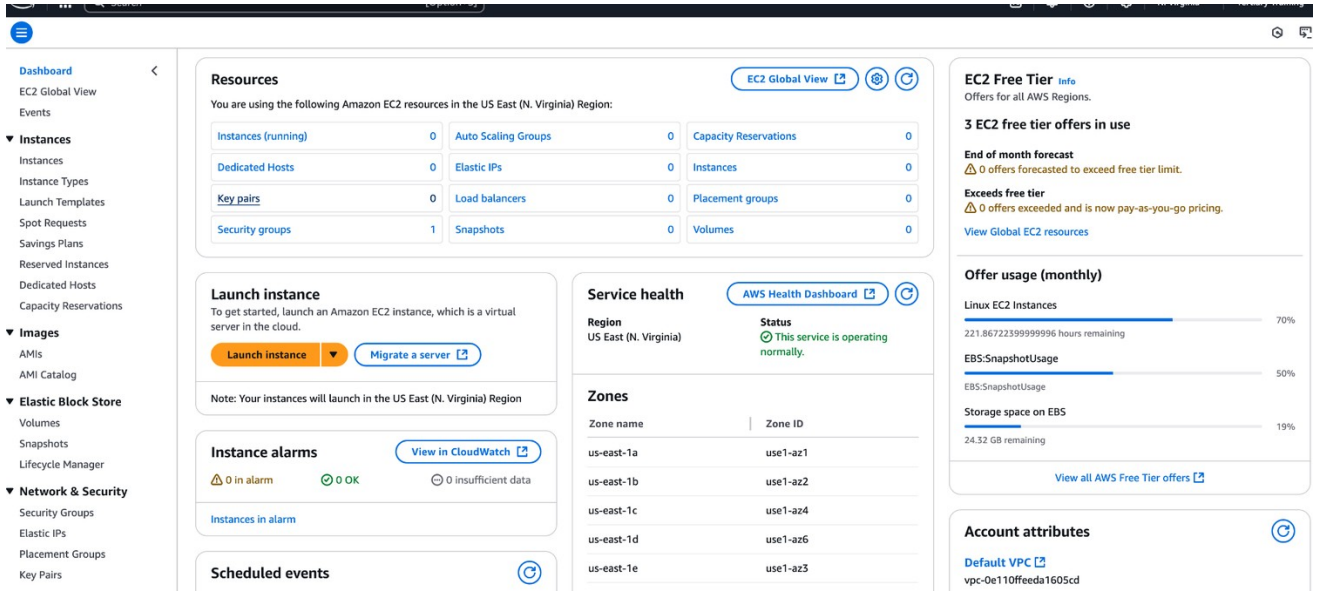



**WHAT THE DATA SHOWS**

The control chain weakens at verification: successful execution is not the same as confirmed service health.


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# EC2 Interface




**1** **Mechanism: launch templates…**

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state…

**2** **Failure mode**

Failure mode: configuration drift, unmanaged nodes, stale AMIs and unbounded Run Command concurrency turn a…

**3** **Evidence: inventory coverage…**

Evidence: inventory coverage, patch compliance, command output, instance status checks, application health and…

**CURRENT OPERATIONS LENS**

Manage compute as a governed fleet: immutable inputs, tagged targets, bounded concurrency, health evidence and replacement. Validate service limits and regional availability against current AWS documentation.


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# Enhanced Networking

### 1 EC2 Enhanced Networking (SR-IOV)

EC2 Enhanced Networking (SR-IOV)

Works for newer generation EC2 Instances

Leverages Message Passing Interface (MPI) standard

### 2 Higher bandwidth, higher PPS…

Higher bandwidth, higher PPS (packet per second), lower latency

Elastic Fabric Adapter (EFA)

Bypasses the underlying Linux OS to provide low-latency reliable transport

### 3 Option 1

Option 1: Elastic Network Adapter (ENA) up to 100 Gbps

Improved ENA for HPC, only works for Linux

### 4 Option 2

Option 2: Intel 82599 VF up to 10 Gbps – LEGACY

Great for inter-node communications, tightly coupled workloads.

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# Placement Groups

1

### Sometimes you want control over

Sometimes you want control over the EC2 Instance placement strategy

Spread—spreads instances across underlying hardware (max 7 instances per group per AZ) – critical applications

2

### That strategy can be defined

That strategy can be defined using placement groups

Partition—spreads instances across many different partitions (which rely on different sets of racks) within an AZ. Scales to 100s of EC2 instances…

3

### When you create a placement

When you create a placement group, you specify one of the following strategies for the group

4

### Cluster—clusters instances into a…

Cluster—clusters instances into a low-latency group in a single Availability Zone

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# Same AZ

### 1 Placement Groups Cluster

Placement Groups Cluster

Pros: Great network (10 Gbps bandwidth between instances with Enhanced Networking enabled - recommended)

### 2 Placement group Cluster

Placement group Cluster

Cons: If the AZ fails, all instances fails at the same time

### 3 Low latency

Low latency

Big Data job that needs to complete fast

### 4 10 Gbps network

10 Gbps network

Application that needs extremely low latency and high network throughput

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# Placements Groups Partition

## 1 us-east-1a

us-east-1a

us-east-1b

The instances in a partition do not

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

## 2 Partition 1

Partition 1

Up to 7 partitions per AZ

share racks with the instances in the other partitions

## 3 Partition 2

Partition 2

Can span across multiple AZs in the same region

A partition failure can affect many EC2 but won't affect other partitions

## 4 Partition 3

Partition 3

Up to 100s of EC2 instances

EC2 instances get access to the partition information as metadata

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# Shutdown Behaviour: How should the instance react when shutdown is done using the OS?

## 1 Stop (default)

Stop (default)

Shutdown Behavior

$ shutdown

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

## 2 This is not applicable when

This is not applicable when shutting down from AWS console.

EC2 Instance (running)

## 3 CLI Attribute

CLI Attribute

OS shutdown

Failure mode: configuration drift, unmanaged nodes, stale AMIs and unbounded Run Command concurrency turn a routine change into fleet-wide impact.

## 4 InstanceInitiatedShutdownBehavior

InstanceInitiatedShutdownBehavior

EC2 Instance (stopped)

Evidence: inventory coverage, patch compliance, command output, instance status checks, application health and replacement success must be correlated.

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# Termination Protection

### 1 Enable termination protection

Enable termination protection

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

### 2 To protect against accidental…

To protect against accidental termination in AWS Console or CLI

### 3 We have an instance where

We have an instance where shutdown behavior = terminate and enable terminate protection is ticked

### 4 We shutdown the instance from

We shutdown the instance from the OS, what will happen? The instance will still be terminated!

**CURRENT OPERATIONS LENS**

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# EC2 Launch Troubleshooting

### 1 # InstanceLimitExceeded if you get

# InstanceLimitExceeded if you get this error, it means that you have reached your limit of max number of vCPUs per region

NOTE: vCPU-based limits only apply to running On-Demand instances and Spot instances

### 2 On-Demand instance limits are set

On-Demand instance limits are set on a per-region basis

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

### 3 Example: If you run on-demand

Example: If you run on-demand (A, C, D, H, I, M, R,T, Z) instance types you'll have 64 vCPUs (default)

### 4 Resolution: Either launch the…

Resolution: Either launch the instance in a different region or request AWS to increase your limit of the region

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# EC2 Launch Troubleshooting

### 1 # InsufficientInstanceCapacity

# InsufficientInstanceCapacity: if you get this error, it means AWS does not have that enough On-Demand capacity in the particular AZ where the…

If urgent, submit a request for a different instance type now, which can be resized later.

### 2 Resolution

Resolution

Also, can request the EC2 instance in a different AZ

### 3 Wait for few mins before

Wait for few mins before requesting again.

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

### 4 If requesting more than 1

If requesting more than 1 requests, break down the requests. If you need 5 instances, rather than a single request of 5, request one by one.

**CURRENT OPERATIONS LENS**

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# EC2 Launch Troubleshooting

### 1 # Instance Terminates Immediately…

# Instance Terminates Immediately (goes from pending to terminated)

The instance store-backed AMI that you used to launch the instance is missing a required part (an image.part.xx file).

### 2 You've reached your EBS volume

You've reached your EBS volume limit.

To find the exact reason, check out the EC2 console of AWS - instances - Description tab, note the reason next to the State transition reason label.

### 3 An EBS snapshot is corrupt.

An EBS snapshot is corrupt.

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

### 4 The root EBS volume is

The root EBS volume is encrypted and you do not have permissions to access the KMS key for decryption.

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# EC2 SSH troubleshooting

### 1 Make sure the private key

Make sure the private key (pem file) on your linux machine has 400 permissions, else you will get "Unprotected private key file" error

SG is not configured correctly

CPU load of the instance is high

### 2 Make sure the username for

Make sure the username for the OS is given correctly when logging via SSH,

NACL is not configured correctly

### 3 else you will get "Host

else you will get "Host key not found, "Permission denied", or "Connection closed by [instance] port 22" error

Check the route table for the subnet (routes traffic destined outside VPC to IGW)

### 4 Possible reasons for "Connection…

Possible reasons for "Connection timed out" to EC2 instance via SSH

Instance doesn’t have a public IPv4

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# SSH vs. EC2 Instance Connect

1

**Connect using SSH Connect using**

Connect using SSH Connect using EC2 Instance Connect

"ip_prefix": "18.206.107.24/29", "region": "us-east-1",

push one-time SSH public key (valid for 60 secs)

2

**Inbound Rules**

Inbound Rules

"service": "EC2_INSTANCE_CONNECT",

EC2 Console,

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

3

**(IPv4: 1.2.3.4)**

(IPv4: 1.2.3.4)

"network_border_group": "us-east-1"

EC2 Instance Connect CLI

Failure mode: configuration drift, unmanaged nodes, stale AMIs and unbounded Run Command concurrency turn a routine change into fleet-wide impact.

4

**(IPv4: 5.6.7.8)**

(IPv4: 5.6.7.8)

EC2 Instance Connect API

Allows AWS IP Range

**CURRENT OPERATIONS LENS**

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# EC2 Instances Purchasing Options

### 1 On-Demand Instances

On-Demand Instances - short workload, predictable pricing, pay by second

Savings Plans (1 & 3 years) - commitment to an amount of usage, long workload

Capacity reservations - reserve capacity in a specific AZ for any duration

### 2 Reserved (1 & 3 years)

Reserved (1 & 3 years)

Spot Instances - short workloads, cheap, can lose instances (less reliable)

### 3 Reversed Instances

Reversed Instances - long workloads

Dedicated Hosts - book an entire physical server, control instance placement

### 4 Convertible Reserved Instances

Convertible Reserved Instances - long workloads with flexible instances

Dedicated Instances - no other customers will share your hardware

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# EC2 On Demand

### 1 Pay for what you use

Pay for what you use

No long-term commitment

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

### 2 Linux or Windows

Linux or Windows - billing per second, after the first minute

Recommended for short term and un-interrupted workloads, where you can't predict how the application will behave

### 3 All other operating systems

All other operating systems - billing per hour

Failure mode: configuration drift, unmanaged nodes, stale AMIs and unbounded Run Command concurrency turn a routine change into fleet-wide impact.

### 4 Has the highest cost but

Has the highest cost but no upfront payment

Evidence: inventory coverage, patch compliance, command output, instance status checks, application health and replacement success must be correlated.

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# Up to 72% discount compared to On-demand

### 1 You reserve a specific instance

You reserve a specific instance attributes (Instance Type, Region, Tenancy, OS)

Recommended for steady-state usage applications (think database)

Up to 66% discount

### 2 Reservation Period

Reservation Period - 1 year (+discount) or 3 years (++ +discount)

You can buy and sell in the Reserved Instance Marketplace

EC2 Reserved Instances

### 3 Payment options

Payment options - No Upfront (+), Partial Upfront (++), All Upfront (+++)

Convertible Reserved Instance

Note: the % discounts are different from the video as AWS change them over time – the exact numbers are not needed for the exam. This is just for…

### 4 Reserved Instance's Scope

Reserved Instance's Scope - Regional or Zonal (reserve capacity in an AZ)

Can change the EC2 instance type, instance family, OS, scope and tenancy

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# EC2 Savings Plans

### 1 Get a discount based on

Get a discount based on long-term usage (up to 72% - same as RIs)

Flexible across

### 2 Commit to a certain type

Commit to a certain type of usage ($10/hour for 1 or 3 years)

Instance Size (e.g., m5.xlarge, m5.2xlarge)

### 3 Usage beyond EC2 Savings Plans

Usage beyond EC2 Savings Plans is billed at the On-Demand price

OS (e.g., Linux, Windows)

### 4 Locked to a specific instance

Locked to a specific instance family & AWS region (e.g., M5 in us-east-1)

Tenancy (Host, Dedicated, Default)

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# EC2 Spot Instances

### 1 Can get a discount of

Can get a discount of up to 90% compared to On-demand

Batch jobs

Workloads with a flexible start and end time

### 2 Instances that you can “lose”

Instances that you can “lose” at any point of time if your max price is less than the current spot price

Data analysis

Not suitable for critical jobs or databases

### 3 The MOST cost-efficient instances…

The MOST cost-efficient instances in AWS

Image processing

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

### 4 Useful for workloads that are

Useful for workloads that are resilient to failure

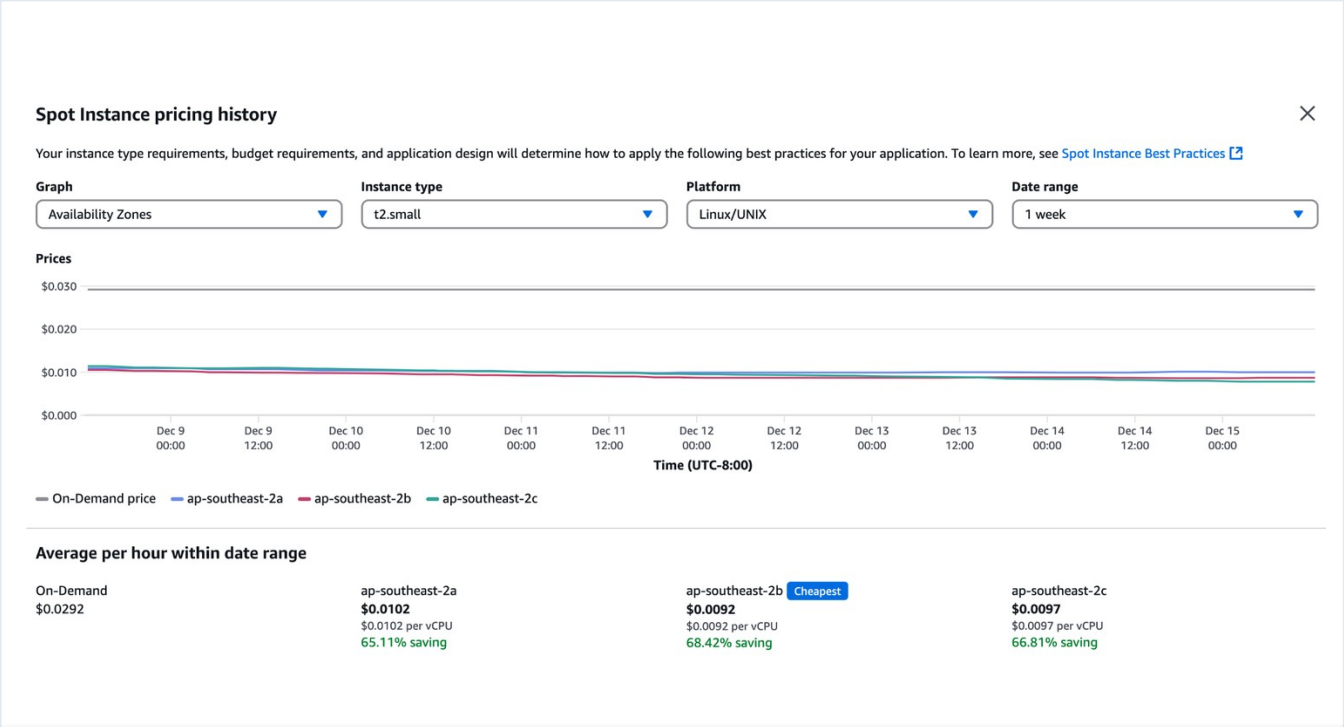
Any distributed workloads

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# EC2 Spot Instance Pricing History




**1** **Mechanism: launch templates…**

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state…

**2** **Failure mode**

Failure mode: configuration drift, unmanaged nodes, stale AMIs and unbounded Run Command concurrency turn a…

**3** **Evidence: inventory coverage…**

Evidence: inventory coverage, patch compliance, command output, instance status checks, application health and…

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# EC2 Dedicated Hosts

### 1 A physical server with EC2

A physical server with EC2 instance capacity fully dedicated to your use

Reserved - 1 or 3 years (No Upfront, Partial Upfront, All Upfront) The most expensive option

### 2 Allows you address compliance…

Allows you address compliance requirements and use your existing server - bound software licenses (per-socket, per-core, pe—VM software licenses)

Useful for software that have complicated licensing model (BYOL – Bring Your Own License)

### 3 Purchasing Options

Purchasing Options

Or for companies that have strong regulatory or compliance needs

### 4 On-demand - pay per second

On-demand - pay per second for active Dedicated Host

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

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# EC2 Dedicated Instances

| Characteristic | Dedicated Instances | Dedicated Hosts |
|---|---|---|
| Enables the use of dedicated physical servers | X | X |
| Per instance billing (subject to a $2 per region fee) | X | |
| Per host billing | | X |
| Visibility of sockets, cores, host ID | | X |
| Affinity between a host and instance | | X |
| Targeted instance placement | | X |
| Automatic instance placement | X | X |
| Add capacity using an allocation request | | X |

1

### Instances run on hardware that's

Instances run on hardware that's dedicated to you Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state…

2

### Failure mode

Failure mode: configuration drift, unmanaged nodes, stale AMIs and unbounded Run Command concurrency turn a…

3

### Evidence: inventory coverage…

Evidence: inventory coverage, patch compliance, command output, instance status checks, application health and…

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# EC2 Capacity Reservations

### 1 Reserve on-Demand instances…

Reserve on-Demand instances capacity in a specific AZ for any duration

Suitable for short-term, uninterrupted workloads that needs to be in a specific AZ

### 2 You always have access to

You always have access to EC2 capacity when you need it

### 3 No time commitment (create/cancel…

No time commitment (create/cancel anytime), no billing discounts Combine with Regional Reserved Instances and Savings Plans to benefit from billing…

### 4 You’re charged at On-Demand rate

You’re charged at On-Demand rate whether you run instances or not

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# Which purchasing option is right for me?



**1 On demand**

On demand: coming and staying in resort whenever we like, we pay the full price

**2 Mechanism: launch templates…**

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state…

**3 Failure mode**

Failure mode: configuration drift, unmanaged nodes, stale AMIs and unbounded Run Command concurrency turn a…

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# EC2 Purchase Options — Cost Model Comparison

### 1 On-Demand: no term commitment…

On-Demand: no term commitment; suitable for short-lived, irregular or uncertain demand.

### 2 Savings Plans

Savings Plans: commit to consistent compute spend in exchange for discounted eligible usage.

### 3 Reserved Instances

Reserved Instances: apply a term-based discount and, for zonal reservations, optional capacity assurance.

### 4 Spot Instances

Spot Instances: use spare capacity at deep discount with interruption-aware, fault-tolerant workloads.

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# AWS charges for IPv4 addresses

### 1 IPv4 created in your account

IPv4 created in your account

Starting February 1st 2024, there's a charge for all Public

One Public IPv4

### 2 $0.005 per hour of Public

$0.005 per hour of Public IPv4 (~ $3.6 per month)

EC2 Instance

per AZ, no Free Tier

### 3 For new accounts in AWS,

For new accounts in AWS, you have a free tier for the EC2 service: 750 hours of Public IPv4 per month for the first 12 months

Free until 750 hours/month of usage

RDS Database

### 4 For all other services there

For all other services there is no free tier

Load Balancer

One Public IPv4 No Free Tier

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# Public IPv4 Addressing — Cost and Operational Controls

### 1 Billing scope

Billing scope: public IPv4 addresses are chargeable whether attached to a running resource or retained while idle.

### 2 Visibility: use AWS Public IP

Visibility: use AWS Public IP Insights and cost-allocation evidence to identify address ownership, attachment and utilisation.

### 3 Control: release unused…

Control: release unused addresses, minimise public exposure and prefer private connectivity, load balancing or managed endpoints where appropriate.

### 4 IPv6 decision

IPv6 decision: validate client reachability, dual-stack routing, security rules, DNS records and application support before migration.

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# EC2 Spot Instance Requests

### 1 Can get a discount of

Can get a discount of up to 90% compared to On-demand

Other strategy: Spot Block

Not great for critical jobs or databases

### 2 Define max spot price and

Define max spot price and get the instance while current spot price < max

“block” spot instance during a specified time frame (1 to 6 hours) without interruptions

### 3 The hourly spot price varies

The hourly spot price varies based on offer and capacity

In rare situations, the instance may be reclaimed

### 4 If the current spot price

If the current spot price > your max price you can choose to stop or terminate your instance with a 2 minutes grace period.

Used for batch jobs, data analysis, or workloads that are resilient to failures.

**CURRENT OPERATIONS LENS**

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# EC2 Spot Instances Pricing

### 1 User-defined max price

User-defined max price

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

### 2 Failure mode

Failure mode: configuration drift, unmanaged nodes, stale AMIs and unbounded Run Command concurrency turn a routine change into fleet-wide impact.

### 3 Evidence: inventory coverage…

Evidence: inventory coverage, patch compliance, command output, instance status checks, application health and replacement success must be correlated.

### 4 Decision point

Decision point: prefer canary and wave deployment with explicit stop thresholds; replace inconsistent instances rather than repairing them…

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# Spot Fleets

### 1 Spot Fleets = set of

Spot Fleets = set of Spot Instances + (optional) On-Demand Instances

Spot Fleet stops launching instances when reaching capacity or max cost

capacityOptimized: pool with the optimal capacity for the number of instances

### 2 The Spot Fleet will try

The Spot Fleet will try to meet the target capacity with price constraints

Strategies to allocate Spot Instances

priceCapacityOptimized (recommended): pools with highe

### 3 Define possible launch pools

Define possible launch pools: instance type (m5.large), OS, Availability Zone

lowestPrice: rom the pool with the lowest price (cost optimization, short workload)

st capacity available, then select the pool with the lowest price (best choice for most workloads)

### 4 Can have multiple launch pools,

Can have multiple launch pools, so that the fleet can choose

diversified: distributed across all pools (great for availability, long workloads)

Spot Fleets allow us to automatically request Spot Instances with the lowest price

**CURRENT OPERATIONS LENS**

Manage compute as a governed fleet: immutable inputs, tagged targets, bounded concurrency, health evidence and replacement. Validate service limits and regional availability against current AWS documentation.


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# Burstable Instances (T2/T3)

1

### AWS has the concept of

AWS has the concept of burstable instances (T2/T3 machines)

If all the credits are gone, the CPU becomes BAD

2

### Burst means that overall, the

Burst means that overall, the instance has OK CPU performance.

If the machine stops bursting, credits are accumulated over time

3

### When the machine needs to

When the machine needs to process something unexpected (a spike in load for example), it can burst, and CPU can be VERY good.

4

### If the machine bursts, it

If the machine bursts, it utilizes “burst credits”

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

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# Burstable Instances (T2/T3)

### 1 Burstable instances can be amazing

Burstable instances can be amazing to handle unexpected traffic and getting the insurance that it will be handled correctly

### 2 If your instance consistently runs

If your instance consistently runs low on credit, you need to move to a different kind of non-burstable instance

### 3 Credit usage

Credit usage

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

### 4 Credit balance

Credit balance

Failure mode: configuration drift, unmanaged nodes, stale AMIs and unbounded Run Command concurrency turn a routine change into fleet-wide impact.

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# What happens when credit ar exhausted?

### 1 Experiment: run a CPU stress

Experiment: run a CPU stress command (to peak at 100%)

Credit exhaustion

### 2 After the credits are exhausted,

After the credits are exhausted, the measured CPU utilization drops

Lowered CPU Utilization

### 3 100% CPU Utilization

100% CPU Utilization

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

### 4 Credit Usage

Credit Usage

Failure mode: configuration drift, unmanaged nodes, stale AMIs and unbounded Run Command concurrency turn a routine change into fleet-wide impact.

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# T2/T3 Unlimited

1

### It is possible to have

It is possible to have an “unlimited burst credit balance”

2

### You pay extra money if

You pay extra money if you go over your credit balance, but you don’t lose in performance

3

### If average CPU usage over

If average CPU usage over a 24-hour period exceeds the baseline, the instance is billed for additional usage per vCPU/hour

4

### Be careful, costs could go

Be careful, costs could go high if you’re not monitoring the CPU health of your instances

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# Public subnet

1

### Elastic IPs

Elastic IPs

You can attach it to one instance at a time

EC2 Instance

2

### When you stop and then

When you stop and then start an EC2 instance, it changes its public IP

You can remap it across instances\

(IPv4: 1.2.3.4)

3

### If you need to have

If you need to have a fixed public IP, you need an Elastic IP

You don’t pay for the Elastic IP if it’s attached to a server

4

### An Elastic IP is a

An Elastic IP is a public IPv4 you own as long as you don’t delete it

You pay for the Elastic IP if it’s not attached to a server

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# Elastic IPs

### 1 With an Elastic IP address,

With an Elastic IP address, you can mask the failure of an instance or software by rapidly remapping the address to another instance in your account.

You could use a random public IP and register a DNS name to it

### 2 You can only have 5

You can only have 5 Elastic IP in your account (you can ask AWS to increase that).

Or use a Load Balancer with a static hostname

### 3 How you can avoid using

How you can avoid using Elastic IP

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

### 4 Always think if other alternatives

Always think if other alternatives are available to you

Failure mode: configuration drift, unmanaged nodes, stale AMIs and unbounded Run Command concurrency turn a routine change into fleet-wide impact.

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# CloudWatch Metrics for EC2

### 1 AWS Provided metrics (AWS pushes

AWS Provided metrics (AWS pushes them)

Custom metric (yours to push)

Make sure the IAM permissions on the EC2 instance role are correct !

### 2 Basic Monitoring (default)

Basic Monitoring (default): metrics are collected at a 5 minute internal

Basic Resolution: 1 minute resolution

### 3 Detailed Monitoring (paid)

Detailed Monitoring (paid): metrics are collected at a 1 minute interval

High Resolution: all the way to 1 second resolution

### 4 Includes CPU, Network, Disk and

Includes CPU, Network, Disk and Status Check Metrics

Include RAM, application level metrics

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# EC2 included metrics

### 1 CPU: CPU Utilization + Credit

CPU: CPU Utilization + Credit Usage / Balance

System status = check the underlying hardware

### 2 Network: Network In / Out

Network: Network In / Out

Attached EBS status = check attached EBS volumes

### 3 Status Check

Status Check

Disk: Read / Write for Ops / Bytes (only for instance store)

### 4 Instance status = check the

Instance status = check the EC2 VM

RAM is NOT included in the AWS EC2 metrics

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# Unified CloudWatch Agent

### 1 For virtual servers (EC2…

For virtual servers (EC2 instances, on-premises servers, …)

Centralized configuration using SSM Parameter Store

CloudWatch

### 2 Collect additional system-level…

Collect additional system-level metrics such as RAM, processes, used disk space, etc.

Make sure IAM permissions are correct

Server with Unified CW Agent

### 3 Collect logs to send to

Collect logs to send to CloudWatch Logs

Default namespace for metrics collected by the Unified CloudWatch agent is CWAgent (can be configured/changed)

EC2 Instance with Unified CW Agent

### 4 No logs from inside your

No logs from inside your EC2 instance will be sent to CloudWatch Logs without using an agent

Corporate Data Center

push metrics

**CURRENT OPERATIONS LENS**

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# Unified CloudWatch Agent – procstat Plugin

### 1 Collect metrics and monitor system

Collect metrics and monitor system utilization of individual processes

pid_file: name of process identification number (PID) files they create

### 2 Supports both Linux and Windows

Supports both Linux and Windows servers

exe: process name that match string you specify (RegEx)

### 3 Example: amount of time the

Example: amount of time the process uses CPU, amount of memory the process uses, …

pattern: command lines used to start the processes (RegEx)

### 4 Select which processes to monitor

Select which processes to monitor by

Metrics collected by procstat plugin begins with "procstat" prefix (e.g., procstat_cpu_time, procstat_cpu_usage, …)

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# Status Checks

### 1 Automated checks to identify…

Automated checks to identify hardware and software issues

Resolution: stop and start the instance (instance migrated to a new host)

Attached EBS Status Checks

### 2 System status Checks

System status Checks

Instance Status Checks

Monitors EBS volumes attached to your instance (reachable & complete I/O Operations)

### 3 Monitors problems with AWS systems

Monitors problems with AWS systems (software/hardware issues on the physical host, loss of system power, …)

Monitors software/network configuration of your instance (invalid network configuration, exhausted memory, …)

Resolution: reboot the instance or replace affected EBS volumes

### 4 Check Personal Health Dashboard…

Check Personal Health Dashboard for any scheduled critical maintenance by AWS to your instance's host

Resolution: reboot the instance or change instance configuration

Stop & Start

**CURRENT OPERATIONS LENS**

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# Status Checks - CW Metrics & Recovery

### 1 CloudWatch Metrics (1 minute…

CloudWatch Metrics (1 minute interval)

StatusCheckFailed (for any)

Option 2: Auto Scaling Group

### 2 StatusCheckFailed_System

StatusCheckFailed_System

Option 1: CloudWatch Alarm

Set min/max/desired 1 to recover an instance but won't keep the same private and elastic IP

### 3 StatusCheckFailed_Instance

StatusCheckFailed_Instance

Recover EC2 instance with the same private/public IP, EIP, metadata, and Placement Group

EC2 Instance

### 4 StatusCheckFailed_AttachedEBS

StatusCheckFailed_AttachedEBS

Send notifications using SNS

CloudWatch

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

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# EC2 Hibernate

1

### We know we can stop,

We know we can stop, terminate instances

First start: the OS boots & the EC2 User Data script is run

2

### Stop - the data on

Stop - the data on disk (EBS) is kept intact in the next start

Following starts: the OS boots up

3

### Terminate - any EBS volumes

Terminate - any EBS volumes (root) also set-up to be destroyed is lost

Then your application starts, caches get warmed up, and that can take time!

4

### On start, the following happens

On start, the following happens

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

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# EC2 Hibernate

### 1 Introducing EC2 Hibernate

Introducing EC2 Hibernate

The root EBS volume must be encrypted

EC2 Instance

### 2 The in-memory (RAM) state is

The in-memory (RAM) state is preserved

Long-running processing

(Encrypted)

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

### 3 The instance boot is much

The instance boot is much faster! (the OS is not stopped / restarted)

Saving the RAM state

Hibernation

### 4 Under the hood

Under the hood: the RAM state is written to a file in the root EBS volume

Services that take time to initialize

**CURRENT OPERATIONS LENS**

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# EC2 Hibernate – Good to know

### 1 Supported Instance Families

Supported Instance Families - C3, C4, C5, I3, M3, M4, R3, R4,T2,T3, …

Root Volume - must be EBS, encrypted, not instance store, and large.

### 2 instance RAM Size

instance RAM Size - must be less than 150 GB.

Available for On-Demand, Reserved and Spot Instances.

### 3 Instance Size

Instance Size - not supposed for bare metal instances.

An instance can NOT be hibernated for more than 60 days.

### 4 AMI - Amazon Linux 2,

AMI - Amazon Linux 2, Linux AMI, Ubuntu, RHEL, CentOS & Windows…

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# AMI Overview

### 1 AMI = Amazon Machine Image

AMI = Amazon Machine Image

AMI are built for a specific region (and can be copied across regions)

An AWS Marketplace AMI: an AMI someone else made (and potentially sells)

### 2 AMI is a customization of

AMI is a customization of an EC2 instance

You can launch EC2 instances from

### 3 You add your own software,

You add your own software, configuration, operating system, monitoring…

A Public AMI: AWS provided

### 4 Faster boot / configuration time

Faster boot / configuration time because all your software is pre-packaged

Your own AMI: you make and maintain them yourself

**CURRENT OPERATIONS LENS**

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# AMI Process (from an EC2 instance)

### 1 Start an EC2 instance and

Start an EC2 instance and customize it

US-EAST-1A

Launch from AMI

### 2 Stop the instance (for data

Stop the instance (for data integrity)

US-EAST-1B

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

### 3 Build an AMI

Build an AMI – this will also create EBS snapshots

Custom AMI

### 4 Launch instances from other AMIs

Launch instances from other AMIs

Create AMI

Failure mode: configuration drift, unmanaged nodes, stale AMIs and unbounded Run Command concurrency turn a routine change into fleet-wide impact.

**CURRENT OPERATIONS LENS**

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# AMI No-Reboot Option

1

### Enables you to create an

Enables you to create an AMI without shutting down your instance

Attached EBS Volume

create image

2

### By default, it's not selected

By default, it's not selected (AWS will shut down the instance before creating an AMI to maintain the file system integrity)

EBS Snapshot

With No-Reboot Enabled

3

### With No-Reboot Disabled (default)

With No-Reboot Disabled (default)

EC2 Instance (stopped)

EC2 Instance

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

4

### EC2 Instance (running)

EC2 Instance (running)

EBS Volume

Note: OS Buffers are not flushed to disk before the snapshot is created

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# AWS Backup Plans to create AMI

1

**AWS Backup doesn't reboot the**

AWS Backup doesn't reboot the instances while taking EBS snapshots (no-reboot behavior)

Create AMI

Amazon EC2

2

**This won't help you to**

This won't help you to create an AMI that guarantees file system integrity since you need to reboot the instance

Amazon EC2 (rebooting)

AWS Amazon

3

**To maintain integrity you need**

To maintain integrity you need to provide the reboot parameter while taking images (EventBridge

EventBridge Rule (schedule)

4

**+ Lambda + CreateImage API**

+ Lambda + CreateImage API with reboot)

Lambda Function

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

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# EC2 Instance Migration between AZ

### 1 us-east-1a

us-east-1a

create image launch/restore

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

### 2 us-east-1b

us-east-1b

Failure mode: configuration drift, unmanaged nodes, stale AMIs and unbounded Run Command concurrency turn a routine change into fleet-wide impact.

### 3 EC2 Instance

EC2 Instance

Evidence: inventory coverage, patch compliance, command output, instance status checks, application health and replacement success must be correlated.

### 4 EBS Volume

EBS Volume

Decision point: prefer canary and wave deployment with explicit stop thresholds; replace inconsistent instances rather than repairing them…

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# Cross-Account AMI Sharing

### 1 You can share an AMI

You can share an AMI with another AWS account

Source AMI

### 2 Sharing an AMI does not

Sharing an AMI does not affect the ownership of the AMI

Share with Account B

### 3 You can only share AMIs

You can only share AMIs that have unencrypted volumes and volumes that are encrypted with a customer managed key

EC2 Instance

### 4 If you share an AMI

If you share an AMI with encrypted volumes, you must also share any customer managed keys used to encrypt them.

(AMI: Source AMI)

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# AMI Sharing with KMS Encryption

**1 Custom AMI**

Custom AMI

EC2 Instance

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

**2 (encrypted)**

(encrypted)

(AMI: Custom AMI)

Failure mode: configuration drift, unmanaged nodes, stale AMIs and unbounded Run Command concurrency turn a routine change into fleet-wide impact.

**3 IAM Permissions**

IAM Permissions

Evidence: inventory coverage, patch compliance, command output, instance status checks, application health and replacement success must be correlated.

**4 kms:DescribeKey kms:CreateGrant …**

kms:DescribeKey kms:CreateGrant kms:Decrypt kms:GenerateDataKey kms:ReEncrypt

Decision point: prefer canary and wave deployment with explicit stop thresholds; replace inconsistent instances rather than repairing them…

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# Cross-Account AMI Copy

### 1 If you copy an AMI

If you copy an AMI that has been shared with your account, you are the owner of the target AMI in your account

Source AMI

copy image

### 2 The owner of the source

The owner of the source AMI must grant you read permissions for the storage that backs the AMI (EBS Snapshot)

EBS Snapshot

### 3 If the shared AMI has

If the shared AMI has encrypted snapshots, the owner must share the key or keys with you as well

Copy of Source AMI

### 4 Can encrypt the AMI with

Can encrypt the AMI with your own CMK while copying

IAM Permissions (read snapshot)

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# AMI Copy with KMS Encryption

### 1 Custom AMI

Custom AMI

Decrypt: CMK – A Encrypt: CMK – B

us-east-1 us-east-1

### 2 (encrypted)

(encrypted)

Cross-Region / Cross-Account Encrypted AMI Copy

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

### 3 IAM Permissions

IAM Permissions

Source Target

Failure mode: configuration drift, unmanaged nodes, stale AMIs and unbounded Run Command concurrency turn a routine change into fleet-wide impact.

### 4 kms:DescribeKey kms:CreateGrant …

kms:DescribeKey kms:CreateGrant kms:Decrypt

Account A Account B

Evidence: inventory coverage, patch compliance, command output, instance status checks, application health and replacement success must be correlated.

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# EC2 Image Builder

1

**Used to automate the creation**

Used to automate the creation of Virtual Machines or container images

Builder EC2 Instance

(is the AMI working, secure?)

2

**=> Automate the creation…**

=> Automate the creation, maintain, validate and test EC2 AMIs

Build Components applied (customize software on instance)

AMI is distributed

3

**Can be run on a**

Can be run on a schedule (weekly, whenever packages are updated, etc…)

Test EC2 Instance

(can be multiple regions)

4

**Free service (only pay for**

Free service (only pay for the underlying resources)

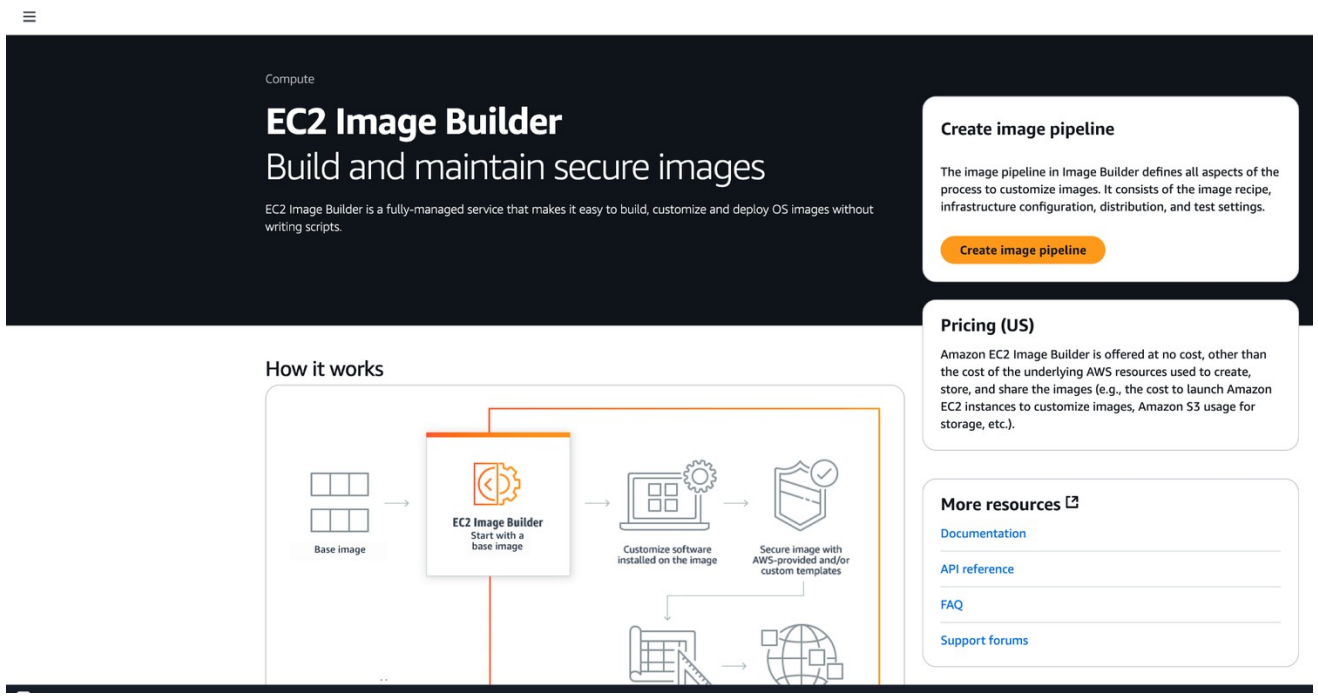
Test suite is run

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# Image Builder




**1** **Mechanism: launch templates…**

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state…

**2** **Failure mode**

Failure mode: configuration drift, unmanaged nodes, stale AMIs and unbounded Run Command concurrency turn a…

**3** **Evidence: inventory coverage…**

Evidence: inventory coverage, patch compliance, command output, instance status checks, application health and…

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# AMI in Production

**1 You can force users to**

You can force users to only launch EC2 instances from pre-approved AMIs (AMIs tagged with specific tags) using IAM policies

IAM Permissions

AWS Config

**2 Combine with AWS Config to**

Combine with AWS Config to find non- compliant EC2 instance (instances launched with non-approved AMIs)

… "Condition": {

NON_COMPLIANT

**3 (not approved)**

(not approved)

"StringEquals": {

EC2 Instance

Mechanism: launch templates, AMIs, instance metadata, IAM roles and Systems Manager define the repeatable state and management channel for a fleet.

**4 (approved)**

(approved)

"ec2:ResourceTag/ Environment": "Prod"

Failure mode: configuration drift, unmanaged nodes, stale AMIs and unbounded Run Command concurrency turn a routine change into fleet-wide impact.

**CURRENT OPERATIONS LENS**

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# Original PPT Coverage Retained — Topic 03

1

### Modernised slides

55 technical slides extracted from the original v4 PowerPoint and rebuilt in the current visual system.

2

### Traceability

Every blended slide names its original v4 source slide number in the kicker.

3

### Current validation

Terminology is normalised; service limits and Region availability remain explicit current-documentation checks.

4

### Procedure boundary

Click-by-click sequences, activities and quizzes are kept out of the deck and remain in the Learner Guide or lab package.


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TOPIC 03 · CONCEPT-LED LAB BRIEF

# Operate an EC2 Fleet with Systems Manager

**LAB 03**

Apply K3, K5, A3 and A5 by onboarding a managed node, using Session Manager and Run Command, assessing patch compliance and executing a controlled Automation runbook.

Build a least-privilege managed-node baseline and produce patch, command and change evidence without exposing inbound administration ports.

### YOU'LL BUILD

An EC2 managed node, SSM document, patch-compliance report, automation execution record and rollback evidence.

### TOOLCHAIN

EC2 · IAM · Systems Manager · Session Manager · Run Command · Patch Manager · Automation

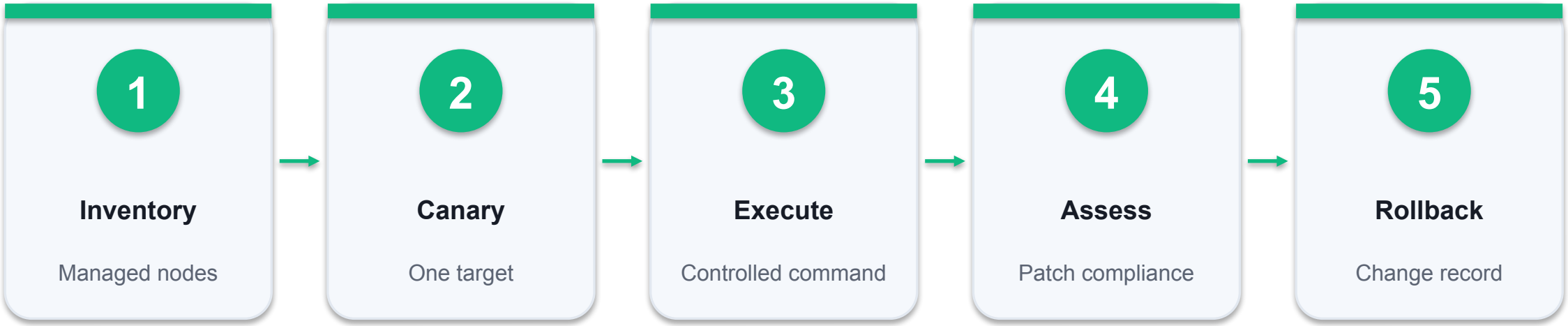
### DONE WHEN

The instance is managed without public SSH, the custom document validates, command and compliance evidence are present, the change record includes verification and rollback, and validation emits LAB 03 PASS.


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LAB 03 · CONCEPTUAL WORKFLOW

# Lab 03 Evidence Model




**PROCEDURE BOUNDARY**

Detailed commands and numbered steps are in the Learner Guide and lab folder—not on these slides.


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LAB 03 · ACCEPTANCE EVIDENCE

# Operate an EC2 Fleet with Systems Manager

## ✅ Expected result

The instance is managed without public SSH, the custom document validates, command and compliance evidence are present, the change record includes verification and rollback, and validation emits LAB 03 PASS.

IF IT DOESN'T WORK

### Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

### Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

### Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.


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# Recap — Compute, Virtualisation and Systems Management

1

**Explain**

Inventory, patch, access and automate consistently across a fleet.

2

**Distinguish**

Build, scan, test and version AMIs instead of hand-configuring servers.

3

**Apply**

K3, K5, A3 and A5 by onboarding a managed node, using Session Manager and Run Command, assessing patch compliance and executing a controlled…

4

**Evidence**

The instance is managed without public SSH, the custom document validates, command and compliance evidence are present, the change record includes verification and…


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04

TOPIC 04 · LO1 · K4 · K5 · A3 · A5

# Reliability, Elasticity and High Availability

Elastic Load Balancing · Auto Scaling · health checks · containers · graceful degradation


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# Key Ideas — Reliability, Elasticity and High Availability

1

### High availability is an end-to-end property

Compute redundancy fails if identity, data or network dependencies remain single points.

2

### Elasticity follows demand

Auto Scaling changes capacity using a measurable signal and bounded policy.

3

### Health checks drive routing

A check must represent the user's ability to receive a correct response.

4

### Resilience requires recovery

Design degradation, retry, failover and rollback before an incident.

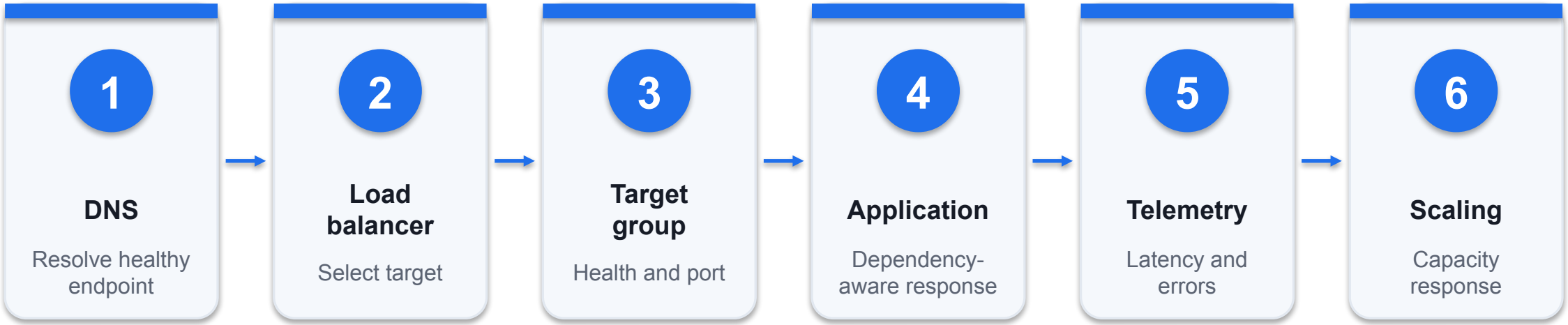

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# Load Balancer Selection

| Type | Layer and strength | Typical workload |
| --- | --- | --- |
| **ALB** | Layer 7 HTTP routing, host/path rules | Web, APIs and containers |
| **NLB** | Layer 4 performance and static IP | TCP/UDP/TLS and extreme scale |
| **GWLB** | Transparent appliance insertion | Firewalls and inspection fleets |


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# Request Path and Health

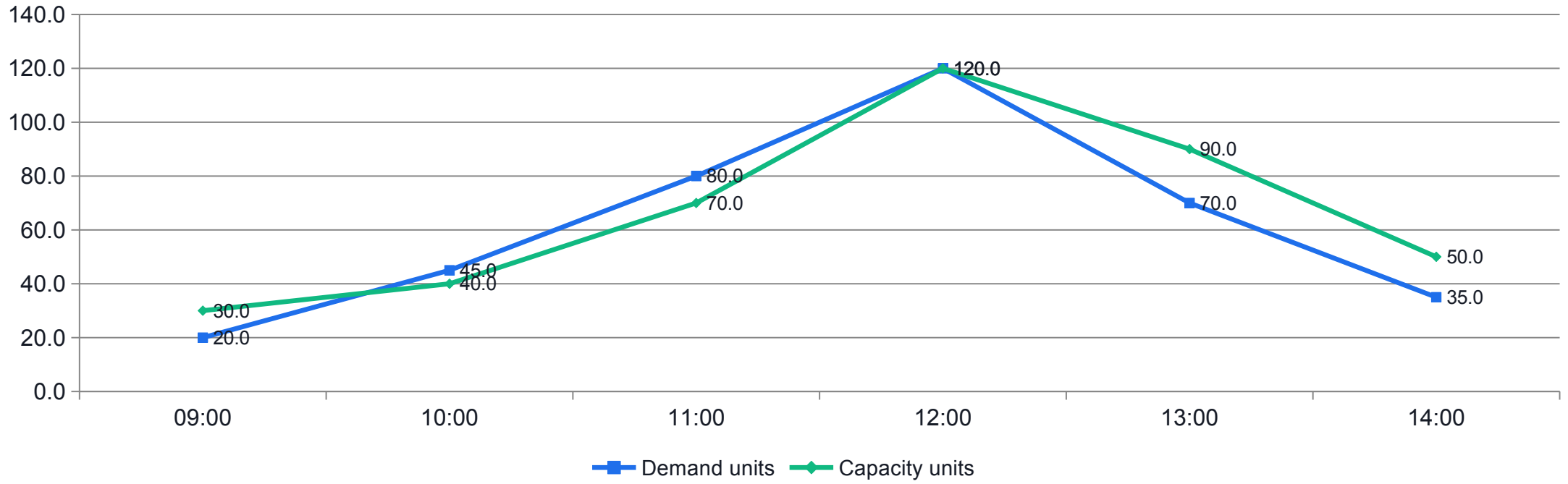



**TROUBLESHOOTING**

Follow the request path in order; do not jump directly to the instance.


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# Scaling Policy Behaviour




**WHAT THE DATA SHOWS**

Target tracking follows demand with delay; warm-up and cooldown prevent unstable oscillation.


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# Auto Scaling Controls

1

## Desired capacity

The current fleet target between minimum and maximum bounds.

2

## Target tracking

Adjusts capacity to hold a metric near a chosen target.

3

## Step scaling

Uses breach magnitude to change capacity by defined steps.

4

## Lifecycle hook

Pauses launch or termination for controlled work.


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# Failure Handling

| Pattern | Response | Caution |
| --- | --- | --- |
| **Retry with backoff** | Recover from transient dependency failure | Bound attempts and add jitter |
| **Circuit breaker** | Stop calling an unhealthy dependency | Define recovery probe |
| **Bulkhead** | Isolate resource pools | Capacity must reflect priority |
| **Graceful degradation** | Preserve critical function | Communicate reduced service |


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# Container Operations in SOA-C03

## ECS / EKS

- Service desired count
- Task or pod health and placement

## Images

- Versioned and scanned
- Immutable deployment input

## Observability

- Container Insights
- Logs, metrics and traces by workload


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Availability Test

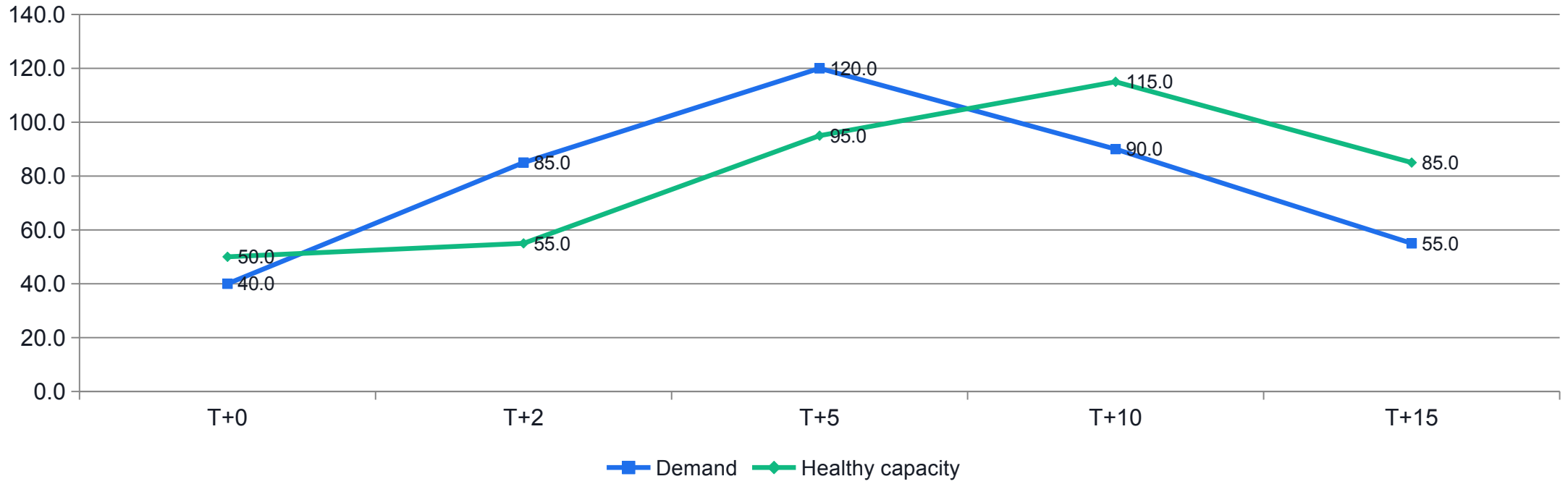
# Redundancy is not resilience until failover is exercised.

Measure detection, decision, traffic shift, recovery, data integrity and return to steady state.


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TOPIC 04 · NATIVE EDITABLE TECHNICAL CHART

# Scaling Response with Warm-Up




**WHAT THE DATA SHOWS**

Warm-up prevents premature scaling decisions, but it also delays usable capacity; validate boot time and health-check behaviour together.


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# Scalability & High Availability

### 1 Scalability means that an…

Scalability means that an application / system can handle greater loads by adapting.

Scalability is linked but different to High Availability

### 2 There are two kinds of

There are two kinds of scalability

Let’s deep dive into the distinction, using a call center as an example

### 3 Vertical Scalability

Vertical Scalability

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

### 4 Horizontal Scalability (=…

Horizontal Scalability (= elasticity)

Failure mode: shallow health checks, incorrect warm-up or coupled dependencies can report healthy capacity while users experience failure.

**CURRENT OPERATIONS LENS**

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# Vertical Scalability

### 1 Vertically scalability means…

Vertically scalability means increasing the size of the instance

RDS, ElastiCache are services that can scale vertically.

### 2 For example, your application runs

For example, your application runs on a t2.micro

There's usually a limit to how much you can vertically scale (hardware limit)

### 3 Scaling that application…

Scaling that application vertically means running it on a t2.large

junior operator

### 4 Vertical scalability is very…

Vertical scalability is very common for non distributed systems, such as a database.

senior operator

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# Horizontal Scalability

### 1 Horizontal Scalability means…

Horizontal Scalability means increasing the number of instances / systems for your application

### 2 Horizontal scaling implies…

Horizontal scaling implies distributed systems.

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

### 3 This is very common for

This is very common for web applications / modern applications

### 4 It’s easy to horizontally scale

It’s easy to horizontally scale thanks the cloud offerings such as Amazon EC2

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# second building in San Francisco

### 1 first building in New York

first building in New York

application / system in at least 2 data centers (== Availability Zones)

### 2 High Availability

High Availability

The goal of high availability is to survive a data center loss

### 3 High Availability usually goes…

High Availability usually goes hand in hand with horizontal scaling

The high availability can be passive (for RDS Multi AZ for example)

### 4 High availability means running…

High availability means running your

The high availability can be active (for horizontal scaling)

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# High Availability & Scalability For EC2

1 **Vertical Scaling**

Vertical Scaling: Increase instance size (= scale up / down)

Auto Scaling Group

Load Balancer multi-AZ

2 **From: t2.nano**

From: t2.nano - 0.5G of RAM, 1 vCPU

Load Balancer

3 **To: u-12tb1.metal**

To: u-12tb1.metal – 12.3 TB of RAM, 448 vCPUs

High Availability: Run instances for the same application across multi-AZ

4 **Horizontal Scaling**

Horizontal Scaling: Increase number of instances (= scale out / in)

Auto Scaling Group multi-AZ

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# What is load balancing?

1

## Elastic Load Balancer

Elastic Load Balancer

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

2

## EC2 Instance

EC2 Instance

Failure mode: shallow health checks, incorrect warm-up or coupled dependencies can report healthy capacity while users experience failure.

3

## Load Balances are servers that

Load Balances are servers that forward traffic to multiple servers (e.g., EC2 instances) downstream

4

## Evidence: request count, target…

Evidence: request count, target response time, HTTP error rate, scaling activities, lifecycle-hook events and recovery time expose the complete…

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# Why use a load balancer?

### 1 Spread load across multiple…

Spread load across multiple downstream instances

Provide SSL termination (HTTPS) for your websites

### 2 Expose a single point of

Expose a single point of access (DNS) to your application

Enforce stickiness with cookies

### 3 Seamlessly handle failures of…

Seamlessly handle failures of downstream instances

High availability across zones

### 4 Do regular health checks to

Do regular health checks to your instances

Separate public traffic from private traffic

**CURRENT OPERATIONS LENS**

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# Why use an Elastic Load Balancer?

### 1 An Elastic Load Balancer is

An Elastic Load Balancer is a managed load balancer

It costs less to setup your own load balancer but it will be a lot more effort on your end

Route 53, AWS WAF, AWS Global Accelerator

### 2 AWS guarantees that it will

AWS guarantees that it will be working

It is integrated with many AWS offerings / services

### 3 AWS takes care of upgrades,

AWS takes care of upgrades, maintenance, high availability

EC2, EC2 Auto Scaling Groups, Amazon ECS

### 4 AWS provides only a few

AWS provides only a few configuration knobs

AWS Certificate Manager (ACM), CloudWatch

**CURRENT OPERATIONS LENS**

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# Health Checks

### 1 Health Checks are crucial for

Health Checks are crucial for Load Balancers

Elastic Load Balancer

### 2 They enable the load balancer

They enable the load balancer to know if instances it forwards traffic to are available to reply to requests

EC2 Instance

### 3 The health check is done

The health check is done on a port and a route (/health is common)

Protocol: HTTP Port: 4567
Endpoint: /health

### 4 If the response is not

If the response is not 200 (OK), then the instance is unhealthy

**CURRENT OPERATIONS LENS**

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# Types of load balancer on AWS

### 1 AWS has 4 kinds of

AWS has 4 kinds of managed Load Balancers

HTTP, HTTPS, WebSocket

Operates at layer 3 (Network layer) - IP Protocol

### 2 Classic Load Balancers (v1

Classic Load Balancers (v1 - old generation) - 2009 - CLB

Network Load Balancer (v2 - new generation) - 2017 - NLB

Overall, it is recommended to use the newer generation load balancers as they provide more features

### 3 HTTP, HTTPS,TCP, SSL (secure TCP)

HTTP, HTTPS,TCP, SSL (secure TCP)

TCP, TLS (secure TCP), UDP

Some load balancers can be setup as internal (private) or external (public) ELB

### 4 Application Load Balancer v2

Application Load Balancer v2 - new generation) - 2016 - ALB

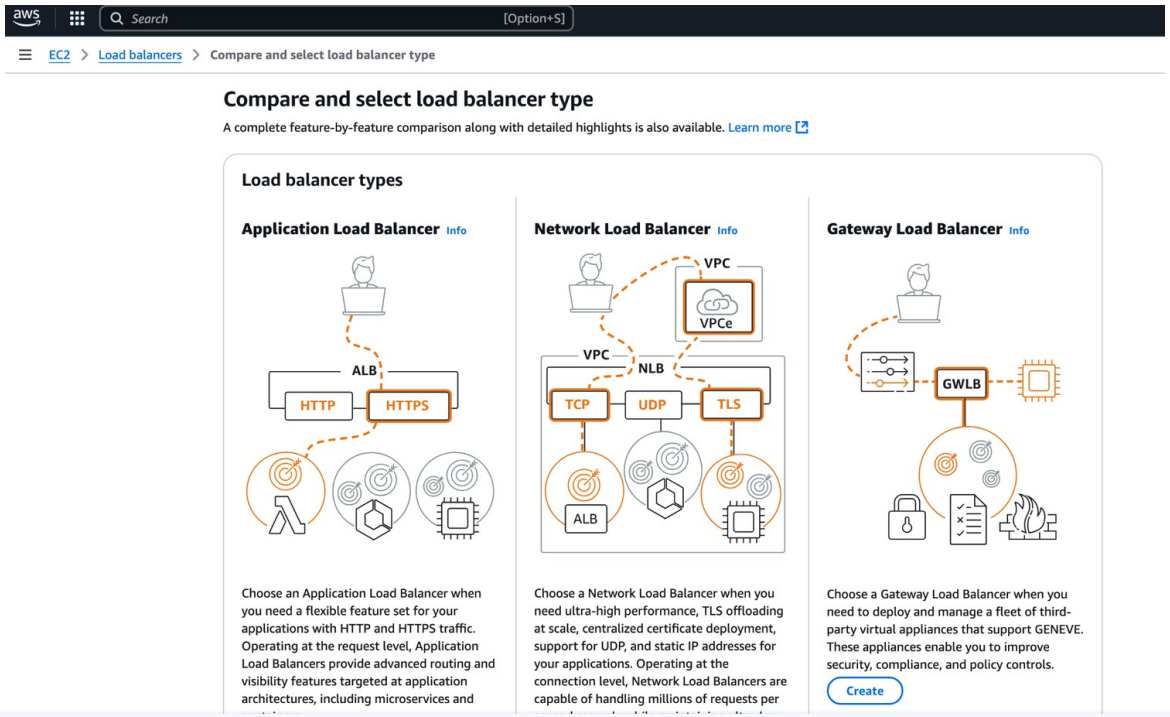
Gateway Load Balancer - 2020 - GWLB

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# Types of load balancer on AWS




**1** **Mechanism: DNS, load-balancer…**

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks…

**2** **Failure mode**

Failure mode: shallow health checks, incorrect warm-up or coupled dependencies can report healthy capacity while…

**3** **Evidence: request count, target…**

Evidence: request count, target response time, HTTP error rate, scaling activities, lifecycle-hook events and…

**CURRENT OPERATIONS LENS**

Evaluate the complete request path, capacity response, health model, failure domain and recovery behaviour—not one component in isolation. Validate service limits and regional availability against current AWS documentation.


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# Users

| Type | Protocol | Port Range | Source | Description |
|---|---|---|---|---|
| HTTP | TCP | 80 | 0.0.0.0/0 | Allow HTTP from an... |
| HTTPS | TCP | 443 | 0.0.0.0/0 | Allow HTTPS from a... |

1

### HTTPS / HTTP

HTTPS / HTTP
Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks…

2

### Failure mode

Failure mode: shallow health checks, incorrect warm-up or coupled dependencies can report healthy capacity while…

3

### Evidence: request count, target…

Evidence: request count, target response time, HTTP error rate, scaling activities, lifecycle-hook events and…

**CURRENT OPERATIONS LENS**

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# Classic Load Balancers (v1)

### 1 Supports TCP (Layer 4), HTTP

Supports TCP (Layer 4), HTTP & HTTPS (Layer 7)

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

### 2 Health checks are TCP or

Health checks are TCP or HTTP based

Failure mode: shallow health checks, incorrect warm-up or coupled dependencies can report healthy capacity while users experience failure.

### 3 Fixed hostname XXX.region.elb.amaz…

Fixed hostname XXX.region.elb.amazonaws.com

Evidence: request count, target response time, HTTP error rate, scaling activities, lifecycle-hook events and recovery time expose the complete…

### 4 Decision point

Decision point: distinguish capacity shortage, unhealthy deployment and downstream dependency failure before changing scaling thresholds.

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# Application Load Balancer (v2)

### 1 Application load balancers is…

Application load balancers is Layer 7 (HTTP)

Support redirects (from HTTP to HTTPS for example)

### 2 Load balancing to multiple HTTP

Load balancing to multiple HTTP applications across machines (target groups)

### 3 Load balancing to multiple…

Load balancing to multiple applications on the same machine (ex: containers)

### 4 Support for HTTP/2 and WebSocket

Support for HTTP/2 and WebSocket

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

**CURRENT OPERATIONS LENS**

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# Application Load Balancer (v2)

### 1 Routing tables to different target

Routing tables to different target groups

ALB are a great fit for micro services & container-based application (example: Docker & Amazon ECS)

### 2 Routing based on path in

Routing based on path in URL (example.com/users & example.com/posts)

Has a port mapping feature to redirect to a dynamic port in ECS

### 3 Routing based on hostname in

Routing based on hostname in URL ( & other.example.com)

In comparison, we'd need multiple Classic Load Balancer per application

### 4 Routing based on Query String,

Routing based on Query String, Headers (example.com/users? id=123=false)

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

**CURRENT OPERATIONS LENS**

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# Target Group for Users application

### 1 Application Load Balancer (v2)…

Application Load Balancer (v2) HTTP Based Traffic

Route /search

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

### 2 External Application Load…

External Application Load Balancer (v2)

Health Check

Failure mode: shallow health checks, incorrect warm-up or coupled dependencies can report healthy capacity while users experience failure.

### 3 Route /user

Route /user

Evidence: request count, target response time, HTTP error rate, scaling activities, lifecycle-hook events and recovery time expose the complete…

### 4 Target Group for Search…

Target Group for Search application

Decision point: distinguish capacity shortage, unhealthy deployment and downstream dependency failure before changing scaling thresholds.

**CURRENT OPERATIONS LENS**

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# Application Load Balancer (v2) Target Groups

### 1 EC2 instances (can be managed

EC2 instances (can be managed by an Auto Scaling Group) - HTTP

ALB can route to multiple target groups

### 2 ECS tasks (managed by ECS itself)

ECS tasks (managed by ECS itself) - HTTP

Health checks are at the target group level

### 3 Lambda functions

Lambda functions - HTTP request is translated into a JSON event

### 4 IP Addresses

IP Addresses - must be private IPs

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

**CURRENT OPERATIONS LENS**

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# Target Group 1

### 1 AWS – EC2 based

AWS – EC2 based

Target Group 2

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

### 2 Application Load Balancer (v2)…

Application Load Balancer (v2)
Query Strings/Parameters Routing

On-premises – Private IP routing

### 3 External Application Load…

External Application Load Balancer (v2)

?Platform=Desktop

Failure mode: shallow health checks, incorrect warm-up or coupled dependencies can report healthy capacity while users experience failure.

### 4 ?Platform=Mobile

?Platform=Mobile

Evidence: request count, target response time, HTTP error rate, scaling activities, lifecycle-hook events and recovery time expose the complete…

**CURRENT OPERATIONS LENS**

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# Application Load Balancer (v2) Good to Know

### 1 Fixed hostname (XXX.region.elb.a ma…

Fixed hostname (XXX.region.elb.amazonaws.com)

Client IP 12.34.56.78

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

### 2 The application servers don’t see

The application servers don’t see the IP of the client directly

Load Balancer IP (Private IP)

### 3 The true IP of the

The true IP of the client is inserted in the header X-Forwarded-For

Connection termination

### 4 We can also get Port

We can also get Port (X-Forwarded-Port) and proto (X-Forwarded-Proto)

Failure mode: shallow health checks, incorrect warm-up or coupled dependencies can report healthy capacity while users experience failure.

**CURRENT OPERATIONS LENS**

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# Network Load Balancer (v2)

1

### Network load balancers (Layer 4)

Network load balancers (Layer 4) allow to

NLB has one static IP per AZ and supports assigning Elastic IP

2

### Forward TCP & UDP traffic

Forward TCP & UDP traffic to your instances

(helpful for whitelisting specific IP)

3

### Handle millions of request per

Handle millions of request per seconds

NLB are used for extreme performance,TCP or UDP traffic

4

### Ultra-low latency

Ultra-low latency

Not included in the AWS free tier

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

**CURRENT OPERATIONS LENS**

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# Network Load Balancer (v2) TCP (Layer 4) Based Traffic

### 1 Target Group for Users application

Target Group for Users application

Health Check

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

### 2 External Network Load Balancer…

External Network Load Balancer (v2)

Failure mode: shallow health checks, incorrect warm-up or coupled dependencies can report healthy capacity while users experience failure.

### 3 TCP + Rules

TCP + Rules

Evidence: request count, target response time, HTTP error rate, scaling activities, lifecycle-hook events and recovery time expose the complete…

### 4 Target Group for Search…

Target Group for Search application

Decision point: distinguish capacity shortage, unhealthy deployment and downstream dependency failure before changing scaling thresholds.

**CURRENT OPERATIONS LENS**

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# Network Load Balancer - Target Groups

### 1 EC2 Instances

EC2 Instances

i-1234567890abcdef0
i-1234567890abcdef0

192.168.1.118

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

### 2 IP Addresses

IP Addresses - must be private IPs

Target Group

Failure mode: shallow health checks, incorrect warm-up or coupled dependencies can report healthy capacity while users experience failure.

### 3 Application Load Balancer

Application Load Balancer

Load Balancer

Evidence: request count, target response time, HTTP error rate, scaling activities, lifecycle-hook events and recovery time expose the complete…

### 4 Health Checks support the TCP,

Health Checks support the TCP, HTTP, HTTPS Protocols

(IP Addresses)

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# Gateway Load Balancer

1

**Deploy, scale, and manage a**

Deploy, scale, and manage a fleet of 3rd party network virtual appliances in AWS

Combines the following functions

Application

2

**Example: Firewalls, Intrusion…**

Example: Firewalls, Intrusion Detection and

Transparent Network Gateway - single entry/exit for all traffic

(destination)

3

**Prevention Systems, Deep Packet…**

Prevention Systems, Deep Packet Inspection Systems, payload manipulation, …

Load Balancer - distributes traffic to your virtual appliances

Target Group

4

**Operates at Layer 3 (Network…**

Operates at Layer 3 (Network Layer) – IP Packets

Uses the GENEVE protocol on port 60

3rd Party Security Virtual Appliances

**CURRENT OPERATIONS LENS**

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# Gateway Load Balancer - Target Groups

### 1 EC2 Instances

EC2 Instances

Gateway Load Balancer

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

### 2 IP Addresses

IP Addresses - must be private IPs

(IP Addresses)

Failure mode: shallow health checks, incorrect warm-up or coupled dependencies can report healthy capacity while users experience failure.

### 3 i-1234567890abcdef0…

i-1234567890abcdef0
i-1234567890abcdef0

192.168.1.118

Evidence: request count, target response time, HTTP error rate, scaling activities, lifecycle-hook events and recovery time expose the complete…

### 4 Target Group

Target Group

Decision point: distinguish capacity shortage, unhealthy deployment and downstream dependency failure before changing scaling thresholds.

**CURRENT OPERATIONS LENS**

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# Sticky Sessions (Session Affinity)

### 1 It is possible to implement

It is possible to implement stickiness so that the

For both CLB & ALB, the “cookie” used for stickiness has an expiration date you control

### 2 same client is always redirected

same client is always redirected to the same instance behind a load balancer

Use case: make sure the user doesn’t lose his session data

### 3 This works for Classic Load

This works for Classic Load Balancer

Enabling stickiness may bring imbalance to the load over the backend EC2 instances

### 4 Application Load Balancer, and…

Application Load Balancer, and Network Load Balancer

EC2 Instance

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

**CURRENT OPERATIONS LENS**

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# Sticky Sessions – Cookie Names

### 1 Application based cookies

Application based cookies

Cookie name must be specified individually for each target group

Cookie name is AWSALBAPP

### 2 Custom cookie

Custom cookie

Don’t use AWSALB, AWSALBAPP or AWSALBTG reserved for use by the ELB)

Duration based cookies

### 3 Generated by the target

Generated by the target

Application cookie

Cookie generated by the load balancer

### 4 Can include any custom attributes

Can include any custom attributes required by the application

Generated by the load balancer

Cookie name is AWSALB for ALB, AWSELB for CLB

**CURRENT OPERATIONS LENS**

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# Cross-Zone Load Balancing

1

### Availability Zone 1

Availability Zone 1

Without Cross Zone Load Balancing

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

2

### Availability Zone 2

Availability Zone 2

Requests are distributed in the instances of the node of the Elastic Load Balancer

3

### With Cross Zone Load Balancing

With Cross Zone Load Balancing

Failure mode: shallow health checks, incorrect warm-up or coupled dependencies can report healthy capacity while users experience failure.

4

### each load balancer instance…

each load balancer instance distributes evenly across all registered instances in all AZ

**CURRENT OPERATIONS LENS**

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# Cross-Zone Load Balancing

### 1 Application Load Balancer

Application Load Balancer

Disabled by default

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

### 2 Enabled by default (can be

Enabled by default (can be disabled at the Target Group level)

You pay charges ($) for inter AZ data if enabled

### 3 No charges for inter AZ

No charges for inter AZ data

Classic Load Balancer

Failure mode: shallow health checks, incorrect warm-up or coupled dependencies can report healthy capacity while users experience failure.

### 4 Network Load Balancer $ Gateway

Network Load Balancer $ Gateway Load Balancer

No charges for inter AZ data if enabled

**CURRENT OPERATIONS LENS**

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# SSL/TLS - Basics

### 1 An SSL Certificate allows traffic

An SSL Certificate allows traffic between your clients and your load balancer to be encrypted in transit (in-flight encryption)

Public SSL certificates are issued by Certificate Authorities (CA)

### 2 SSL refers to Secure Sockets

SSL refers to Secure Sockets Layer, used to encrypt connections

Comodo, Symantec, GoDaddy, GlobalSign, Digicert, Letsencrypt, etc…

### 3 TLS refers to Transport Layer

TLS refers to Transport Layer Security, which is a newer version

SSL certificates have an expiration date (you set) and must be renewed

### 4 Nowadays, TLS certificates are…

Nowadays, TLS certificates are mainly used, but people still refer as SSL

**CURRENT OPERATIONS LENS**

Evaluate the complete request path, capacity response, health model, failure domain and recovery behaviour—not one component in isolation. Validate service limits and regional availability against current AWS documentation.


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# The load balancer uses an X.509 certificate (SSL/TLS server certificate)

### 1 You can manage certificates using

You can manage certificates using ACM (AWS Certificate Manager)

You can add an optional list of certs to support multiple domains

Over private VPC

### 2 You can create upload your

You can create upload your own certificates alternatively

Cilents can use SNI (Server Name Indication) to specify the hostname they reach

Load Balancer - SSL Certificates

### 3 HTTPS listener

HTTPS listener

Ability to specify a security policy to support older versions of SSL / TLS (legacy clients)

LOAD BALANCER

### 4 You must specify a default

You must specify a default certificate

HTTPS (encrypted) Over www

**CURRENT OPERATIONS LENS**

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# SSL - Server Name Indication (SNI)

1 **SNI solves the problem of**

SNI solves the problem of loading multiple SSL certificates onto multiple web servers (to serve multiple websites)

Only works for ALB & NLB (newer generation), CloudFront

I would like www.mycorp.com

2 **It's a "newer" protocol, and**

It's a "newer" protocol, and requires the client

Does not work for CLB (older gen)

Target group for www.mycorp.com

3 **to indicate the hostname of**

to indicate the hostname of the target server in the initial SSL handshake

SSL Cert: Domain1.example.com

Target group for Domain1.example.com

4 **The server will then find**

The server will then find the correct certificate, or return the default one

SSL Cert: www.mycorp.com

Use the correct SSL cert

**CURRENT OPERATIONS LENS**

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# Elastic Load Balancers - SSL Certificates

### 1 Classic Load Balancer (v1)

Classic Load Balancer (v1)

Supports multiple listeners with multiple SSL certificates

### 2 Support only one SSL certificate

Support only one SSL certificate

Uses Server Name Indication (SNI) to make it work

### 3 Must use multiple CLB for

Must use multiple CLB for multiple hostname with multiple SSL certificates

Network Load Balancer (v2)

### 4 Application Load Balancer (v2)

Application Load Balancer (v2)

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

**CURRENT OPERATIONS LENS**

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# Connection Draining

### 1 Feature naming

Feature naming

Stops sending new requests to the EC2 instance which is de-registering

waiting for existing

### 2 Connection Draining

Connection Draining – for CLB

Between 1 to 3600 seconds (default: 300 seconds)

connections to complete

### 3 Deregistration Delay

Deregistration Delay – for ALB & NLB

Can be disabled (set value to 0)

new connections

### 4 Time to complete “in-flight…

Time to complete “in-flight requests” while the instance is de-registering or unhealthy

Set to a low value if your requests are short

established to all other instances

**CURRENT OPERATIONS LENS**

Evaluate the complete request path, capacity response, health model, failure domain and recovery behaviour—not one component in isolation. Validate service limits and regional availability against current AWS documentation.


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# ELB Health Checks

### 1 Target Health Status

Target Health Status

unavailable: health checks disabled

Health Check

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

### 2 Initial: registering the target

Initial: registering the target

If a group contains only unhealthy targets, ELB routes requests across its unhealthy targets

Description

### 3 Unused: target is not registered

Unused: target is not registered

Target Group

HealthCheckProtocol

Failure mode: shallow health checks, incorrect warm-up or coupled dependencies can report healthy capacity while users experience failure.

### 4 Draining: de-registering the…

Draining: de-registering the target

Application Load Balancer

Protocol used to perform health checks

**CURRENT OPERATIONS LENS**

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# Load Balancer Error codes

### 1 Successful request

Successful request : Code 200.

Error 403 : Forbidden

An error 500 / Internal server error would mean some error on the ELB itself.

### 2 Unsuccessful at client side

Unsuccessful at client side : 4XX code.

Error 460 : Client closed connection.

Error 502 : Bad Gateway

### 3 Error 400

Error 400 : Bad Request

Error 463 : X-Forwarded For header with >30 IP (Similar to malformed request).

An error 503/ Service Unavailable

### 4 Error 401

Error 401 : Unauthorized

Unsuccessful at server side : 5xx code.

Error 504 / Gateway timeout : probably an issue within the server.

**CURRENT OPERATIONS LENS**

Evaluate the complete request path, capacity response, health model, failure domain and recovery behaviour—not one component in isolation. Validate service limits and regional availability against current AWS documentation.


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# Load Balancers Monitoring

### 1 All Load Balancer metrics are

All Load Balancer metrics are directly pushed to CloudWatch metrics

HTTPCode_Backend_3XX, redirected request

RequestCount

### 2 BackendConnectionErrors

BackendConnectionErrors

HTTPCode_ELB_4XX: Client error codes

ReuqestCountPerTarget

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

### 3 HealthyHostCount / UnHealthyHostCo…

HealthyHostCount / UnHealthyHostCount

HTTPCode_ELB_5XX

SurgeQueueLength: the total

Failure mode: shallow health checks, incorrect warm-up or coupled dependencies can report healthy capacity while users experience failure.

### 4 HTTPCode_Backend_2XX: Successful…

HTTPCode_Backend_2XX: Successful request.

Server error codes generated by the load balancer.

number of requests (HTTP listener) or connections (TCP

**CURRENT OPERATIONS LENS**

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# Load Balancer troubleshooting using metrics

### 1 HTTP 400

HTTP 400: BAD_REQUEST=> The client sent a malformed request that does not meet HTTP specifications.

instances are enabled and make sure that the keep-alive timeout is greater than the idle timeout settings of load balancer.

### 2 HTTP 503

HTTP 503: Service Unavailable => Ensure that you have healthy instances in

Set alarms & look at the documentation for troubleshooting: message.html

### 3 every Availability Zone that your

every Availability Zone that your load balancer is configured to respond in. Look for HealthyHostCount in CloudWatch

### 4 HTTP 504

HTTP 504: Gateway Timeout => Check if keep-alive settings on your EC2

**CURRENT OPERATIONS LENS**

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# Load Balancers Access Logs

### 1 Access logs from Load Balancers

Access logs from Load Balancers can be stored in S3 and contain

Only pay for the S3 storage

### 2 Client IP address

Client IP address

Helpful for compliance reason

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

### 3 Request paths

Request paths

Helpful for keeping access data even after ELB or EC2 instances are terminated

### 4 Server response

Server response

Access Logs are already encrypted

Failure mode: shallow health checks, incorrect warm-up or coupled dependencies can report healthy capacity while users experience failure.

**CURRENT OPERATIONS LENS**

Evaluate the complete request path, capacity response, health model, failure domain and recovery behaviour—not one component in isolation. Validate service limits and regional availability against current AWS documentation.


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# Application Load Balancer Request Tracing

### 1 Request tracing

Request tracing – Each HTTP request has an added custom header

Application Load Balancer is not (yet) integrated with X-Ray

### 2 'X-Amzn-Trace-Id'

'X-Amzn-Trace-Id'

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

### 3 X-Amzn-Trace-Id: Root=1-67891233-a…

X-Amzn-Trace-Id: Root=1-67891233-abcdef012345678912345678

Failure mode: shallow health checks, incorrect warm-up or coupled dependencies can report healthy capacity while users experience failure.

### 4 This is useful in logs

This is useful in logs / distributed tracing platform to track a single request

**CURRENT OPERATIONS LENS**

Evaluate the complete request path, capacity response, health model, failure domain and recovery behaviour—not one component in isolation. Validate service limits and regional availability against current AWS documentation.


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# Target Groups Settings

### 1 Deregistration_delay.timeout_secon…

Deregistration_delay.timeout_seconds time the load balancer waits before deregistering a target

Stickiness.type: application-based or duration-based cookie

### 2 slow_start.duration_seconds: (see…

slow_start.duration_seconds: (see next slide)

Stickiness.app_cookie.cookie_name: name of the application cookie

### 3 load_balancing.algorithm.type: ow…

load_balancing.algorithm.type: ow the load balancer selects targets when routing requests (Round Robin, Least Outstanding Requests)

Stickiness.app_cookie.duration_seconds: application-based cookie expiration period

### 4 Stickiness.enabled

Stickiness.enabled

Stickiness.lb_cookie.duration_seconds: duration-based cookie expiration period

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

**CURRENT OPERATIONS LENS**

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# Slow Start Mode

1

**By default, a target receives**

By default, a target receives its full share of requests once it's registered with the target group

The duration period elapses

without Slow Start mode

2

**slow start mode gives healthy**

slow start mode gives healthy targets time to warm-up before the load balancer sends them a full share of requests

The target becomes unhealthy

with Slow Start mode

3

**The load balancer linearly…**

The load balancer linearly increases the number of requests that it sends to the target

To disable, set slow start duration value to 0

EC2 Instance EC2 Instance

4

**A target exits Slow Start**

A target exits Slow Start Mode when

Slow Start duration

Target Group

**CURRENT OPERATIONS LENS**

Evaluate the complete request path, capacity response, health model, failure domain and recovery behaviour—not one component in isolation. Validate service limits and regional availability against current AWS documentation.


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# Request Routing Algorithms - Least Outstanding Requests

### 1 The next instance to receive

The next instance to receive the request is the instance that has the lowest number of pending/unfinished requests

(HTTP/HTTPS Listener)

### 2 Works with Application Load…

Works with Application Load Balancer and Classic Load Balancer (HTTP/HTTPS)

### 3 EC2 Instance

EC2 Instance

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

### 4 Registered Instances

Registered Instances

Failure mode: shallow health checks, incorrect warm-up or coupled dependencies can report healthy capacity while users experience failure.

**CURRENT OPERATIONS LENS**

Evaluate the complete request path, capacity response, health model, failure domain and recovery behaviour—not one component in isolation. Validate service limits and regional availability against current AWS documentation.


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# Request Routing Algorithms - Round Robin

### 1 Equally choose the targets from

Equally choose the targets from the target group

Target Group (ALB) / Registered Instances (CLB)

### 2 Works with Application Load…

Works with Application Load Balancer and Classic Load Balancer (TCP)

### 3 CLB (TCP Listener)

CLB (TCP Listener)

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

### 4 EC2 Instance

EC2 Instance

Failure mode: shallow health checks, incorrect warm-up or coupled dependencies can report healthy capacity while users experience failure.

**CURRENT OPERATIONS LENS**

Evaluate the complete request path, capacity response, health model, failure domain and recovery behaviour—not one component in isolation. Validate service limits and regional availability against current AWS documentation.


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# Request Routing Algorithms - Flow Hash

### 1 Selects a target based on

Selects a target based on the protocol, source/destination IP address, source/destination port, and TCP sequence number

Flow Hash Algorithm

Target Group

### 2 Each TCP/UDP connection is routed

Each TCP/UDP connection is routed to a single target for the life of the connection

Source & destination IP
Source & destination Port TCP sequence no.

### 3 Works with Network Load Balancer

Works with Network Load Balancer

Hash 8743b…

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

### 4 Network Load Balancer

Network Load Balancer

EC2 Instance

Failure mode: shallow health checks, incorrect warm-up or coupled dependencies can report healthy capacity while users experience failure.

**CURRENT OPERATIONS LENS**

Evaluate the complete request path, capacity response, health model, failure domain and recovery behaviour—not one component in isolation. Validate service limits and regional availability against current AWS documentation.


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# ALB - Listener Rules

### 1 Processed in order (with Default

Processed in order (with Default Rule)

http-request-method

Application Load Balancer

### 2 Supported Actions (forward…

Supported Actions (forward, redirect, fixed-response)

path-pattern

Rule (Default)

Mechanism: DNS, load-balancer health, target-group health, Auto Scaling policy and application dependency checks cooperate to keep capacity available.

### 3 Rule Conditions

Rule Conditions

http-header

Target Group 2

Failure mode: shallow health checks, incorrect warm-up or coupled dependencies can report healthy capacity while users experience failure.

### 4 host-header

host-header

query-string

Target Group 3

Evidence: request count, target response time, HTTP error rate, scaling activities, lifecycle-hook events and recovery time expose the complete…

**CURRENT OPERATIONS LENS**

Evaluate the complete request path, capacity response, health model, failure domain and recovery behaviour—not one component in isolation. Validate service limits and regional availability against current AWS documentation.


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# Original PPT Coverage Retained — Topic 04

1

### Modernised slides

45 technical slides extracted from the original v4 PowerPoint and rebuilt in the current visual system.

2

### Traceability

Every blended slide names its original v4 source slide number in the kicker.

3

### Current validation

Terminology is normalised; service limits and Region availability remain explicit current-documentation checks.

4

### Procedure boundary

Click-by-click sequences, activities and quizzes are kept out of the deck and remain in the Learner Guide or lab package.


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TOPIC 04 · CONCEPT-LED LAB BRIEF

# Deploy and Test a Multi-AZ Highly Available Service

**LAB 04**

Apply K4, K5, A3 and A5 by deploying an ALB and Auto Scaling service across two Availability Zones, testing health-based recovery and recording a controlled change.

Deploy a small Multi-AZ web tier, verify health and scaling signals, then run a bounded failure exercise.

### YOU'LL BUILD

A two-AZ VPC web tier, ALB, target group, launch template, Auto Scaling group, failover evidence and cleanup record.

### TOOLCHAIN

VPC · EC2 · ALB · Auto Scaling · CloudWatch · CloudFormation

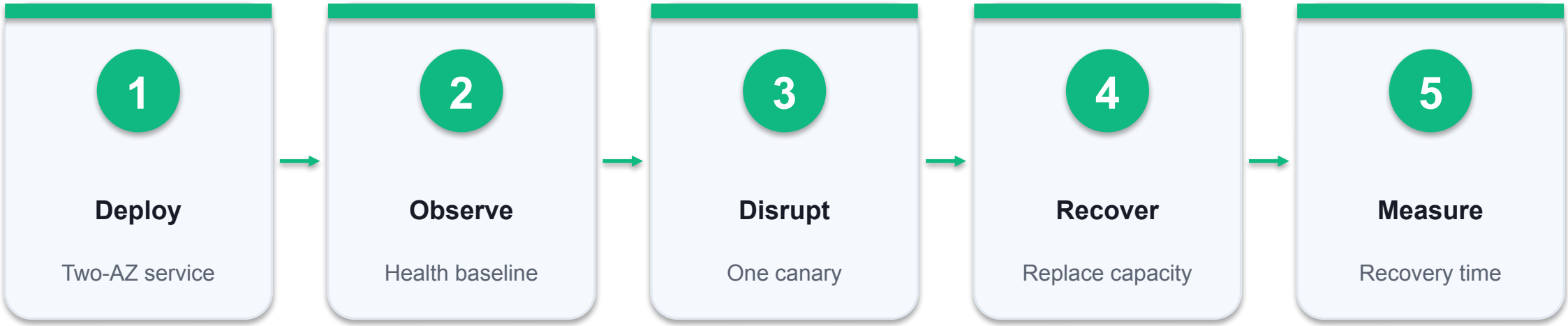
### DONE WHEN

The service spans two Availability Zones, returns a healthy response during one-instance replacement, recovers desired capacity, records measured recovery time, and validation emits LAB 04 PASS.


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LAB 04 · CONCEPTUAL WORKFLOW

# Lab 04 Evidence Model




**PROCEDURE BOUNDARY**

Detailed commands and numbered steps are in the Learner Guide and lab folder—not on these slides.


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LAB 04 · ACCEPTANCE EVIDENCE

# Deploy and Test a Multi-AZ Highly Available Service

## ✅ Expected result

The service spans two Availability Zones, returns a healthy response during one-instance replacement, recovers desired capacity, records measured recovery time, and validation emits LAB 04 PASS.

IF IT DOESN'T WORK

**Input mismatch**

Reconcile identifiers, required fields and units with the supplied data dictionary.

**Validation stop**

Inspect the named rule and source row; preserve the check rather than bypassing it.

**Evidence differs**

Check the denominator, time window and assumptions before rerunning from clean inputs.


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# Recap — Reliability, Elasticity and High Availability

1

**Explain**

Compute redundancy fails if identity, data or network dependencies remain single points.

2

**Distinguish**

Auto Scaling changes capacity using a measurable signal and bounded policy.

3

**Apply**

K4, K5, A3 and A5 by deploying an ALB and Auto Scaling service across two Availability Zones, testing health-based recovery and recording a…

4

**Evidence**

The service spans two Availability Zones, returns a healthy response during one-instance replacement, recovers desired capacity, records measured recovery time, and…


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05

TOPIC 05 · LO1 · LO4 · K4 · A1 · A4 · A6

# Backup, Restore and Business Continuity

RTO/RPO · AWS Backup · snapshots · replication · pilot light · warm standby


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# Key Ideas — Backup, Restore and Business Continuity

1

## Backup is not recovery

A protected copy has value only when restore time and data integrity are proven.

2

## RTO and RPO drive design

Recovery time and acceptable data loss determine technology and cost.

3

## Cross-account protection limits blast radius

Separate vault ownership and controls protect against compromised workloads.

4

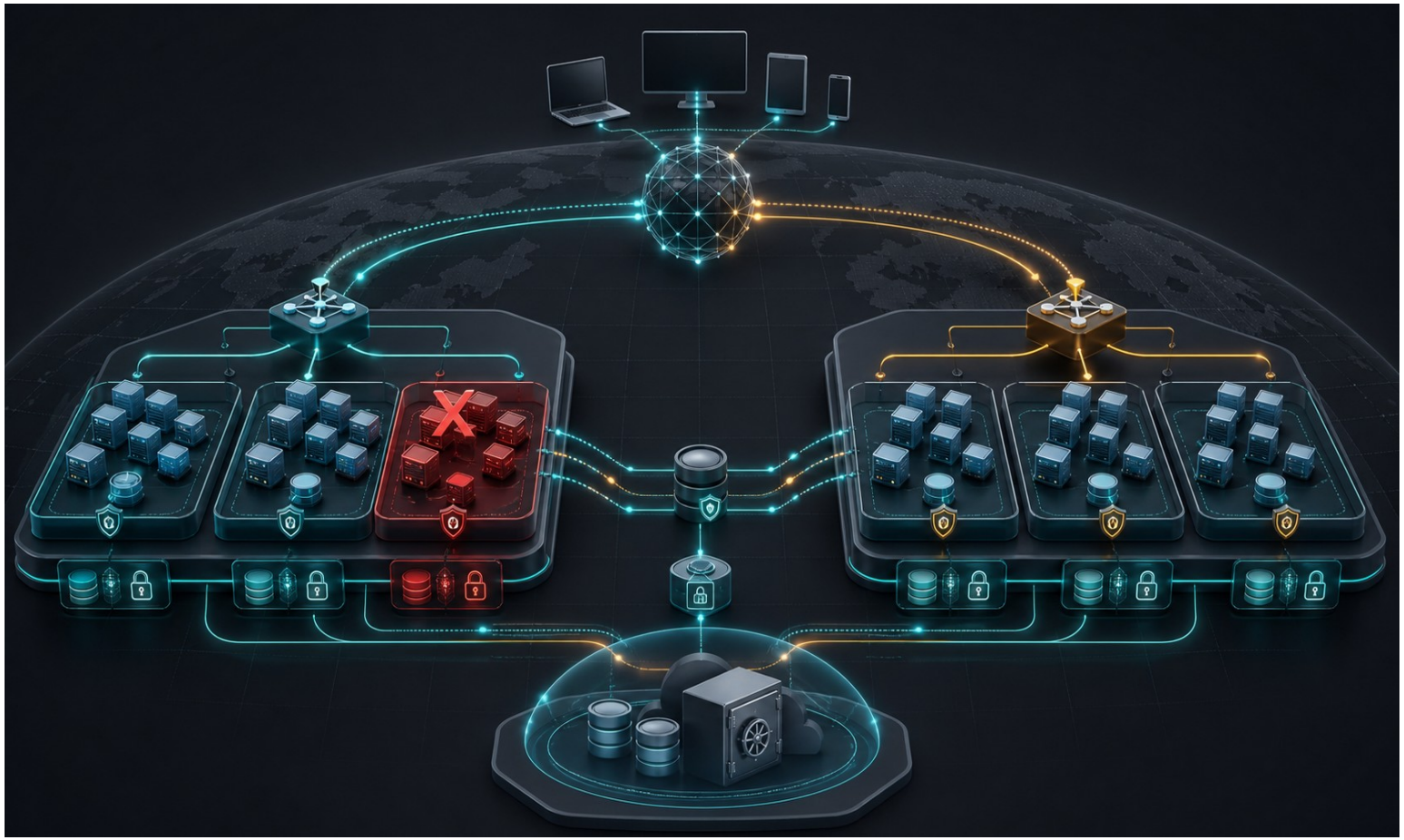
## Continuity is a rehearsed capability

People, runbooks, dependencies and evidence matter as much as replication.


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IMAGEGEN VISUAL · TOPIC 05

# Resilience Is a Tested Capability



READ THE SYSTEM

Identify independent failure domains, recovery paths and the evidence needed to prove continuity.


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# Air-Gapped Recovery Is an Operating Capability



READ THE SYSTEM

Separate production compromise from protected recovery points and an isolated restore environment; verify access, retention, restore time and application integrity.

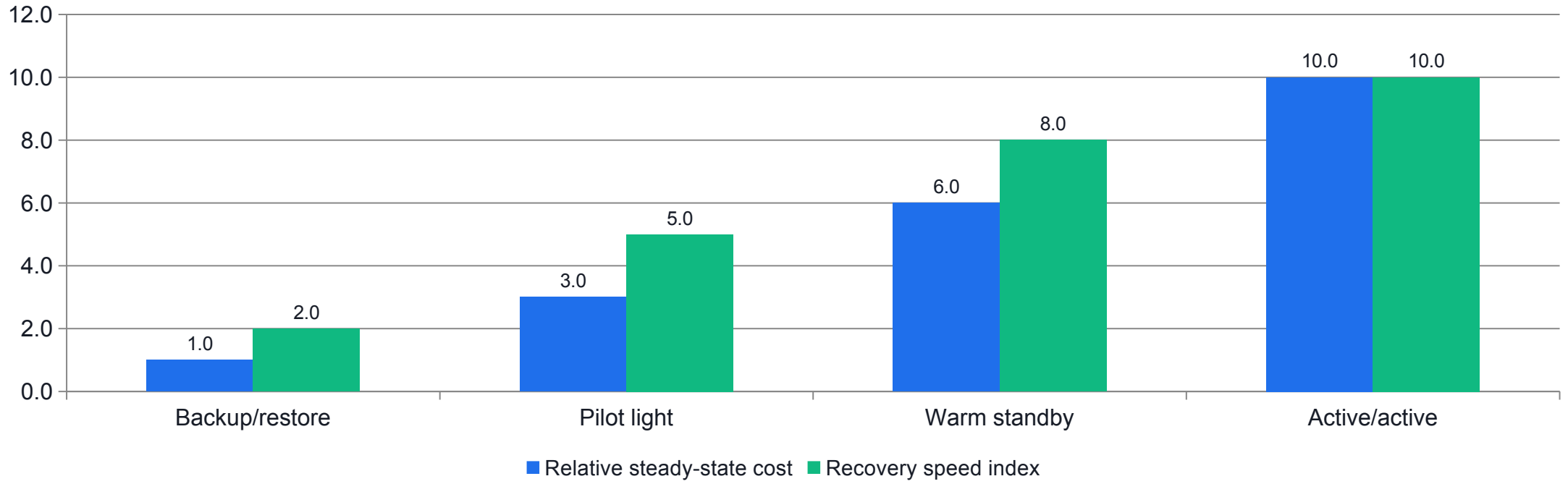

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# RTO and RPO

| Objective | Question | Design influence |
| --- | --- | --- |
| **RTO** | How quickly must service return? | Automation, standby capacity and failover |
| **RPO** | How much recent data may be lost? | Backup frequency and replication |
| **Retention** | How far back must recovery reach? | Lifecycle, vault and compliance |
| **Integrity** | How do we know restored data is correct? | Checksums and application validation |


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# Continuity Patterns Trade Cost for Recovery

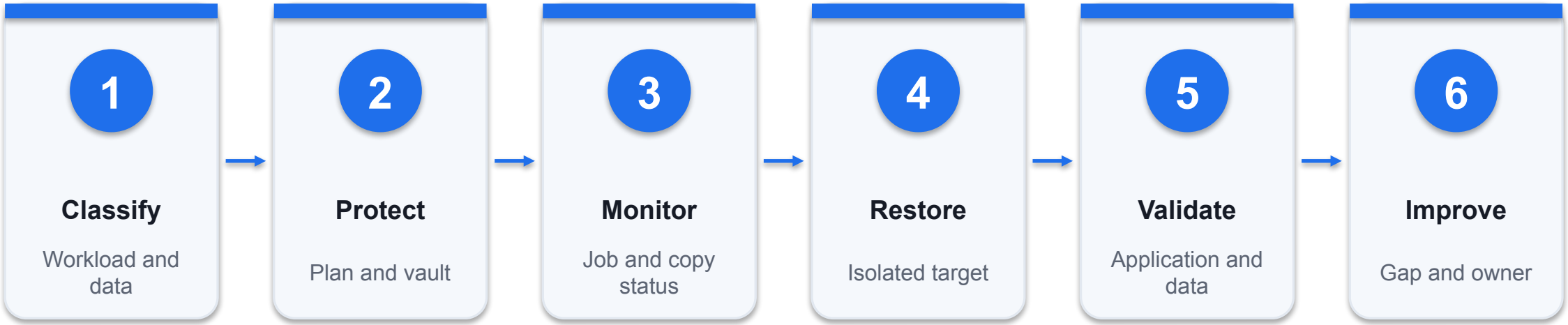



**WHAT THE DATA SHOWS**

Choose the least costly pattern that demonstrably meets business RTO, RPO and compliance requirements.


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# Recovery Assurance Loop




**A6 EVIDENCE**

A completed backup job is an input; a tested restore with measured recovery is evidence.


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# AWS Backup Controls

1

**Plan**

Schedules, windows, lifecycle and copy actions.

2

**Vault**

Access policy, encryption and logical separation.

3

**Vault Lock**

Retention controls designed to resist early deletion.

4

**Audit Manager**

Evidence against backup policy and compliance frameworks.


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# Replication Choices

| Mechanism | Strength | Operational concern |
| --- | --- | --- |
| **EBS snapshot copy** | Point-in-time regional protection | Restore and launch workflow |
| **S3 replication** | Object-level asynchronous copy | Existing objects and delete markers |
| **RDS cross-Region replica** | Lower RPO read copy | Promotion and endpoint change |
| **DynamoDB global table** | Multi-Region active data | Conflict and application behaviour |


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# Continuity Runbook

## Decision

- Declared trigger and authority
- Communication path

## Execution

- Dependency order
- Automated and manual steps

## Evidence

- Timestamps and validation
- Actual RTO/RPO and gaps


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Recovery Principle

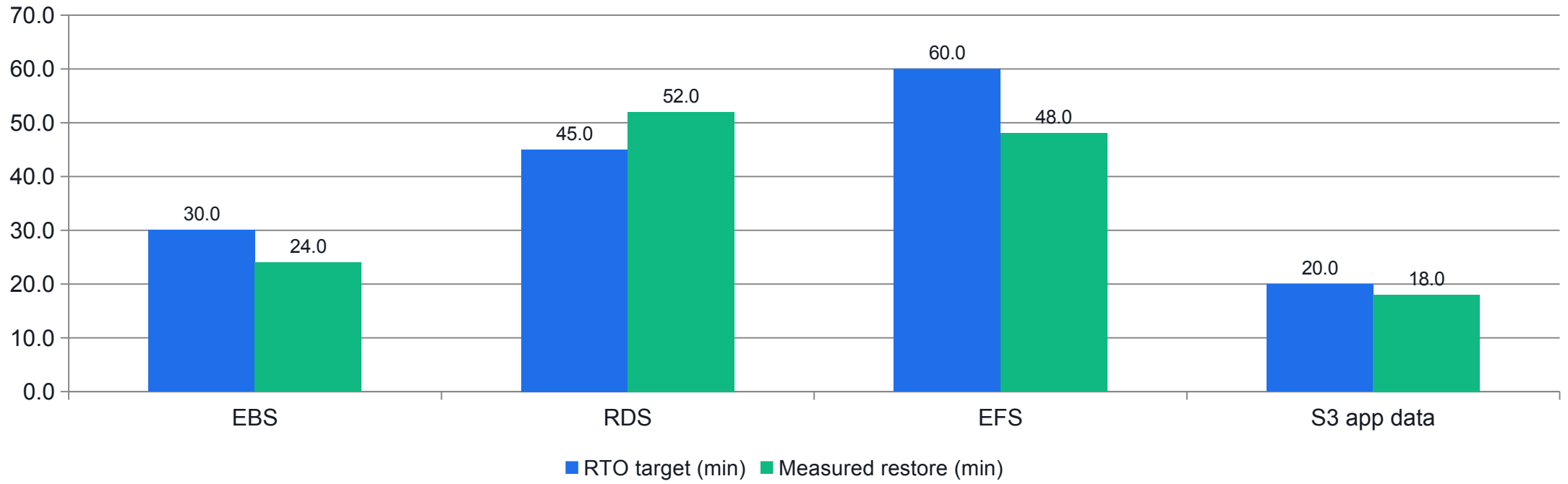
# A restore you have not tested is only a hypothesis.

Schedule recovery exercises, isolate tests from production, measure actual results and close every gap with an owner.


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TOPIC 05 · NATIVE EDITABLE TECHNICAL CHART

# Recovery Objectives versus Test Results




**WHAT THE DATA SHOWS**

A passed backup can still miss the service objective: the RDS test exceeds RTO and requires a corrective action with an owner.


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# AWS DataSync

### 1 Move large amount of data

Move large amount of data to and from

Amazon S3 (any storage classes – including Glacier)

File permissions and metadata are preserved (NFS POSIX, SMB…)

### 2 On-premises / other cloud to

On-premises / other cloud to AWS (NFS, SMB, HDFS, S3 API…) - needs agent

Amazon EFS

One agent task can use 10 Gbps, can setup a bandwidth limit

### 3 AWS to AWS (different storage…

AWS to AWS (different storage services) – no agent needed

Amazon FSx (Windows, Lustre, NetApp, OpenZFS...)

### 4 Can synchronize to

Can synchronize to

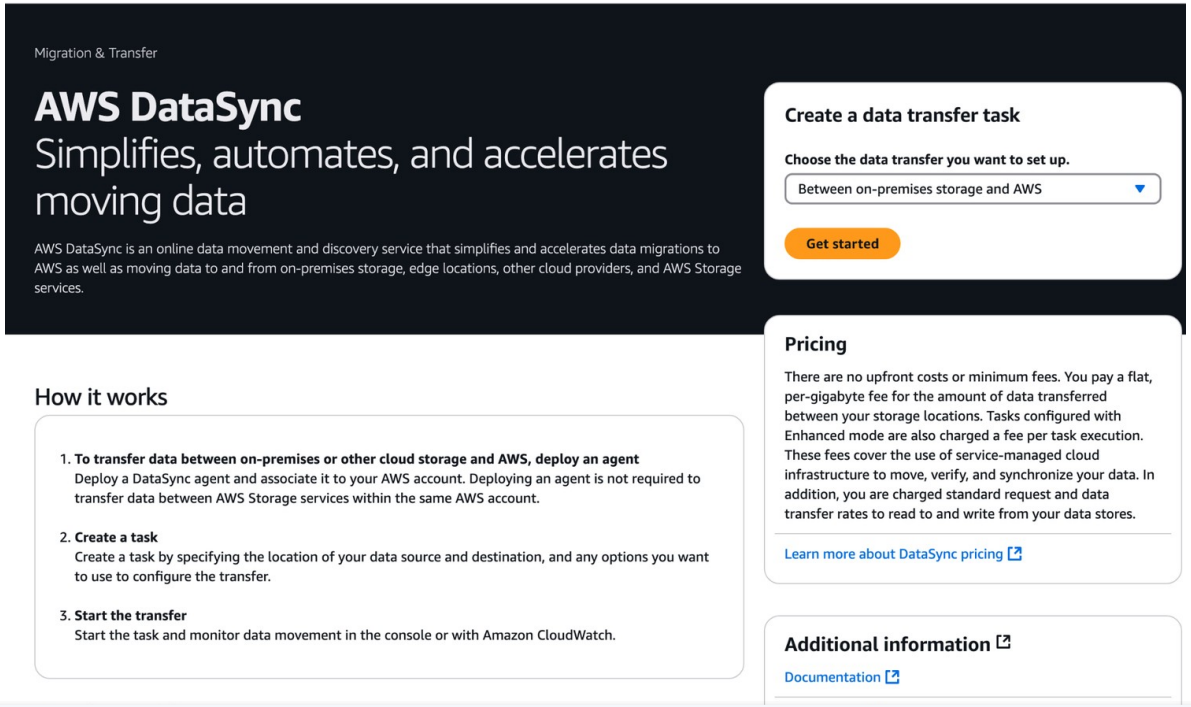
Replication tasks can be scheduled hourly, daily, weekly

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# AWS DataSync




**1 Mechanism: backup plans, vault…**

Mechanism: backup plans, vault policies, Vault Lock, cross-account copies and restore-testing plans separate…

**2 Failure mode**

Failure mode: an immutable recovery point can still be unusable when keys, IAM roles, network dependencies…

**3 Evidence: job completion…**

Evidence: job completion, recovery-point age, restore duration, validation workflow, data checksum and actual…

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# AWS DataSync

### 1 NFS / SMB to AWS

NFS / SMB to AWS (S3, EFS, FSx…)

AWS Storage Resources

S3 One Zone-IA

### 2 On-Premises

On-Premises

S3 Standard

S3 Glacier

Mechanism: backup plans, vault policies, Vault Lock, cross-account copies and restore-testing plans separate protection from operational recovery.

### 3 NFS or SMB

NFS or SMB

S3 Intelligent- Tiering

S3 Glacier Deep Archive

### 4 AWS DataSync Agent

AWS DataSync Agent

S3 Standard-IA

Amazon FSx

Failure mode: an immutable recovery point can still be unusable when keys, IAM roles, network dependencies, metadata or application validation are…

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# AWS DataSync

### 1 Transfer between AWS storage…

Transfer between AWS storage services

Mechanism: backup plans, vault policies, Vault Lock, cross-account copies and restore-testing plans separate protection from operational recovery.

### 2 copy data and metadata between

copy data and metadata between AWS Storage Services

Failure mode: an immutable recovery point can still be unusable when keys, IAM roles, network dependencies, metadata or application validation are…

### 3 Amazon EFS

Amazon EFS

Evidence: job completion, recovery-point age, restore duration, validation workflow, data checksum and actual RTO/RPO prove recoverability.

### 4 Amazon FSx

Amazon FSx

Decision point: choose backup/restore, pilot light, warm standby or active/active from measured business objectives—not from service familiarity.

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# AWS Backup

### 1 Fully managed service

Fully managed service

Amazon EC2 / Amazon EBS

AWS Storage Gateway (Volume Gateway)

### 2 Centrally manage and automate…

Centrally manage and automate backups across AWS services

Amazon RDS (all DBs engines) / Amazon Aurora / Amazon DynamoDB

Supports cross-region backups

### 3 No need to create custom

No need to create custom scripts and manual processes

Amazon DocumentDB / Amazon Neptune

Supports cross-account backups

### 4 Supported services

Supported services

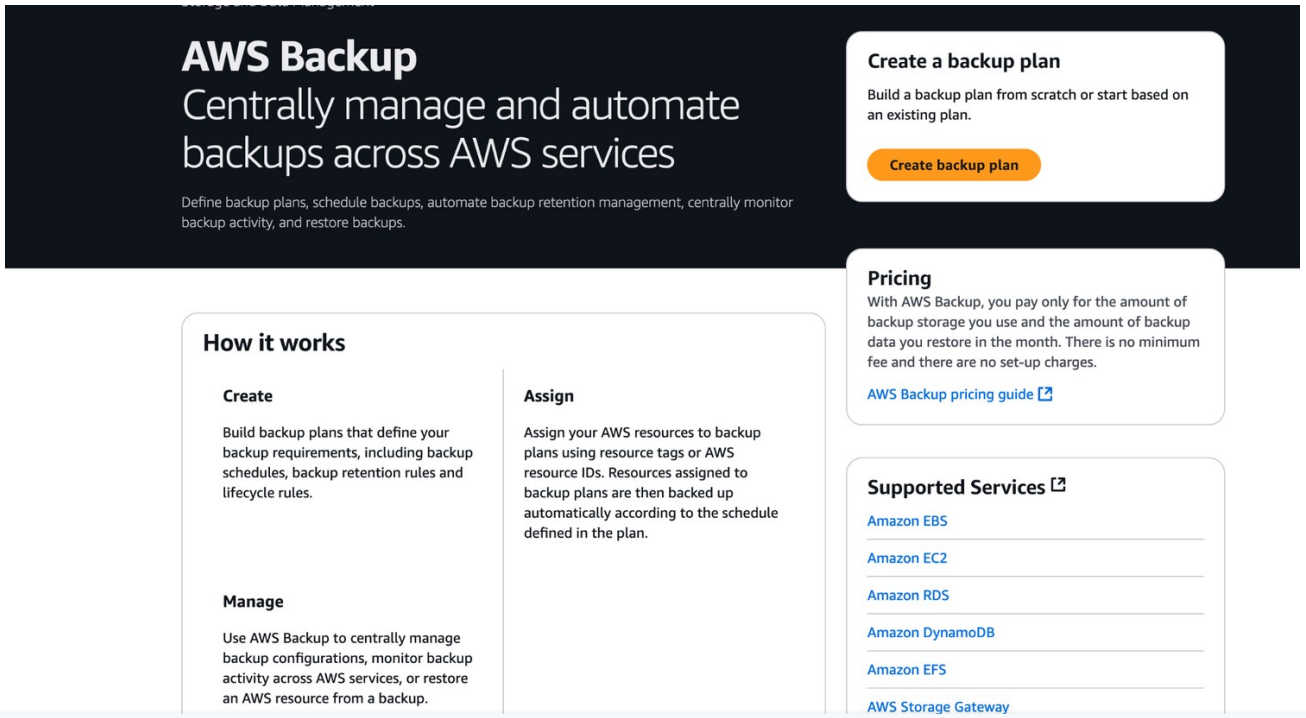
Amazon EFS / Amazon FSx (Lustre & Windows File Server)

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# AWS Backup




**1** **Mechanism: backup plans, vault…**

Mechanism: backup plans, vault policies, Vault Lock, cross-account copies and restore-testing plans separate…

**2** **Failure mode**

Failure mode: an immutable recovery point can still be unusable when keys, IAM roles, network dependencies…

**3** **Evidence: job completion…**

Evidence: job completion, recovery-point age, restore duration, validation workflow, data checksum and actual…

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# AWS Backup

### 1 Supports PITR for supported…

Supports PITR for supported services

Backup frequency (every 12 hours, daily, weekly, monthly, cron expression)

### 2 On-Demand and Scheduled backups

On-Demand and Scheduled backups

Backup window

Mechanism: backup plans, vault policies, Vault Lock, cross-account copies and restore-testing plans separate protection from operational recovery.

### 3 Tag-based backup policies

Tag-based backup policies

Transition to Cold Storage (Never, Days, Weeks, Months,Years)

### 4 You create backup policies known

You create backup policies known as Backup Plans

Retention Period (Always, Days, Weeks, Months,Years)

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# AWS Backup

## 1 Create Backup Plan

Create Backup Plan

Automatically backed up to

Mechanism: backup plans, vault policies, Vault Lock, cross-account copies and restore-testing plans separate protection from operational recovery.

## 2 (frequency, retention policy)

(frequency, retention policy)

DynamoDB DocumentDB

Failure mode: an immutable recovery point can still be unusable when keys, IAM roles, network dependencies, metadata or application validation are…

## 3 Assign AWS Resources

Assign AWS Resources

Evidence: job completion, recovery-point age, restore duration, validation workflow, data checksum and actual RTO/RPO prove recoverability.

## 4 Storage Gateway

Storage Gateway

Decision point: choose backup/restore, pilot light, warm standby or active/active from measured business objectives—not from service familiarity.

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# AWS Backup Vault Lock

### 1 Enforce a WORM (Write Once

Enforce a WORM (Write Once Read Many) state for all the backups that you store in your AWS Backup Vault

Even the root user cannot delete backups when enabled

### 2 Additional layer of defense to

Additional layer of defense to protect your backups against

Backup Vault Lock Policy

### 3 Inadvertent or malicious delete…

Inadvertent or malicious delete operations

Backups can’t be deleted

Mechanism: backup plans, vault policies, Vault Lock, cross-account copies and restore-testing plans separate protection from operational recovery.

### 4 Updates that shorten or alter

Updates that shorten or alter retention periods

Failure mode: an immutable recovery point can still be unusable when keys, IAM roles, network dependencies, metadata or application validation are…

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# What’s an EBS Volume?

### 1 An EBS (Elastic Block Store)

An EBS (Elastic Block Store) Volume is a network drive you can attach to your instances while they run

Analogy:Think of them as a “network USB stick”

### 2 It allows your instances to

It allows your instances to persist data, even after their termination

Free tier: 30 GB of free EBS storage of type General Purpose (SSD) or Magnetic per month

### 3 They can only be mounted

They can only be mounted to one instance at a time (at the CCP level)

### 4 They are bound to a

They are bound to a specific availability zone

Mechanism: backup plans, vault policies, Vault Lock, cross-account copies and restore-testing plans separate protection from operational recovery.

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# EBS Volume

### 1 It's a network drive (i.e.

It's a network drive (i.e. not a physical drive)

An EBS Volume in us-east-1a cannot be attached to us-east-1b

You can increase the capacity of the drive over time

### 2 It uses the network to

It uses the network to communicate the instance, which means there might be a bit of latency

To move a volume across, you first need to snapshot it

### 3 It can be detached from

It can be detached from an EC2 instance and attached to another one quickly

Have a provisioned capacity (size in GBs, and IOPS)

### 4 It's locked to an Availability

It's locked to an Availability Zone (AZ)

You get billed for all the provisioned capacity

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# EC2 Instance Store

### 1 EBS volumes are network drives

EBS volumes are network drives with good but “limited” performance

Good for buffer / cache / scratch data / temporary content

### 2 If you need a high-performance

If you need a high-performance hardware disk, use EC2 Instance store

Risk of data loss if hardware fails

### 3 Better I/O performance

Better I/O performance

Backups and Replication are your responsibility

Mechanism: backup plans, vault policies, Vault Lock, cross-account copies and restore-testing plans separate protection from operational recovery.

### 4 EC2 Instance Store lose their

EC2 Instance Store lose their storage if they’re stopped (ephemeral)

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# EBS Volume Types

### 1 EBS Volumes come in 6

EBS Volumes come in 6 types

sc1 (HDD): Lowest cost HDD volume designed for less frequently accessed workloads

### 2 gp2 / gp3 (SSD)

gp2 / gp3 (SSD): General purpose SSD volume that balances price and performance for a wide variety of workloads

EBS Volumes are characterized in Size | Throughput | IOPS (I/O Ops Per Sec)

### 3 io1 / io2 Block Express (SSD)

io1 / io2 Block Express (SSD): Highest-performance SSD volume for mission-critical low-latency or high-throughput workloads

When in doubt always consult the AWS documentation – it's good!

### 4 st1 (HDD)

st1 (HDD): Low cost HDD volume designed for frequently accessed, throughput-intensive workloads

Only gp2/gp3 and io1/io2 Block Express can be used as boot volumes

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# EBS Volume Types Use cases Provisioned IOPS (PIOPS) SSD

### 1 Critical business applications…

Critical business applications with sustained IOPS performance

Max PIOPS: 64,000 for Nitro EC2 instances & 32,000 for other

Max PIOPS: 256,000 with an IOPS:GiB ratio of 1,000:1

### 2 Or applications that need more

Or applications that need more than 16,000 IOPS

Can increase PIOPS independently from storage size

Supports EBS Multi-attach

### 3 Great for databases workloads…

Great for databases workloads (sensitive to storage perf and consistency)

io2 Block Express (4 GiB – 64 TiB)

### 4 io1 (4 GiB

io1 (4 GiB - 16 TiB)

Sub-millisecond latency

Mechanism: backup plans, vault policies, Vault Lock, cross-account copies and restore-testing plans separate protection from operational recovery.

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# EBS Volume Types Use cases

### 1 Hard Disk Drives (HDD)

Hard Disk Drives (HDD)

Big Data, Data Warehouses, Log Processing

Scenarios where lowest cost is important

### 2 Cannot be a boot volume

Cannot be a boot volume

Max through put500 MiB/s – max IOPS 500

Max throughput 250 MiB/s – max IOPS 250

### 3 125 GiB to 16 TiB

125 GiB to 16 TiB

Cold HDD (sc1)

Mechanism: backup plans, vault policies, Vault Lock, cross-account copies and restore-testing plans separate protection from operational recovery.

### 4 Throughput Optimized HDD (st1)

Throughput Optimized HDD (st1)

For data that is infrequently accessed

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# EBS - Volume Types Summary

| | General Purpose SSD volumes | | Provisioned IOPS SSD volumes | |
|---|---|---|---|---|
| **Volume type** | gp3 | gp2 | io2 Block Express [3] | io1 |
| **Durability** | 99.8% - 99.9% durability (0.1% - 0.2% annual failure rate) | | 99.999% durability (0.001% annual failure rate) | 99.8% - 99.9% durability (0.1% - 0.2% annual failure rate) |
| **Use cases** | • Transactional workloads<br>• Virtual desktops<br>• Medium-sized, single-instance databases<br>• Low-latency interactive applications<br>• Boot volumes<br>• Development and test environments | | Workloads that require:<br>• Sub-millisecond latency<br>• Sustained IOPS performance<br>• More than 64,000 IOPS or 1,000 MiB/s of throughput | • Workloads that require sustained IOPS performance or more than 16,000 IOPS<br>• I/O-intensive database workloads |
| **Volume size** | 1 GiB - 16 TiB | | 4 GiB - 64 TiB [4] | 4 GiB - 16 TiB |
| **Max IOPS per volume** (16 KiB I/O) | 16,000 | | 256,000 [5] | 64,000 |
| **Max throughput per volume** | 1,000 MiB/s | 250 MiB/s [1] | 4,000 MiB/s | 1,000 MiB/s [2] |
| **Amazon EBS Multi-attach** | Not supported | | Supported | |
| **NVMe reservations** | Not supported | | Supported | Not supported |
| **Boot volume** | Supported | | | |

**1** **Mechanism: backup plans, vault…**

Mechanism: backup plans, vault policies, Vault Lock, cross-account copies and restore-testing plans separate…

**2** **Failure mode**

Failure mode: an immutable recovery point can still be unusable when keys, IAM roles, network dependencies…

**3** **Evidence: job completion…**

Evidence: job completion, recovery-point age, restore duration, validation workflow, data checksum and actual…

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# EBS Multi-Attach - io1/io2 family

### 1 Attach the same EBS volume

Attach the same EBS volume to multiple EC2 instances in the same AZ

Up to 16 EC2 Instances at a time

### 2 Each instance has full read

Each instance has full read & write permissions to the high-performance volume

Must use a file system that's cluster-aware (not XFS, EXT4, etc…)

### 3 Achieve higher application…

Achieve higher application availability in clustered Linux applications (ex:Teradata)

Availability Zone 1

### 4 Applications must manage…

Applications must manage concurrent write operations

io2 volume with Multi-Attach

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# EBS Volume Resizing

### 1 You can only increase the

You can only increase the EBS volumes

After increasing the size, it's possible for the volume to be in a long time in the "optimisation" phase.The volume is still usable

repartition

### 2 Size (any volume type)

Size (any volume type)

You can't decrease the size of your EBS volume (create another smaller volume then migrate data)

### 3 IOPS (only in IO1)

IOPS (only in IO1)

EBS Volume

Mechanism: backup plans, vault policies, Vault Lock, cross-account copies and restore-testing plans separate protection from operational recovery.

### 4 After resizing an EBS volume,

After resizing an EBS volume, you need to repartition your drive

Optimizing

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# EBS Snapshots

1

### Make a backup (snapshot) of

Make a backup (snapshot) of your EBS volume at a point in time

US-EAST-1B

2

### Not necessary to detach volume

Not necessary to detach volume to do snapshot, but recommended

EBS Snapshot

3

### Can copy snapshots across AZ

Can copy snapshots across AZ or Region

Mechanism: backup plans, vault policies, Vault Lock, cross-account copies and restore-testing plans separate protection from operational recovery.

4

### US-EAST-1A

US-EAST-1A

Failure mode: an immutable recovery point can still be unusable when keys, IAM roles, network dependencies, metadata or application validation are…

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# Amazon Data Lifecycle Manager

**1 Automate the creation, retention…**

Automate the creation, retention, and deletion of EBS snapshots and EBS-backed AMIs

Can’t be used to manage instance-store backed AMIs

**2 Schedule backups, cross-account…**

Schedule backups, cross-account snapshot copies, delete outdated backups, …

EBS Volume

**3 Uses resource tags to identify**

Uses resource tags to identify the resources (EC2 instances, EBS volumes)

EBS Volumes

**4 Can’t be used to manage**

Can’t be used to manage snapshots/AMIs created outside DLM

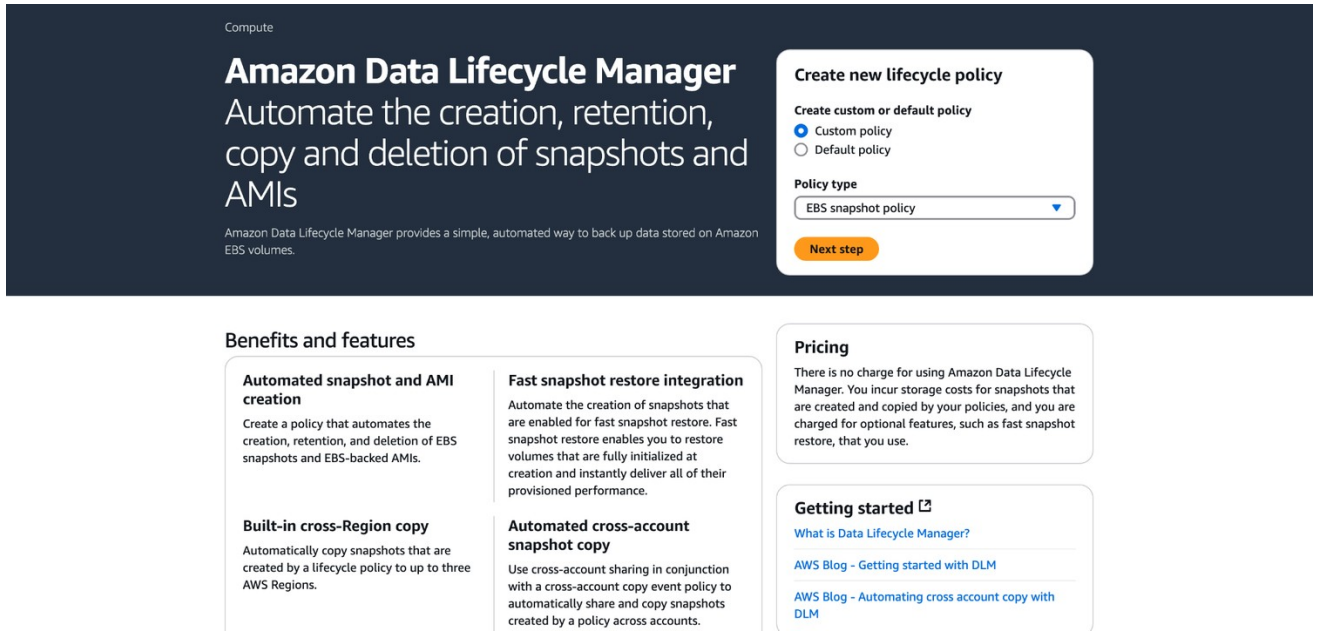
EC2 Instance

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# Amazon Data Lifecycle Manager




**1** **Mechanism: backup plans, vault…**

Mechanism: backup plans, vault policies, Vault Lock, cross-account copies and restore-testing plans separate…

**2** **Failure mode**

Failure mode: an immutable recovery point can still be unusable when keys, IAM roles, network dependencies…

**3** **Evidence: job completion…**

Evidence: job completion, recovery-point age, restore duration, validation workflow, data checksum and actual…

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# EBS Snapshots - Fast Snapshot Restore (FSR)

### 1 EBS Snapshots stored in S3

EBS Snapshots stored in S3

FSR helps you to create a volume from a snapshot that is fully initialized at creation (no I/O latency)

(FSR enabled)

### 2 By default, there's a latency

By default, there's a latency of I/O operations the first time each block is accessed (block must be pulled from S3)

Enabled for a snapshot in a particular AZ (billed per minute – very expensive $$$)

EBS Volume

### 3 Solution: force the…

Solution: force the initialization of the entire

Can be enabled on snapshots created by Data Lifecycle Manager

storage blocks

### 4 volume (using the dd or

volume (using the dd or fio command), or you can enable FSR

EBS Snapshot

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# EBS Snapshots Features

### 1 EBS Snapshot Archive

EBS Snapshot Archive

Setup rules to retain deleted snapshots so you can recover them after an accidental deletion

### 2 Move a Snapshot to an

Move a Snapshot to an ”archive tier” that is 75% cheaper

Specify retention (from 1 day to 1 year)

### 3 Takes within 24 to 72

Takes within 24 to 72 hours for restoring the archive

EBS Snapshot

### 4 Recycle Bin for EBS Snapshots

Recycle Bin for EBS Snapshots

Recycle Bin

Mechanism: backup plans, vault policies, Vault Lock, cross-account copies and restore-testing plans separate protection from operational recovery.

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# EBS Migration

1

### EBS Volumes are only locked

EBS Volumes are only locked to a specific AZ

Create a volume from the snapshot in the AZ of your choice

2

### To migrate it to a

To migrate it to a different AZ (or region)

Let's practice!

3

### Snapshot the volume

Snapshot the volume

Mechanism: backup plans, vault policies, Vault Lock, cross-account copies and restore-testing plans separate protection from operational recovery.

4

### (optional) Copy the volume to

(optional) Copy the volume to a different region

Failure mode: an immutable recovery point can still be unusable when keys, IAM roles, network dependencies, metadata or application validation are…

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# EBS Encryption

### 1 When you create an encrypted

When you create an encrypted EBS volume, you get the following

All volumes created from the snapshot

Copying an unencrypted snapshot allows encryption

### 2 Data at rest is encrypted

Data at rest is encrypted inside the volume

Encryption and decryption are handled transparently (you have nothing to do)

Snapshots of encrypted volumes are encrypted

### 3 All the data in flight

All the data in flight moving between the instance and the volume is encrypted

Encryption has a minimal impact on latency

### 4 All snapshots are encrypted

All snapshots are encrypted

EBS Encryption leverages keys from KMS (AES-256)

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# Amazon EFS - Elastic File System

### 1 Managed NFS (network file system)

Managed NFS (network file system) that can be mounted on many EC2

EC2 Instances

EFS FileSystem

### 2 EFS works with EC2 instances

EFS works with EC2 instances in multi-AZ

us-east-1b

Mechanism: backup plans, vault policies, Vault Lock, cross-account copies and restore-testing plans separate protection from operational recovery.

### 3 Highly available, scalable…

Highly available, scalable, expensive (3x gp2), pay per use

us-east-1c

### 4 us-east-1a

us-east-1a

Security Group

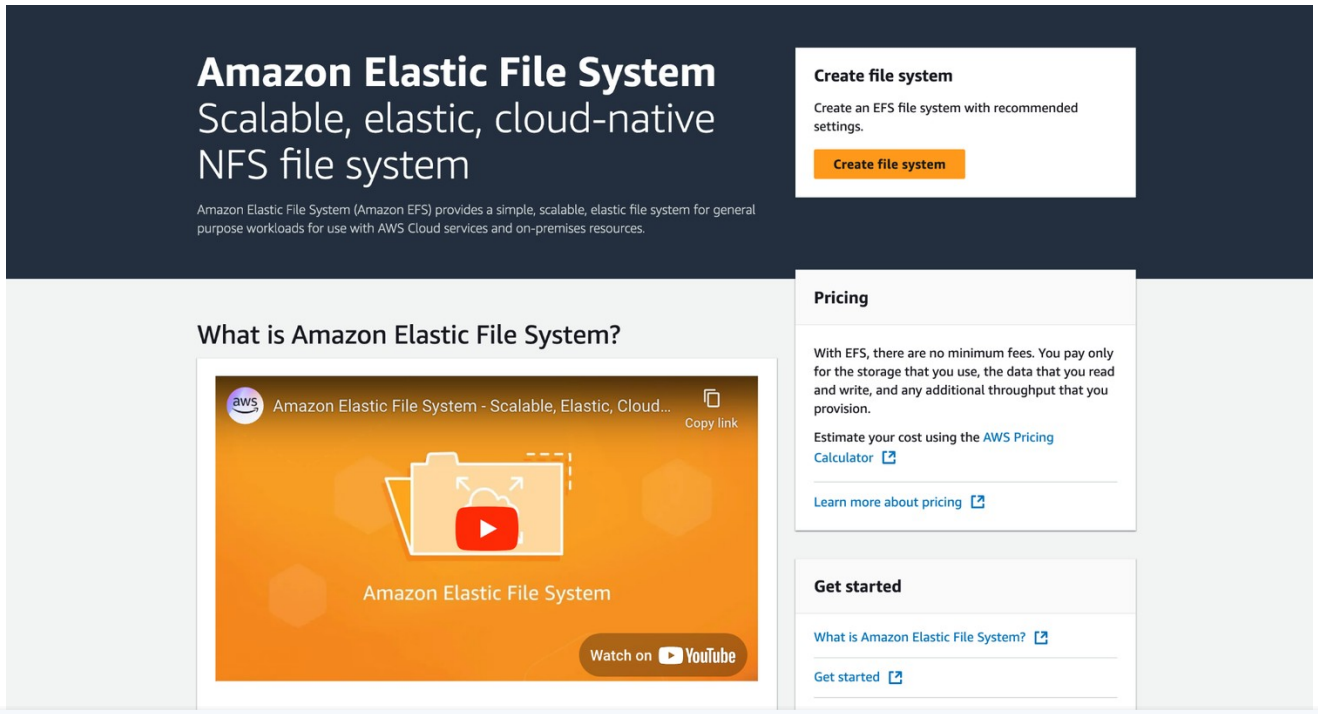
Failure mode: an immutable recovery point can still be unusable when keys, IAM roles, network dependencies, metadata or application validation are…

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# Amazon EFS - Elastic File System




**1** **Mechanism: backup plans, vault…**

Mechanism: backup plans, vault policies, Vault Lock, cross-account copies and restore-testing plans separate…

**2** **Failure mode**

Failure mode: an immutable recovery point can still be unusable when keys, IAM roles, network dependencies…

**3** **Evidence: job completion…**

Evidence: job completion, recovery-point age, restore duration, validation workflow, data checksum and actual…

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# Amazon EFS - Elastic File System

### 1 Use cases

Use cases: content management, web serving, data sharing, Wordpress

Encryption at rest using KMS

### 2 Uses NFSv4.1 protocol

Uses NFSv4.1 protocol

POSIX file system (~Linux) that has a standard file API

### 3 Uses security group to control

Uses security group to control access to EFS

File system scales automatically, pay-per-use, no capacity planning!

### 4 Compatible with Linux based AMI

Compatible with Linux based AMI (not windows)

Mechanism: backup plans, vault policies, Vault Lock, cross-account copies and restore-testing plans separate protection from operational recovery.

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# EFS - Performance & Storage Classes

### 1 1000s of concurrent NFS clients,

1000s of concurrent NFS clients, 10 GB+ /s throughput

Max I/O - higher latecy, throughput, highly parallel (big data, media processing)

Elastic - automatically scales throughput up or down based on your workloads

### 2 Grow to Petabyte-scale network…

Grow to Petabyte-scale network file system, automatically

Throughput Mode

Up to 3GiB/s for reads and 1GiB/s for writes

### 3 Performance Mode (set at EFS

Performance Mode (set at EFS creation time)

Bursting - 1TB = 50MiB/s + burst of up to 100MiB/s

Used for unpredictable workloads

### 4 General Purpose (default)

General Purpose (default) - latency-sensitive use cases (web server, CMS, etc…)

Provisioned - set your throughput regardless of storage size, ex: 1 GiB/s for 1 TB storage

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# EFS - Storage Classes

### 1 Storage Tiers (lifecycle…

Storage Tiers (lifecycle management feature move file after N days)

Implement Lifecycle policies to move files between storage tiers

Over 90% in cost savings

### 2 Standard: for frequently accessed…

Standard: for frequently accessed files

Availability and durability

Amazon EFS File System

### 3 Infrequent access (EFS-IA)

Infrequent access (EFS-IA): cost to retrieve files, lower price to store.

Standard: Multi-AZ, great for prod

Lifecycle Policy

### 4 Archive: rarely accessed data (few

Archive: rarely accessed data (few times each year), 50% cheaper

One Zone: One AZ, great for dev, backup enabled by default, compatible with IA (EFS One Zone-IA)

no access for 60 days

**CURRENT OPERATIONS LENS**

Distinguish backup completion from recovery assurance; measure restore time, data integrity, dependency order and actual RPO. Validate service limits and regional availability against current AWS documentation.


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# Original PPT Coverage Retained — Topic 05

1

### Modernised slides

30 technical slides extracted from the original v4 PowerPoint and rebuilt in the current visual system.

2

### Traceability

Every blended slide names its original v4 source slide number in the kicker.

3

### Current validation

Terminology is normalised; service limits and Region availability remain explicit current-documentation checks.

4

### Procedure boundary

Click-by-click sequences, activities and quizzes are kept out of the deck and remain in the Learner Guide or lab package.


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TOPIC 05 · CONCEPT-LED LAB BRIEF

# Prove Backup and Restore Readiness

**LAB 06**

Apply K4, A1, A4 and A6 by protecting an S3/EBS-style workload, restoring into isolation and measuring actual recovery evidence.

Create a backup plan and vault configuration, run a protected copy, restore into isolation and validate integrity and recovery time.

### YOU'LL BUILD

A backup plan, selection, restore checklist, recovery evidence and measured RTO/RPO report.

### TOOLCHAIN

AWS Backup · S3 · EBS · KMS · CloudWatch Events

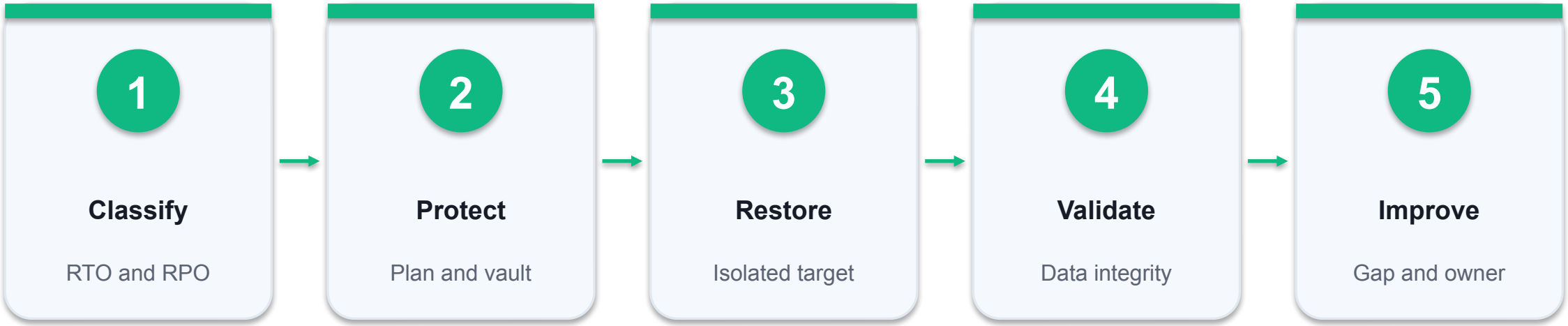
### DONE WHEN

A completed recovery point is restored to an isolated target, integrity is checked, actual recovery is compared with RTO/RPO, gaps have owners, and validation emits LAB 06 PASS.


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LAB 06 · CONCEPTUAL WORKFLOW

# Lab 06 Evidence Model




**PROCEDURE BOUNDARY**

Detailed commands and numbered steps are in the Learner Guide and lab folder—not on these slides.


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LAB 06 · ACCEPTANCE EVIDENCE

# Prove Backup and Restore Readiness

## ✅ Expected result

A completed recovery point is restored to an isolated target, integrity is checked, actual recovery is compared with RTO/RPO, gaps have owners, and validation emits LAB 06 PASS.

IF IT DOESN'T WORK

### Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

### Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

### Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.


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# Recap — Backup, Restore and Business Continuity

1

**Explain**

A protected copy has value only when restore time and data integrity are proven.

2

**Distinguish**

Recovery time and acceptable data loss determine technology and cost.

3

**Apply**

K4, A1, A4 and A6 by protecting an S3/EBS-style workload, restoring into isolation and measuring actual recovery evidence.

4

**Evidence**

A completed recovery point is restored to an isolated target, integrity is checked, actual recovery is compared with RTO/RPO, gaps have owners, and validation emits…


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06

TOPIC 06 · LO2 · LO4 · K3 · K4 · A4 · A5

# Deployment, Provisioning and Automation

CloudFormation · StackSets · change sets · drift · Systems Manager Automation · Git and IaC


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# Key Ideas — Deployment, Provisioning and Automation

1

### Infrastructure is versioned

Templates and policies create a reviewable desired state.

2

### Preview before execution

Change sets reveal creates, replacements and deletions before the stack changes.

3

### Drift is a control signal

Difference between desired and actual state requires investigation, not blind overwrite.

4

### Automation must be idempotent

Repeated execution should converge safely to the intended state.


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# Infrastructure Change Path




**A5 CONNECTION**

The template, plan, approval, execution and verification form one auditable release record.

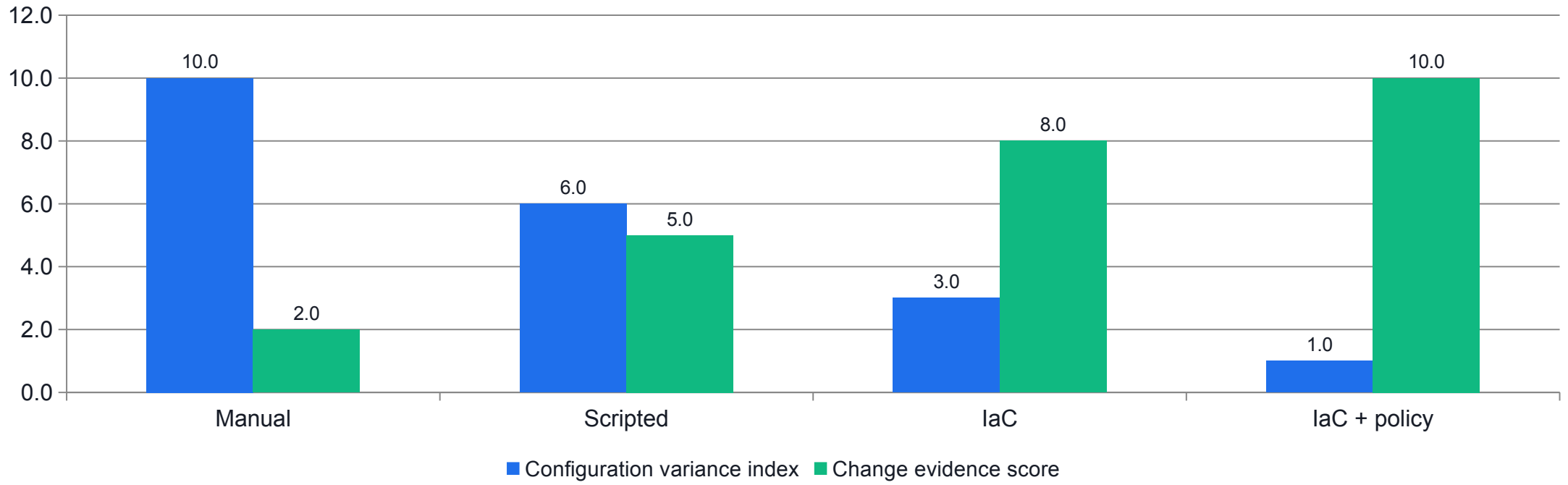

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# CloudFormation Update Controls

| Control | Purpose | Operational use |
| --- | --- | --- |
| **Change set** | Preview resource actions | Review replacements and deletions |
| **Stack policy** | Protect critical resources | Restrict updates during normal change |
| **Rollback** | Return after failed update | Preserve events for diagnosis |
| **Termination protection** | Prevent accidental stack deletion | Use on persistent environments |


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# Automation Reduces Configuration Variance




**WHAT THE DATA SHOWS**

Automation improves consistency; policy and version control make the change explainable and governable.


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# Template Engineering

1

## Parameters

Expose safe environment-specific values without duplicating templates.

2

## Mappings and conditions

Represent controlled variation, not arbitrary complexity.

3

## Outputs and exports

Publish stable integration values with dependency awareness.

4

## Deletion and update policies

Protect or snapshot stateful resources during change.


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# StackSets Deployment Models

| Model | Best fit | Trade-off |
| --- | --- | --- |
| **Self-managed** | Explicit administration and target roles | More role setup and control |
| **Service-managed** | AWS Organizations integration | Requires trusted access and OU planning |
| **Automatic deployment** | New accounts inherit baseline | Test policy and Region scope carefully |


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# Automation Runbook Design

## Inputs

- Validated parameters
- No embedded secrets

## Control

- Retries and timeouts
- Approval for destructive steps

## Output

- Structured evidence
- Explicit success and rollback


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IaC Discipline

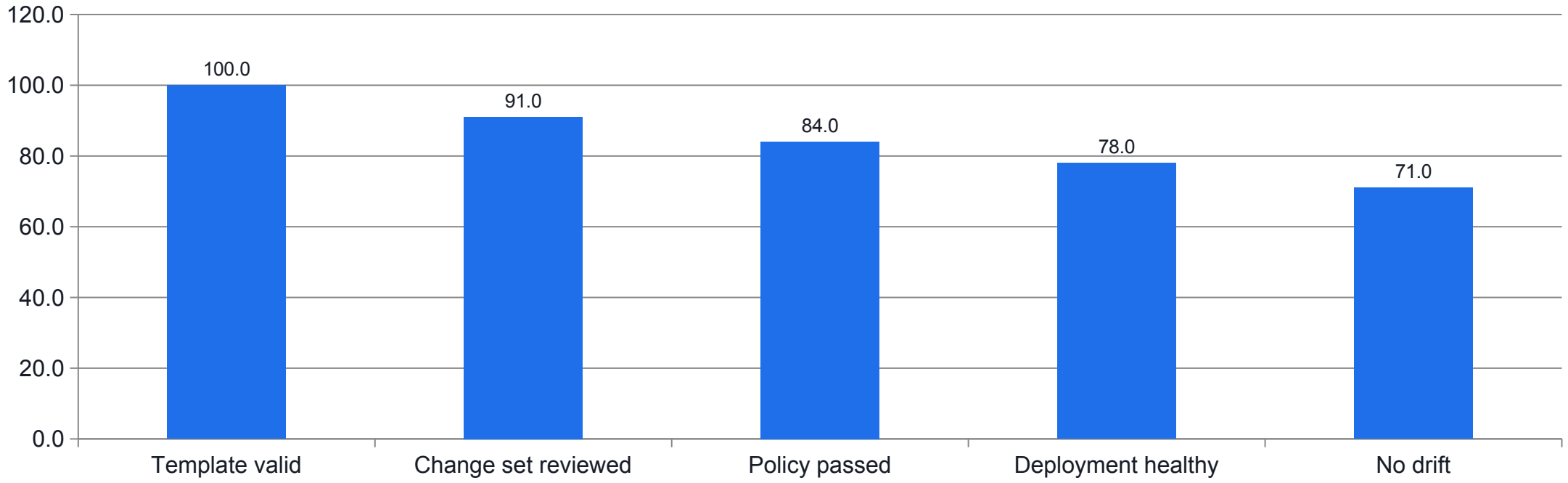
# Do not fix drift by hiding it.

Identify who changed what and why, decide whether actual or desired state is correct, then reconcile through the controlled source.


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TOPIC 06 · NATIVE EDITABLE TECHNICAL CHART

# Change Assurance Funnel




**WHAT THE DATA SHOWS**

Each gate removes a different failure class; skipping drift reconciliation leaves successful deployments operationally inconsistent.


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# Elastic Beanstalk - Overview

### 1 Elastic Beanstalk is a developer

Elastic Beanstalk is a developer centric view of deploying an application on AWS

Just the application code is the responsibility of the developer

### 2 It uses all the component's

It uses all the component's we've seen before: EC2, ASG, ELB, RDS, …

We still have full control over the configuration

### 3 Managed service

Managed service

Beanstalk is free but you pay for the underlying instances

### 4 Automatically handles capacity…

Automatically handles capacity provisioning, load balancing, scaling, application health monitoring, instance configuration, …

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Treat infrastructure automation as controlled change: validate, preview, approve, deploy, observe, reconcile drift and retain rollback evidence. Validate service limits and regional availability against current AWS documentation.


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# Multi AZ

1

## Availability zone 1

Availability zone 1

ElastiCache

PUBLIC SUBNET

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

2

## Auto Scaling group

Auto Scaling group

Store / retrieve

PRIVATE SUBNET

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

3

## Availability zone 2

Availability zone 2

session data

Amazon RDS Read / write data

4

## Availability zone 3

Availability zone 3

+ Cached data

DATA SUBNET

Evidence: validation result, reviewed change set, policy decision, stack events, health alarms, drift detail and reconciled source must all be…

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# Developer problems on AWS

### 1 Managing infrastructure

Managing infrastructure

Most web apps have the same architecture (ALB + ASG)

### 2 Deploying Code

Deploying Code

All the developers want is for their code to run!

### 3 Configuring all the databases…

Configuring all the databases, load balancers, etc

Possibly, consistently across different applications and environments

### 4 Scaling concerns

Scaling concerns

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

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# Elastic Beanstalk - Components

### 1 Application: collection of…

Application: collection of Elastic Beanstalk components (environments, versions, configurations, …)

Tiers: Web Server Environment Tier & Worker Environment Tier

Upload Version

### 2 Application Version

Application Version: an iteration of your application code

You can create multiple environments (dev, test, prod, …)

Launch Environment

### 3 Environment

Environment

update version

Manage Environment

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 4 Collection of AWS resources…

Collection of AWS resources running an application version (only one application version at a time)

Create Application

deploy new version

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# Elastic Beanstalk - Supported Platforms

### 1 Java with Tomcat

Java with Tomcat

Single Container Docker

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 2 .NET Core on Linux

.NET Core on Linux

Multi-container Docker

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

### 3 .NET on Windows Server

.NET on Windows Server

Preconfigured Docker

Evidence: validation result, reviewed change set, policy decision, stack events, health alarms, drift detail and reconciled source must all be…

### 4 Packer Builder

Packer Builder

Decision point: reconcile an emergency change by deciding whether the template or the live state is authoritative, then restore one controlled source…

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# Web Server Tier vs Worker Tier

### 1 Web Environment

Web Environment

Availability Zone 1

EC2 Instance (Worker)

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 2 (myapp.us-east-1.elasticbeanstalk…

(myapp.us-east-1.elasticbeanstalk.com)

Availability Zone 2

Scale based on the number of SQS messages

### 3 Auto Scaling group

Auto Scaling group

SQS message

Can push messages to SQS queue from another Web Server Tier

### 4 Worker Environment

Worker Environment

pull SQS message

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

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# Elastic Beanstalk Deployment Modes

### 1 Single Instance Great for dev

Single Instance Great for dev

Availability Zone 2

Elastic IP

### 2 High Availability with Load…

High Availability with Load Balancer Great for prod

RDS Master

### 3 Availability Zone 1

Availability Zone 1

RDS Standby

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 4 Auto Scaling Group

Auto Scaling Group

EC2 Instance

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

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# AWS CloudFormation

### 1 CloudFormation is a declarative…

CloudFormation is a declarative way of outlining your AWS Infrastructure, for any resources (most of them are supported)

I want two Elastic IPs for these EC2 instances

### 2 For example, within a…

For example, within a CloudFormation template, you say

I want an S3 bucket

### 3 I want a security group

I want a security group

I want a load balancer (ELB) in front of these EC2 instances

### 4 I want two EC2 instances

I want two EC2 instances using this security group

Then CloudFormation creates those for you, in the right order, with the exact configuration that you specify

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# CloudFormation - Template Example



```
AWSTemplateFormatVersion: '2010-09-09'
Description: 'AWS CloudFormation Sample Template LAMP_Multi_AZ: C
Parameters: …
Mappings: …
Resources:
  ApplicationLoadBalancer:
    Type: AWS::ElasticLoadBalancingV2::LoadBalancer
    Properties:
      Subnets: !Ref Subnets
  ALBListener:
    Type: AWS::ElasticLoadBalancingV2::Listener
    Properties:
      DefaultActions:
        - Type: forward
          TargetGroupArn: !Ref ALBTargetGroup
      LoadBalancerArn: !Ref ApplicationLoadBalancer
      Port: '80'
      Protocol: HTTP
  ALBTargetGroup:
    Type: AWS::ElasticLoadBalancingV2::TargetGroup
    Properties:
      HealthCheckIntervalSeconds: 10
      HealthCheckTimeoutSeconds: 5
      HealthyThresholdCount: 2
      Port: 80
      Protocol: HTTP
      UnhealthyThresholdCount: 5
      VpcId: !Ref VpcId
      TargetGroupAttributes:
        - Key: stickiness.enabled
          Value: 'true'
```

**1** **Application Composer**

Application Composer
Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events…

**2** **Failure mode**

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback…

**3** **Evidence: validation result…**

Evidence: validation result, reviewed change set, policy decision, stack events, health alarms, drift detail and…

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# Benefits of AWS CloudFormation (1/2)

### 1 Infrastructure as code

Infrastructure as code

Each resources within the stack is tagged with an identifier so you can easily see how much a stack costs you

### 2 No resources are manually created,

No resources are manually created, which is excellent for control

You can estimate the costs of your resources using the CloudFormation template

### 3 The code can be version

The code can be version controlled for example using Git

Savings strategy: In Dev, you could automation deletion of templates at 5 PM and recreated at 8 AM, safely

### 4 Changes to the infrastructure are

Changes to the infrastructure are reviewed through code

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

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# Benefits of AWS CloudFormation (2/2)

### 1 Productivity

Productivity

Separation of concern: create many stacks for many apps and many layers. Eg.

Don't re-invent the wheel

### 2 Ability to destroy and re-create

Ability to destroy and re-create an infrastructure on the cloud on the fly

VPC stacks

Leverage existing templates on the web!

### 3 Automated generation of Diagram…

Automated generation of Diagram for your templates!

Network stacks

Leverage the documentation

### 4 Declarative programming (no need…

Declarative programming (no need to figure out ordering and orchestration)

App stacks

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# How CloudFormation Works

### 1 Templates must be uploaded in

Templates must be uploaded in S3 and then referenced in CloudFormation

AWS CloudFormation

### 2 To update a template, we

To update a template, we can’t edit previous ones. We have to re- upload a new version of the template to AWS

AWS Resources

### 3 Stacks are identified by a

Stacks are identified by a name

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 4 Deleting a stack deletes every

Deleting a stack deletes every single artifact that was created by CloudFormation.

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# Deploying CloudFormation Templates

### 1 Manual way

Manual way

Automated way

CloudFormation Console

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 2 Editing templates in Application…

Editing templates in Application Composer or code editor

Editing templates in a YAML file

Application Composer / Code Editor

### 3 Using the console to input

Using the console to input parameters, etc…

Using the AWS CLI (Command Line Interface) to deploy the templates, or using a Continuous Delivery (CD) tool

create-stack

### 4 We'll mostly do this way

We'll mostly do this way in the course for learning purposes

Recommended way when you fully want to automate your flow

CloudFormation

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# CloudFormation - Building Blocks

### 1 Template's Components

Template's Components

Parameters - the dynamic inputs for your template

Template's Helpers

### 2 AWSTemplateFormatVersion…

AWSTemplateFormatVersion - identifies the capabilities of the template “2010-09-09”

Mappings - the static variables for your template

### 3 Description - comments about the

Description - comments about the template

Outputs - references to what has been created

### 4 Resources (MANDATORY)

Resources (MANDATORY) - your AWS resources declared in the template

Conditionals - list of conditions to perform resource creation

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# Introductory Example

### 1 We're going to create a

We're going to create a simple EC2 instance

We'll see how in no-time, we are able to get started with CloudFormation!

### 2 And we're going to add

And we're going to add security group to it

Web Server

### 3 For now, forget about the

For now, forget about the code syntax

WebServerSecurityGroup

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 4 We'll look at the structure

We'll look at the structure of the files later

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

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# YAML Crash Course

```yaml
invoice: 34843
date: 2001-01-23
bill-to:
  given: Chris
  family: Dumars
  address:
    lines: |
      458 Walkman Dr.
      Suite #292
    city: Royal Oak
    state: MI
    postal: 48046
products:
  - sku: BL394D
    quantity: 4
    description: Basketball
    price: 450.00
  - sku: BL4438H
    quantity: 1
    description: Super Hoop
    price: 2392.00
```

1

### YAML and JSON are the

YAML and JSON are the languages you can use for CloudFormation

2

### Mechanism: templates define…

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events…

3

### Failure mode

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback…

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# CloudFormation - Resources

### 1 Resources are the core of

Resources are the core of your CloudFormation template (MANDATORY)

There are over 700 types of resources (!)

### 2 They represent the different AWS

They represent the different AWS Components that will be created and configured

Resource types identifiers are of the form

### 3 Resources are declared and can

Resources are declared and can reference each other

service-provider::service-name::data-type-name

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 4 AWS figures out creation, updates

AWS figures out creation, updates and deletes of resources for us

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# How do I find Resources documentation?

### 1 I can’t teach you all

I can’t teach you all the 700+ resources, but I can teach you how to learn how to use them

### 2 All the resources can be

All the resources can be found here

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 3 Then, we just read the

Then, we just read the docs ©

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

### 4 Example here (for an EC2

Example here (for an EC2 instance)

Evidence: validation result, reviewed change set, policy decision, stack events, health alarms, drift detail and reconciled source must all be…

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# Analysis of CloudFormation Template

### 1 Going back to the example

Going back to the example of the introductory lecture, let's learn why it was written this way.

### 2 Relevant documentation can be…

Relevant documentation can be found here

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 3 Failure mode

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

### 4 Evidence: validation result…

Evidence: validation result, reviewed change set, policy decision, stack events, health alarms, drift detail and reconciled source must all be…

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# CloudFormation - Resources FAQ

### 1 Can I create a dynamic

Can I create a dynamic number of resources?

Almost. Only a select few niches are not there yet

### 2 Yes, you can by using

Yes, you can by using CloudFormation Macros and Transform

You can work around that using CloudFormation Custom Resources

### 3 It is not in the

It is not in the scope of this course

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 4 Is every AWS Service supported?

Is every AWS Service supported?

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

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# CloudFormation - Parameters

### 1 Parameters are a way to

Parameters are a way to provide inputs to your AWS CloudFormation template

Parameters are extremely powerful, controlled, and can prevent errors from happening in your templates, thanks to types

Parameter1: value1

### 2 They’re important to know about

They’re important to know about if

AWS CloudFormation

+ Parameter2: valu

### 3 You want to reuse your

You want to reuse your templates across the company

upload template

CF template Parameters

### 4 Some inputs can not be

Some inputs can not be determined ahead of time

provide parameter values

create stack

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# When should you use a Parameter?

```
Parameters:
  SecurityGroupDescription:
    Description: Security Group Description
    Type: String
```

1

### Ask yourself this

Ask yourself this
Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events…

2

### Failure mode

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback…

3

### Evidence: validation result…

Evidence: validation result, reviewed change set, policy decision, stack events, health alarms, drift detail and…

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# CloudFormation - Parameters Settings

### 1 CommaDelimited List

CommaDelimitedList

List<AWS-Specific Parameter>

Min/MaxLength

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 2 List<Number>

List<Number>

SSM Parameter (get parameter value from SSM Parameter store)

Min/MaxValue

### 3 AWS-Specific Parameter

AWS-Specific Parameter

Description

Allowed Values (array)

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

### 4 (to help catch invalid values

(to help catch invalid values - match against existing values in the AWS account)

Constraint Description (String)

AllowedPattern (regex)

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# CloudFormation - Parameters Example

### 1 AllowedValues NoEcho

AllowedValues NoEcho

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 2 Failure mode

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

### 3 Evidence: validation result…

Evidence: validation result, reviewed change set, policy decision, stack events, health alarms, drift detail and reconciled source must all be…

### 4 Decision point

Decision point: reconcile an emergency change by deciding whether the template or the live state is authoritative, then restore one controlled source…

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# How to Reference a Parameter?

### 1 The Fn

The Fn::Ref function can be leveraged to reference parameters

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 2 Parameters can be used anywhere

Parameters can be used anywhere in a template

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

### 3 The shorthand for this in

The shorthand for this in YAML is !Ref

Evidence: validation result, reviewed change set, policy decision, stack events, health alarms, drift detail and reconciled source must all be…

### 4 The function can also reference

The function can also reference other elements within the template

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# CloudFormation - Pseudo Parameters

### 1 AWS offers us Pseudo Parameters

AWS offers us Pseudo Parameters in any CloudFormation template

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 2 These can be used at

These can be used at any time and are enabled by default

### 3 Important pseudo parameters

Important pseudo parameters

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

### 4 Evidence: validation result…

Evidence: validation result, reviewed change set, policy decision, stack events, health alarms, drift detail and reconciled source must all be…

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# CloudFormation - Mappings

### 1 Mappings are fixed variables…

Mappings are fixed variables within your CloudFormation template

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 2 They're very handy to…

They're very handy to differentiate between different environments (dev vs prod), regions (AWS regions), AMI types…

### 3 All the values are hardcoded

All the values are hardcoded within the template

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

### 4 Evidence: validation result…

Evidence: validation result, reviewed change set, policy decision, stack events, health alarms, drift detail and reconciled source must all be…

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# Accessing Mapping Values (Fn::FindInMap)

### 1 We use Fn

We use Fn::FindInMap to return a named value from a specific key

### 2 !FindInMap [ MapName…

!FindInMap [ MapName, TopLevelKey, SecondLevelKey ]

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 3 Mappings work great for AMIs

Mappings work great for AMIs Because AMIs are region-specific!

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

### 4 Evidence: validation result…

Evidence: validation result, reviewed change set, policy decision, stack events, health alarms, drift detail and reconciled source must all be…

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# When would you use Mappings vs Parameters?

### 1 Mappings are great when you

Mappings are great when you know in advance all the values that can be taken and that they can be deduced from variables such as

They allow safer control over the template

### 2 Availability Zone

Availability Zone

Use parameters when the values are really user specific

### 3 AWS Account

AWS Account

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 4 Environment (dev vs prod)

Environment (dev vs prod)

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

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# CloudFormation - Outputs

### 1 The Outputs section declares…

The Outputs section declares optional outputs values that we can import into other stacks (if you export them first)!

Application Stack

### 2 You can also view the

You can also view the outputs in the AWS Console or in using the AWS CLI

Network Stack

### 3 They’re very useful for example

They’re very useful for example if you define a network CloudFormation, and output the variables such as VPC ID and your Subnet IDs

VPC ID: vpc-2f09a348

### 4 It’s the best way to

It’s the best way to perform some collaboration cross stack, as you let expert handle their own part of the stack

Exported Output

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# CloudFormation - Outputs

### 1 Creating a SSH Security Group

Creating a SSH Security Group as part of one template

### 2 We create an output that

We create an output that references that security group

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 3 Failure mode

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

### 4 Evidence: validation result…

Evidence: validation result, reviewed change set, policy decision, stack events, health alarms, drift detail and reconciled source must all be…

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# CloudFormation - Outputs Cross-Stack Reference

### 1 We then create a second

We then create a second template that leverages that security group

### 2 For this, we use the Fn

For this, we use the Fn::ImportValue function

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 3 You can’t delete the underlying

You can’t delete the underlying stack until all the references are deleted

### 4 Failure mode

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

**CURRENT OPERATIONS LENS**

Treat infrastructure automation as controlled change: validate, preview, approve, deploy, observe, reconcile drift and retain rollback evidence. Validate service limits and regional availability against current AWS documentation.


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# CloudFormation - Conditions

### 1 Conditions are used to control

Conditions are used to control the creation of resources or outputs based on a condition

Any parameter value

Prod Stack

### 2 Conditions can be whatever you

Conditions can be whatever you want them to be, but common ones are

Each condition can reference another condition, parameter value or mapping

Parameters

### 3 Environment (dev / test /

Environment (dev / test / prod)

EC2 Instance

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 4 AWS Region

AWS Region

EBS Volume

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

**CURRENT OPERATIONS LENS**

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# How to define a Condition

```
Conditions:
  CreateProdResources: !Equals [ !Ref EnvType, prod ]
```

**1 The logical ID is for**

The logical ID is for you to choose. It's how you name condition

**2 Mechanism: templates define…**

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events…

**3 Failure mode**

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback…

**CURRENT OPERATIONS LENS**

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# How to use a Condition

### 1 Conditions can be applied to

Conditions can be applied to resources / outputs / etc…

### 2 Mechanism: templates define…

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 3 Failure mode

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

### 4 Evidence: validation result…

Evidence: validation result, reviewed change set, policy decision, stack events, health alarms, drift detail and reconciled source must all be…

**CURRENT OPERATIONS LENS**

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# CloudFormation - Intrinsic Functions

## 1 Fn::GetAtt

Fn::GetAtt

Fn::ToJsonString

Fn::Select

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

## 2 Fn::FindInMap

Fn::FindInMap

Condition Functions (Fn::If, Fn::Not, Fn::Equals, etc…)

Fn::Transform

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

## 3 Fn::ImportValue

Fn::ImportValue

Fn::Base64

Fn::Length

Evidence: validation result, reviewed change set, policy decision, stack events, health alarms, drift detail and reconciled source must all be…

## 4 Fn::ForEach

Fn::ForEach

Fn::GetAZs

Blue = must know

Decision point: reconcile an emergency change by deciding whether the template or the live state is authoritative, then restore one controlled source…

**CURRENT OPERATIONS LENS**

Treat infrastructure automation as controlled change: validate, preview, approve, deploy, observe, reconcile drift and retain rollback evidence. Validate service limits and regional availability against current AWS documentation.


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# Intrinsic Functions - Fn::Ref

### 1 The Fn

The Fn::Ref function can be leveraged to reference

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 2 Parameters - returns the value

Parameters - returns the value of the parameter

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

### 3 Resources - returns the physical

Resources - returns the physical ID of the underlying resource (e.g., EC2 ID)

### 4 The shorthand for this in

The shorthand for this in YAML is !Ref

Evidence: validation result, reviewed change set, policy decision, stack events, health alarms, drift detail and reconciled source must all be…

**CURRENT OPERATIONS LENS**

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# Intrinsic Functions - Fn::GetAtt

### 1 Attributes are attached to any

Attributes are attached to any resources you create

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 2 To know the attributes of

To know the attributes of your resources, the best place to look at is the documentation

### 3 Example: the AZ of an

Example: the AZ of an EC2 instance!

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

### 4 Evidence: validation result…

Evidence: validation result, reviewed change set, policy decision, stack events, health alarms, drift detail and reconciled source must all be…

**CURRENT OPERATIONS LENS**

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# Intrinsic Functions - Fn::FindInMap

### 1 We use Fn

We use Fn::FindInMap to return a named value from a specific key

### 2 !FindInMap [ MapName…

!FindInMap [ MapName, TopLevelKey, SecondLevelKey ]

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 3 Failure mode

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

### 4 Evidence: validation result…

Evidence: validation result, reviewed change set, policy decision, stack events, health alarms, drift detail and reconciled source must all be…

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# Intrinsic Functions - Fn::ImportValue

### 1 Import values that are exported

Import values that are exported in other stacks

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 2 For this, we use the Fn

For this, we use the Fn::ImportValue function

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

### 3 Evidence: validation result…

Evidence: validation result, reviewed change set, policy decision, stack events, health alarms, drift detail and reconciled source must all be…

### 4 Decision point

Decision point: reconcile an emergency change by deciding whether the template or the live state is authoritative, then restore one controlled source…

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# Intrinsic Functions - Fn::Base64

### 1 Convert String to it's Base64

Convert String to it's Base64 representation

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 2 Example: pass encoded data to

Example: pass encoded data to EC2 Instance's UserData property

Failure mode: unmanaged console changes, implicit defaults, replacement-sensitive resources and missing rollback signals create difficult recovery…

### 3 Evidence: validation result…

Evidence: validation result, reviewed change set, policy decision, stack events, health alarms, drift detail and reconciled source must all be…

### 4 Decision point

Decision point: reconcile an emergency change by deciding whether the template or the live state is authoritative, then restore one controlled source…

**CURRENT OPERATIONS LENS**

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# Intrinsic Functions -Condition Functions

```
Conditions:
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```

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**CURRENT OPERATIONS LENS**

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# CloudFormation - Rollbacks

### 1 Stack Creation Fails

Stack Creation Fails

The stack automatically rolls back to the previous known working state

Or from the CLI using continue-update-rollback API call20

### 2 Default: everything rolls back…

Default: everything rolls back (gets deleted). We can look at the log

Ability to see in the log what happened and error messages

### 3 Option to disable rollback and

Option to disable rollback and troubleshoot what happened

Rollback Failure? Fix resources manually then issue

### 4 Stack Update Fails

Stack Update Fails

ContinueUpdateRollback API from Console

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

**CURRENT OPERATIONS LENS**

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# CloudFormation - Service Role

**1 IAM role that allows…**

IAM role that allows CloudFormation to create/update/delete stack resources on your behalf

But you don't want to give the user all the required permissions to create the stack resources

Permissions

**2 Give ability to users to**

Give ability to users to create/update/delete the

User must have iam:PassRole permissions

s3:*Bucket

**3 stack resources even if they**

stack resources even if they don't have permissions to work with the resources in the stack

cloudformation:*

Service Role

**4 You want to achieve the**

You want to achieve the least privilege principle

iam:PassRole

S3 bucket Stack

**CURRENT OPERATIONS LENS**

Treat infrastructure automation as controlled change: validate, preview, approve, deploy, observe, reconcile drift and retain rollback evidence. Validate service limits and regional availability against current AWS documentation.


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# CloudFormation Capabilities

### 1 CAPABILITY_NAMED_IAM and…

CAPABILITY_NAMED_IAM and CAPABILITY_IAM Necessary to enable when you CloudFormation template is creating or updating IAM resources (IAM User, Role…

You're acknowledging that your template may change before deploying

### 2 Specify CAPABILITY_NAMED_IAM if…

Specify CAPABILITY_NAMED_IAM if the resources are named

InsufficientCapabilitiesExcepti on

Mechanism: templates define desired state; change sets expose impact; policies and hooks gate risk; stack events and drift detection expose outcome.

### 3 CAPABILITY_AUTO_EXPAND

CAPABILITY_AUTO_EXPAND

Exception that will be thrown by CloudFormation if the capabilities haven't been acknowledged when deploying a template (security measure)

### 4 Necessary when your…

Necessary when your CloudFormation template includes Macros or Nested Stacks (stacks within stacks) to perform dynamic transformations

**CURRENT OPERATIONS LENS**

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# Original PPT Coverage Retained — Topic 06

1

### Modernised slides

45 technical slides extracted from the original v4 PowerPoint and rebuilt in the current visual system.

2

### Traceability

Every blended slide names its original v4 source slide number in the kicker.

3

### Current validation

Terminology is normalised; service limits and Region availability remain explicit current-documentation checks.

4

### Procedure boundary

Click-by-click sequences, activities and quizzes are kept out of the deck and remain in the Learner Guide or lab package.


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TOPIC 06 · CONCEPT-LED LAB BRIEF

# Control Infrastructure Change and Configuration Drift

**LAB 05**

Apply K3, K4, A4 and A5 by validating a CloudFormation template, reviewing a change set, detecting drift and using a Systems Manager Automation runbook with verification.

Use desired-state evidence to identify drift, choose the correct state, reconcile through code and verify the application outcome.

### YOU'LL BUILD

A validated CloudFormation stack, reviewed change set, drift report, reconciliation decision, Automation verification and cleanup record.

### TOOLCHAIN

CloudFormation · Change Sets · Drift Detection · Systems Manager Automation · AWS CLI

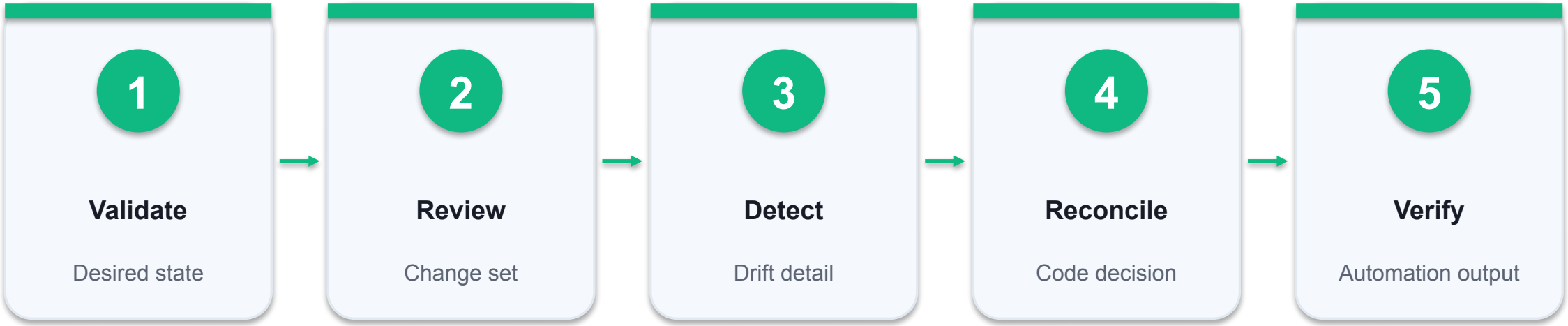
### DONE WHEN

The change set is reviewed before execution, drift evidence identifies the changed property, the decision record explains reconciliation, the Automation runbook verifies state, and validation emits LAB 05 PASS.


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LAB 05 · CONCEPTUAL WORKFLOW

# Lab 05 Evidence Model




**PROCEDURE BOUNDARY**

Detailed commands and numbered steps are in the Learner Guide and lab folder—not on these slides.


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LAB 05 · ACCEPTANCE EVIDENCE

# Control Infrastructure Change and Configuration Drift

## ✅ Expected result

The change set is reviewed before execution, drift evidence identifies the changed property, the decision record explains reconciliation, the Automation runbook verifies state, and validation emits LAB 05 PASS.

IF IT DOESN'T WORK

**Input mismatch**

Reconcile identifiers, required fields and units with the supplied data dictionary.

**Validation stop**

Inspect the named rule and source row; preserve the check rather than bypassing it.

**Evidence differs**

Check the denominator, time window and assumptions before rerunning from clean inputs.


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# Recap — Deployment, Provisioning and Automation

1

**Explain**

Templates and policies create a reviewable desired state.

2

**Distinguish**

Change sets reveal creates, replacements and deletions before the stack changes.

3

**Apply**

K3, K4, A4 and A5 by validating a CloudFormation template, reviewing a change set, detecting drift and using a Systems Manager Automation…

4

**Evidence**

The change set is reviewed before execution, drift evidence identifies the changed property, the decision record explains reconciliation, the Automation runbook…


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07

TOPIC 07 · LO2 · LO3 · K1 · K3 · A1 · A7

# Security, Identity and Compliance

IAM · Organizations · KMS · Config · Security Hub · GuardDuty · Inspector · Macie


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# Key Ideas — Security, Identity and Compliance

1

## Identity is the control plane

Short-lived credentials, least privilege and explicit trust bound administrative power.

2

## Prevent, detect and respond

Policies prevent known bad states; telemetry detects deviation; runbooks contain and recover.

3

## Encryption includes key operations

Access, rotation, grants, aliases and recovery must be governed—not only enabled.

4

## Compliance requires evidence

A rule result is useful when linked to scope, owner, remediation and exception process.

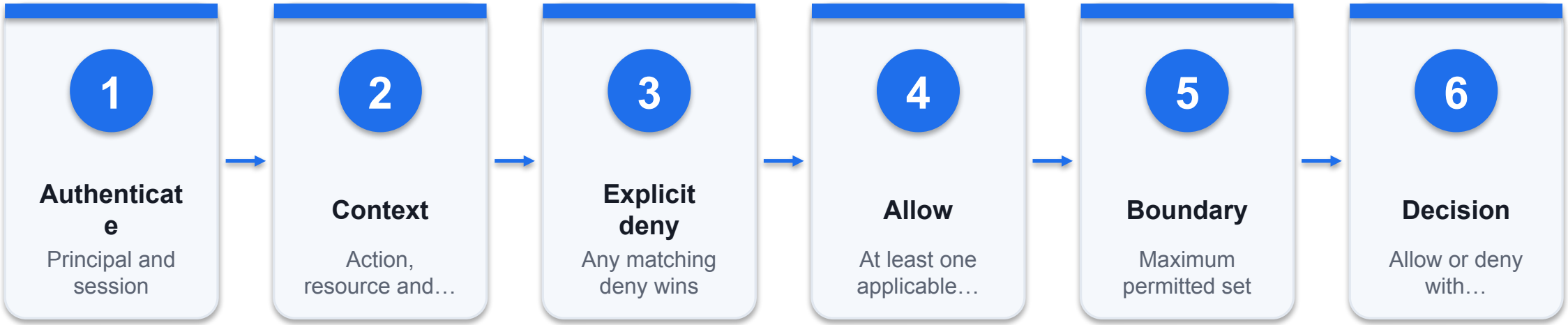

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# Identity Policy Layers

| Layer | What it controls | Common failure |
| --- | --- | --- |
| **Identity policy** | Permissions granted to a principal | Overbroad action or resource |
| **Resource policy** | Who may access a resource | Unintended external principal |
| **Permission boundary** | Maximum identity permissions | Assuming it grants access |
| **SCP** | Maximum permissions in an organisation | Blocking required service operation |


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# Access Evaluation

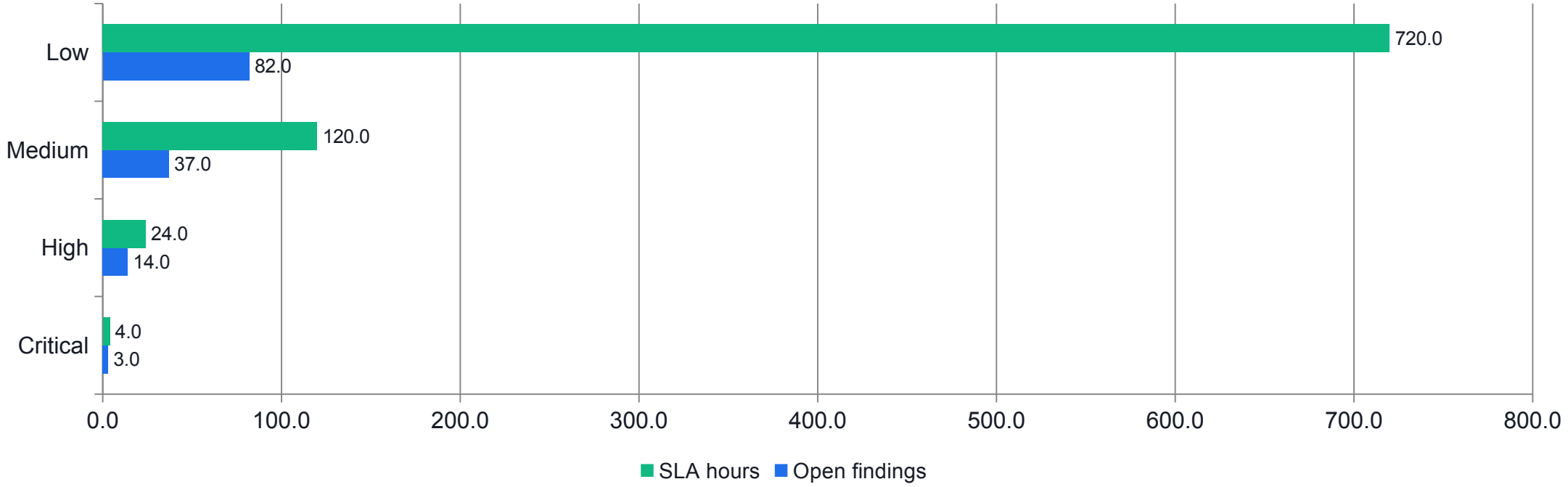



**TROUBLESHOOTING**

Use CloudTrail, policy simulator and access analysis to explain the decision path.


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# Security Findings Need Prioritisation




**WHAT THE DATA SHOWS**

Severity, exposure, asset value and exploitability drive response; high volume alone does not determine priority.


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# Security Services by Question

1

## GuardDuty

What suspicious account, workload or network activity is occurring?

2

## Inspector

Which compute workloads have vulnerabilities or exposure?

3

## Macie

Where is sensitive data in S3 and how is it exposed?

4

## Security Hub

How do findings and standards combine across accounts?

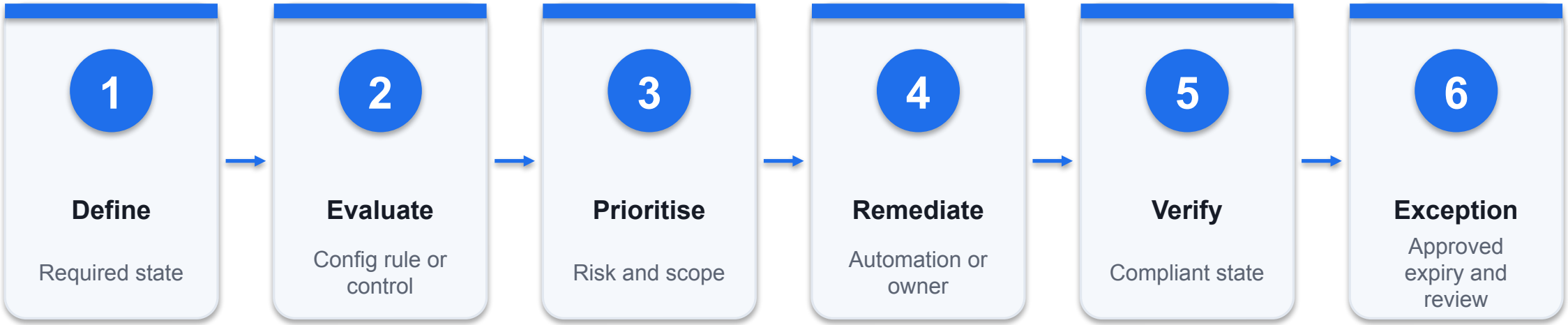

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# KMS Key Choices

| Choice | Operational control | Use carefully |
| --- | --- | --- |
| **AWS owned** | Service manages lifecycle | Limited customer control |
| **AWS managed** | Visible service key | Policy is service-oriented |
| **Customer managed** | Policy, grants, rotation and aliases | Availability and deletion governance |
| **Multi-Region key** | Related key material across Regions | Application and recovery design |


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# Compliance Remediation




**A7 OUTPUT**

Service quality is a measurable standard with evidence, not a one-time checkbox.


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Security Operations

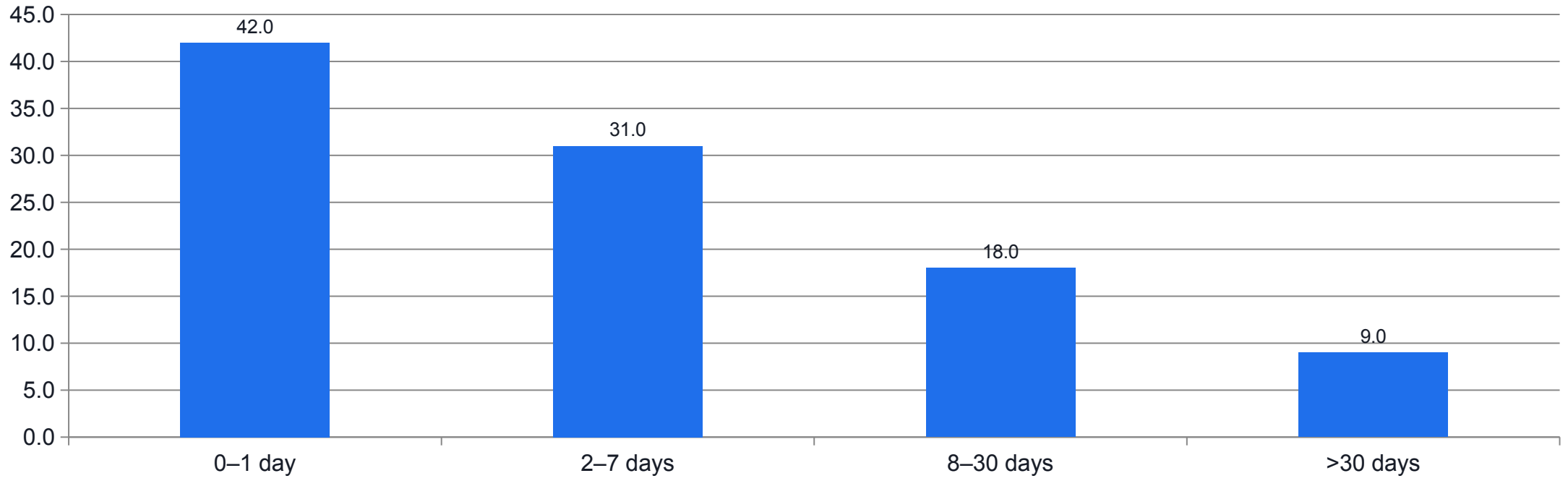
# Least privilege is a lifecycle, not a launch-day setting.

Observe actual use, remove unused permissions, constrain trust, investigate denied actions and revalidate after organisational change.


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TOPIC 07 · NATIVE EDITABLE TECHNICAL CHART

# Security Finding Age Profile

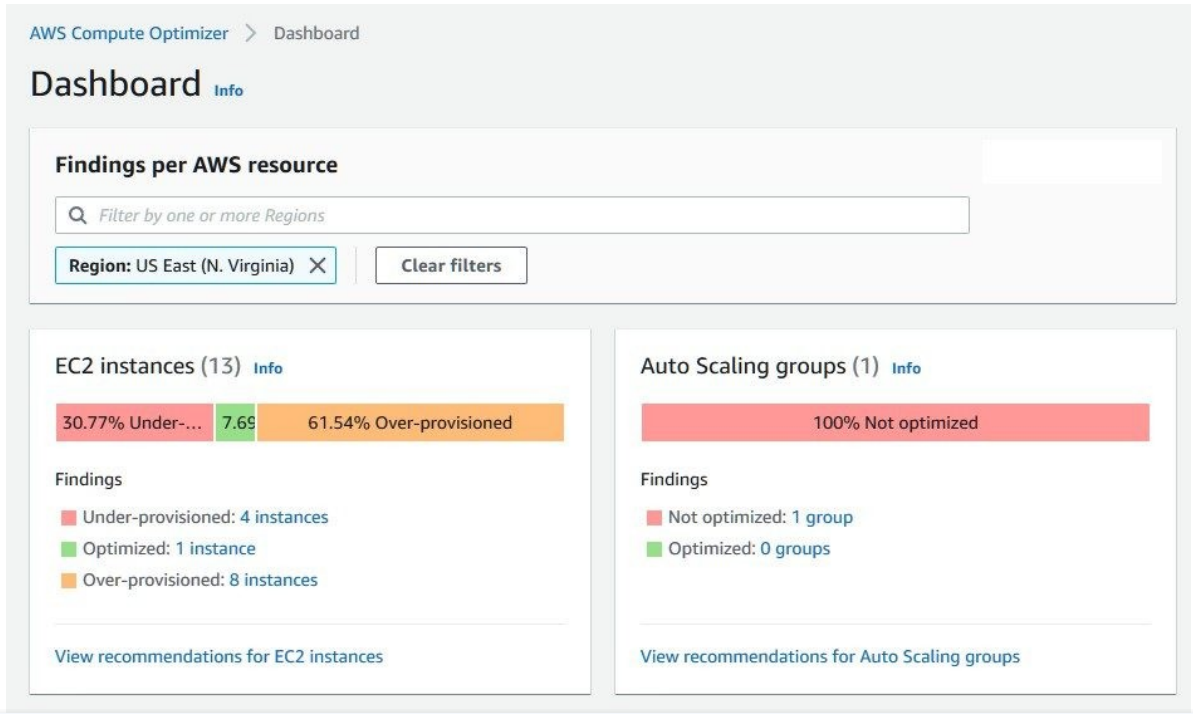



**WHAT THE DATA SHOWS**

Finding age is a service-quality signal: exceptions need owners and expiry dates, while overdue high-risk findings require escalation.


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# AWS Compute Optimizer




**1 Reduce costs and improve…**

Reduce costs and improve performance by Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session…

**2 Failure mode**

Failure mode: broad principals, unscoped PassRole, stale grants, public resource policies and long-lived…

**3 Evidence: IAM Access Analyzer…**

Evidence: IAM Access Analyzer findings, CloudTrail identity context, KMS key/grant policy, Config evaluations and…

**CURRENT OPERATIONS LENS**

Start with the trust boundary, evaluate effective access, protect data, record exceptions and verify that remediation reduced risk. Validate service limits and regional availability against current AWS documentation.


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# Cost and Usage Reports

| | M | N | O | P | R | S | T |
|---|---|---|---|---|---|---|---|
| 1 | lineItem/ProductCode | lineItem/UsageType | lineItem/Operation | lineItem/AvailabilityZone | lineItem/UsageAmount | lineItem/CurrencyCode | lineItem/LineItemDescription |
| 2 | AmazonEC2 | CW:AlarmMonitorUsage | Unknown | | 0.00134409 | USD | $0.00 per alarm-month - first 10 alarms |
| 3 | AmazonS3 | Requests-Tier1 | ListAllMyBuckets | | 2 | USD | $0.00 per request - PUT, COPY, POST, or LIST requests under the monthly global free tier |
| 4 | AmazonEC2 | CW:AlarmMonitorUsage | Unknown | | 0.00134409 | USD | $0.00 per alarm-month - first 10 alarms |
| 5 | AmazonEC2 | APS2-EBS:VolumeUsage.gp2 | CreateVolume-Gp2 | | 0.01344086 | USD | $0.00 per GB-month of General Purpose (SSD) provisioned storage under monthly free tier |
| 6 | AmazonEC2 | APS2-EBS:VolumeUsage.gp2 | CreateVolume-Gp2 | | 0.01344086 | USD | $0.00 per GB-month of General Purpose (SSD) provisioned storage under monthly free tier |
| 7 | AmazonEC2 | USW2-BoxUsage:t2.micro | RunInstances:0002 | us-west-2a | 1 | USD | $0.00 per Windows t2.micro instance-hour (or partial hour) under monthly free tier |
| 8 | AmazonEC2 | USW2-USE1-AWS-Out-Bytes | PublicIP-Out | | 0.00000174 | USD | $0.000 per GB - data transfer out under the monthly global free tier |
| 9 | AmazonEC2 | USW2-USE1-AWS-In-Bytes | PublicIP-In | | 0.00000138 | USD | $0.00 per GB - US West (Oregon) data transfer from US East (Northern Virginia) |
| 10 | AmazonEC2 | USW2-USW1-AWS-In-Bytes | PublicIP-In | | 0.00000149 | USD | $0.00 per GB - US West (Oregon) data transfer from US West (Northern California) |
| 11 | AmazonS3 | Requests-Tier1 | ListAllMyBuckets | | 2 | USD | $0.00 per request - PUT, COPY, POST, or LIST requests under the monthly global free tier |
| 12 | AmazonEC2 | USW2-DataTransfer-Out-Bytes | RunInstances | | 0.00038144 | USD | $0.000 per GB - data transfer out under the monthly global free tier |
| 13 | AmazonEC2 | USW2-USW1-AWS-Out-Bytes | PublicIP-Out | | 0.00000174 | USD | $0.000 per GB - data transfer out under the monthly global free tier |
| 14 | AmazonEC2 | USW2-DataTransfer-In-Bytes | RunInstances | | 0.00030951 | USD | $0.000 per GB - data transfer in per month |
| 15 | AmazonEC2 | USW2-BoxUsage:t2.micro | RunInstances:0002 | us-west-2a | 1 | USD | $0.00 per Windows t2.micro instance-hour (or partial hour) under monthly free tier |
| 16 | AmazonEC2 | USW2-USW1-AWS-Out-Bytes | PublicIP-Out | | 0.00000349 | USD | $0.000 per GB - data transfer out under the monthly global free tier |
| 17 | AmazonEC2 | USW2-USW1-AWS-In-Bytes | PublicIP-In | | 0.00000276 | USD | $0.00 per GB - US West (Oregon) data transfer from US West (Northern California) |
| 18 | AmazonEC2 | APS2-EBS:VolumeUsage.gp2 | CreateVolume-Gp2 | | 0.01344086 | USD | $0.00 per GB-month of General Purpose (SSD) provisioned storage under monthly free tier |
| 19 | AmazonEC2 | CW:AlarmMonitorUsage | Unknown | | 0.00134409 | USD | $0.00 per alarm-month - first 10 alarms |
| 20 | AmazonEC2 | USW2-BoxUsage:t2.micro | RunInstances:0002 | us-west-2a | 1 | USD | $0.00 per Windows t2.micro instance-hour (or partial hour) under monthly free tier |
| 21 | AmazonEC2 | USW2-DataTransfer-Regional-Bytes | PublicIP-Out | | 0.00000349 | USD | $0.000 per GB - regional data transfer under the monthly global free tier |
| 22 | AmazonEC2 | USW2-DataTransfer-In-Bytes | RunInstances | | 0.00032071 | USD | $0.000 per GB - data transfer in per month |
| 23 | AmazonEC2 | USW2-DataTransfer-Regional-Bytes | PublicIP-In | | 0.00000302 | USD | $0.000 per GB - regional data transfer under the monthly global free tier |
| 24 | AmazonEC2 | USW2-USE1-AWS-Out-Bytes | PublicIP-Out | | 0.00000174 | USD | $0.000 per GB - data transfer out under the monthly global free tier |
| 25 | AmazonEC2 | USW2-DataTransfer-Out-Bytes | RunInstances | | 0.00045736 | USD | $0.000 per GB - data transfer out under the monthly global free tier |
| 26 | AmazonEC2 | USW2-DataTransfer-In-Bytes | RunInstances | | 0.00036737 | USD | $0.000 per GB - data transfer in per month |
| 27 | AmazonEC2 | USW2-APN2-AWS-In-Bytes | PublicIP-In | | 0.00000005 | USD | $0.00 per GB - US West (Oregon) data transfer from Asia Pacific (Seoul) |
| 28 | AmazonEC2 | USW2-APN2-AWS-Out-Bytes | PublicIP-Out | | 0.00000018 | USD | $0.000 per GB - data transfer out under the monthly global free tier |
| 29 | AmazonEC2 | USW2-USE1-AWS-In-Bytes | PublicIP-In | | 0.00000153 | USD | $0.00 per GB - US West (Oregon) data transfer from US East (Northern Virginia) |
| 30 | AmazonEC2 | USW2-DataTransfer-Out-Bytes | RunInstances | | 0.00039945 | USD | $0.000 per GB - data transfer out under the monthly global free tier |
| 31 | AmazonEC2 | CW:AlarmMonitorUsage | Unknown | | 0.00134409 | USD | $0.00 per alarm-month - first 10 alarms |

1

### Mechanism: effective access is the

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session…

2

### Failure mode

Failure mode: broad principals, unscoped PassRole, stale grants, public resource policies and long-lived…

3

### Evidence: IAM Access Analyzer…

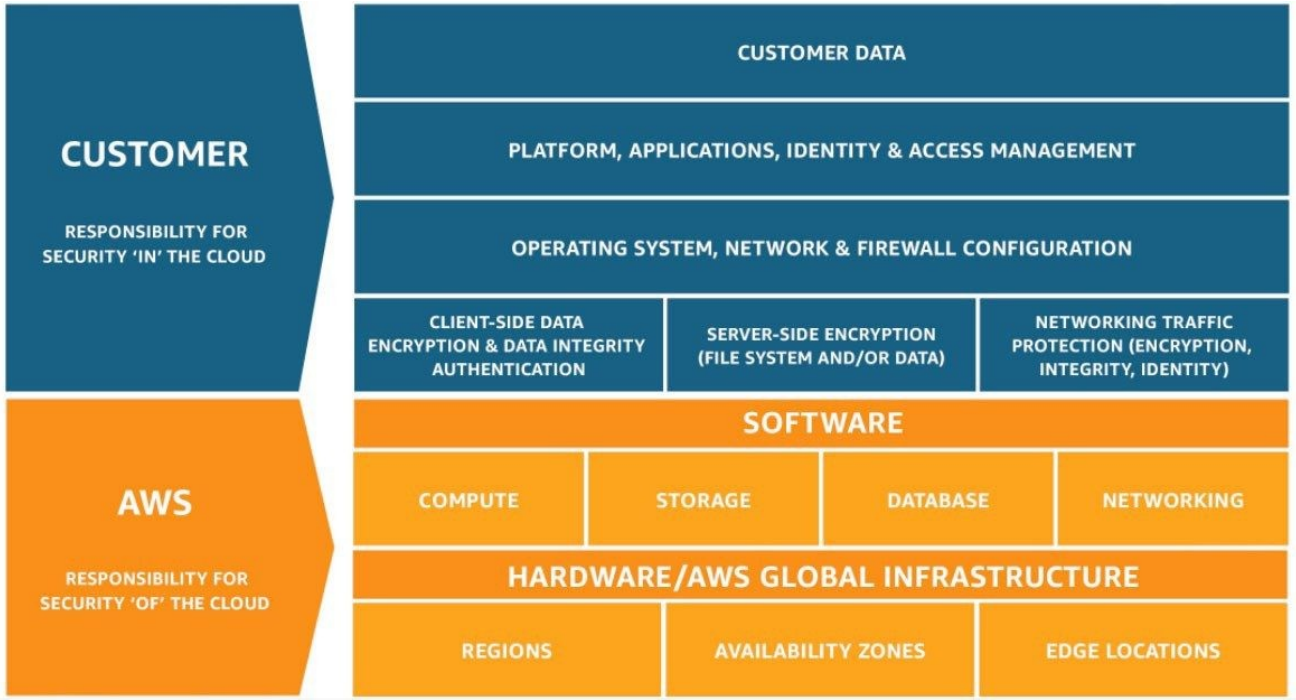
Evidence: IAM Access Analyzer findings, CloudTrail identity context, KMS key/grant policy, Config evaluations and…

**CURRENT OPERATIONS LENS**

Start with the trust boundary, evaluate effective access, protect data, record exceptions and verify that remediation reduced risk. Validate service limits and regional availability against current AWS documentation.


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# AWS Shared Responsibility Model




1

### Mechanism: effective access is the

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session…

2

### Failure mode

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# AWS Shared Responsibility Model

### 1 AWS responsibility

AWS responsibility - Security of the Cloud

For EC2 instance, customer is responsible for management of the guest OS (including security patches and updates), firewall & network configuration…

### 2 Protecting infrastructure…

Protecting infrastructure (hardware, software, facilities, and networking) that runs all the AWS services

Encrypting application data

### 3 Managed services like S3…

Managed services like S3, DynamoDB, RDS, etc.

Shared controls

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session policy, SCP/RCP and explicit denies.

### 4 Customer responsibility

Customer responsibility - Security in the Cloud

Patch Management, Configuration Management, Awareness & Training

**CURRENT OPERATIONS LENS**

Start with the trust boundary, evaluate effective access, protect data, record exceptions and verify that remediation reduced risk. Validate service limits and regional availability against current AWS documentation.


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# attacker

## 1 normal users

normal users

*Distributed Denial-of-Service

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session policy, SCP/RCP and explicit denies.

## 2 Not accessible Not responsive

Not accessible Not responsive

Failure mode: broad principals, unscoped PassRole, stale grants, public resource policies and long-lived credentials expand the trust boundary…

## 3 application server

application server

Evidence: IAM Access Analyzer findings, CloudTrail identity context, KMS key/grant policy, Config evaluations and exception records prove control…

## 4 What's a DDOS* Attack?

What's a DDOS* Attack?

Decision point: remediate with the smallest policy change, validate new access before deployment and verify that the finding resolves without…

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# DDoS Protection on AWS

### 1 AWS Shield Standard

AWS Shield Standard: protects against DDoS attack for your website and applications, for all customers at no additional costs

Availability protection using global edge network

### 2 AWS Shield Advanced

AWS Shield Advanced: 24/7 premium DDoS protection

Combined with AWS Shield, provides attack mitigation at the edge

### 3 AWS WAF

AWS WAF: filter specific requests based on rules

Be ready to scale - leverage AWS Auto Scaling

### 4 CloudFront and Route 53

CloudFront and Route 53

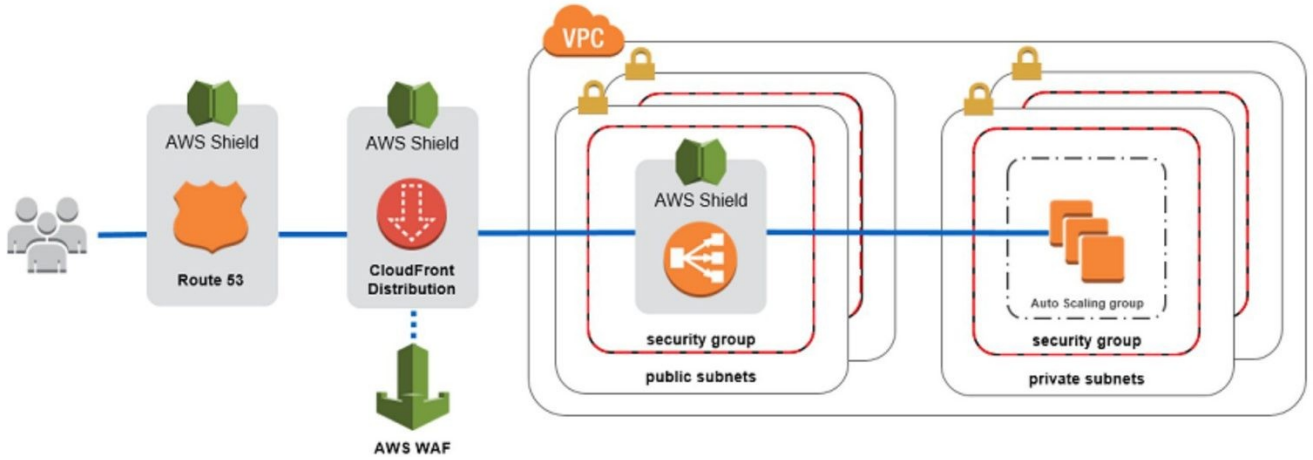
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# Sample Reference Architecture




1

### Mechanism: effective access is the

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session…

2

### Failure mode

Failure mode: broad principals, unscoped PassRole, stale grants, public resource policies and long-lived…

3

### Evidence: IAM Access Analyzer…

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# AWS Shield

### 1 AWS Shield Standard

AWS Shield Standard

Optional DDoS mitigation service ($3,000 per month per organization)

### 2 Free service that is activated

Free service that is activated for every AWS customer

Protect against more sophisticated attack on Amazon EC2, Elastic Load Balancing (ELB), Amazon CloudFront, AWS Global Accelerator, and Route 53

### 3 Provides protection from attacks…

Provides protection from attacks such as SYN/UDP Floods, Reflection attacks and other layer 3/layer 4 attacks

24/7 access to AWS DDoS response team (DRP)

### 4 AWS Shield Advanced

AWS Shield Advanced

Protect against higher fees during usage spikes due to DDoS

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# AWS Shield




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# AWS WAF - Web Application Firewall

### 1 Protects your web applications…

Protects your web applications from common web exploits (Layer 7)

Rules can include: IP addresses, HTTP headers, HTTP body, or URI strings

### 2 Layer 7 is HTTP (vs

Layer 7 is HTTP (vs Layer 4 is TCP)

Protects from common attack - SQL injection and Cross-Site (XSS)

### 3 Deploy on Application Load…

Deploy on Application Load Balancer, API Gateway, CloudFront

Size constraints, geo-match (block countries)

### 4 Define Web ACL (Web Access

Define Web ACL (Web Access Control List)

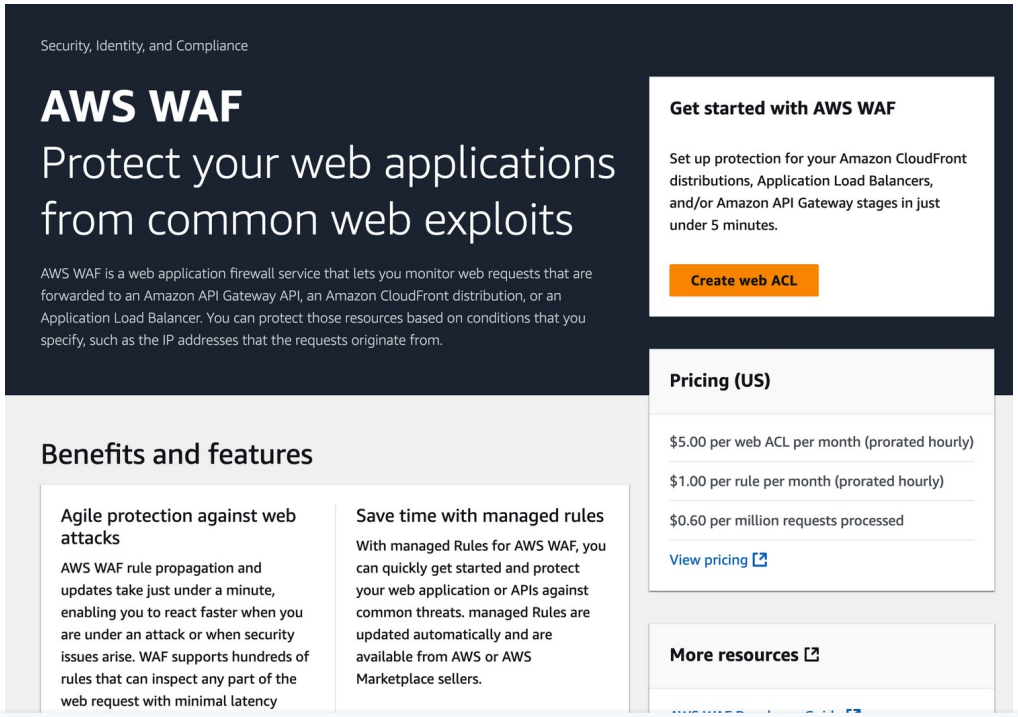
Rate-based rules (to count occurrences of events) - for DDOS protection

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# AWS WAF - Web Application Firewall




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# Penetration Testing on AWS Cloud

### 1 AWS customers are welcome to

AWS customers are welcome to carry out security assessments or penetration tests against their AWS infrastructures without prior approval for 8…

Amazon Aurora

Amazon Elastic Beanstalk environments

### 2 Amazon EC2 instances, NAT…

Amazon EC2 instances, NAT Gateways, and Elastic Load Balancers

Amazon API Gateways

List can increase over time (you won't be tested on that at the exam)

### 3 Amazon RDS

Amazon RDS

AWS Lambda and Lambda Edge functions

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session policy, SCP/RCP and explicit denies.

### 4 Amazon CloudFront

Amazon CloudFront

Amazon Lightsail resources

Failure mode: broad principals, unscoped PassRole, stale grants, public resource policies and long-lived credentials expand the trust boundary…

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# Penetration Testing on your AWS Cloud

### 1 Prohibited Activities

Prohibited Activities

Protocol flooding

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session policy, SCP/RCP and explicit denies.

### 2 DNS zone walking via Amazon

DNS zone walking via Amazon Route 53 Hosted Zones

Request flooding (login request flooding, API request flooding)

### 3 Denial of Service (DoS)…

Denial of Service (DoS), Distributed Denial of Service (DDoS), Simulated DoS, Simulated DDoS

For any other simulated events, contact aws-security-simulated-event@amazon.com

### 4 Port flooding

Port flooding

Failure mode: broad principals, unscoped PassRole, stale grants, public resource policies and long-lived credentials expand the trust boundary…

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# Amazon Inspector

### 1 Automated Security Assessments

Automated Security Assessments

Analyze the running OS against known vulnerabilities

Identifies software vulnerabilities in function code and package dependencies

### 2 For EC2 instances

For EC2 instances

For Container Images push to Amazon ECR

Assessment of functions as they are deployed

### 3 Leveraging the AWS System Manager

Leveraging the AWS System Manager (SSM) agent

Assessment of Container Images as they are pushed

Reporting & integration with AWS Security Hub

### 4 Analyze against unintended…

Analyze against unintended network accessibility

For Lambda Functions

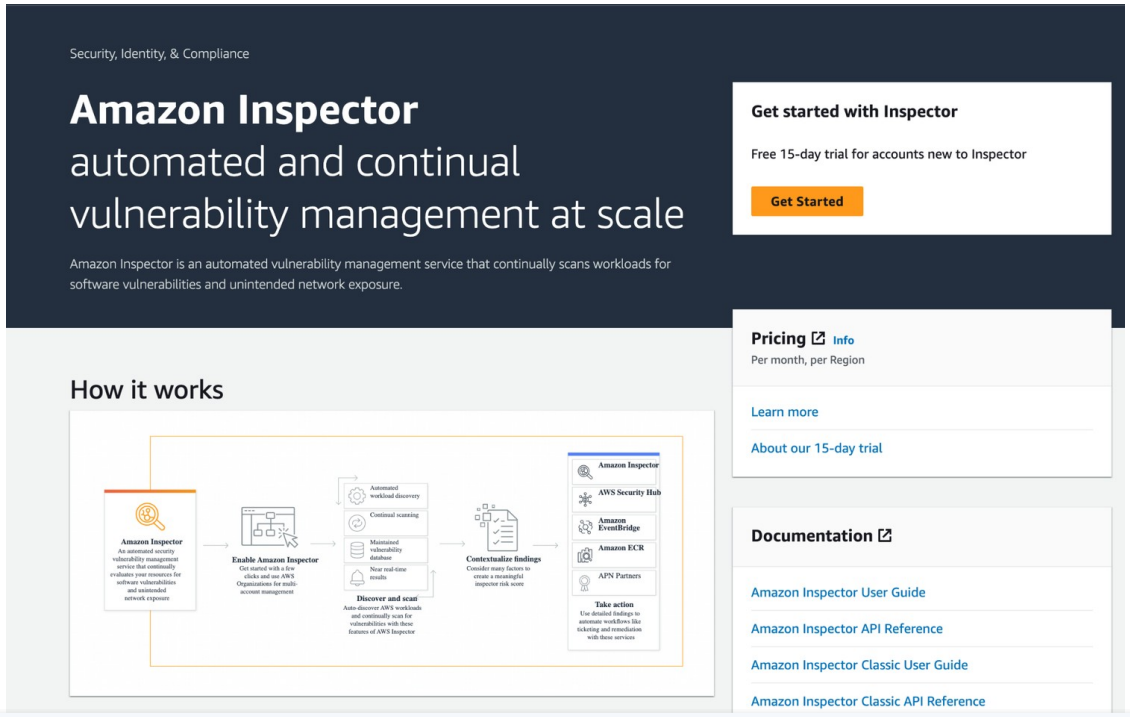
Send findings to Amazon Event Bridge

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# Amazon Inspector




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# What does Amazon Inspector evaluate?

### 1 Remember: only for EC2 instances,

Remember: only for EC2 instances, Container Images & Lambda functions

A risk score is associated with all vulnerabilities for prioritization

### 2 Continuous scanning of the…

Continuous scanning of the infrastructure, only when needed

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session policy, SCP/RCP and explicit denies.

### 3 Package vulnerabilities (EC2, ECR…

Package vulnerabilities (EC2, ECR & Lambda) - database of CVE

### 4 Network reachability (EC2)

Network reachability (EC2)

Failure mode: broad principals, unscoped PassRole, stale grants, public resource policies and long-lived credentials expand the trust boundary…

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# Logging in AWS for security and compliance

### 1 To help compliance requirements…

To help compliance requirements, AWS provides many service-specific security and audit logs

CloudWatch Logs - for full data retention

WAF Logs - full logging of all requests analyzed by the service

### 2 Service Logs include

Service Logs include

VPC Flow Logs - IP traffic within your VPC

Logs can be analyzed using AWS Athena if they're stored in S3

### 3 CloudTrail trails

CloudTrail trails - trace all API calls

ELB Access Logs - metadata of requests made to your load balancers

You should encrypt logs in S3, control access using IAM & Bucket Policies, MFA

### 4 Config Rules

Config Rules - for config & compliance over time

CloudFront Logs - web distribution access logs

Move Logs to Glacier for cost savings

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# Intelligent Threat discovery to protect your AWS Account

### 1 Uses Machine Learning algorithms…

Uses Machine Learning algorithms, anomaly detection, 3rd party data

CloudTrail Management Events – create VPC subnet, create trail, …

Optional Feature - EKS Audit Logs, RDS & Aurora, EBS, Lambda, S3 Data Events…

### 2 Can be enabled at account

Can be enabled at account or organization scope (30 days trial), no need to install software

CloudTrail S3 Data Events – get object, list objects, delete object, …

Can setup EventBridge rules to be notified in case of findings

### 3 Input data includes

Input data includes

VPC Flow Logs - unusual internal traffic, unusual IP address

EventBridge rules can target AWS Lambda or SNS

### 4 CloudTrail Events Logs

CloudTrail Events Logs - unusual API calls, unauthorized deployments

DNS Logs - compromised EC2 instances sending encoded data within DNS queries

Can protect against CryptoCurrency attacks (has a dedicated "finding" for it)

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# Amazon GuardDuty

## 1 VPC Flow Logs

VPC Flow Logs

Optional Features

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session policy, SCP/RCP and explicit denies.

## 2 CloudTrail Logs

CloudTrail Logs

EBS Volumes

Failure mode: broad principals, unscoped PassRole, stale grants, public resource policies and long-lived credentials expand the trust boundary…

## 3 DNS Logs (AWS DNS)

DNS Logs (AWS DNS)

Lambda Network RDS & Aurora Activity Login Activity

## 4 EventBridge

EventBridge

EKS Audit Logs & Runtime Monitoring

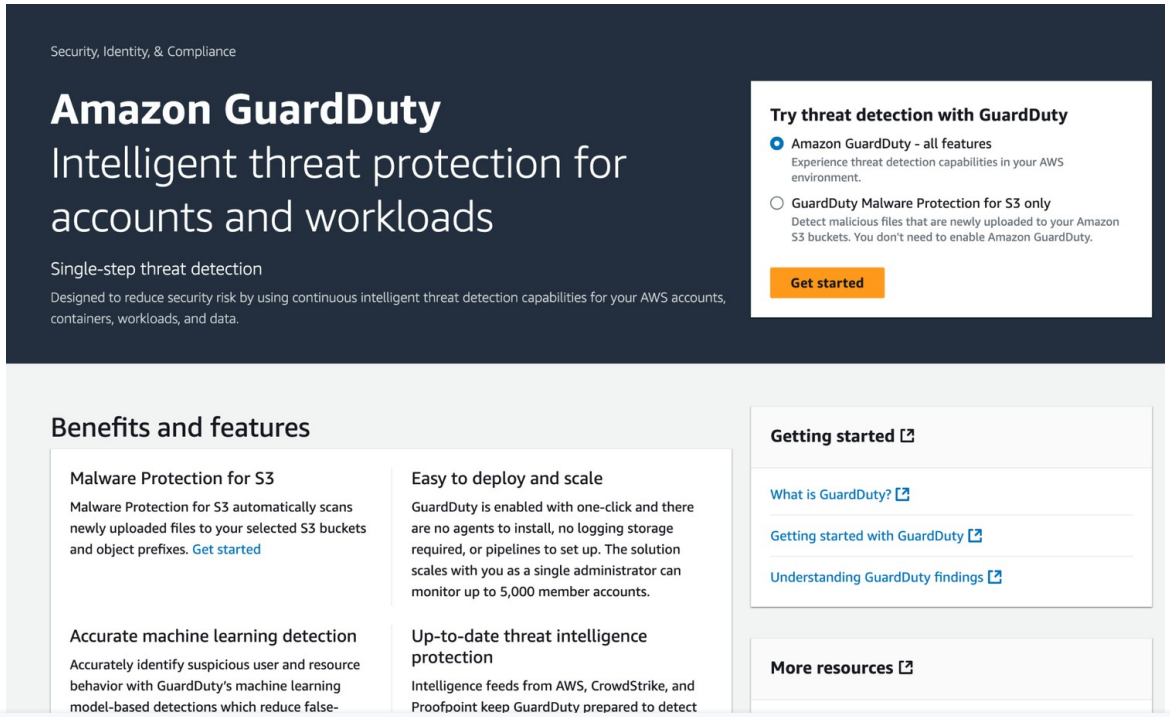
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# Amazon GuardDuty




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# AWS Macie

### 1 Amazon Macie is a fully

Amazon Macie is a fully managed data security and data privacy service that uses machine learning and pattern matching to discover and protect your…

Amazon EventBridge

### 2 Macie helps identify and alert

Macie helps identify and alert you to sensitive data, such as personally identifiable information (PII)

integrations

### 3 S3 Buckets

S3 Buckets

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session policy, SCP/RCP and explicit denies.

### 4 Discover Sensitive Data (PII)

Discover Sensitive Data (PII)

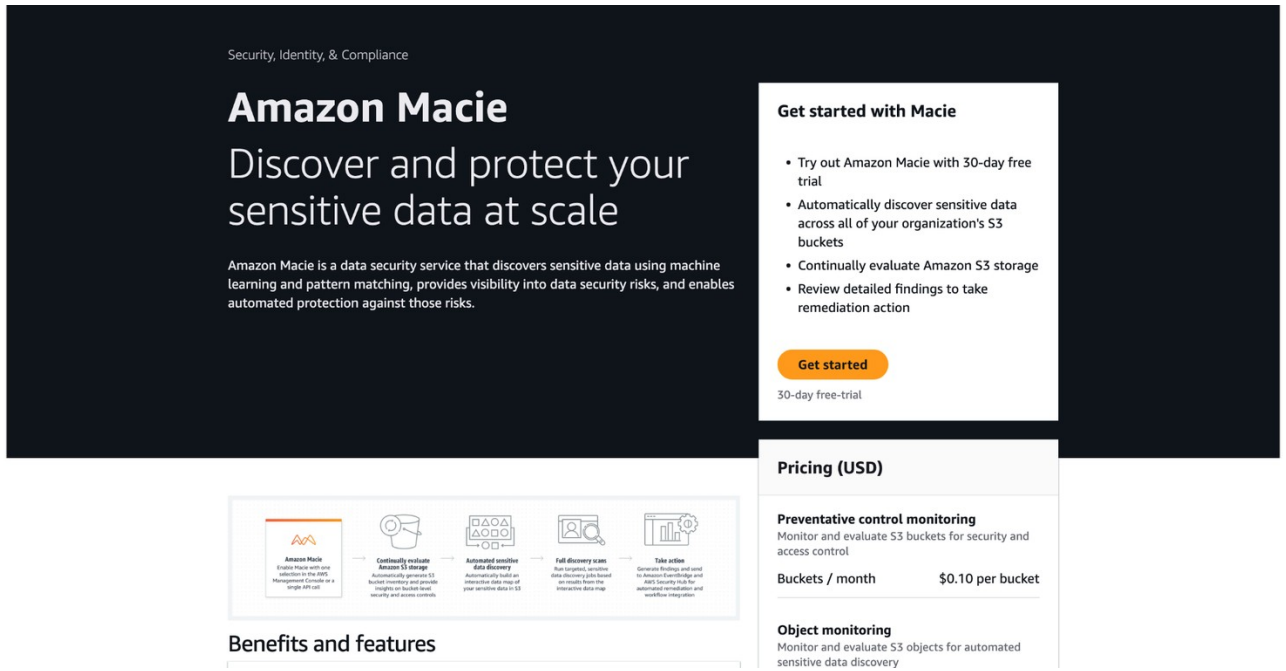
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# Trusted Advisor

### 1 No need to install anything

No need to install anything - high level AWS account assessment

Fault tolerance

Full Set of Checks

### 2 Analyze your AWS accounts and

Analyze your AWS accounts and provides recommendation on 6 categories

Service limits

Programmatic Access using AWS Support API

### 3 Cost optimization

Cost optimization

Operational Excellence

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session policy, SCP/RCP and explicit denies.

### 4 Performance

Performance

Business & Enterprise Support plan

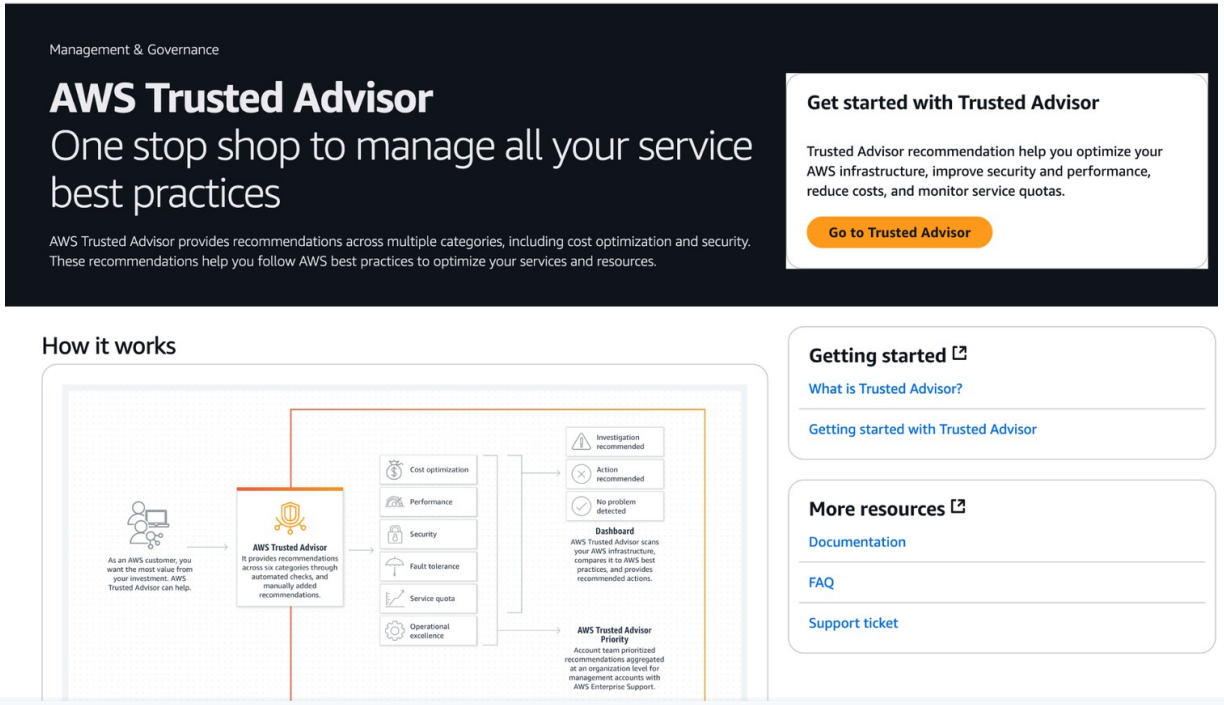
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# Why encryption?

### 1 Encryption in flight (TLS /

Encryption in flight (TLS / SSL)

HTTPS Website

aGVsbG8gd29 ybGQgZWh…

### 2 Data is encrypted before sending

Data is encrypted before sending and decrypted after receiving

TLS Encryption

### 3 TLS certificates help with…

TLS certificates help with encryption (HTTPS)

TLS Decryption

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session policy, SCP/RCP and explicit denies.

### 4 Encryption in flight ensures no

Encryption in flight ensures no MITM (man in the middle attack) can happen

Username: admin Password: supersecret

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# Why encryption?

1

**Server-side encryption at rest**

Server-side encryption at rest

The encryption / decryption keys must be managed somewhere, and the server must have access to it

2

**Data is encrypted after being**

Data is encrypted after being received by the server

AWS Service (e.g., S3)

3

**Data is decrypted before being**

Data is decrypted before being sent

Encryption

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session policy, SCP/RCP and explicit denies.

4

**It is stored in an**

It is stored in an encrypted form thanks to a key (usually a data key)

Decryption

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# Why encryption?

### 1 Client-side encryption

Client-side encryption

Could leverage Envelope Encryption

Any storage service (FTP, S3, …)

### 2 Data is encrypted by the

Data is encrypted by the client and never decrypted by the server

Data key (client-side)

Decryption

### 3 Data will be decrypted by

Data will be decrypted by a receiving client

Encryption

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session policy, SCP/RCP and explicit denies.

### 4 The server should not be

The server should not be able to decrypt the data

Encrypted object

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# AWS KMS (Key Management Service)

### 1 Anytime you hear “encryption” for

Anytime you hear “encryption” for an AWS service, it’s most likely KMS

Able to audit KMS Key usage using CloudTrail

Encrypted secrets can be stored in the code / environment variables

### 2 AWS manages encryption keys for

AWS manages encryption keys for us

Seamlessly integrated into most AWS services (EBS, S3, RDS, SSM…)

### 3 Fully integrated with IAM for

Fully integrated with IAM for authorization

Never ever store your secrets in plaintext, especially in your code!

### 4 Easy way to control access

Easy way to control access to your data

KMS Key Encryption also available through API calls (SDK, CLI)

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# KMS Keys Types

### 1 KMs Keys is the new

KMs Keys is the new name of KMS Customer Master Key

You never get access to the KMS Key unencrypted (must call KMS API to use)

The public key is downloadable, but you can't access the Private Key unencrypted

### 2 Symmetric (AES-256 keys)

Symmetric (AES-256 keys)

Asymmetric (RSA & ECC key pairs)

Use case: encryption outside of AWS by users who can't call the KMS API

### 3 Single encryption key that is

Single encryption key that is used to Encrypt and Decrypt

Public (Encrypt) and Private Key (Decrypt) pair

### 4 AWS services that are integrated

AWS services that are integrated with KMS use Symmetric CMKs

Used for Encrypt/Decrypt, or Sign/Verify operations

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# AWS KMS (Key Management Service)

## 1 Types of KMS Keys

Types of KMS Keys

Customer managed keys imported: $1 / month

Customer-managed KMS Key: (must be enabled) automatic & on-demand

## 2 AWS Owned Keys (free)

AWS Owned Keys (free): SSE-S3, SSE-SQS, SSE-DDB (default key)

+ pay for API call to KMS ($0.03 / 10000 calls)

Imported KMS Key: only manual rotation possible using alias

## 3 AWS Managed Key

AWS Managed Key: free (aws/service-name, example: aws/rds or aws/ebs)

Automatic Key rotation

## 4 Customer managed keys created in…

Customer managed keys created in KMS: $1 / month

AWS-managed KMS Key: automatic every 1 year

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# Copying Snapshots across regions

### 1 EBS Volume Encrypted With KMS

EBS Volume Encrypted With KMS

KMS ReEncrypt with KMS Key B

### 2 EBS Snapshot Encrypted With KMS

EBS Snapshot Encrypted With KMS

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session policy, SCP/RCP and explicit denies.

### 3 Region eu-west-2

Region eu-west-2

Failure mode: broad principals, unscoped PassRole, stale grants, public resource policies and long-lived credentials expand the trust boundary…

### 4 Region ap-southeast-2

Region ap-southeast-2

Evidence: IAM Access Analyzer findings, CloudTrail identity context, KMS key/grant policy, Config evaluations and exception records prove control…

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# KMS Key Policies

### 1 Control access to KMS keys,

Control access to KMS keys, “similar” to S3 bucket policies

Complete access to the key to the root user = entire AWS account

Useful for cross-account access of your KMS key

### 2 Difference: you cannot control…

Difference: you cannot control access without them

Custom KMS Key Policy

### 3 Default KMS Key Policy

Default KMS Key Policy

Define users, roles that can access the KMS key

### 4 Created if you don’t provide

Created if you don’t provide a specific KMS Key Policy

Define who can administer the key

**CURRENT OPERATIONS LENS**

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# KMS Automatic Key Rotation

1

**111122223333**

111122223333

AWS-managed KMS Keys: automatically rotated every 1 year

Previous key is kept active so you can decrypt old data

2

**KMS Key ID = 1234abcd-12ab-34cd-56…**

KMS Key ID = 1234abcd-12ab-34cd-56ef-1234567890ab

For Customer-Managed Symmetric KMS Key

New Key has the same KMS Key ID (only the backing key is changed)

3

**444455556666**

444455556666

Automatic key rotation is optionally enabled

Backing Key Backing Key Saved Backing Key

4

**Automatic Key rotation**

Automatic Key rotation

Customize Rotation Period between 90 and 2560 days (default: 365 days)

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# KMS On-Demand Key Rotation

### 1 For Customer-Managed Symmetric…

For Customer-Managed Symmetric KMS Key (not AWS managed CMK)

On Demand Key rotation

Backing Key Backing Key Saved Backing Key

### 2 Does NOT require Automatic Key

Does NOT require Automatic Key Rotation to be enabled

111122223333

### 3 Does NOT change existing Automatic

Does NOT change existing Automatic Rotation schedules

KMS Key ID = 1234abcd-12ab-34cd-56ef-1234567890ab

### 4 Limit to how many times

Limit to how many times you can trigger an on-demand key rotation

444455556666

**CURRENT OPERATIONS LENS**

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# KMS Manual Key Rotation

1

**(Customer-Managed Symmetric KMS…**

(Customer-Managed Symmetric KMS Key & Imports)

When you want to rotate key (example: every month)

Good solution to rotate KMS Key that are not eligible for automatic rotation (like asymmetric KMS key)

2

**111122223333**

111122223333

New Key has a different KMS Key ID

Rotate Key Manually

3

**Alias = MyCustomKey**

Alias = MyCustomKey

Keep the previous key active so you can decrypt old data

111122223333 444455556666

4

**KMS Key ID = 1234abcd-12ab-34cd-56…**

KMS Key ID = 1234abcd-12ab-34cd-56ef-1234567890ab

Better to use aliases in this case (to hide the change of key for the application)

KMS Key ID = 1234abcd-12ab-34cd-56ef-1234567890ab KMS Key ID = 0978dcba-09fe-65ba-ab0987654321

**CURRENT OPERATIONS LENS**

Start with the trust boundary, evaluate effective access, protect data, record exceptions and verify that remediation reduced risk. Validate service limits and regional availability against current AWS documentation.


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# KMS Alias Updating

### 1 Key Alias = MyAppKey

Key Alias = MyAppKey

Better to use aliases in this case (to hide the change of key for the application)

### 2 UpdateAlias API

UpdateAlias API

Application

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session policy, SCP/RCP and explicit denies.

### 3 KMS Key ID = eba96e8f-7f66-4d5a-ae…

KMS Key ID = eba96e8f-7f66-4d5a-aeb6-f32ae8d65d85

Failure mode: broad principals, unscoped PassRole, stale grants, public resource policies and long-lived credentials expand the trust boundary…

### 4 KMS Key ID = 881cd7f5-b3eb-43e2-82…

KMS Key ID = 881cd7f5-b3eb-43e2-82fc-d6f5bda81e00

Evidence: IAM Access Analyzer findings, CloudTrail identity context, KMS key/grant policy, Config evaluations and exception records prove control…

**CURRENT OPERATIONS LENS**

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# Sharing KMS Encrypted RDS DB Snapshots

### 1 You can share RDS DB

You can share RDS DB snapshots encrypted with KMS CMK with other accounts but must first share the KMS CMK with the target account using Key Policy

DB Instance

### 2 Account A (111122223333)

Account A (111122223333)

(Snapshot: mydbsnapshot)

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session policy, SCP/RCP and explicit denies.

### 3 Account B (444455556666)

Account B (444455556666)

(encrypted)

Failure mode: broad principals, unscoped PassRole, stale grants, public resource policies and long-lived credentials expand the trust boundary…

### 4 access through CMK Key Policy

access through CMK Key Policy

Key Policy

Evidence: IAM Access Analyzer findings, CloudTrail identity context, KMS key/grant policy, Config evaluations and exception records prove control…

**CURRENT OPERATIONS LENS**

Start with the trust boundary, evaluate effective access, protect data, record exceptions and verify that remediation reduced risk. Validate service limits and regional availability against current AWS documentation.


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# KMS Key Deletion Considerations

### 1 Schedule CMK for deletion with

Schedule CMK for deletion with a waiting period of 7 to 30 days

The key is not rotated even if planned

### 2 CMK’s status is "Pending deletion"

CMK’s status is "Pending deletion" during the waiting period

You can cancel the key deletion during the waiting period

### 3 During the CMK’s deletion waiting

During the CMK’s deletion waiting period

Consider disabling your key instead of deleting it if you’re not sure!

### 4 The CMK can't be used

The CMK can't be used for cryptographic operations (e.g., can’t decrypt KMS- encrypted objects in S3 - SSE-KMS)

**CURRENT OPERATIONS LENS**

Start with the trust boundary, evaluate effective access, protect data, record exceptions and verify that remediation reduced risk. Validate service limits and regional availability against current AWS documentation.


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# KMS Key Deletion - CloudWatch Alarm

### 1 Use CloudTrail, CloudWatch Logs…

Use CloudTrail, CloudWatch Logs, CloudWatch Alarms and SNS to be notified when someone tries to use a CMK that's "Pending deletion" in a…

"* is pending deletion"

log API calls (DENIED)

### 2 Amazon SNS

Amazon SNS

(Pending deletion)

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session policy, SCP/RCP and explicit denies.

### 3 CloudWatch

CloudWatch

CloudTrail

Failure mode: broad principals, unscoped PassRole, stale grants, public resource policies and long-lived credentials expand the trust boundary…

### 4 Metric Filter

Metric Filter

cryptographic operation

Evidence: IAM Access Analyzer findings, CloudTrail identity context, KMS key/grant policy, Config evaluations and exception records prove control…

**CURRENT OPERATIONS LENS**

Start with the trust boundary, evaluate effective access, protect data, record exceptions and verify that remediation reduced risk. Validate service limits and regional availability against current AWS documentation.


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# CloudHSM

1

### KMS => AWS manages the

KMS => AWS manages the software for encryption

HSM device is tamper resistant, FIPS 140-2 Level 3 compliance

Redshift supports CloudHSM for database encryption and key management

2

### CloudHSM => AWS provisions…

CloudHSM => AWS provisions encryption hardware

Supports both symmetric and asymmetric encryption (SSL/TLS keys)

Good option to use with SSE-C encryption

3

### Dedicated Hardware (HSM = Hardware

Dedicated Hardware (HSM = Hardware Security Module)

No free tier available

4

### You manage your own encryption

You manage your own encryption keys entirely (not AWS)

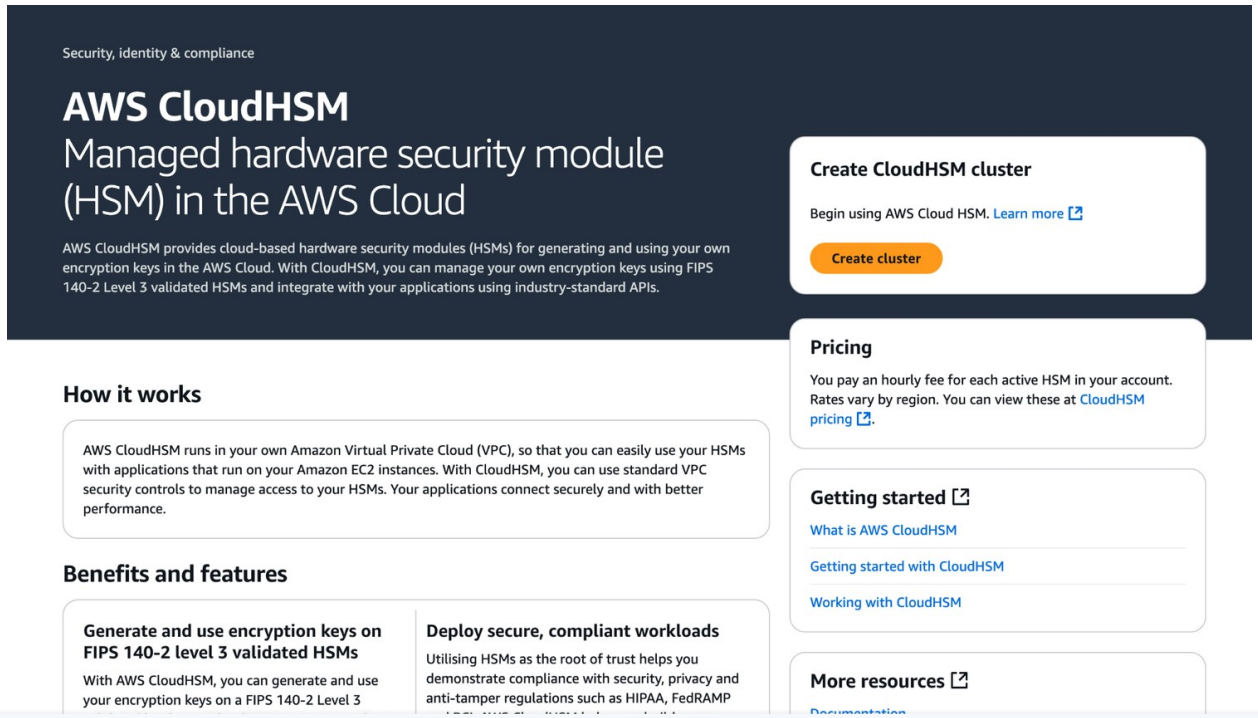
Must use the CloudHSM Client Software

**CURRENT OPERATIONS LENS**

Start with the trust boundary, evaluate effective access, protect data, record exceptions and verify that remediation reduced risk. Validate service limits and regional availability against current AWS documentation.


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# CloudHSM




1

### Mechanism: effective access is the

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session…

2

### Failure mode

Failure mode: broad principals, unscoped PassRole, stale grants, public resource policies and long-lived…

3

### Evidence: IAM Access Analyzer…

Evidence: IAM Access Analyzer findings, CloudTrail identity context, KMS key/grant policy, Config evaluations and…

**CURRENT OPERATIONS LENS**

Start with the trust boundary, evaluate effective access, protect data, record exceptions and verify that remediation reduced risk. Validate service limits and regional availability against current AWS documentation.


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# CloudHSM Diagram

## 1 AWS CloudHSM

AWS CloudHSM

User manages the Keys

Manage the Keys

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session policy, SCP/RCP and explicit denies.

## 2 CloudHSM Client

CloudHSM Client

IAM permissions

Manage the Users

Failure mode: broad principals, unscoped PassRole, stale grants, public resource policies and long-lived credentials expand the trust boundary…

## 3 AWS manages the Hardware

AWS manages the Hardware

CRUD an HSM Cluster

Evidence: IAM Access Analyzer findings, CloudTrail identity context, KMS key/grant policy, Config evaluations and exception records prove control…

## 4 SSL Connection

SSL Connection

CloudHSM Software

Decision point: remediate with the smallest policy change, validate new access before deployment and verify that the finding resolves without…

**CURRENT OPERATIONS LENS**

Start with the trust boundary, evaluate effective access, protect data, record exceptions and verify that remediation reduced risk. Validate service limits and regional availability against current AWS documentation.


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# CloudHSM - High Availability

### 1 CloudHSM clusters are spread…

CloudHSM clusters are spread across Multi AZ (HA)

Availability Zone 1

### 2 Great for availability and…

Great for availability and durability

CloudHSM 2

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session policy, SCP/RCP and explicit denies.

### 3 CloudHSM 1

CloudHSM 1

Availability Zone 2

Failure mode: broad principals, unscoped PassRole, stale grants, public resource policies and long-lived credentials expand the trust boundary…

### 4 CloudHSM Client

CloudHSM Client

Evidence: IAM Access Analyzer findings, CloudTrail identity context, KMS key/grant policy, Config evaluations and exception records prove control…

**CURRENT OPERATIONS LENS**

Start with the trust boundary, evaluate effective access, protect data, record exceptions and verify that remediation reduced risk. Validate service limits and regional availability against current AWS documentation.


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# CloudHSM - Integration with AWS Services

### 1 Through integration with AWS KMS

Through integration with AWS KMS

(Custom Key Store)

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session policy, SCP/RCP and explicit denies.

### 2 Configure KMS Custom Key Store

Configure KMS Custom Key Store with CloudHSM

KMS Encryption

Failure mode: broad principals, unscoped PassRole, stale grants, public resource policies and long-lived credentials expand the trust boundary…

### 3 Example: EBS, S3, RDS …

Example: EBS, S3, RDS …

CloudTrail

Evidence: IAM Access Analyzer findings, CloudTrail identity context, KMS key/grant policy, Config evaluations and exception records prove control…

### 4 EBS Volume

EBS Volume

keys usage logs

Decision point: remediate with the smallest policy change, validate new access before deployment and verify that the finding resolves without…

**CURRENT OPERATIONS LENS**

Start with the trust boundary, evaluate effective access, protect data, record exceptions and verify that remediation reduced risk. Validate service limits and regional availability against current AWS documentation.


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# AWS Artifact (not really a service)

### 1 Portal that provides customers…

Portal that provides customers with on-demand access to AWS compliance documentation and AWS agreements

### 2 Artifact Reports

Artifact Reports - Allows you to download AWS security and compliance documents from third-party auditors, like AWS ISO certifications, Payment Card…

### 3 Artifact Agreements

Artifact Agreements - Allows you to review, accept, and track the status of AWS agreements such as the Business Associate Addendum (BAA) or the…

### 4 Can be used to support

Can be used to support internal audit or compliance

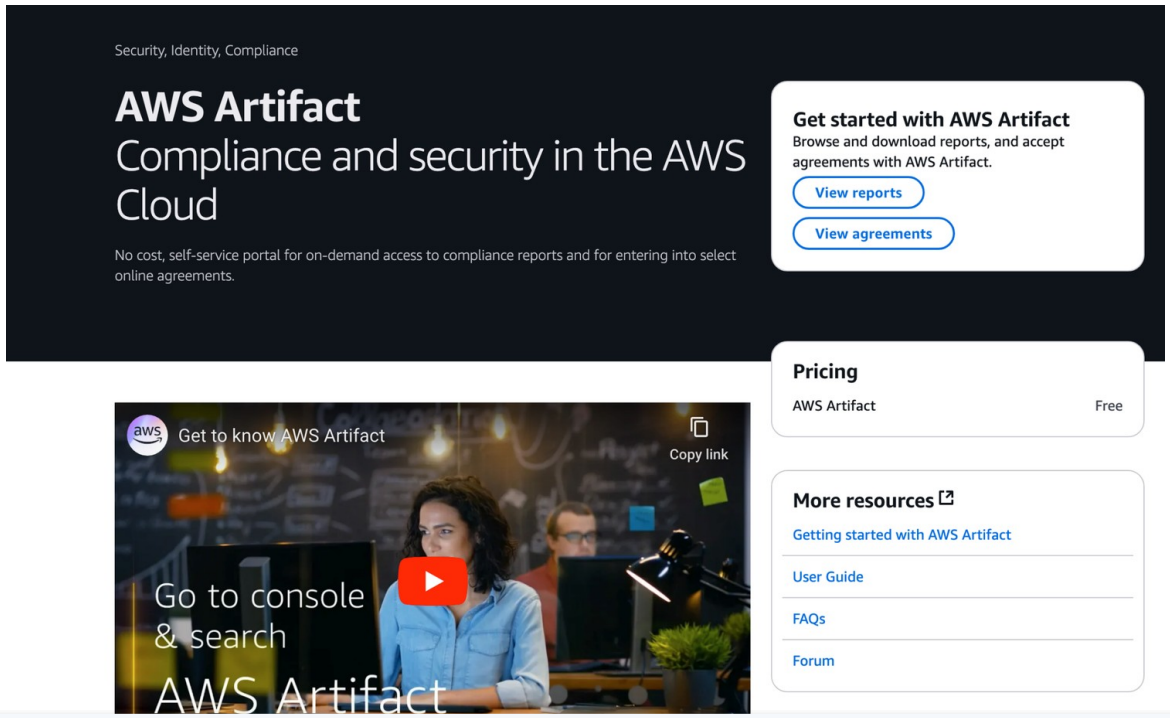
Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session policy, SCP/RCP and explicit denies.

**CURRENT OPERATIONS LENS**

Start with the trust boundary, evaluate effective access, protect data, record exceptions and verify that remediation reduced risk. Validate service limits and regional availability against current AWS documentation.


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# AWS Artifact (not really a service)




**1** **Mechanism: effective access is the**

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session…

**2** **Failure mode**

Failure mode: broad principals, unscoped PassRole, stale grants, public resource policies and long-lived…

**3** **Evidence: IAM Access Analyzer…**

Evidence: IAM Access Analyzer findings, CloudTrail identity context, KMS key/grant policy, Config evaluations and…

**CURRENT OPERATIONS LENS**

Start with the trust boundary, evaluate effective access, protect data, record exceptions and verify that remediation reduced risk. Validate service limits and regional availability against current AWS documentation.


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# AWS Certificate Manager (ACM)

### 1 Easily provision, manage, and…

Easily provision, manage, and deploy TLS Certificates

Automatic TLS certificate renewal

APIs on API Gateway

### 2 Provide in-flight encryption for…

Provide in-flight encryption for websites (HTTPS)

Integrations with (load TLS certificates on)

Cannot use ACM with EC2 (can't be extracted)

### 3 Supports both public and private

Supports both public and private TLS certificates

Elastic Load Balancers (CLB, ALB, NLB)

Auto Scaling group

### 4 Free of charge for public

Free of charge for public TLS certificates

CloudFront Distributions

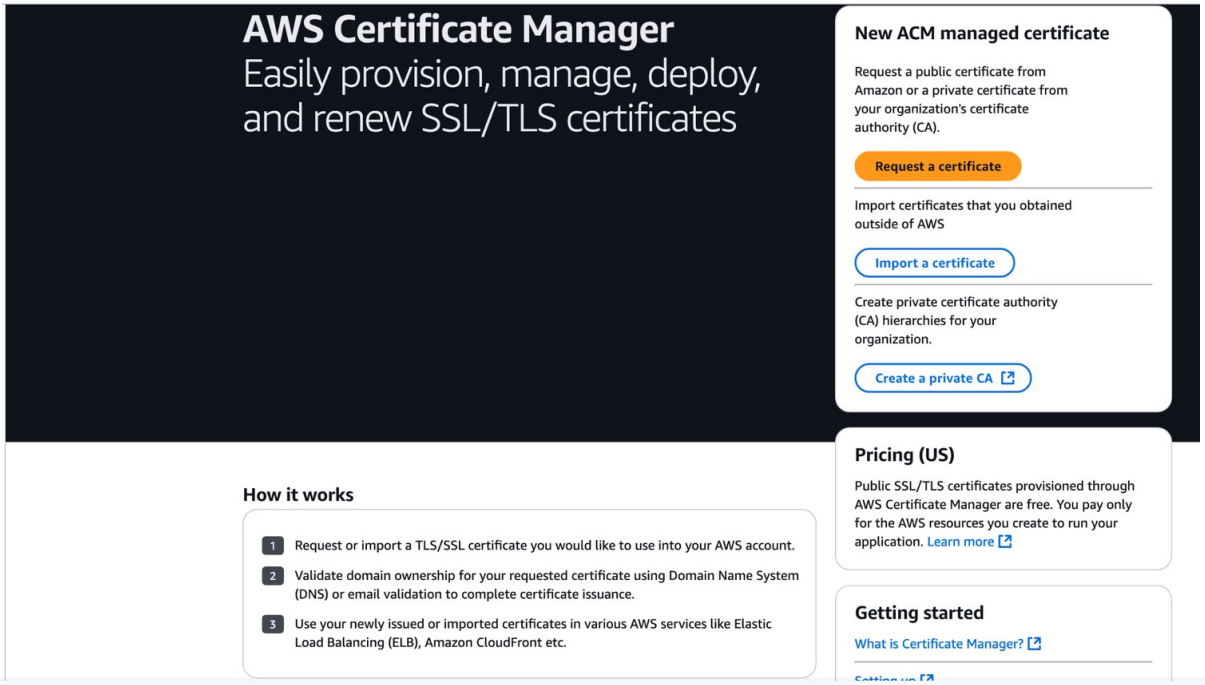
EC2 Instance

**CURRENT OPERATIONS LENS**

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# AWS Certificate Manager (ACM)




**1** **Mechanism: effective access is the**

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session…

**2** **Failure mode**

Failure mode: broad principals, unscoped PassRole, stale grants, public resource policies and long-lived…

**3** **Evidence: IAM Access Analyzer…**

Evidence: IAM Access Analyzer findings, CloudTrail identity context, KMS key/grant policy, Config evaluations and…

**CURRENT OPERATIONS LENS**

Start with the trust boundary, evaluate effective access, protect data, record exceptions and verify that remediation reduced risk. Validate service limits and regional availability against current AWS documentation.


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# ACM - Requesting Public Certificates

### 1 List domain names to be

List domain names to be included in the certificate

DNS Validation is preferred for automation purposes

The Public Certificate will be enrolled for automatic renewal

### 2 Fully Qualified Domain Name (FQDN)

Fully Qualified Domain Name (FQDN): corp.example.com

Email validation will send emails to contact addresses in the WHOIS database

ACM automatically renews ACM-generated certificates 60 days before expiry

### 3 Wildcard Domain

Wildcard Domain: *.example.com

DNS Validation will leverage a CNAME record to DNS config (ex: Route 53)

### 4 Select Validation Method

Select Validation Method: DNS Validation or Email validation

It will take a few hours to get verified

**CURRENT OPERATIONS LENS**

Start with the trust boundary, evaluate effective access, protect data, record exceptions and verify that remediation reduced risk. Validate service limits and regional availability against current AWS documentation.


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# ACM - Importing Public Certificates

### 1 Option to generate the certificate

Option to generate the certificate

starting 45 days prior to expiration

AWS Config

### 2 outside of ACM and then

outside of ACM and then import it

The # of days can be configured

EventBridge

### 3 No automatic renewal, must import

No automatic renewal, must import a new certificate before expiry

Events are appearing in EventBridge

Rule events: Non-compliance

### 4 ACM sends daily expiration events

ACM sends daily expiration events

AWS Config was a managed rule named acm-certificate-expiration-check to check for expiring certificates (configurable number of days)

ACM Events

**CURRENT OPERATIONS LENS**

Start with the trust boundary, evaluate effective access, protect data, record exceptions and verify that remediation reduced risk. Validate service limits and regional availability against current AWS documentation.


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# ACM - Integration with ALB

## 1 Auto Scaling group

Auto Scaling group

With HTTP -> HTTPS redirect rule

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session policy, SCP/RCP and explicit denies.

## 2 EC2 Instance

EC2 Instance

provision and maintain TLS certs

Failure mode: broad principals, unscoped PassRole, stale grants, public resource policies and long-lived credentials expand the trust boundary…

## 3 AWS Certificate Manager

AWS Certificate Manager

Redirect to HTTPS HTTPS

Evidence: IAM Access Analyzer findings, CloudTrail identity context, KMS key/grant policy, Config evaluations and exception records prove control…

## 4 Application Load Balancer

Application Load Balancer

Decision point: remediate with the smallest policy change, validate new access before deployment and verify that the finding resolves without…

**CURRENT OPERATIONS LENS**

Start with the trust boundary, evaluate effective access, protect data, record exceptions and verify that remediation reduced risk. Validate service limits and regional availability against current AWS documentation.


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# ACM - Integration with API Gateway

### 1 Create a Custom Domain Name

Create a Custom Domain Name in API Gateway

The TLS Certificate must be in the same region as CloudFront, in us-east-1

CloudFront

### 2 Edge-Optimized (default)

Edge-Optimized (default): For global clients

Then setup CNAME or (better) A-Alias record in Route 53

linked certificate

### 3 Requests are routed through the

Requests are routed through the CloudFront Edge locations (improves latency)

For clients within the same region

API Gateway

### 4 The API Gateway still lives

The API Gateway still lives in only one region

The TLS Certificate must be imported on API Gateway, in the same region as the API Stage

Edge-Optimized

**CURRENT OPERATIONS LENS**

Start with the trust boundary, evaluate effective access, protect data, record exceptions and verify that remediation reduced risk. Validate service limits and regional availability against current AWS documentation.


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# API Gateway - Endpoint Types

### 1 Edge-Optimized (default)

Edge-Optimized (default): For global clients

Could manually combine with CloudFront (more control over the caching strategies and the distribution)

### 2 Requests are routed through the

Requests are routed through the CloudFront Edge locations (improves latency)

Can only be accessed from your VPC using an interface VPC endpoint (ENI)

### 3 The API Gateway still lives

The API Gateway still lives in only one region

Use a resource policy to define access

### 4 For clients within the same

For clients within the same region

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session policy, SCP/RCP and explicit denies.

**CURRENT OPERATIONS LENS**

Start with the trust boundary, evaluate effective access, protect data, record exceptions and verify that remediation reduced risk. Validate service limits and regional availability against current AWS documentation.


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# SSM Parameter Store

### 1 Secure storage for configuration…

Secure storage for configuration and secrets

Configuration management using path & IAM

Applications

### 2 Optional Seamless Encryption…

Optional Seamless Encryption using KMS

Notifications with Amazon EventBridge

Plaintext configuration

### 3 Serverless, scalable, durable…

Serverless, scalable, durable, easy SDK

Integration with CloudFormation

Encrypted configuration

### 4 Version tracking of…

Version tracking of configurations / secrets

SSM Parameter

Check IAM permissions

**CURRENT OPERATIONS LENS**

Start with the trust boundary, evaluate effective access, protect data, record exceptions and verify that remediation reduced risk. Validate service limits and regional availability against current AWS documentation.


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# SSM Parameter Store Hierarchy

### 1 /my-department/

/my-department/

/aws/reference/secretsmanager/secret_ID_in_Secrets_Manager

Prod Lambda Function

Mechanism: effective access is the intersection of identity policy, resource policy, permission boundary, session policy, SCP/RCP and explicit denies.

### 2 db-password

db-password

/aws/service/ami-amazon-linux-latest/amzn2-ami-hvm-x86_64-gp2

Failure mode: broad principals, unscoped PassRole, stale grants, public resource policies and long-lived credentials expand the trust boundary…

### 3 other-app/

other-app/

Dev Lambda Function

Evidence: IAM Access Analyzer findings, CloudTrail identity context, KMS key/grant policy, Config evaluations and exception records prove control…

### 4 /other-department/

/other-department/

GetParameters or GetParametersByPath API

Decision point: remediate with the smallest policy change, validate new access before deployment and verify that the finding resolves without…

**CURRENT OPERATIONS LENS**

Start with the trust boundary, evaluate effective access, protect data, record exceptions and verify that remediation reduced risk. Validate service limits and regional availability against current AWS documentation.


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# Original PPT Coverage Retained — Topic 07

1

### Modernised slides

55 technical slides extracted from the original v4 PowerPoint and rebuilt in the current visual system.

2

### Traceability

Every blended slide names its original v4 source slide number in the kicker.

3

### Current validation

Terminology is normalised; service limits and Region availability remain explicit current-documentation checks.

4

### Procedure boundary

Click-by-click sequences, activities and quizzes are kept out of the deck and remain in the Learner Guide or lab package.


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TOPIC 07 · CONCEPT-LED LAB BRIEF

# Build a Multi-Account Compliance Evidence Flow

**LAB 07**

Apply K1, K3, A1 and A7 by evaluating required tags and encryption, aggregating findings and documenting a controlled remediation exception.

Deploy Config-style controls, review findings and produce auditable remediation and exception evidence.

### YOU'LL BUILD

A Config rule template, tag policy, findings register, remediation decision and service-quality scorecard.

### TOOLCHAIN

AWS Config · Organizations · Security Hub · EventBridge · Systems Manager Automation

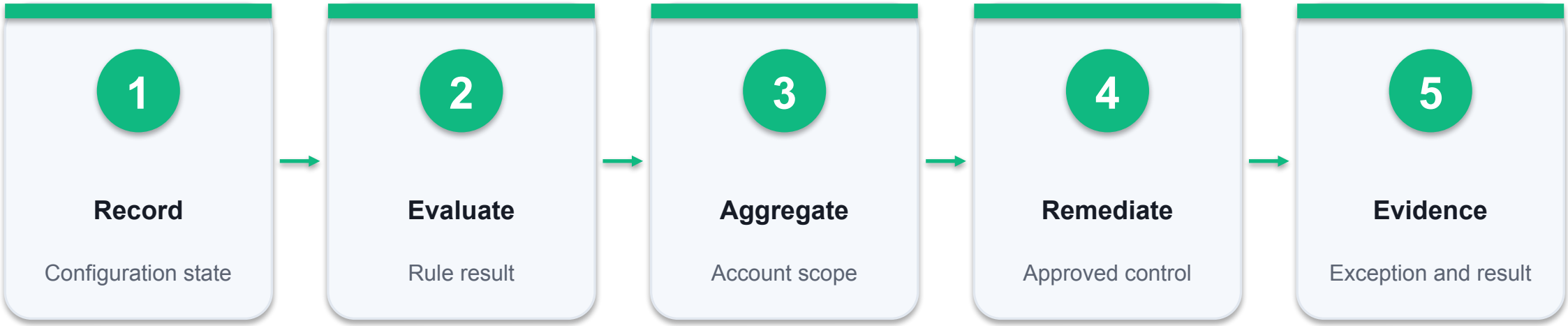
### DONE WHEN

Required controls evaluate learner-owned resources, findings are prioritised with context, remediation has owner/SLA/verification, the exception expires, and validation emits LAB 07 PASS.


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LAB 07 · CONCEPTUAL WORKFLOW

# Lab 07 Evidence Model




**PROCEDURE BOUNDARY**

Detailed commands and numbered steps are in the Learner Guide and lab folder—not on these slides.


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LAB 07 · ACCEPTANCE EVIDENCE

# Build a Multi-Account Compliance Evidence Flow

## ✅ Expected result

Required controls evaluate learner-owned resources, findings are prioritised with context, remediation has owner/SLA/verification, the exception expires, and validation emits LAB 07 PASS.

IF IT DOESN'T WORK

### Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

### Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

### Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.


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TOPIC 07 · CONCEPT-LED LAB BRIEF

# Diagnose IAM Access and Protect Data with KMS

**LAB 08**

Apply K1, K3, A1 and A7 by explaining an access decision, narrowing permissions and verifying encryption and key-policy behaviour.

Use policy evidence to distinguish identity, resource and KMS decisions, then implement and verify least privilege.

### YOU'LL BUILD

A policy-evaluation worksheet, corrected least-privilege policy, KMS key-policy review and encrypted-object evidence.

### TOOLCHAIN

IAM · Access Analyzer · Policy Simulator · KMS · S3 · CloudTrail

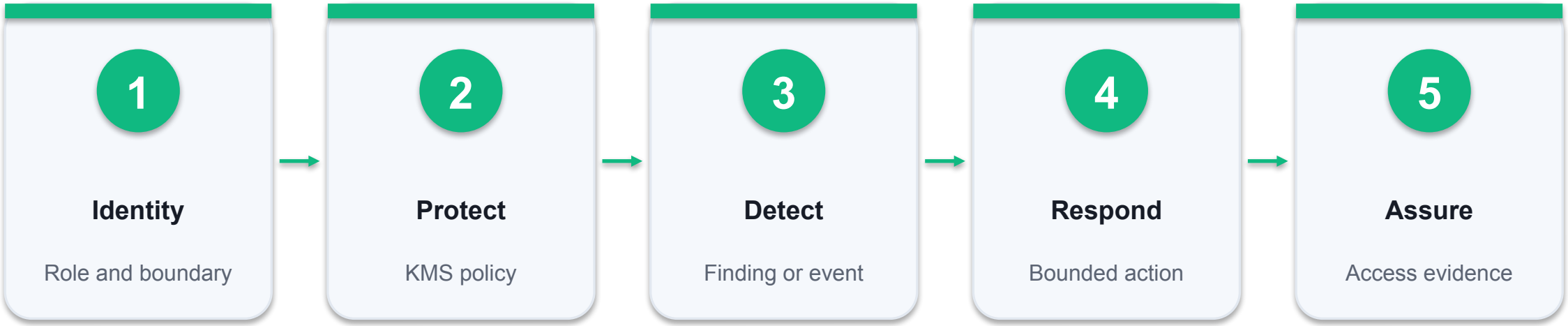
### DONE WHEN

The decision path explains the deny, corrected policy uses exact actions/resources, encrypted-object metadata is verified, Access Analyzer reports no ERROR, and validation emits LAB 08 PASS.


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LAB 08 · CONCEPTUAL WORKFLOW

# Lab 08 Evidence Model




**PROCEDURE BOUNDARY**

Detailed commands and numbered steps are in the Learner Guide and lab folder—not on these slides.


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LAB 08 · ACCEPTANCE EVIDENCE

# Diagnose IAM Access and Protect Data with KMS

## ✅ Expected result

The decision path explains the deny, corrected policy uses exact actions/resources, encrypted-object metadata is verified, Access Analyzer reports no ERROR, and validation emits LAB 08 PASS.

IF IT DOESN'T WORK

### Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

### Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

### Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.


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# Recap — Security, Identity and Compliance

1

**Explain**

Short-lived credentials, least privilege and explicit trust bound administrative power.

2

**Distinguish**

Policies prevent known bad states; telemetry detects deviation; runbooks contain and recover.

3

**Apply**

K1, K3, A1 and A7 by evaluating required tags and encryption, aggregating findings and documenting a controlled remediation exception.

4

**Evidence**

Required controls evaluate learner-owned resources, findings are prioritised with context, remediation has owner/SLA/verification, the exception expires, and…


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08

TOPIC 08 · LO5 · K2 · A2

# Networking, DNS and Content Delivery

VPC · routes · security groups · NACLs · endpoints · Route 53 · CloudFront


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# Key Ideas — Networking, DNS and Content Delivery

1

### Connectivity is a path

DNS, route, policy, state, endpoint and application health must all succeed.

2

### Security groups are stateful

Return traffic is automatically allowed; network ACLs evaluate each direction.

3

### Private connectivity reduces exposure

VPC endpoints, Transit Gateway and hybrid links change routing and policy boundaries.

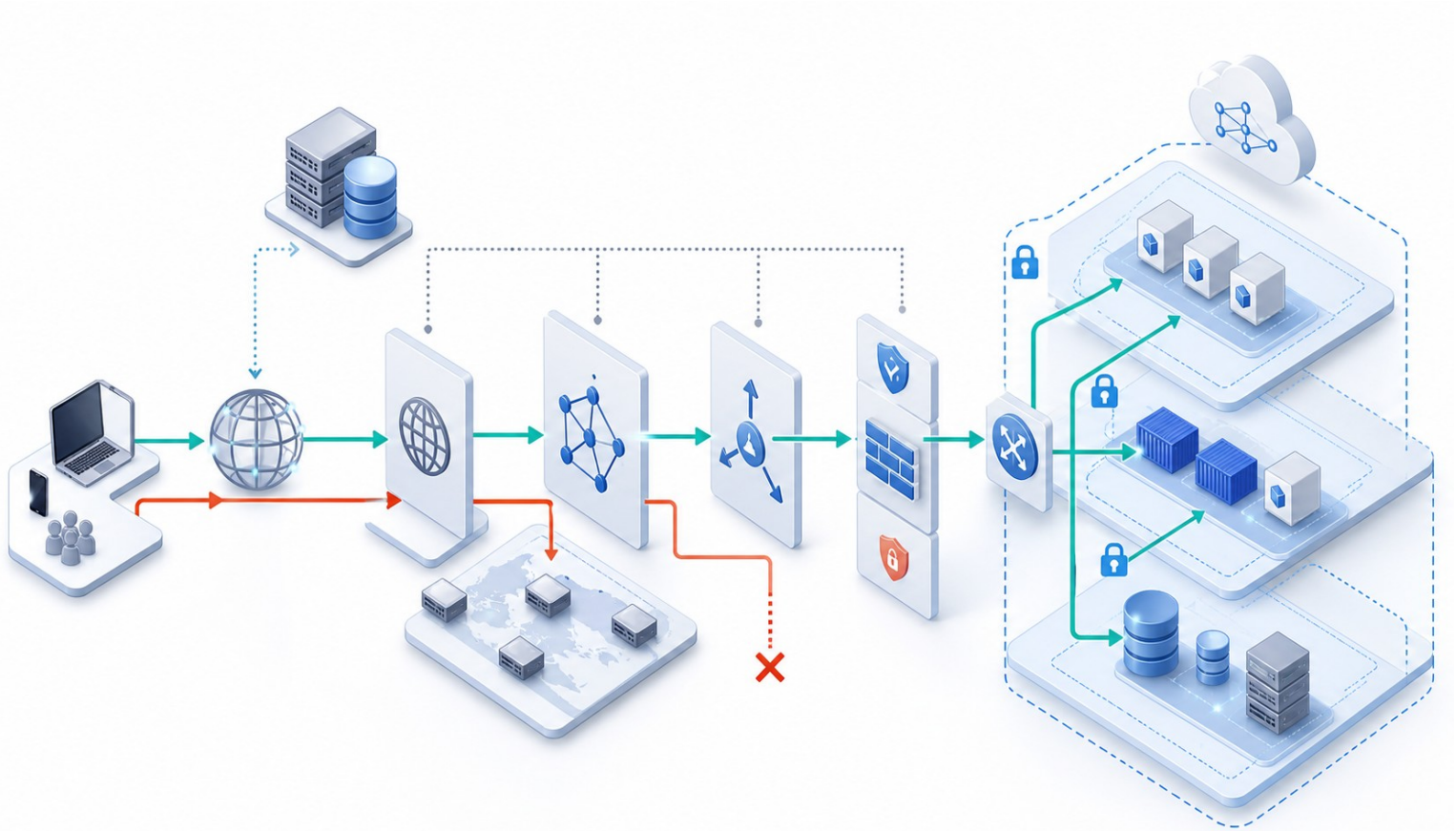
4

### DNS is a control surface

Routing policy, health evaluation, TTL and resolver behaviour affect recovery.


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# Trace the Complete Request Path

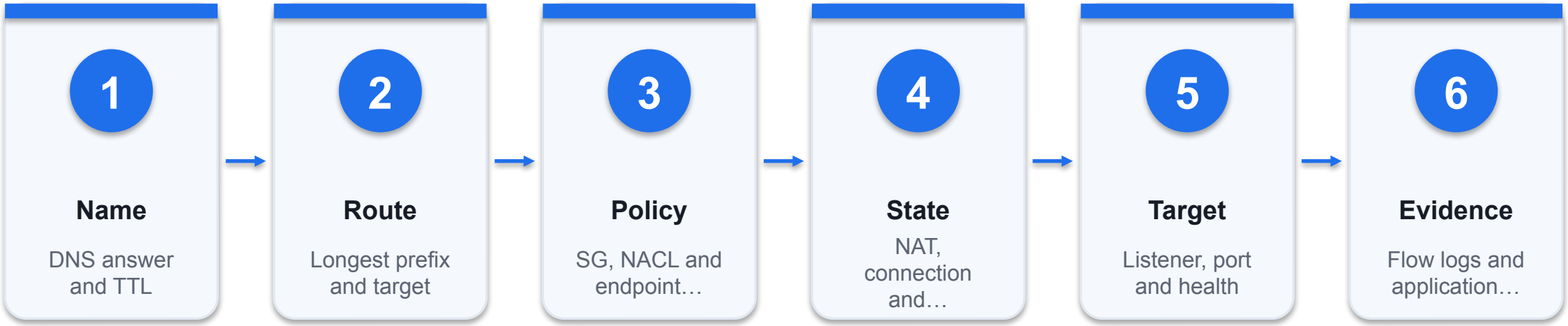


READ THE SYSTEM

Follow DNS, edge, routing, policy, load balancing and target health in order; compare the intended path with the blocked path before changing configuration.


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# Connectivity Troubleshooting Order




**A2 STANDARD**

Test from source to destination and record the first failed control boundary.

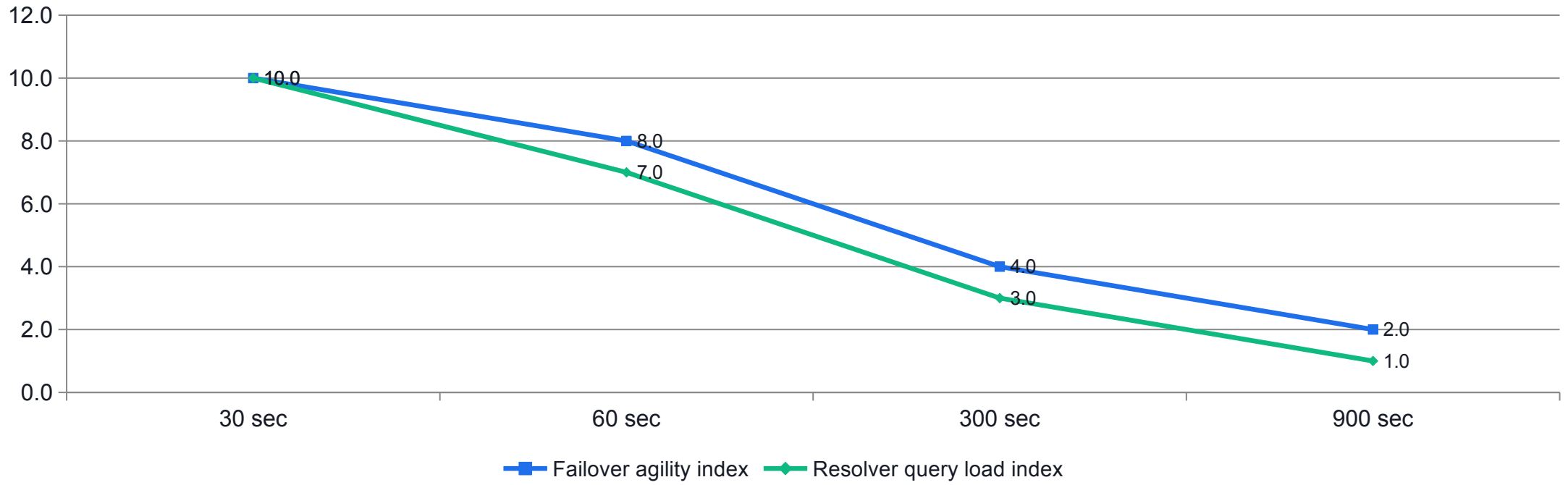

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# Security Group vs Network ACL

| Property | Security group | Network ACL |
| --- | --- | --- |
| **Scope** | Elastic network interface | Subnet |
| **State** | Stateful | Stateless |
| **Rules** | Allow only | Numbered allow and deny |
| **Evaluation** | All rules together | Lowest rule number first |


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# DNS TTL Trades Agility for Query Load




**WHAT THE DATA SHOWS**

Lower TTL can speed client refresh but increases query volume and cannot force every resolver to behave identically.


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# Route 53 Routing Policies

1

## Simple

One or more values without health-based policy logic.

2

## Weighted

Controlled traffic split for migration or experiments.

3

## Latency

Direct users to the Region with lowest measured latency.

4

## Failover

Primary and secondary routing driven by health.


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# VPC Endpoint Types

| Endpoint | Connectivity | Common use |
| --- | --- | --- |
| **Gateway** | Route-table target | S3 and DynamoDB |
| **Interface** | Private IP through PrivateLink | Many AWS and partner services |
| **Gateway Load Balancer** | Inspection appliance insertion | Central security services |
| **Resource / service network** | VPC Lattice private application access | Cross-VPC service connectivity |


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# CloudFront Operations

## Cache

- Cache key and TTL
- Invalidation and versioned objects

## Origin

- Origin access control
- Origin health and failover

## Security

- TLS certificate and WAF
- Logs and real-time metrics


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Network Diagnosis

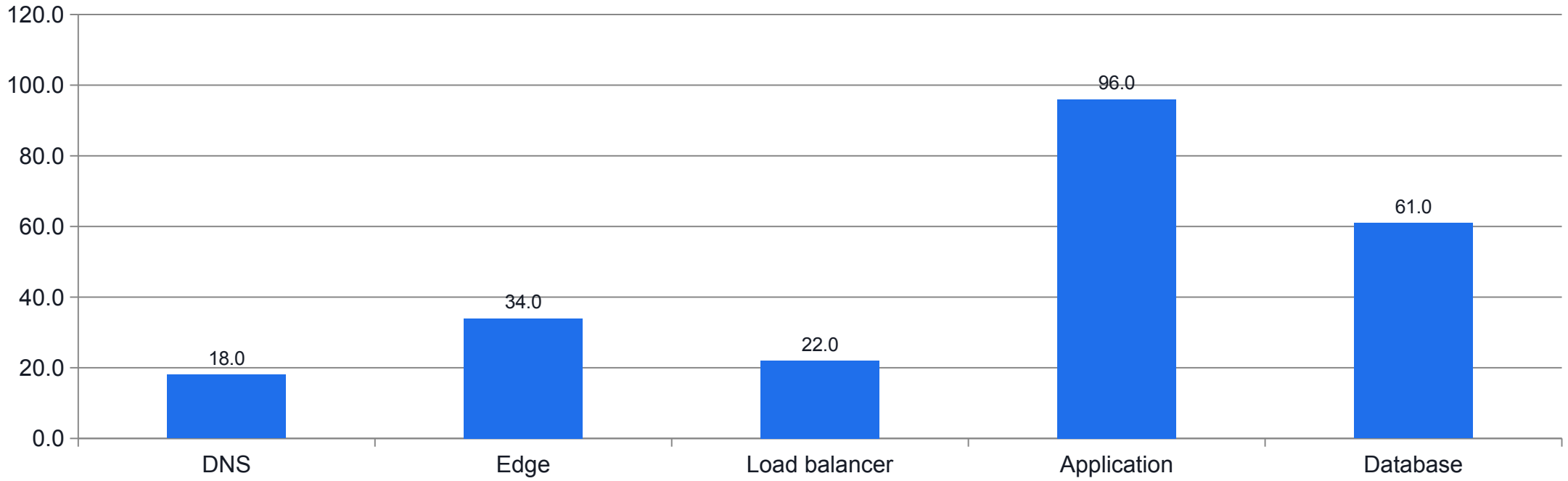
# A timeout is a symptom, not a location.

Use DNS results, route tables, reachability analysis, flow logs, load-balancer health and application evidence to isolate the boundary.


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TOPIC 08 · NATIVE EDITABLE TECHNICAL CHART

# Request-Path Latency Budget




**WHAT THE DATA SHOWS**

The application tier dominates p95, but network evidence must confirm that retries, path asymmetry or packet loss are not inflating it.


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# Amazon CloudFront

### 1 Content Delivery Network (CDN)

Content Delivery Network (CDN)

DDoS protection (because worldwide), integration with Shield, AWS Web Application Firewall

### 2 Improves read performance…

Improves read performance, content is cached at the edge

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 3 Improves users experience

Improves users experience

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

### 4 216 Point of Presence globally

216 Point of Presence globally (edge locations)

Evidence: resolver result, route tables, Reachability Analyzer, Flow Logs, load-balancer logs, target health and application traces locate the…

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# CloudFront - Origins

### 1 For distributing files and caching

For distributing files and caching them at the edge

Custom Origin (HTTP)

Any HTTP backend you want

### 2 Enhanced security with CloudFront…

Enhanced security with CloudFront Origin Access Control (OAC)

Application Load Balancer

### 3 OAC is replacing Origin Access

OAC is replacing Origin Access Identity (OAI)

EC2 instance

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 4 CloudFront can be used as

CloudFront can be used as an ingress (to upload files to S3)

S3 website (must first enable the bucket as a static S3 website)

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# CloudFront at a high level

### 1 CloudFront Edge Location

CloudFront Edge Location

User-Agent: Mozilla/4.0 (compatible; MSIE5.01; Windows NT) Host: www.example.com

### 2 Forward Request to your Origin

Forward Request to your Origin

Accept-Encoding: gzip, deflate

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 3 Local Cache

Local Cache

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

### 4 GET /beach.jpg? size=300x300…

GET /beach.jpg? size=300x300 HTTP/1.1

Evidence: resolver result, route tables, Reachability Analyzer, Flow Logs, load-balancer logs, target health and application traces locate the…

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# CloudFront - S3 as an Origin

### 1 Los Angeles

Los Angeles

+ S3 bucket policy

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 2 Edge Mumbai

Edge Mumbai

Origin (S3 bucket)

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

### 3 Public www

Public www

Private AWS

Evidence: resolver result, route tables, Reachability Analyzer, Flow Logs, load-balancer logs, target health and application traces locate the…

### 4 Origin Access Control

Origin Access Control

Decision point: static analyzers prove configured reachability but send no packets; combine them with live data-plane evidence before declaring the…

**CURRENT OPERATIONS LENS**

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# CloudFront vs S3 Cross Region Replication

### 1 CloudFront

CloudFront

S3 Cross Region Replication

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 2 Global Edge network

Global Edge network

Must be setup for each region you want replication to happen

### 3 Files are cached for a

Files are cached for a TTL (maybe a day)

Files are updated in near real-time

### 4 Great for static content that

Great for static content that must be available everything

Great for dynamic content that needs to be available at low-latency in few regions

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# CloudFront - ALB or EC2 as an origin

### 1 Edge Location

Edge Location

Security group

Must be Public

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 2 Public IPs

Public IPs

EC2 Instances

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

### 3 Allow Security Group of Load

Allow Security Group of Load Balancer

Can be Private

Evidence: resolver result, route tables, Reachability Analyzer, Flow Logs, load-balancer logs, target health and application traces locate the…

### 4 Allow Public IP of Edge

Allow Public IP of Edge Locations

Application Load Balancer

Decision point: static analyzers prove configured reachability but send no packets; combine them with live data-plane evidence before declaring the…

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# CloudFront Geo Restriction

### 1 You can restrict who can

You can restrict who can access your distribution

Use case: Copyright Laws to control access to content

### 2 Allowlist: Allow your users to

Allowlist: Allow your users to access your content only if they're in one of the countries on a list of approved countries.

### 3 Blocklist: Prevent your users from

Blocklist: Prevent your users from accessing your content if they're in one of the countries on a list of banned countries.

### 4 The “country” is determined using

The “country” is determined using a 3rd party Geo-IP database

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# CloudFront Access Logs

### 1 Logs every request made to

Logs every request made to CloudFront into a logging S3 bucket

Log File (Distribution A)

### 2 Edge Location

Edge Location

Log File (Distribution B)

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 3 Origin Bucket

Origin Bucket

data about requests

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

### 4 Logs Bucket

Logs Bucket

Evidence: resolver result, route tables, Reachability Analyzer, Flow Logs, load-balancer logs, target health and application traces locate the…

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# CloudFront Reports

### 1 It’s possible to generate reports

It’s possible to generate reports on

Usage Reports

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 2 Cache Statistics Report

Cache Statistics Report

Viewers Report

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

### 3 Popular Objects Report

Popular Objects Report

These reports are based on the data from the Access Logs.

### 4 Top Referrers Report

Top Referrers Report

Evidence: resolver result, route tables, Reachability Analyzer, Flow Logs, load-balancer logs, target health and application traces locate the…

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# CloudFront troubleshooting

### 1 CloudFront caches HTTP 4xx and

CloudFront caches HTTP 4xx and 5xx status codes returned by S3 ( or the origin server)

### 2 4xx error code indicates that

4xx error code indicates that user doesn’t have access to the underlying bucket (403) or the object user is requesting is not found (404)

### 3 5xx error codes indicates Gateway

5xx error codes indicates Gateway issues

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 4 Failure mode

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

**CURRENT OPERATIONS LENS**

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# CloudFront Caching

### 1 Cache based on

Cache based on

You want to maximize the cache hit rate to minimize requests on the origin

Expire based on TTL

### 2 Session Cookies

Session Cookies

Control the TTL (0 seconds to 1 year), can

Edge Location

### 3 Query String Parameters

Query String Parameters

be set by the origin using the Cache- Control header, Expires header…

Check / Update cache Based on Headers / Cookies

### 4 The cache lives at each

The cache lives at each CloudFront Edge Location

You can invalidate part of the cache using the CreateInvalidation API

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# CloudFront Cache Behaviour for Headers

### 1 Forward all headers to your

Forward all headers to your origin

caching based on values in all the specified headers

GET /image/cat.jpg HTTP/1.1

### 2 => no caching, every request

=> no caching, every request to origin

(None) == Forward only the default headers

Host: pics.mywebsite.com

### 3 => TTL must be set

=> TTL must be set to 0

no caching based on request headers

User-Agent: Mozilla/5.0 (Mac OS X 10_15_2….) Date: Tue, 28 Jan 2020 17:01:57 GMT

### 4 Forward a whitelist of headers

Forward a whitelist of headers

Best caching performance

Authorization: SAPISIDHASH fdd00ecee39fe….

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# CloudFront Caching – Whitelist Headers

### 1 Edge Location

Edge Location

Authorization: SAPISIDHASH fdd00ecee39fe….

Better caching because less headers values

### 2 GET /image/cat.jpg HTTP/1.1

GET /image/cat.jpg HTTP/1.1

Keep-Alive: 300 Accept-Ranges: bytes

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 3 Host: pics.mywebsite.com

Host: pics.mywebsite.com

whitelisting

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

### 4 User-Agent: Mozilla/5.0 (Mac OS X

User-Agent: Mozilla/5.0 (Mac OS X 10_15_2….) Date: Tue, 28 Jan 2020 17:01:57 GMT

Cache based on Host & Authorization

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# CloudFront Origin Headers vs Cache Behavior

### 1 Origin Custom Headers

Origin Custom Headers

Cache-related settings

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 2 Origin-level setting

Origin-level setting

Contains the whitelist of headers to forward

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

### 3 Set a constant header /

Set a constant header / header value for all requests to origin

### 4 Behavior setting

Behavior setting

Evidence: resolver result, route tables, Reachability Analyzer, Flow Logs, load-balancer logs, target health and application traces locate the…

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# CloudFront Caching TTL

### 1 “Cache-Control: max-age” is…

“Cache-Control: max-age” is preferred to “Expires” header

Default TTL

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 2 If the origin always sends

If the origin always sends back the header Cache-Control then you can set the TTL to be controlled only by that header

Cache-Control (optional)

### 3 In case you want to

In case you want to set min/max boundaries, you choose “customize” for the Object Caching setting

### 4 In case the Cache-Control header

In case the Cache-Control header is missing, it will default to “default value”

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# CloudFront Cache Behaviour for Cookies

### 1 Cookies is a specific request

Cookies is a specific request header

Forward a whitelist of cookies

GET /image/cat.jpg HTTP/1.1

### 2 Default: do not process the

Default: do not process the cookies

caching based on values in all the specified cookies

Cookie: username=John Doe; location=uk; lang = eng; user_id = 12342

### 3 Caching is not based on

Caching is not based on cookies

Forward all cookies

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 4 Cookies are not forwarded

Cookies are not forwarded

Worst caching performance

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# CloudFront Caching - Cookies

### 1 Edge Location

Edge Location

Whitelisting on user_id

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 2 GET /image/cat.jpg HTTP/1.1

GET /image/cat.jpg HTTP/1.1

Cache based on yser_id

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

### 3 Cookie: username=John Doe…

Cookie: username=John Doe; location=uk; lang = eng; user_id = 12342

Better caching because less cookies values

### 4 Cookie: user_id = 12342

Cookie: user_id = 12342

Evidence: resolver result, route tables, Reachability Analyzer, Flow Logs, load-balancer logs, target health and application traces locate the…

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# CloudFront Cache Behaviour for Query Strings

### 1 Query Strings Parameters are in

Query Strings Parameters are in the URL

Forward a whitelist of query strings

GET /image/cat.jpg?border=red&size=large HTTP/1.1

### 2 Default: do not process the

Default: do not process the query strings

Caching based on the parameter whitelist

### 3 Caching is not based on

Caching is not based on query strings

Forward all query strings

### 4 Parameters are not forwarded

Parameters are not forwarded

Caching based on all parameters

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

**CURRENT OPERATIONS LENS**

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# CloudFront Caching - Query String Parameters

1

## Edge Location

Edge Location

Cache based on size

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

2

## GET /image/cat.jpg?border=red&size…

GET /image/cat.jpg?border=red&size=large HTTP/1.1

Better caching because less query string values

3

## GET /image/cat.jpg?size=large…

GET /image/cat.jpg?size=large HTTP/1.1

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

4

## Whitelisting on size

Whitelisting on size

Evidence: resolver result, route tables, Reachability Analyzer, Flow Logs, load-balancer logs, target health and application traces locate the…

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# CloudFront - Maximize cache hits by separating static and dynamic distributions

### 1 Dynamic Content (REST, HTTP…

Dynamic Content (REST, HTTP server)

Static content

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 2 API Gateway + Lambda

API Gateway + Lambda

Cache based on correct headers and cookie

### 3 CloudFront

CloudFront

No headers / session caching rules Required for maximizing cache hits

### 4 Static requests

Static requests

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

**CURRENT OPERATIONS LENS**

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# CloudFront - Increasing Cache Ratio

### 1 Monitor the CloudWatch metric…

Monitor the CloudWatch metric CacheHitRate

Specify none or the minimally required query string parameters

### 2 Specify how long to cache your…

Specify how long to cache your objects: Cache-Control max-age header

Separate static and dynamic distributions (two origins)

### 3 Specify none or the minimally

Specify none or the minimally required headers

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 4 Specify none or the minimally

Specify none or the minimally required cookies

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

**CURRENT OPERATIONS LENS**

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# CloudFront with ALB sticky sessions

### 1 Must forward / whitelist the

Must forward / whitelist the cookie that controls the session affinity to the origin to allow the session affinity to work

GET /orders/all HTTP/1.1

CloudFront

### 2 Set a TTL to a

Set a TTL to a value lesser than when the authentication cookie expires

Cookie: AWSALB=…

Target Group

### 3 forward cookies

forward cookies

Whitelisting on

Application Load Balancer

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 4 Edge Location

Edge Location

AWSALB cookie

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# What is DNS?

### 1 Domain Name System which…

Domain Name System which translates the human friendly hostnames into the machine IP addresses

example.com
www.example.com
api.example.com

### 2 www.google.com => 172.217.18.36

www.google.com => 172.217.18.36

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 3 DNS is the backbone of

DNS is the backbone of the Internet

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

### 4 DNS uses hierarchical naming…

DNS uses hierarchical naming structure

Evidence: resolver result, route tables, Reachability Analyzer, Flow Logs, load-balancer logs, target health and application traces locate the…

**CURRENT OPERATIONS LENS**

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# DNS Terminologies

### 1 Sub Domain

Sub Domain

Zone File: contains DNS records

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 2 FQDN (Fully Qualified Domain Name)

FQDN (Fully Qualified Domain Name)

Name Server: resolves DNS queries (Authoritative or Non-Authoritative)

### 3 Domain Registrar

Domain Registrar: Amazon Route 53, GoDaddy, …

Top Level Domain (TLD): .com, .us, .in, .gov, .org , …

### 4 DNS Records

DNS Records: A, AAAA, CNAME, NS, …

Second Level Domain (STD): amazon.com, google.com, …

**CURRENT OPERATIONS LENS**

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# How DNS Works

### 1 Local DNS Server

Local DNS Server

TLD DNS Server

Managed by ICANN

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 2 Assigned and Managed by your

Assigned and Managed by your company or assigned by your ISP dynamically

.com NS 1.2.3.4

Managed by Domain Registrar (e.g., Amazon Registrar, Inc.)

### 3 example.com?

example.com?

example.com NS 5.6.7.8

SLD DNS Server

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

### 4 Root DNS Server

Root DNS Server

example.com IP 9.10.11.12

Managed by IANA (Branch of ICANN)

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Amazon Route 53

### 1 A highly available, scalable…

A highly available, scalable, fully managed and Authoritative DNS

The only AWS service which provides 100% availability SLA

EC2 Instance

### 2 Authoritative = the customer (you)

Authoritative = the customer (you) can update the DNS records

Why Route 53? 53 is a reference to the traditional DNS port

### 3 Route 53 is also a

Route 53 is also a Domain Registrar

example.com?

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 4 Ability to check the health

Ability to check the health of your resources

54.22.33.44

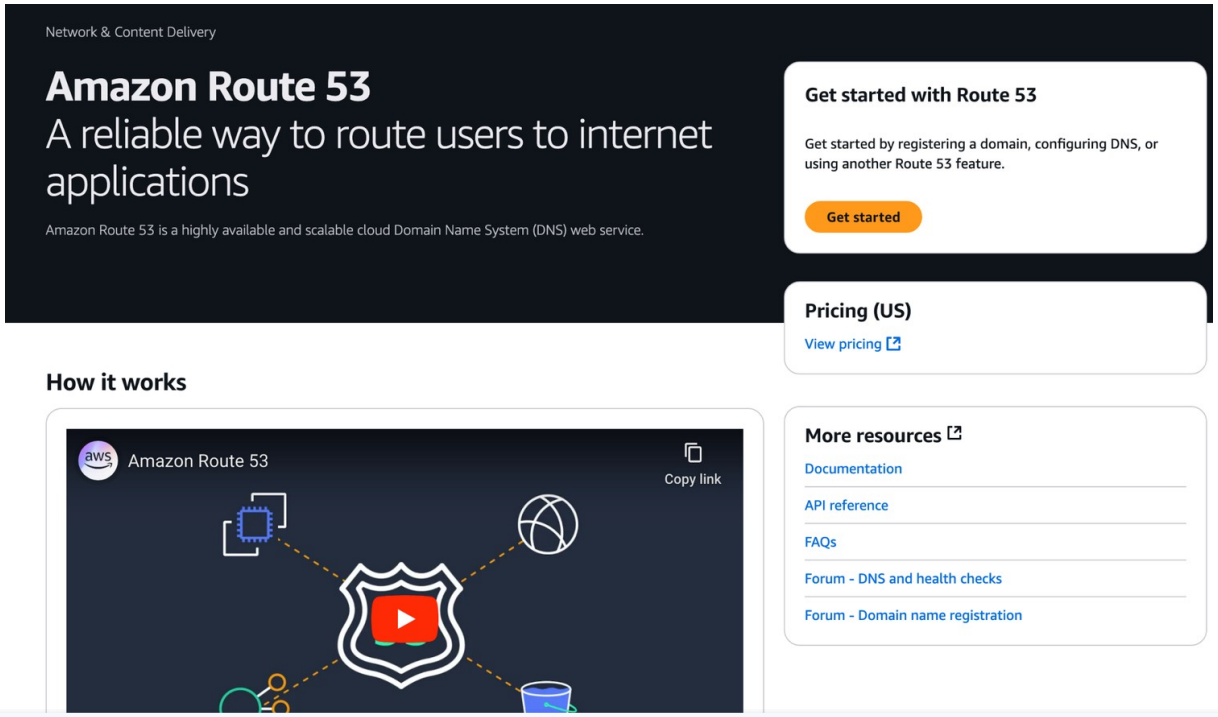
Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

**CURRENT OPERATIONS LENS**

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# Amazon Route 53




**1** **Mechanism: Route 53 selects an**

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and…

**2** **Failure mode**

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule…

**3** **Evidence: resolver result, route…**

Evidence: resolver result, route tables, Reachability Analyzer, Flow Logs, load-balancer logs, target health and…

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Route 53 - Records

### 1 How you want to route

How you want to route traffic for a domain

Value - e.g., 12.34.56.78

(must know) A / AAAA / CNAME / NS

### 2 Each record contains

Each record contains

Routing Policy - how Route 53 responds to queries

(advanced) CAA / DS / MX / NAPTR / PTR / SOA / TXT / SPF / SRV

### 3 Domain/ SubDomain namer

Domain/SubDomain namer - e.g., example.com

TTL - amount of time the record cached at DNS Resolvers

### 4 Record Type

Record Type - e.g., A or AAAA

Route 53 supports the following DNS record types

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Route 53 - Record Types

### 1 A - maps a hostname

A - maps a hostname to IPv4

Can’t create a CNAME record for the top node of a DNS namespace (Zone Apex)

### 2 AAAA - maps a hostname

AAAA - maps a hostname to IPv6

Example: you can’t create for example.com, but you can create for www.example.com

### 3 CNAME - maps a hostname

CNAME - maps a hostname to another hostname

NS - Name Servers for the Hosted Zone

### 4 The target is a domain

The target is a domain name which must have an A or AAAA record

Control how traffic is routed for a domain

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Route 53 - Hosted Zones

### 1 A container for records that

A container for records that define how to route traffic to a domain and its subdomains

### 2 Puiblic Hosted Zones

Puiblic Hosted Zones - contains records that specify how to route traffic on the Internet (public domain names) application1.mypublicdomain.com

### 3 Private Hosted Zones

Private Hosted Zones - contain records that specify how you route traffic within one or more VPCs (private domain names) application1.company.internal

### 4 You pay $0.50 per month

You pay $0.50 per month per hosted zone

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Route 53 - Public vs Private Hosted Zones

### 1 EC2 Instance

EC2 Instance

Public Hosted Zone

EC2 Instance EC2 Instance

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 2 (Public IP)

(Public IP)

Amazon CloudFront

(webapp.example.internal) (api.example.internal)

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

### 3 Application

Application

54.22.33.44

(Private IP) (Private IP)

Evidence: resolver result, route tables, Reachability Analyzer, Flow Logs, load-balancer logs, target health and application traces locate the…

### 4 Load Balancer

Load Balancer

Private Hosted Zone

DB Instance

Decision point: static analyzers prove configured reachability but send no packets; combine them with live data-plane evidence before declaring the…

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Route 53 - Records TTL (Time To Live)

### 1 Web Server

Web Server

A 12.34.56.78

High TTL - e.g. 24 hr

### 2 DNS Request

DNS Request

HTTP Request

Less traffic on Route 53

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 3 myapp.example.com?

myapp.example.com?

HTTP Response

Possibly outdated records

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

### 4 Amazon Route 53

Amazon Route 53

Will cache the result for The TTL of the record

Low TTL - e.g. 60 sec

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Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# CNAME vs Alias

### 1 AWS Resources (Load Balancer…

AWS Resources (Load Balancer, CloudFront...) expose an AWS hostname

Points a hostname to an AWS Resource (app.mydomain.com => blabla.amazonaws.com)

### 2 lb1-1234.us-east2.elb.amazonaws.co…

lb1-1234.us-east2.elb.amazonaws.com and you want myapp.mydomain.com

Works for ROOT DOMAIN and NON ROOT DOMAIN (aka mydomain.com)

### 3 Points a hostname to any

Points a hostname to any other hostname. (app.mydomain.com => blabla.anything.com)

Free of charge

### 4 ONLY FOR NON ROOT DOMAIN

ONLY FOR NON ROOT DOMAIN (aka something.mydomain.com)

Native health check

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Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Route 53 - Alias Records

### 1 Maps a hostname to an

Maps a hostname to an AWS resource

Alias Record is always of type A/AAAA for

Amazon Route 53

### 2 An extension to DNS functionality

An extension to DNS functionality

AWS resources (IPv4 / IPv6)

Alias Record (Enabled)

### 3 Automatically recognizes changes…

Automatically recognizes changes in the resource's IP addresses

You can't set the TTL

AWS-Managed

### 4 Unlike CNAME, it can be

Unlike CNAME, it can be used for the top node of a DNS namespace (Zone Apex), e.g.: example.com

MyALB-123456789.us-east-1.elb.amazonaws.com

(IP Addresses might change)

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Route 53 - Alias Records Targets

### 1 Elastic Load Balancers

Elastic Load Balancers

S3 Websites

You cannot set an ALIAS record for an EC2 DNS name

### 2 CloudFront Distributions

CloudFront Distributions

VPC Interface Endpoints

Global Accelerator

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 3 API Gateway

API Gateway

Global Accelerator accelerator

Elastic Load Balancer

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

### 4 Elastic Beanstalk environments

Elastic Beanstalk environments

Route 53 record in the same hosted zone

Amazon CloudFront

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Route 53 - Routing Policies

### 1 Define how Route 53 responds

Define how Route 53 responds to DNS queries

Route 53 Supports the following Routing Policies

Geoproximity (using Route 53 Traffic Flow feature)

### 2 Don’t get confused by the

Don’t get confused by the word “Routing”

Latency based

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 3 It’s not the same as

It’s not the same as Load balancer routing which routes the traffic

Geolocation

### 4 DNS does not route any

DNS does not route any traffic, it only responds to the DNS queries

Multi-Value Answer

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Routing Policies - Simple

1

### Typically, route traffic to a

Typically, route traffic to a single resource

When Alias enabled, specify only one AWS resource

Amazon Route 53

2

### Can specify multiple values in

Can specify multiple values in the same record

Can’t be associated with Health Checks

Single Value

3

### If mutliple values are returned

If mutliple values are returned a

foo.example.com

A 55.66.77.88

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

4

### random one is chosen by

random one is chosen by the client

A 11.22.33.44

A 99.11.22.33

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Route 53 Weighted Routing — Selection Logic

1

### Traffic share = record weight

Traffic share = record weight divided by the sum of weights for eligible records.

2

### Weights are relative values and

Weights are relative values and do not need to total 100.

3

### A weight of zero stops

A weight of zero stops normal selection unless every eligible record has zero weight.

4

### Health checks remove unhealthy…

Health checks remove unhealthy endpoints before the remaining weights are evaluated.

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Route 53 Latency-Based Routing — Decision Model

### 1 Selection: Route 53 chooses the

Selection: Route 53 chooses the record associated with the AWS Region expected to provide the lowest network latency for the requester.

### 2 Measurement: the decision uses AWS

Measurement: the decision uses AWS latency measurements between users and Regions, not geographic distance alone.

### 3 Health: associate health checks so

Health: associate health checks so unhealthy endpoints are excluded before a latency-based answer is returned.

### 4 Evidence: compare resolver…

Evidence: compare resolver answers, endpoint health and application latency from representative user locations before changing policy.

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Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Route 53 - Health Checks

### 1 HTTP Health Checks are only

HTTP Health Checks are only for public resources

Health checks that monitor CloudWatch Alarms (full control !!) – e.g., throttles of DynamoDB, alarms on RDS, custom metrics,

DNS Record (latency, geoproximity, …)

### 2 Health Check => Automated DNS

Health Check => Automated DNS Failover

… (helpful for private resources)

Auto Scaling group

### 3 Health checks that monitor an

Health checks that monitor an endpoint (application, server, other AWS resource)

Health Checks are integrated with CW metrics

EC2 Instance

### 4 Health checks that monitor other

Health checks that monitor other health checks (Calculated Health Checks)

Amazon Route 53

Health Check

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# eu-west-1

## 1 Health Checker (us-east-1)

Health Checker (us-east-1)

Interval – 30 sec (can set to 10 sec – higher cost)

Health Checks pass only when the endpoint responds with the 2xx and 3xx status codes

## 2 Health Checks

Health Checks - Monitor an Endpoint

Supported protocol: HTTP, HTTPS and TCP

Health Checks can be setup to pass / fail based on the text in the first 5120 bytes of the response

## 3 about 15 global health checkers

about 15 global health checkers will check the endpoint health

If> 18% of health checkers report the endpoint is healthy, Route 53 considers it Healthy. Otherwise, it's Unhealthy

Configure your router/firewall to allow incoming requests from Route 53 Health Checkers

## 4 Healthy/ Unhealthy Threshold

Healthy/Unhealthy Threshold – 3 (default)

Ability to choose which locations you want Route 53 to use

HTTP Request

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Amazon Route 53

## 1 Route 53

Route 53 - Calculated Health Checks

Specify how many of the health checks need to pass to make the parent pass

Check Check

## 2 Combine the results of multiple

Combine the results of multiple Health Checks into a single Health Check

Usage: perform maintenance to your website without causing all health checks to fail

(Child) (Child)

## 3 You can use OR, AND

You can use OR, AND or NOT

Health Check

EC2 Instance

## 4 Can monitor up to 256

Can monitor up to 256 Child Health Checks

Health Health

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Health Checks - Private Hosted Zones

### 1 Route 53 health checkers are

Route 53 health checkers are outside the VPC

Private subnet

### 2 They can’t access private…

They can’t access private endpoints (private VPC or on-premises resource)

CloudWatch Alarm

### 3 You can create a CloudWatch

You can create a CloudWatch Metric and associate a CloudWatch Alarm, then create a Health Check that checks the alarm itself

### 4 Health Checker (us-east-1)

Health Checker (us-east-1)

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Routing Policies - Failover (Active-Passive)

### 1 EC2 Instance

EC2 Instance

(Secondary – Disaster Recovery)

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 2 Health Check

Health Check

DNS Requests

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

### 3 (mandatory)

(mandatory)

Evidence: resolver result, route tables, Reachability Analyzer, Flow Logs, load-balancer logs, target health and application traces locate the…

### 4 Amazon Route 53

Amazon Route 53

Decision point: static analyzers prove configured reachability but send no packets; combine them with live data-plane evidence before declaring the…

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Routing Policies - Geolocation

### 1 Different from Latency-based!

Different from Latency-based!

Use cases: website localization, restrict content distribution, load balancing …

A 99.11.22.33

### 2 This routing is based on

This routing is based on user location

Can be associated with Health Checks

### 3 Specify location by Continent…

Specify location by Continent, Country or by US State (if there's overlapping, most precise location selected)

A 11.22.33.44

### 4 Should create a "Default" record

Should create a "Default" record (in case there's no match on location)

A 55.66.77.88

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Routing Policies - Geoproximity

### 1 Route traffic to your resources

Route traffic to your resources based on the geographic location of users and resources

To shrink (-1 to -99) – less traffic to the resource

You must use Route 53 Traffic Flow to use this feature

### 2 Ability to shift more traffic

Ability to shift more traffic to resources based on the defined bias

Resources can be

### 3 To change the size of

To change the size of the geographic region, specify bias values

AWS resources (specify AWS region)

### 4 To expand (1 to 99)

To expand (1 to 99) – more traffic to the resource

Non-AWS resources (specify Latitude and Longitude)

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Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Routing Policies - Geoproximity

1

## us-west-1 Bias

us-west-1 Bias: 0

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

2

## us-east-1 Bias

us-east-1 Bias: 0

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

3

## Evidence: resolver result, route…

Evidence: resolver result, route tables, Reachability Analyzer, Flow Logs, load-balancer logs, target health and application traces locate the…

4

## Decision point

Decision point: static analyzers prove configured reachability but send no packets; combine them with live data-plane evidence before declaring the…

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Routing Policies - Geoproximity

### 1 us-west-1 Bias

us-west-1 Bias: 0

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 2 us-east-1 Bias

us-east-1 Bias: 50

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

### 3 Higher bias in us-east-1

Higher bias in us-east-1

Evidence: resolver result, route tables, Reachability Analyzer, Flow Logs, load-balancer logs, target health and application traces locate the…

### 4 Decision point

Decision point: static analyzers prove configured reachability but send no packets; combine them with live data-plane evidence before declaring the…

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Route 53 - Traffic flow

### 1 Simplify the process of creating

Simplify the process of creating and maintaining records in large and complex configurations

Can be applied to different Route 53 Hosted Zones (different domain names)

### 2 Visual editor to manage complex

Visual editor to manage complex routing decision trees

Supports versioning

### 3 Configurations can be saved as

Configurations can be saved as

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 4 Traffic Flow Policy

Traffic Flow Policy

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Routing Policies - IP-based Routing

### 1 Routing is based on client's

Routing is based on client's IP addresses

CIDR Collection

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 2 You provide a list of

You provide a list of CIDRs for your clients and the corresponding endpoints/locations (user-IP-to-endpoint mappings)

EC2 Instance

### 3 Use cases

Use cases: Optimize performance, reduce network costs…

(203.0.113.56)

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

### 4 Example: route end users from

Example: route end users from a particular ISP to a specific endpoint

(200.5.4.100)

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Routing Policies - Multi-Value

| Name | Type | Value | TTL | Set ID | Health Check |
|---|---|---|---|---|---|
| www.example.com | A Record | 192.0.2.2 | 60 | Web1 | A |
| www.example.com | A Record | 198.51.100.2 | 60 | Web2 | B |
| www.example.com | A Record | 203.0.113.2 | 60 | Web3 | C |

1

### Use when routing traffic to

Use when routing traffic to multiple resources
Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and…

2

### Failure mode

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule…

3

### Evidence: resolver result, route…

Evidence: resolver result, route tables, Reachability Analyzer, Flow Logs, load-balancer logs, target health and…

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Domain Registrar vs DNS Service

1

**You buy or register your**

You buy or register your domain name with a Domain Registrar typically by paying annual charges (e.g., GoDaddy, Amazon Registrar Inc., …)

Amazon Route 53

2

**The Domain Registrar usually…**

The Domain Registrar usually provides you with a DNS service to manage your DNS records

example.com

3

**But you can use another**

But you can use another DNS service to manage your DNS records

manage DNS records

4

**Example: purchase the domain from**

Example: purchase the domain from GoDaddy and use Route 53 to manage your DNS records

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# GoDaddy as Registrar & Route 53 as DNS Service

### 1 Amazon Route 53

Amazon Route 53

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 2 Public Hosted Zone

Public Hosted Zone

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

### 3 stephanetheteacher.com

stephanetheteacher.com

Evidence: resolver result, route tables, Reachability Analyzer, Flow Logs, load-balancer logs, target health and application traces locate the…

### 4 Decision point

Decision point: static analyzers prove configured reachability but send no packets; combine them with live data-plane evidence before declaring the…

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# 3rd Party Registrar with Amazon Route 53

### 1 If you buy your domain

If you buy your domain on a 3rd party registrar, you can still use route 53 as the DNS Service provider

But every Domain Registrar usually comes with some DNS features

### 2 Create a Hosted Zone in

Create a Hosted Zone in Route 53

Mechanism: Route 53 selects an endpoint, edge services apply policy, routes choose a next hop, stateful and stateless controls filter traffic, and…

### 3 Update NS Records on 3rd

Update NS Records on 3rd party website to use Route 53 Name Servers

### 4 Domain Registrar != DNS Service

Domain Registrar != DNS Service

Failure mode: asymmetric routing, overlapping CIDRs, return-path gaps, health-check mismatch and security-rule scope can produce identical user…

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Route 53 Resolver — Hybrid DNS Architecture

### 1 VPC Resolver

VPC Resolver: answers queries for VPC-local names, private hosted zones and public DNS names available to the workload.

### 2 Inbound endpoints

Inbound endpoints: allow approved on-premises DNS resolvers to query private AWS namespaces through dedicated endpoint IP addresses.

### 3 Outbound endpoints

Outbound endpoints: forward selected domain queries from VPCs to on-premises resolvers using resolver rules.

### 4 Operational evidence

Operational evidence: verify rule association, network path, security-group scope, query logs and authoritative answers in both directions.

**CURRENT OPERATIONS LENS**

Trace name resolution, routing, policy, transport and application health in order; separate static reachability from live data-plane behaviour. Validate service limits and regional availability against current AWS documentation.


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# Original PPT Coverage Retained — Topic 08

1

### Modernised slides

55 technical slides extracted from the original v4 PowerPoint and rebuilt in the current visual system.

2

### Traceability

Every blended slide names its original v4 source slide number in the kicker.

3

### Current validation

Terminology is normalised; service limits and Region availability remain explicit current-documentation checks.

4

### Procedure boundary

Click-by-click sequences, activities and quizzes are kept out of the deck and remain in the Learner Guide or lab package.


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TOPIC 08 · CONCEPT-LED LAB BRIEF

# Troubleshoot a VPC Connectivity Path

**LAB 09**

Apply K2 and A2 by isolating a failed request through DNS, routes, security groups, network ACLs, target health and VPC Flow Logs.

Diagnose the first failed control boundary using a repeatable evidence ladder and implement one verified correction.

### YOU'LL BUILD

A VPC diagnostic worksheet, reachability/flow-log evidence, corrected network control and before/after result.

### TOOLCHAIN

VPC · Reachability Analyzer · Flow Logs · Security Groups · NACLs · Endpoints

### DONE WHEN

The report follows the path in order, names the first failed boundary with evidence, changes one learner-owned control, verifies success with the same test, and validation emits LAB 09 PASS.


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LAB 09 · CONCEPTUAL WORKFLOW

# Lab 09 Evidence Model




**PROCEDURE BOUNDARY**

Detailed commands and numbered steps are in the Learner Guide and lab folder—not on these slides.


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LAB 09 · ACCEPTANCE EVIDENCE

# Troubleshoot a VPC Connectivity Path

## ✅ Expected result

The report follows the path in order, names the first failed boundary with evidence, changes one learner-owned control, verifies success with the same test, and validation emits LAB 09 PASS.

IF IT DOESN'T WORK

### Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

### Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

### Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.


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TOPIC 08 · CONCEPT-LED LAB BRIEF

# Operate DNS and Content-Delivery Failover

**LAB 10**

Apply K2 and A2 by configuring health-aware DNS and CloudFront behaviour, then proving client-visible routing and cache outcomes.

Model primary/secondary routing, inspect health status and TTL, then verify CloudFront origin and cache evidence.

### YOU'LL BUILD

A Route 53 health-check definition, failover record plan, CloudFront operations checklist and measured failover report.

### TOOLCHAIN

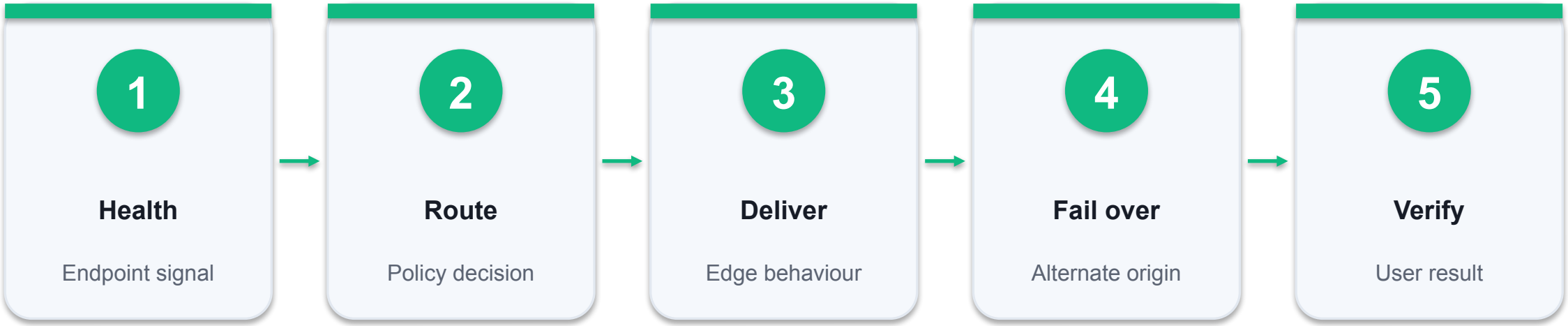
Route 53 · Health Checks · CloudFront · S3 · CloudWatch

### DONE WHEN

Health and record evidence are linked, failover timing is measured rather than assumed, cache and DNS effects are distinguished, baseline is restored, and validation emits LAB 10 PASS.


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# Lab 10 Evidence Model




**PROCEDURE BOUNDARY**

Detailed commands and numbered steps are in the Learner Guide and lab folder—not on these slides.


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LAB 10 · ACCEPTANCE EVIDENCE

# Operate DNS and Content-Delivery Failover

## ✅ Expected result

Health and record evidence are linked, failover timing is measured rather than assumed, cache and DNS effects are distinguished, baseline is restored, and validation emits LAB 10 PASS.

IF IT DOESN'T WORK

**Input mismatch**

Reconcile identifiers, required fields and units with the supplied data dictionary.

**Validation stop**

Inspect the named rule and source row; preserve the check rather than bypassing it.

**Evidence differs**

Check the denominator, time window and assumptions before rerunning from clean inputs.


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# Recap — Networking, DNS and Content Delivery

1

**Explain**

DNS, route, policy, state, endpoint and application health must all succeed.

2

**Distinguish**

Return traffic is automatically allowed; network ACLs evaluate each direction.

3

**Apply**

K2 and A2 by isolating a failed request through DNS, routes, security groups, network ACLs, target health and VPC Flow Logs.

4

**Evidence**

The report follows the path in order, names the first failed boundary with evidence, changes one learner-owned control, verifies success with the same test, and…


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09

TOPIC 09 · LO1 · LO2 · LO4 · K1 · K4 · A1 · A4 · A6

# Storage, Data and Database Operations

S3 · EBS · EFS · RDS · DynamoDB · lifecycle · performance · data protection


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# Key Ideas — Storage, Data and Database Operations

1

## Choose by access pattern

Object, block, file and database services solve different consistency, latency and sharing needs.

2

## Performance has service-specific limits

IOPS, throughput, connections, partitions and caching require different evidence.

3

## Data protection is layered

Encryption, versioning, snapshots, backup and replication address different risks.

4

## Managed does not mean unmanaged

Customers still own schema, configuration, access, capacity and recovery choices.


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# Storage Selection

| Service | Access model | Operational fit |
| --- | --- | --- |
| **S3** | Object and API | Durable data lake, backup and static content |
| **EBS** | Block device to compute | Low-latency instance and database volumes |
| **EFS** | Shared NFS file system | Elastic multi-instance Linux file access |
| **FSx** | Managed specialised file system | Windows, Lustre, NetApp or OpenZFS workloads |


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# S3 Data Protection

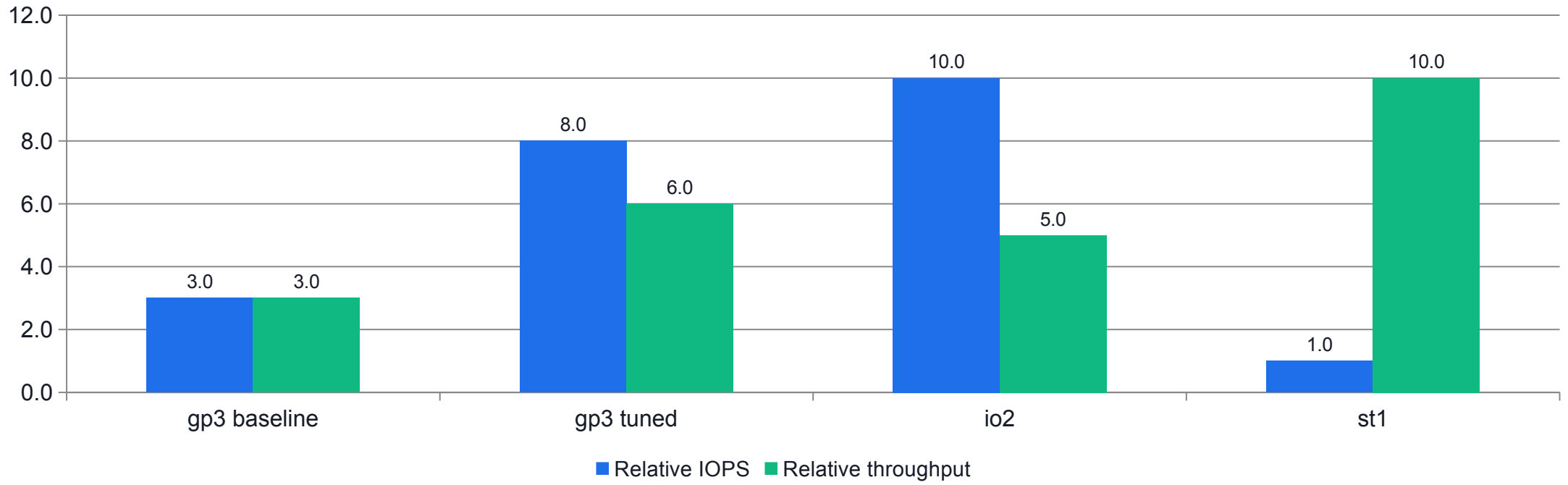



**A1 CONNECTION**

Protection combines preventive access, recoverability, retention and evidence.


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# EBS Performance Must Match Workload




**WHAT THE DATA SHOWS**

IOPS and throughput are different dimensions; select volume type and configuration from measured workload behaviour.


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# RDS Availability and Scale

| Feature | Primary goal | Operational note |
| --- | --- | --- |
| **Multi-AZ** | Availability and failover | Standby is not a read-scaling target |
| **Read replica** | Read scale and reporting | Replication lag and promotion plan |
| **Automated backup** | Point-in-time recovery | Retention and restore testing |
| **RDS Proxy** | Connection pooling and resilience | Application compatibility and limits |


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# DynamoDB Operations

1

## Capacity

On-demand or provisioned with auto scaling and adaptive capacity.

2

## Partition key

Distribute traffic; avoid hot keys and unbounded item growth.

3

## Recovery

Point-in-time recovery, on-demand backup and restore.

4

## Global table

Multi-Region active access with application-aware conflict behaviour.


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# Database Incident Evidence

## Resource

- CPU, memory and storage
- Connections and queue depth

## Engine

- Slow queries and locks
- Wait events and plans

## Change

- Configuration and schema history
- Deployment and failover events


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Data Operations

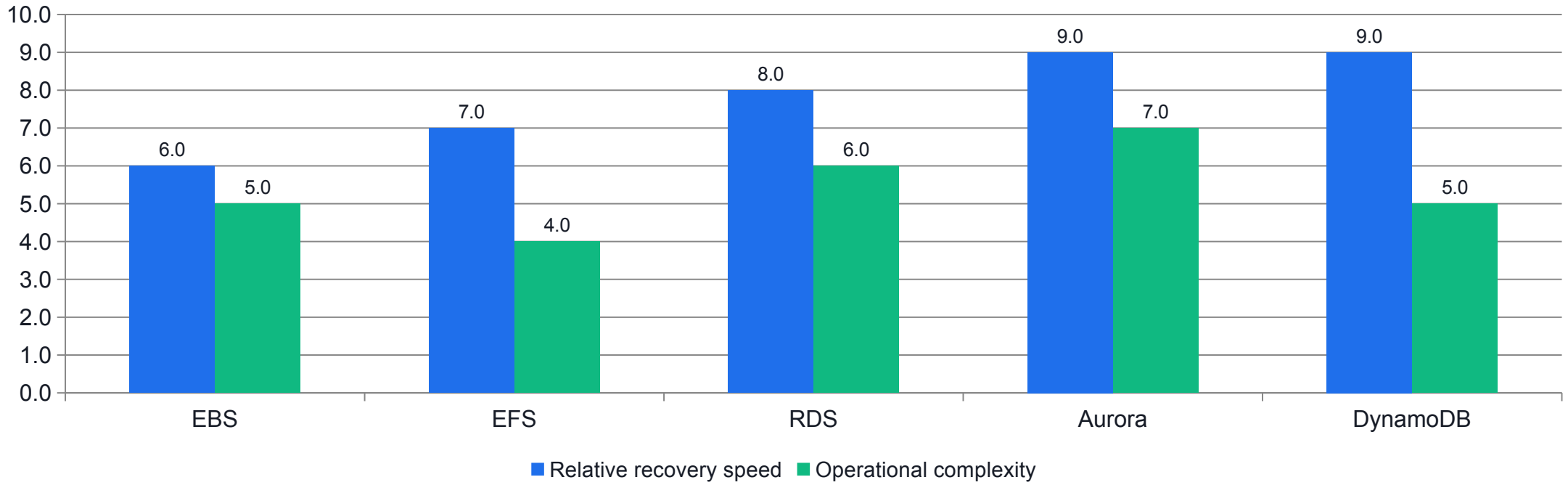
# Durability claims do not replace recovery design.

Protect the correct scope, control access, monitor jobs, test restore, validate application behaviour and measure actual RTO/RPO.


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TOPIC 09 · NATIVE EDITABLE TECHNICAL CHART

# Data Service Recovery Characteristics




**WHAT THE DATA SHOWS**

Recovery speed is not a product ranking; architecture, data size, dependencies and validation depth determine the measured outcome.


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# EBS vs EFS - Elastic Block Storage

### 1 EBS volumes…

EBS volumes…

gp3 & io1: can increase IO independently

EBS backups use IO and you shouldn't run

### 2 one instance (except multi-attach…

one instance (except multi-attach io1/io2)

To migrate an EBS volume across AZ

them while your application is handling a lot of traffic

### 3 are locked at the Availability

are locked at the Availability Zone (AZ) level

Take a snapshot

Root EBS Volumes of instances get terminated by default if the EC2 instance gets terminated. (you can disable that)

### 4 gp2: IO increases if the

gp2: IO increases if the disk size increases

Restore the snapshot to another AZ

Availability Zone 1

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# EBS vs EFS - Elastic File System

### 1 Mounting 100s of instances across

Mounting 100s of instances across AZ

Can leverage Storage Tiers for cost savings

### 2 EFS share website files…

EFS share website files (WordPress)

Remember: EFS vs EBS vs Instance Store

### 3 Only for Linux Instances (POSIX)

Only for Linux Instances (POSIX)

Availability Zone 1

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

### 4 EFS has a higher price

EFS has a higher price point than EBS

Availability Zone 2

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# EFS - Access Points

### 1 Easily manage applications access…

Easily manage applications access to NFS environments

EFS File System

Path: /data

### 2 Enforce a POSIX user and

Enforce a POSIX user and group to use when accessing the file system

Access Point 1

Analytics Users/Groups

### 3 Restrict access to a directory

Restrict access to a directory within the file system and optionally specify a different root directory

Path: /config

Developers Users/Groups

### 4 Can restrict access from NFS

Can restrict access from NFS clients using IAM policies

Access Point 2

Permissions

**CURRENT OPERATIONS LENS**

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# EFS - Operations

### 1 Operations that can be done

Operations that can be done in place

Operations that require a migration using DataSync (replicates all file attributes and metadata)

EFS File System

### 2 Lifecycle Policy (enable IA or

Lifecycle Policy (enable IA or change IA settings)

Migration to encrypted EFS

(destination - encrypted)

### 3 Throughput Mode and Provisioned…

Throughput Mode and Provisioned Throughput Numbers

Performance Mode (e.g. Max IO)

### 4 EFS Access Points

EFS Access Points

AWS DataSync

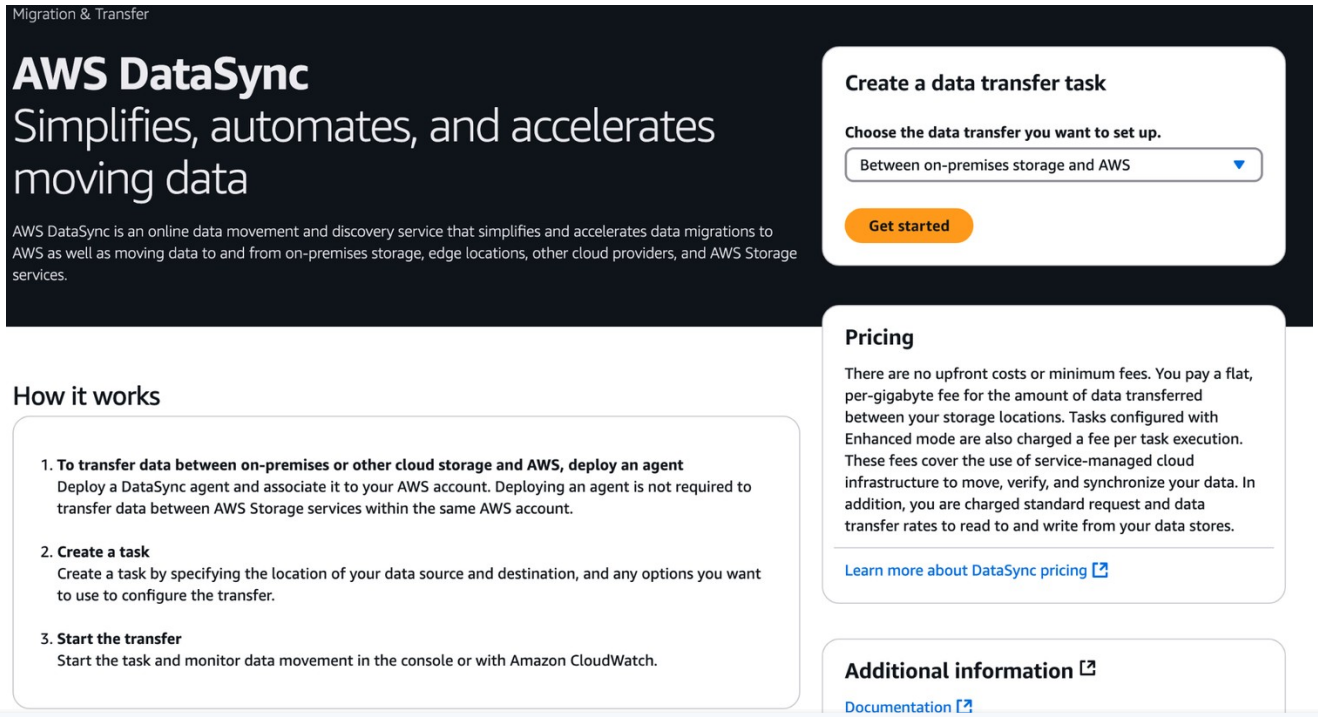
Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

**CURRENT OPERATIONS LENS**

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# AWS DataSync




**1** **Mechanism: block, file, object and**

Mechanism: block, file, object and database services expose different attachment, consistency, scaling…

**2** **Failure mode**

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and…

**3** **Evidence: queue depth…**

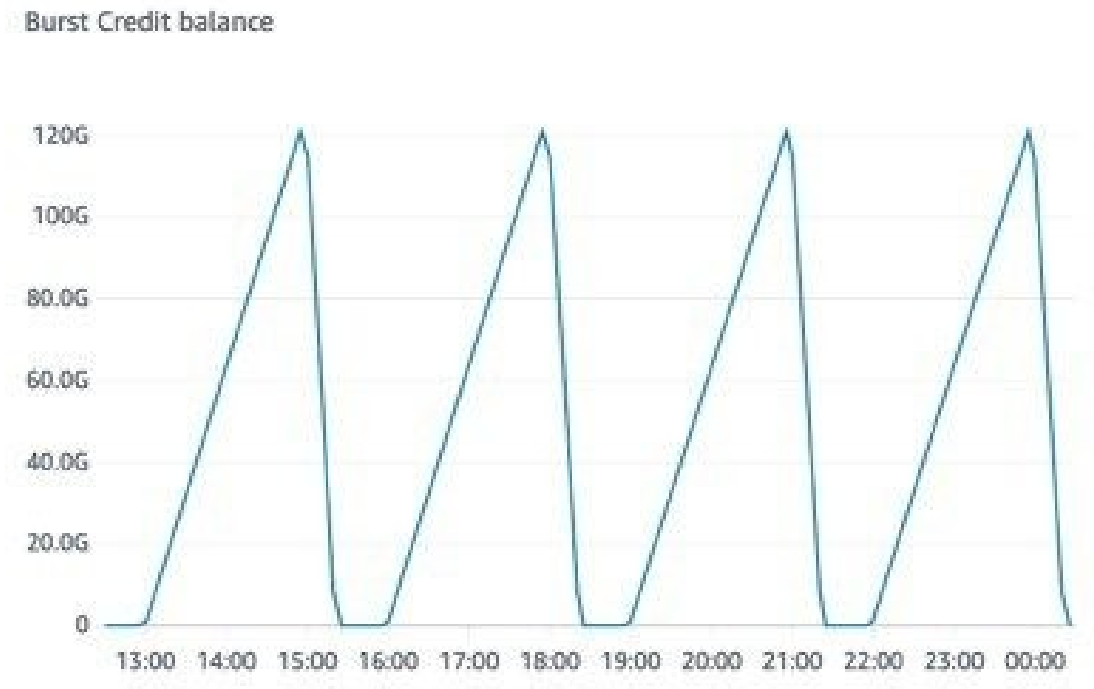
Evidence: queue depth, IOPS/throughput, latency, storage headroom, replica lag, query statistics, backup age and…

**CURRENT OPERATIONS LENS**

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# EFS - CloudWatch Metrics




**1** **PercentIOLimit**

PercentIOLimit
Mechanism: block, file, object and database services expose different attachment, consistency, scaling…

**2** **Failure mode**

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and…

**3** **Evidence: queue depth…**

Evidence: queue depth, IOPS/throughput, latency, storage headroom, replica lag, query statistics, backup age and…

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# Amazon S3

### 1 Amazon S3 is one of

Amazon S3 is one of the main building blocks of AWS

We'll have a step-by-step approach to S3

### 2 It's advertised as "infinitely…

It's advertised as "infinitely scaling" storage

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

### 3 Many websites use Amazon S3

Many websites use Amazon S3 as a backbone

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and key-policy gaps often look like…

### 4 Many AWS services use Amazon

Many AWS services use Amazon S3 as an integration as well

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# Amazon S3 Use cases




1 **Backup and storage**

Backup and storage
Mechanism: block, file, object and database services expose different attachment, consistency, scaling…

2 **Failure mode**

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and…

3 **Evidence: queue depth…**

Evidence: queue depth, IOPS/throughput, latency, storage headroom, replica lag, query statistics, backup age and…

**CURRENT OPERATIONS LENS**

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# Amazon S3 - Objects

### 1 Object (files) have a Key

Object (files) have a Key

They key is composed of prefix + object name

### 2 The key is the FULL

The key is the FULL path

There's no concept of "directories" within buckets (although the UI will trick you to think otherwise)

### 3 s3://my-bucket/ my_file.txt

s3://my-bucket/my_file.txt

Just keys with very long names that contain slashes ("/")

### 4 s3://my-bucket/ my_folder1/ another_…

s3://my-bucket/my_folder1/ another_folder/my_file.txt

S3 Bucket with Objects

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

**CURRENT OPERATIONS LENS**

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# Amazon S3 - Objects (cont.)

### 1 Object values are the content

Object values are the content of the body

Tags (Unicode key / value pair – up to 10) – useful for security / lifecycle

### 2 Max. Object Size is 5TB

Max. Object Size is 5TB (5000GB)

Version ID (if versioning is enabled)

### 3 If uploading more than 5GB,

If uploading more than 5GB, must use “multi-part upload”

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

### 4 Metadata (list of text key

Metadata (list of text key / value pairs – system or user metadata)

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# Amazon S3 - Security

## 1 User-Based

User-Based

Object Access Control List (ACL) - finer grain (can be disabled)

AND there's no explicit DENY

## 2 IAM Policies

IAM Policies - which API calls should be allowed for a specific user from IAM

Bucket Access Control List (ACL) - less common (can be disabled)

Encryption: encrypt objects in Amazon S3 using encryption keys

## 3 Resource-Based

Resource-Based

Note: an IAM principal can access an S3 object if

## 4 Bucket policies

Bucket policies - bucket wide rules from the S3 console - allows cross account

The user IAM permissions ALLOW it OR the resource policy ALLOWS it

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# S3 Bucket Policies

### 1 JSON based policies

JSON based policies

Principal: The account or user to apply the policy to

Grant access to another account (Cross Account)

### 2 Resources: buckets and objects

Resources: buckets and objects

Use S3 bucket for policy to

### 3 Effect: Allow / Deny

Effect: Allow / Deny

Grant public access to the bucket

### 4 Actions: Set of API to

Actions: Set of API to Allow or Deny

Force objects to be encrypted at upload

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# Example: Public Access - Use Bucket Policy

### 1 Anonymous www website visitor

Anonymous www website visitor

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

### 2 S3 Bucket Policy Allows Public

S3 Bucket Policy Allows Public Access

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and key-policy gaps often look like…

### 3 Evidence: queue depth…

Evidence: queue depth, IOPS/throughput, latency, storage headroom, replica lag, query statistics, backup age and restore validation isolate the…

### 4 Decision point

Decision point: choose the data service and recovery method from access pattern, durability, consistency, RTO/RPO and operational ownership—not only…

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# Example: User Access to S3 - IAM permissions

### 1 IAM Policy

IAM Policy

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

### 2 Failure mode

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and key-policy gaps often look like…

### 3 Evidence: queue depth…

Evidence: queue depth, IOPS/throughput, latency, storage headroom, replica lag, query statistics, backup age and restore validation isolate the…

### 4 Decision point

Decision point: choose the data service and recovery method from access pattern, durability, consistency, RTO/RPO and operational ownership—not only…

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# Advanced: Cross-Account Access - Use Bucket Policy

### 1 Other AWS account

Other AWS account

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

### 2 S3 Bucket Policy Allows…

S3 Bucket Policy Allows Cross-Account

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and key-policy gaps often look like…

### 3 Evidence: queue depth…

Evidence: queue depth, IOPS/throughput, latency, storage headroom, replica lag, query statistics, backup age and restore validation isolate the…

### 4 Decision point

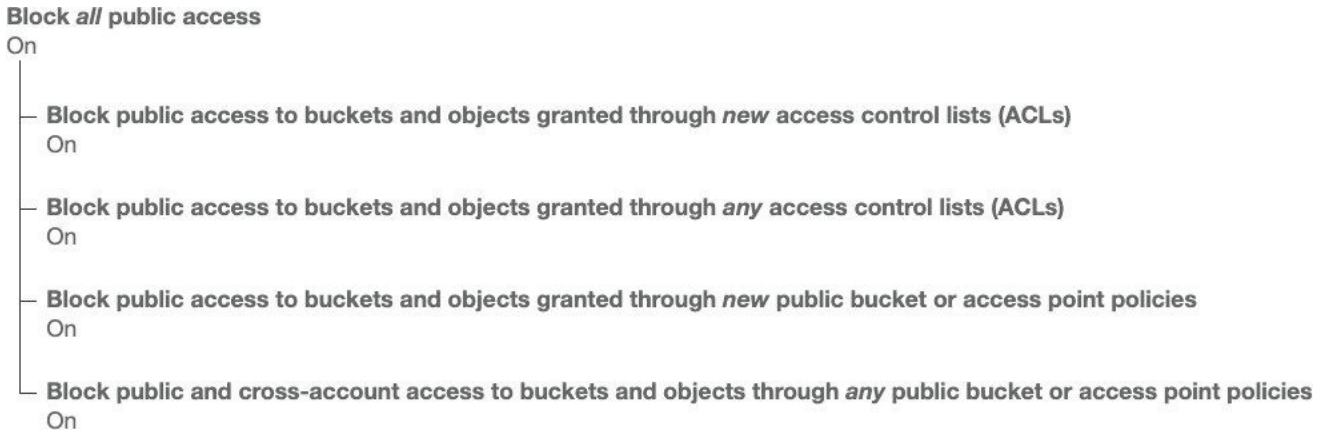
Decision point: choose the data service and recovery method from access pattern, durability, consistency, RTO/RPO and operational ownership—not only…

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# Bucket settings for Block Public Access




**1 These settings were created to**

These settings were created to prevent company data leaks Mechanism: block, file, object and database services expose different attachment, consistency, scaling…

**2 Failure mode**

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and…

**3 Evidence: queue depth…**

Evidence: queue depth, IOPS/throughput, latency, storage headroom, replica lag, query statistics, backup age and…

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# S3 Bucket Policies

### 1 Use S3 bucket for policy

Use S3 bucket for policy to

Optional Conditions on

Examples here: bucket-policies.html

### 2 Grant public access to the

Grant public access to the bucket

Public IP or Elastic IP (not on Private IP)

### 3 Force objects to be encrypted

Force objects to be encrypted at upload

Source VPC or Source VPC Endpoint – only works with VPC Endpoints

### 4 Grant access to another account

Grant access to another account (Cross account)

CloudFront Origin Identity

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# Bucket Policies Examples

1

**"Version": "2012-10-17",**

"Version": "2012-10-17",

"Effect": "Allow",

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

2

**"Id": "Policy1734311494060",**

"Id": "Policy1734311494060",

"Principal": "*",

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and key-policy gaps often look like…

3

**"Statement": [**

"Statement": [

"Action": "s3:GetObject",

Evidence: queue depth, IOPS/throughput, latency, storage headroom, replica lag, query statistics, backup age and restore validation isolate the…

4

**"Sid": "Stmt1734311424063",**

"Sid": "Stmt1734311424063",

"Resource": "arn:aws:s3:::alfred-ang-s3-buckets/*"

Decision point: choose the data service and recovery method from access pattern, durability, consistency, RTO/RPO and operational ownership—not only…

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# Amazon S3 - Static Website Hosting

### 1 S3 can host static websites

S3 can host static websites and have them accessible on the Internet

### 2 If you get a 403

If you get a 403 Forbidden error, make sure the bucket policy allows public reads!

### 3 (demo-bucket)

(demo-bucket)

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

### 4 The website URL will be

The website URL will be (depending on the region)

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and key-policy gaps often look like…

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# Amazon S3 - Versioning

### 1 You can version your files

You can version your files in Amazon S3

Protect against unintended deletes (ability to restore a version)

(my-bucket)

### 2 It is enabled at the

It is enabled at the bucket level

Easy roll back to previous version

s3://my-bucket/my-file.docx

### 3 Same key overwrite will change…

Same key overwrite will change the “version”: 1, 2, 3….

Any file that is not versioned prior to enabling versioning will have version “null”

### 4 It is best practice to

It is best practice to version your buckets

Suspending versioning does not delete the previous versions

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# Amazon S3 - Replication (CRR & SRR)

### 1 Must enable Versioning in source

Must enable Versioning in source and destination buckets

Copying is asynchronous

asynchronous replication

### 2 Cross-Region Replication (CRR)

Cross-Region Replication (CRR)

Must give proper IAM permissions to S3

### 3 Same-Region Replication (SRR)

Same-Region Replication (SRR)

CRR - compliance, lower latency access, replication across accounts

### 4 Buckets can be in different

Buckets can be in different AWS accounts

SRR - log aggregation, live replication between production and test accounts

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# Amazon S3 - Replication (Notes)

### 1 After you enable Replication, only

After you enable Replication, only new objects are replicated

Can replicate delete markers from source to target (optional setting)

Then objects created in bucket 1 are not replicated to bucket 3

### 2 Optionally, you can replicate…

Optionally, you can replicate existing objects using S3 Batch Replication

Deletions with a version ID are not replicated (to avoid malicious deletes)

### 3 Replicates existing objects and…

Replicates existing objects and objects that failed replication

There is no "chaining" of replication

### 4 For DELETE operations

For DELETE operations

If bucket 1 has replication into bucket 2, which has replication into bucket 3

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# S3 Storage Classes

### 1 Amazon S3 Standard

Amazon S3 Standard - General Purpose

Amazon S3 Glacier Flexible Retrieval

### 2 Amazon S3 Standard-Infrequent…

Amazon S3 Standard-Infrequent Access (IA)

Amazon S3 Glacier Deep Archive

### 3 Amazon S3 One Zone-Infrequent…

Amazon S3 One Zone-Infrequent Access

Amazon S3 Intelligent Tiering

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

### 4 Amazon S3 Glacier Instant…

Amazon S3 Glacier Instant Retrieval

Can move between classes manually or using S3 Lifecycle configurations

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# S3 Durability and Availability

### 1 Durability

Durability

Availability

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

### 2 High durability (99.999999999% …

High durability (99.999999999%, 11 9's) of objects across multiple AZ

Measures how readily available a service is

### 3 If you store 10,000,000 objects

If you store 10,000,000 objects with Amazon S3, you can on average expect to incur a loss of a single object once every 10,000 years

Varies depending on storage class

### 4 Same for all storage classes

Same for all storage classes

Example: S3 standard has 99.99% availability = not available 53 minutes a year

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# S3 Standard - General Purpose

### 1 99.99% Availability

99.99% Availability

Use Cases: Big Data analytics, mobile & gaming applications, content distribution…

### 2 Used for frequently accessed data

Used for frequently accessed data

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

### 3 Low latency and high throughput

Low latency and high throughput

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and key-policy gaps often look like…

### 4 Sustain 2 concurrent facility…

Sustain 2 concurrent facility failures

Evidence: queue depth, IOPS/throughput, latency, storage headroom, replica lag, query statistics, backup age and restore validation isolate the…

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# S3 Storage Classes - Infrequent Access

### 1 For data that is less

For data that is less frequently accessed, but requires rapid access when needed

Use cases: Disaster Recovery, backups

Use Cases: Storing secondary backup copies of on-premises data, or data you can recreate

### 2 Lower cost than S3 Standard

Lower cost than S3 Standard

Amazon S3 One Zone-Infrequent Access (S3 Standard-IA)

### 3 Amazon S3 Standard-Infrequent…

Amazon S3 Standard-Infrequent Access (S3 Standard-IA)

High durability (99.999999999%) in a single AZ; data lost when AZ is destroyed

### 4 99.9% Availability

99.9% Availability

99.5% Availability

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# Amazon S3 Glacier Storage Classes

### 1 Low-cost object storage meant for

Low-cost object storage meant for archiving / backup

Minimum storage duration of 90 days

Standard (12 hours), Bulk (48 hours)

### 2 Pricing: price for storage +

Pricing: price for storage + object retrieval cost

Amazon S3 Glacier Flexible Retrieval (formerly Amazon S3 Glacier)

Minimum storage duration of 180 days

### 3 Amazon S3 Glacier Instant…

Amazon S3 Glacier Instant Retrieval

Expedited (1 to 5 minutes), Standard (3 to 5 hours), Bulk (5 to 12 hours) – free

### 4 Millisecond retrieval, great for…

Millisecond retrieval, great for data accessed once a quarter

Amazon S3 Glacier Deep Archive - for long term storage

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# S3 Intelligent-Tiering

### 1 Small monthly monitoring and…

Small monthly monitoring and auto-tiering fee

Infrequent Access tier (automatic): objects not accessed for 30 days

### 2 Moves objects automatically…

Moves objects automatically between Access Tiers based on usage

Archive Instant Access tier (automatic): objects not accessed for 90 days

### 3 There are no retrieval charges

There are no retrieval charges in S3 Intelligent-Tiering

Archive Access tier (optional): configurable from 90 days to 700+ days

### 4 Frequent Access tier (automatic)

Frequent Access tier (automatic): default tier

Deep Archive Access tier (optional): config. from 180 days to 700+ days

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# Amazon S3 - Moving between Storage Classes

1

### You can transition objects between

You can transition objects between storage classes

Standard IA

Glacier Flexible Retrieval

2

### For infrequently accessed object…

For infrequently accessed object, move them to Stanard IA

Intelligent Tiering

Glacier Deep Archive

3

### For archive objects that you

For archive objects that you don’t need fast access to, move them to Glacier or Glacier Deep Archive

One-Zone IA

4

### Moving objects can be automated

Moving objects can be automated using a Lifecycle Rules

Glacier Instant Retrieval

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# Amazon S3 - Lifecycle Rules

### 1 Transition Actions

Transition Actions - configure objects to transition to another storage class

Access log files can be set to delete after a 365 days

Rules can be created for certain objects Tags (example: Department: Finance)

### 2 Move objects to Standard IA

Move objects to Standard IA class 60 days after creation

Can be used to delete old versions of files (if versioning is enabled)

### 3 Move to Glacier for archiving

Move to Glacier for archiving after 6 months

Can be used to delete incomplete Multi-Part uploads

### 4 Expiration Actions

Expiration Actions - configure objects to expire (delete) after some time

Rules can be created for a certain prefix (example: s3://mybucket/mp3/*)

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# Amazon S3 - Lifecycle Rules (Scenario 1)

### 1 Your application on EC2 creates

Your application on EC2 creates images thumbnails after profile

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

### 2 photos are uploaded to Amazon

photos are uploaded to Amazon S3. These thumbnails can be easily recreated, and only need to be kept for 60 days. The source images should be able to…

### 3 S3 source images can be

S3 source images can be on Standard with a lifecycle configuration to trainsition them to Glacier after 60 days

### 4 S3 thumbnails can be on

S3 thumbnails can be on One-Zone IA with a lifecycle configuration to expire them (delete them) after 60 days

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# Amazon S3 - Lifecycle Rules (Scenario 2)

1

### A rule in your company

A rule in your company states that you should be able to recover your deleted S3 objects immediately for 30 days, although this may happen rarely…

Transition afterwards the “noncurrent versions” to Glacier Deep Archive

2

### be recoverable within 48 hours.

be recoverable within 48 hours.

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

3

### Enable S3 Versioning in order

Enable S3 Versioning in order to have object versions, so that “deleted objects” are in fact hidden by a “delete marker” and can be recovered

4

### Transition the “noncurrent…

Transition the “noncurrent versions” of the object to Standard IA

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# Amazon S3 Analytics - Storage Class Analysis

### 1 Help you decide when to

Help you decide when to transition objects to the right storage class

24 to 48 hours to start seeing data analysis

### 2 Recommendations for Standard and…

Recommendations for Standard and Stanard IA

Good first step to put together Lifecycle Rules (or improve them)!

### 3 Does NOT work for One-Zone

Does NOT work for One-Zone IA or Glacier

S3 Analytics

### 4 Report is updated daily

Report is updated daily

.csv report

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

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# S3 Event Notifications

1

## Lambda Function

Lambda Function

Can create as many "S3 events" as desired

2

## S3:ObjectCreated, S3:ObjectRemoved…

S3:ObjectCreated, S3:ObjectRemoved, S3:ObjectRestore, S3:Replication…

S3 event notifications typically deliver events in seconds but can sometimes take a minute or longer

3

## Object name filtering possible…

Object name filtering possible (*.jpg)

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

4

## Use case

Use case: generate thumbnails of images uploaded to S3

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and key-policy gaps often look like…

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# S3 Event Notifications - IAM Permissions

```
{
  "Version": "2012-10-17",
  "Statement": {
    "Effect": "Allow",
    "Action": "SQS:SendMessage",
    "Principal": {
      "Service": "s3.amazonaws.com"
    },
    "Resource": "arn:aws:sqs:us-east-1:123456789012:MyQueue",
    "Condition": {
      "ArnLike": {
        "aws:SourceArn": "arn:aws:s3:::MyBucket"
      }
    }
  }
}
```

1

### Lambda Resource Policy

Lambda Resource Policy
Mechanism: block, file, object and database services expose different attachment, consistency, scaling…

2

### Failure mode

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and…

3

### Evidence: queue depth…

Evidence: queue depth, IOPS/throughput, latency, storage headroom, replica lag, query statistics, backup age and…

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# S3 Event Notifications with Amazon EventBridge

### 1 Amazon S3 bucket

Amazon S3 bucket

as destinations

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

### 2 All events

All events

Advanced filtering options with JSON rules (metadata, object size, name...)

### 3 Amazon EventBridge

Amazon EventBridge

Multiple Destinations - ex Step Functions, Kinesis Streams / Firehose…

### 4 AWS services

AWS services

EventBridge Capabilties - Archive, Replay Events, Reliable delivery

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# S3 - Baseline Performance

### 1 Amazon S3 automatically scales to

Amazon S3 automatically scales to high request rates, latency 100-200 ms

bucket/folder1/sub1/file => /folder1/sub1/

If you spread reads across all four prefixes evenly, you can achieve 22,000 requests per second for GET and HEAD

### 2 Your application can achieve at

Your application can achieve at least 3,500 PUT/COPY/POST/DELETE or 5,500 GET/HEAD requests per second per prefix in a bucket.

bucket/folder1/sub2/file => /folder1/sub2/

### 3 There are no limits to

There are no limits to the number of prefixes in a bucket.

bucket/1/file => /1/

### 4 Example (object path => prefix)

Example (object path => prefix)

bucket/2/file => /2/

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

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# S3 Performance

### 1 Multi-Part upload

Multi-Part upload

Increase transfer speed by transferring file to an AWS edge location which will forward the data to the S3 bucket in the target region

Edge Location

### 2 recommended for files > 100MB,

recommended for files > 100MB, must use for files > 5GB

Compatible with multi-part upload

(public www)

### 3 Can help parallelize uploads…

Can help parallelize uploads (speed up transfers)

Parallel uploads

(private AWS)

### 4 S3 Transfer Acceleration

S3 Transfer Acceleration

Divide In parts

File in USA

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

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# S3 Performance - S3 Byte-Range Fetches

### 1 Can be used to speed

Can be used to speed up downloads

File in S3

### 2 Parallelize GETs by requesting…

Parallelize GETs by requesting specific byte ranges

Byte-range request for header (first XX bytes)

### 3 Better resilience in case of

Better resilience in case of failures

data (for example the head of a file)

### 4 Can be used to retrieve

Can be used to retrieve only partial

Requests in parallel

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# S3 Batch Operations

### 1 Perform bulk operations on…

Perform bulk operations on existing S3 objects with a single request, example

Modify ACLs, tags

S3 Batch Operations manages retries, tracks progress, sends completion notifications, generate reports …

### 2 Modify object metadata &…

Modify object metadata & properties

Restore objects from S3 Glacier

You can use S3 Inventory to get object list to use

### 3 Copy objects between S3 buckets

Copy objects between S3 buckets

Invoke Lambda function to perform custom action on each object

S3 Select to filter your objects

### 4 Encrypt un-encrypted objects

Encrypt un-encrypted objects

A job consists of a list of objects, the action to perform, and optional parameters

S3 Inventory

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# S3 Inventory

### 1 List objects and their…

List objects and their corresponding metadata (alternative to S3 List API operation)

Identify the total storage of previous object versions

You can filter generated report using S3 Select

### 2 Usage examples

Usage examples

Generate daily or weekly reports

Use cases: Business, Compliance, Regulatory needs, …

### 3 Audit and report on the

Audit and report on the replication and encryption status of your objects

Output files: CSV, ORC, or Apache Parquet

### 4 Get the number of objects

Get the number of objects in an S3 bucket

You can query all the data using Amazon Athena, Redshift, Presto, Hive, Spark…

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# Amazon S3 Glacier

### 1 Low-cost object storage meant for

Low-cost object storage meant for archiving / backup

Cost per storage per month ($0.004 / GB – Standard | $0.00099 / GB Deep Archive)

Exam tip: archival from S3 after XXX days => use Glacier

### 2 Data is retained for the

Data is retained for the longer term (10s of years)

Each item in Glacier is called "Archive" (up to 40TB)

### 3 Alternative to on-premises…

Alternative to on-premises magnetic tape storage

Archives are stored in "Vaults"

### 4 Average annual durability is…

Average annual durability is 99.999999999%

By default, data encrypted at rest using AES-256 – keys managed by AWS

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# Amazon S3 Glacier Operations

## 1 Vault Operations

Vault Operations

Glacier Operations

Delete - use Glacier Rest API or AWS SDKs by specifying archive ID

## 2 Create & Delete

Create & Delete - delete only when there's no archives in it

Upload- single operation or by parts (MultiPart upload) for larger archives

Restore links have an expiry date

## 3 Retrieving Metadata

Retrieving Metadata - creation date, number of archives, total size of all archives, …

Download - first initiate a retrieval job for the archive, Glacier then prepares it for download.

Retrieval Options

## 4 Download Inventory

Download Inventory - list of archives in the vault (archive ID, creation date, size, …)

User then has a limited time to download the data from staging server. (optionally, specify a range or portion of bytes to retrieve)

Expedited (1 to 5 minutes retrieval) - $0.03 per GB and $10 per 1000 requests

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# Amazon S3 Glacier - Vault Policies & Vault Lock

### 1 Each Vault has

Each Vault has

Vault Access Policy is like a bucket policy (restrict user / account permissions)

Example 2: implement WORM policy (write once read many)

### 2 ONE vault access policy

ONE vault access policy

Vault Lock Policy is a policy you lock, for regulatory and compliance requirements.

Permissions

### 3 ONE vault lock policy

ONE vault lock policy

The policy is immutable, it can never be changed (that's why it's called a LOCK)

Vault Access Policy

### 4 Vault Policies are written in

Vault Policies are written in JSON

Example 1: forbid deleting an archive if less than 1 year old

Vault Lock Policy

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# Glacier - Notifications for Restore Operations

### 1 Vault Notification Configuration

Vault Notification Configuration

S3 supports the restoration of objects archived to S3 Glacier storage classes

Amazon Glacier

### 2 Configure a vault so that

Configure a vault so that when a job completes a message is sent to SNS

s3:ObjectRestore:Post => notify when object restoration initiated

archive ready restore

### 3 Optionally, specify an SNS topic

Optionally, specify an SNS topic when you initiate a job

s3:ObjectRestore:Completed => notify when object restoration completed

send notification

### 4 S3 Event Notifications

S3 Event Notifications

initiate restore archive job

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

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# S3 Multi Part Upload - Deep Dive

### 1 Upload large objects in parts

Upload large objects in parts (any order)

Failures: restart uploading ONLY failed parts (improves performance)

(max. 10,000)

### 2 Recommended for files > 100MB,

Recommended for files > 100MB, must use for files > 5GB

Use Lifecycle policy to automate old parts

parallel uploads

### 3 Can help parallelize uploads…

Can help parallelize uploads (speed up transfers)

Deletion of unfinished upload after x days (e.g., network outage)

Concatenating parts (COMPLETE request)

### 4 Max. parts

Max. parts: 10,000

Upload using AWS CLI or AWS SDK

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# Amazon Athena

### 1 Serverless query service to…

Serverless query service to analyze data stored in Amazon S3

Commonly used with Amazon Quicksight for reporting/dashboards

Amazon QuickSight

### 2 Uses standard SQL language to

Uses standard SQL language to query the files (built on Presto)

Use cases: Business intelligence / analytics / reporting, analyze & query VPC Flow Logs, ELB Logs, CloudTrail trails, etc...

Reporting & Dashboards

### 3 Supports CSV, JSON, ORC, Avro,

Supports CSV, JSON, ORC, Avro, and Parquet

Exam Tip: analyze data in S3 using serverless SQL, use Athena

### 4 Pricing: $5.00 per TB of

Pricing: $5.00 per TB of data scanned

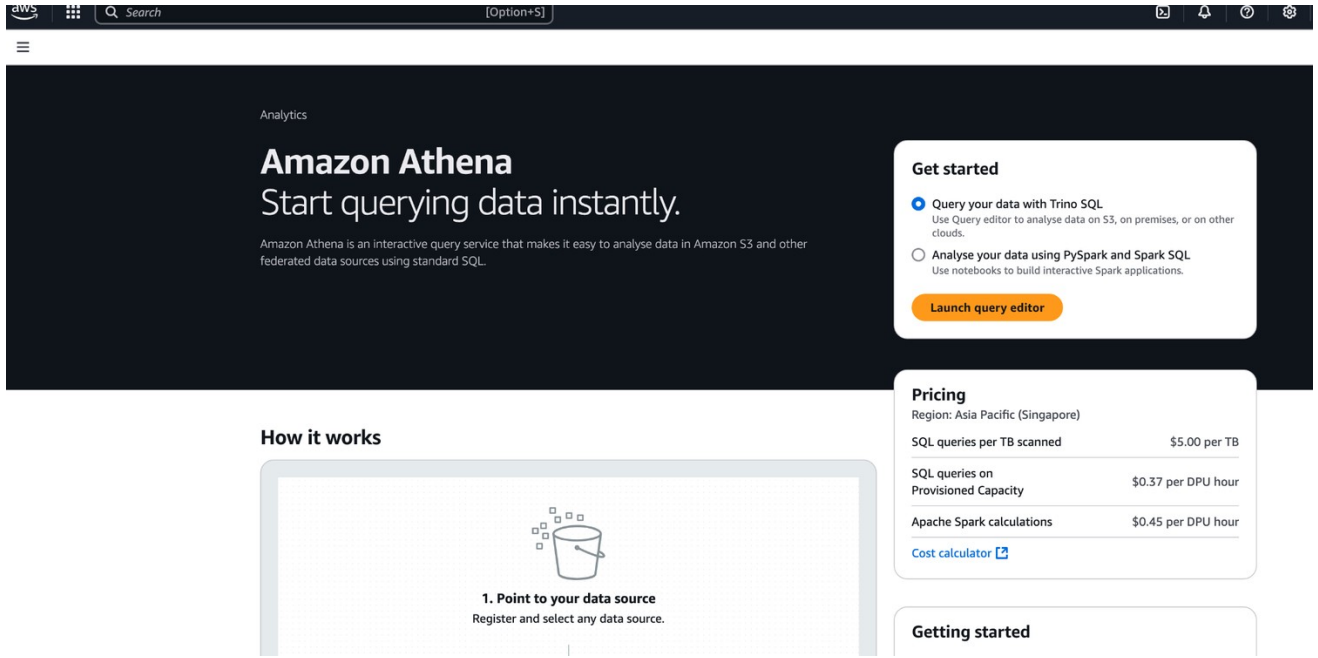
Query & Analyze

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# Amazon Anthena




1

## Mechanism: block, file, object and

Mechanism: block, file, object and database services expose different attachment, consistency, scaling…

2

## Failure mode

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and…

3

## Evidence: queue depth…

Evidence: queue depth, IOPS/throughput, latency, storage headroom, replica lag, query statistics, backup age and…

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# Amazon Athena - Performance Improvement

### 1 Use Columnar data for cost-savings

Use Columnar data for cost-savings (less scan)

Compress data for smaller retrievals (bzip2, gzip, lz4, snappy, zlip, zstd…)

Example: s3://athena-examples/flight/parquet/year=1991/month=1/day=1/

### 2 Apache Parquet or ORC is

Apache Parquet or ORC is recommended

Partition datasets in S3 for easy querying on virtual columns

Use larger files (> 128 MB) to minimize overhead

### 3 Huge performance improvement

Huge performance improvement

s3://yourBucket/pathToTable

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

### 4 Use Glue to convert your

Use Glue to convert your data to Parquet or ORC

/<PARTITION_COLUMN_NAME>=<VALUE>

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# Amazon Athena - Federated Query

### 1 Allows you to run SQL

Allows you to run SQL queries across data stored in relational, non-relational, object, and custom data sources (AWS or on-premises)

Database (On-Premises)

DocumentDB

### 2 Uses Data Source Connectors that

Uses Data Source Connectors that run on AWS Lambda to run Federated Queries (e.g., CloudWatch Logs, DynamoDB, RDS, …)

(Data Source

HBase in EMR

### 3 Store the results back in

Store the results back in Amazon S3

Connector)

SQL Server

### 4 Amazon Athena

Amazon Athena

ElastiCache

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

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# Amazon S3 - Object Encryption

### 1 You can encrypt objects in

You can encrypt objects in S3 buckets using one of 4 methods

Server-Side Encryption with KMS Keys stored in AWS KMS (SSE-KMS)

Cilent-Side Encryption

### 2 Server-Side Encryption (SSE)

Server-Side Encryption (SSE)

Leverage AWS Key Management Service (AWS KMS) to manage encryption keys

It's important to understand which ones are for which situation for the exam

### 3 Server-Side Encryption with…

Server-Side Encryption with Amazon S3-Managed Keys (SSE-S3) Enabled by Default

Server-Side Encryption with Customer-Provided Keys (SSE-C)

### 4 Encrypts S3 objects using keys

Encrypts S3 objects using keys handled, managed, and owned by AWS

When you want to manage your own encryption keys

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# Amazon S3 Encryption - SSE-S3

### 1 HTTP(S) + Header

HTTP(S) + Header

Must set header "x-amz-server-side-encryption":"AES256"

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

### 2 Encryption using keys handled…

Encryption using keys handled, managed, and owned by AWS

Enabled by default for new buckets & new objects

### 3 Object is encrypted server-side

Object is encrypted server-side

S3 Owned Key

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and key-policy gaps often look like…

### 4 Encryption type is AES-256

Encryption type is AES-256

Encryption

Evidence: queue depth, IOPS/throughput, latency, storage headroom, replica lag, query statistics, backup age and restore validation isolate the…

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# Amazon S3 Encryption - SSE-KMS

### 1 HTTP(S) + Header

HTTP(S) + Header

Must set header "x-amz-server-side-encryption":"AWS:KMS"

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

### 2 Encryption using keys handled and

Encryption using keys handled and managed by AWS KMS (Key Management Service)

Encryption

### 3 KMS advantages

KMS advantages: user control + audit key usage using CloudTrail

### 4 Object is encrypted server side

Object is encrypted server side

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and key-policy gaps often look like…

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# SSE-KMS Limitation

### 1 If you use SSE-KMS, you

If you use SSE-KMS, you may be impacted by the KMS limits

Count towards the KMS quota per second (5500, 10000, 30000 req/s based on region)

### 2 When you upload, it calls

When you upload, it calls the

You can request a quota increase using the Service Quotas Console

### 3 GenerateDataKey KMS API

GenerateDataKey KMS API

Upload / download SSE-KMS

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

### 4 When you download, it calls

When you download, it calls the Decrypt KMS API

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and key-policy gaps often look like…

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# Amazon S3 Encryption - SSE-C

### 1 Server-Side Encryption using keys…

Server-Side Encryption using keys fully managed by the customer outside of AWS

HTTPS ONLY

### 2 Amazon S3 does NOT store

Amazon S3 does NOT store the encryption key you provide

+ Key in Header

### 3 HTTPS must be used

HTTPS must be used

Client-Provided Key

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

### 4 Encryption key must provided in

Encryption key must provided in HTTP headers, for every HTTP request made

Encryption

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# Amazon S3 Encryption - Client-Side Encryption

### 1 Use client libraries such as

Use client libraries such as Amazon S3 Cilent-Side Encryption Library

Client Key

### 2 Clients must encrypt data…

Clients must encrypt data themselves before sending to Amazon S3

(encrypted)

### 3 Clients must decrypt data…

Clients must decrypt data themselves when retrieving from Amazon S3

Encryption

### 4 Customer fully manages the keys

Customer fully manages the keys and encryption cycle

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

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# Amazon S3 - Encryption in transit (SSL/TLS)

### 1 Encryption in flight is also

Encryption in flight is also called SSL/TLS

HTTPS is recommended

### 2 Amazon S3 exposes two endpoints

Amazon S3 exposes two endpoints

HTTPS is mandatory for SSE-C

### 3 HTTP Endpoint

HTTP Endpoint - non encrypted

Most clients would use the HTTPS endpoint by default

### 4 HTTPS Endpoint

HTTPS Endpoint - encryption in flight

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

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# Amazon S3 – Force Encryption in Transit aws:SecureTransport

### 1 (my-bucket)

(my-bucket)

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

### 2 Bucket Policy

Bucket Policy

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and key-policy gaps often look like…

### 3 Evidence: queue depth…

Evidence: queue depth, IOPS/throughput, latency, storage headroom, replica lag, query statistics, backup age and restore validation isolate the…

### 4 Decision point

Decision point: choose the data service and recovery method from access pattern, durability, consistency, RTO/RPO and operational ownership—not only…

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# Amazon S3 - Default Encryption vs. Bucket Policies



```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Deny",
            "Action": "s3:PutObject",
            "Principal": "*",
            "Resource": "arn:aws:s3:::my-bucket/*",
            "Condition": {
                "StringNotEquals": {
                    "s3:x-amz-server-side-encryption": "aws:kms"
                }
            }
        }
    ]
}
```

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Deny",
            "Action": "s3:PutObject",
            "Principal": "*",
            "Resource": "arn:aws:s3:::my-bucket/*",
            "Condition": {
                "Null": {
                    "s3:x-amz-server-side-encryption-customer-algorithm": "true"
                }
            }
        }
    ]
}
```

**1** **SSE-S3 encryption is…**

SSE-S3 encryption is automatically applied to new objects stored in S3 bucket

**2** **Mechanism: block, file, object and**

Mechanism: block, file, object and database services expose different attachment, consistency, scaling…

**3** **Failure mode**

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and…

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# What is CORS?

1

### Cross-Origin Resource Sharing…

Cross-Origin Resource Sharing (CORS)

Same origin: &

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

2

### Origin = scheme (protocol) +

Origin = scheme (protocol) + host (domain) + port

Different origins: &

3

### example: (implied port is 443

example: (implied port is 443 for HTTPS, 80 for HTTP)

The requests won't be fulfilled unless the other origin allows for the requests, using CORS Headers (example: Access-Control-Allow-Origin)

4

### Web Browser based mechanism to

Web Browser based mechanism to allow requests to other origins while visiting the main origin

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# What is CORS?

### 1 Web Server

Web Server

Host: www.other.com

CORS Headers received already by the Origin

### 2 (Cross-Origin)

(Cross-Origin)

Preflight Request

The Web Browser can make requests

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

### 3 Web Browser

Web Browser

Access-Control-Allow-Origin: Access-Control-Allow-Methods: GET, PUT, DELETE

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and key-policy gaps often look like…

### 4 HTTPS Request

HTTPS Request

Preflight Response

Evidence: queue depth, IOPS/throughput, latency, storage headroom, replica lag, query statistics, backup age and restore validation isolate the…

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# Amazon S3 - CORS

### 1 If a client makes a

If a client makes a cross-origin request on our S3 bucket, we need to enable the correct CORS headers

(my-bucket-html)

index.html

### 2 It’s a popular exam question

It’s a popular exam question

(Static Website Enabled)

GET /images/coffee.jpg

### 3 You can allow for a

You can allow for a specific origin or for * (all origins)

(my-bucket-assets)

Host: Origin

### 4 Web Browser

Web Browser

GET /index.html

Access-Control-Allow-Origin

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# Amazon S3 - MFA Delete

### 1 MFA (Multi-Factor Authentication)

MFA (Multi-Factor Authentication) - force users to generate a code on a device (usually a mobile phone or hardware) before doing important operations…

MFA won't be required to

Only the bucket owner (root account) can enable/disable MFA Delete

### 2 MFA will be required to

MFA will be required to

Enable Versioning

Google Authenticator

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

### 3 Permanently delete an object…

Permanently delete an object version

List deleted versions

MFA Hardware Device

### 4 Suspend Versioning on the bucket

Suspend Versioning on the bucket

To use MFA Delete, Versioning must be enabled on the bucket

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# S3 Access Logs

1

**For audit purpose, you may**

For audit purpose, you may want to log all access to S3 buckets

The log format is at

2

**Any request made to S3,**

Any request made to S3, from any account, authorized or denied will be logged into another S3 bucket

Log all requests

3

**That data can be analyzed**

That data can be analyzed using data analysis tools…

Logging Bucket

4

**The target logging bucket must**

The target logging bucket must be in the same AWS region

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# S3 Access Logs: Warning

### 1 Do not set your logging

Do not set your logging bucket to be the monitored bucket

Do not try this at home ©

### 2 It will create a logging

It will create a logging loop, and your bucket will grow exponentially

### 3 App Bucket & Logging Bucket

App Bucket & Logging Bucket

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

### 4 Logging loop

Logging loop

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and key-policy gaps often look like…

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# Amazon S3 - Pre-Signed URLs

### 1 Generate pre-signed URLs using the

Generate pre-signed URLs using the S3 Console, AWS CLI or SDK

Users given a pre-signed URL inherit the permissions of the user that generated the URL for GET / PUT

generate pre-signed URL

### 2 URL Expiration

URL Expiration

Allow only logged-in users to download a premium video from your S3 bucket

### 3 S3 Console

S3 Console - 1 min up to 720 mins (12 hours)

Allow an ever-changing list of users to download files by generating URLs dynamically

### 4 AWS CLI

AWS CLI - configure expiration with --expires-in parameter in seconds (default 3600 secs, max. 604800 secs ~ 168 hours)

Allow temporarily a user to upload a file to a precise location in your S3 bucket

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# S3 Glacier Vault Lock

### 1 Adopt a WORM (Write Once

Adopt a WORM (Write Once Read Many) model

Helpful for compliance and data retention

### 2 Create a Vault Lock Policy

Create a Vault Lock Policy

Vault Lock Policy

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

### 3 Lock the policy for future

Lock the policy for future edits

Object can’t be deleted

### 4 (can no longer be changed

(can no longer be changed or deleted)

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and key-policy gaps often look like…

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# S3 Object Lock (versioning must be enabled)

### 1 Adopt a WORM (Write Once

Adopt a WORM (Write Once Read Many) model

Objects retention modes can't be changed, and retention periods can't be shortened

Retention Period: protect the object for a fixed period, it can be extended

### 2 Block an object version deletion

Block an object version deletion for a specified amount of time

Retention mode - Governance

Legal Hold

### 3 Retention mode

Retention mode - Compliance

Most users can't overwrite or delete an object version or alter its lock settings

protect the object indefinitely, independent from retention period

### 4 Object versions can't be…

Object versions can't be overwritten or deleted by any user, including the root user

Some users have special permissions to change the retention or delete the object

can be freely placed and removed using the s3:PutObjectLegalHold IAM permission

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# S3 - Access Points

## 1 Simple Bucket Policy

Simple Bucket Policy

Grant R/W to

entire bucket

Mechanism: block, file, object and database services expose different attachment, consistency, scaling, replication, backup and failover behaviours.

## 2 Finance Access Point

Finance Access Point

/finance prefix

(Analytics)

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and key-policy gaps often look like…

## 3 Sales Access Point

Sales Access Point

/sales prefix

Access Points simplify security management for S3 Buckets

## 4 Analytics Access Point

Analytics Access Point

Grant R to

Each Access Point has

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# S3 - Access Points - VPC Origin

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Principal": "*",
            "Action": [
                "s3:GetObject"
            ],
            "Effect": "Allow",
            "Resource": [
                "arn:aws:s3:::awsexamplebucket1/*",
                "arn:aws:s3:us-west-2:123456789012:accesspoint/example-vpc-ap/object/*"
            ]
        }
    ]
}
```

1

### We can define the access

We can define the access point to be accessible only from within the VPC

2

### Mechanism: block, file, object and

Mechanism: block, file, object and database services expose different attachment, consistency, scaling…

3

### Failure mode

Failure mode: throughput limits, burst-credit exhaustion, hot partitions, replica lag, connection storms and…

**CURRENT OPERATIONS LENS**

Match durability, performance, consistency, scaling and recovery characteristics to the workload's measurable service objective. Validate service limits and regional availability against current AWS documentation.


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# Original PPT Coverage Retained — Topic 09

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70 technical slides extracted from the original v4 PowerPoint and rebuilt in the current visual system.

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### Traceability

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TOPIC 09 · CONCEPT-LED LAB BRIEF

# Diagnose Storage and Database Performance, Then Recover

**LAB 11**

Apply K1, K4, A1, A4 and A6 by correlating EBS/RDS signals, selecting a safe change and validating data recovery.

Analyse a synthetic metric set, choose one evidence-based action and verify a point-in-time recovery plan.

### YOU'LL BUILD

A metric analysis, bottleneck decision, change plan, restore checklist and database recovery evidence.

### TOOLCHAIN

CloudWatch · EBS · RDS · Performance Insights · AWS Backup

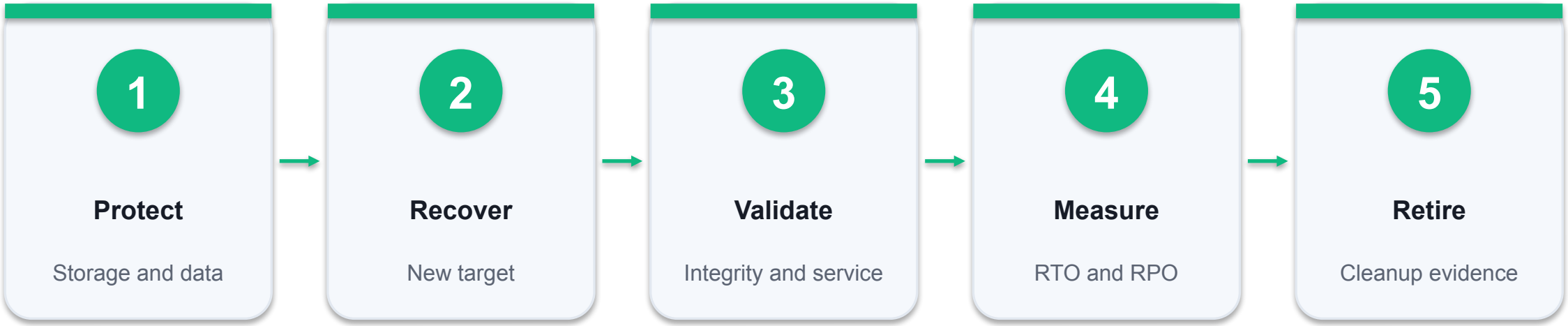
### DONE WHEN

Metric calculations are reproducible, the chosen bottleneck cites evidence, one reversible action is proposed, recovery is isolated and timed, and validation emits LAB 11 PASS.


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LAB 11 · CONCEPTUAL WORKFLOW

# Lab 11 Evidence Model




**PROCEDURE BOUNDARY**

Detailed commands and numbered steps are in the Learner Guide and lab folder—not on these slides.


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LAB 11 · ACCEPTANCE EVIDENCE

# Diagnose Storage and Database Performance, Then Recover

## ✅ Expected result

Metric calculations are reproducible, the chosen bottleneck cites evidence, one reversible action is proposed, recovery is isolated and timed, and validation emits LAB 11 PASS.

IF IT DOESN'T WORK

### Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

### Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

### Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.


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# Recap — Storage, Data and Database Operations

1

**Explain**

Object, block, file and database services solve different consistency, latency and sharing needs.

2

**Distinguish**

IOPS, throughput, connections, partitions and caching require different evidence.

3

**Apply**

K1, K4, A1, A4 and A6 by correlating EBS/RDS signals, selecting a safe change and validating data recovery.

4

**Evidence**

Metric calculations are reproducible, the chosen bottleneck cites evidence, one reversible action is proposed, recovery is isolated and timed, and validation emits…


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10

TOPIC 10 · LO3 · LO4 · LO5 · K2 · K4 · A2 · A4 · A7

# Performance, Cost and Integrated Troubleshooting

Bottleneck analysis · Cost Explorer · Trusted Advisor · runbooks · game days · capstone


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# Key Ideas — Performance, Cost and Integrated Troubleshooting

1

## Optimise after measurement

Right-size and tune using utilisation, saturation, latency and business demand.

2

## Cost anomalies are operational signals

Unexpected spend can reveal leakage, misconfiguration, attack or failed cleanup.

3

## Troubleshoot hypotheses

A runbook narrows evidence systematically instead of changing multiple variables.

4

## Service quality is verified

SLOs, recovery tests and post-incident actions prove operational capability.


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# Evidence-Led Troubleshooting




**A4 STANDARD**

A hypothesis is not a cause until evidence supports it and alternatives are tested.

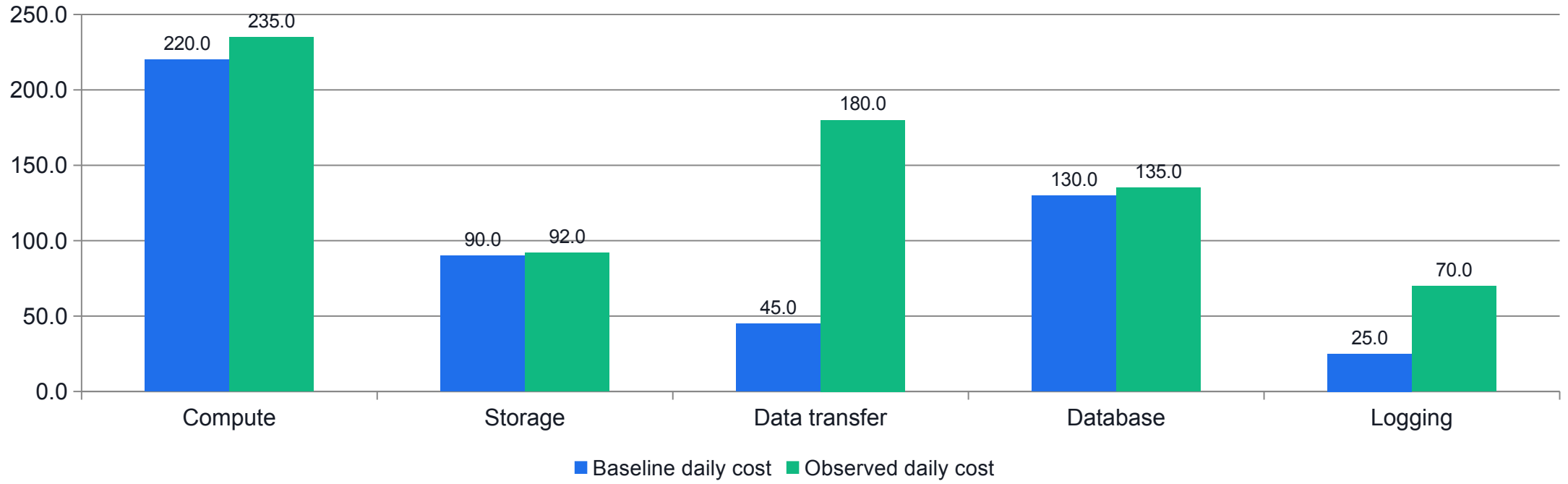

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# Performance Signals

| Signal | Interpretation | Example action |
| --- | --- | --- |
| **Utilisation** | Resource use relative to capacity | Right-size only with sustained evidence |
| **Saturation** | Queued work or constrained resource | Increase capacity or remove bottleneck |
| **Errors** | Failed operations or retries | Trace dependency and change history |
| **Latency** | Time experienced by request | Inspect percentiles and request path |


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# Cost Anomaly Investigation




**WHAT THE DATA SHOWS**

The largest percentage changes are data transfer and logging; investigate traffic path, retention and recent configuration before right-sizing compute.


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# Cost and Optimisation Tools

1

## Cost Explorer

Trend, filter, group and forecast historical spend.

2

## Cost Anomaly Detection

Surface spend patterns that deviate from expected behaviour.

3

## Compute Optimizer

Resource recommendations from utilisation history.

4

## Trusted Advisor

Checks across cost, security, resilience, performance and limits.


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# Runbook vs Playbook

| Artifact | Purpose | Contents |
| --- | --- | --- |
| **Runbook** | Repeatable technical procedure | Inputs, steps, verification and rollback |
| **Playbook** | Coordinated response to a scenario | Roles, decisions, communication and runbooks |
| **Game day** | Exercise assumptions safely | Scenario, injects, observations and improvements |
| **Post-incident review** | Prevent recurrence | Timeline, causes, actions and verification |


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# Service Quality Review

## Reliability

- SLO and error budget
- Recovery test results

## Security

- Findings and exceptions
- Control effectiveness

## Efficiency

- Capacity and cost
- Automation and toil reduction


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Capstone Standard

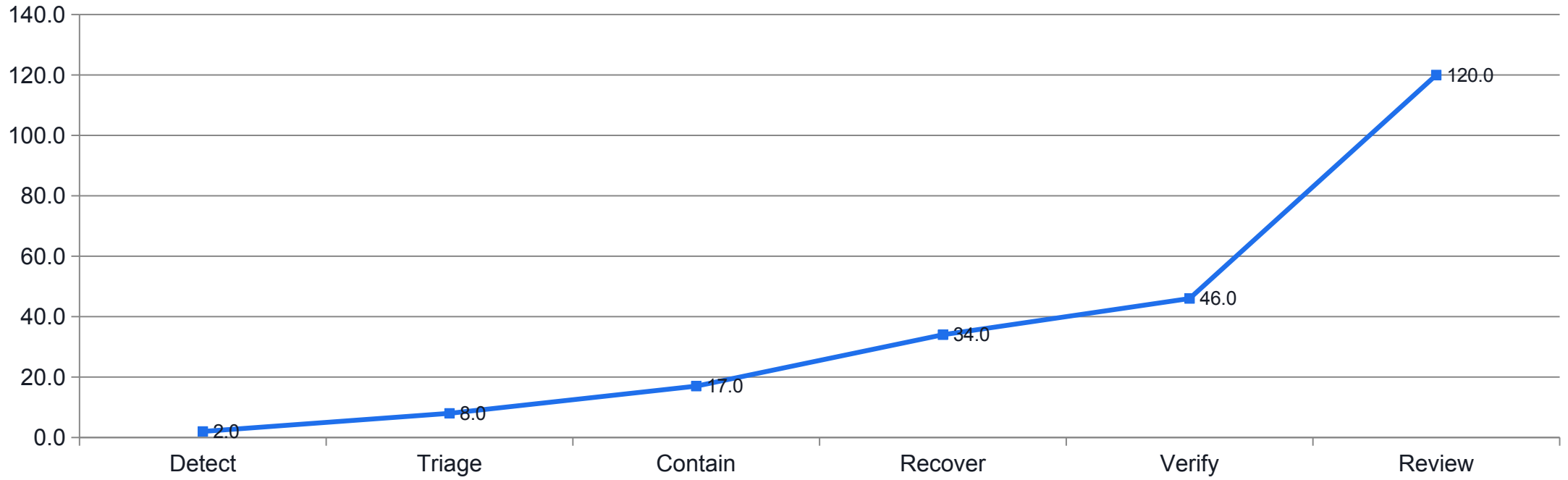
# Restore service, explain the evidence, and improve the system.

A complete CloudOps response connects detection, diagnosis, controlled recovery, validation, communication and recurrence prevention.


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TOPIC 10 · NATIVE EDITABLE TECHNICAL CHART

# Incident Timeline — Detect to Learn




**WHAT THE DATA SHOWS**

Recovery at minute 34 is not completion: verification and recurrence-prevention work extend the operational lifecycle.


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# AWS X-Ray

## 1 Debugging in Production, the good

Debugging in Production, the good old way

Log formats differ across applications and log analysis is hard.

## 2 Test locally

Test locally

Debugging: one big monolith “easy”, distributed services “hard”

## 3 Add log statements everywhere

Add log statements everywhere

No common views of your entire architecture

## 4 Re-deploy in production

Re-deploy in production

Enter… AWS X-Ray!

Mechanism: correlate monitoring, audit, configuration, identity, network and data signals on one timestamped incident timeline.

**CURRENT OPERATIONS LENS**

Synthesize telemetry, configuration, identity, network and data evidence into one incident timeline and one justified recovery decision. Validate service limits and regional availability against current AWS documentation.


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# AWS X-Ray

### 1 Visual analysis of our…

Visual analysis of our applications

Mechanism: correlate monitoring, audit, configuration, identity, network and data signals on one timestamped incident timeline.

### 2 Failure mode

Failure mode: changing multiple layers before preserving evidence destroys causality and expands blast radius during an already uncertain event.

### 3 Evidence: observed facts…

Evidence: observed facts, hypotheses, decisions, actions, verification results and communications remain separate but linked throughout the record.

### 4 Decision point

Decision point: restore the critical service with the smallest reversible action, then verify data integrity and assign recurrence-prevention work…

**CURRENT OPERATIONS LENS**

Synthesize telemetry, configuration, identity, network and data evidence into one incident timeline and one justified recovery decision. Validate service limits and regional availability against current AWS documentation.


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# AWS X-Ray advantages

### 1 Troubleshooting performance…

Troubleshooting performance (bottlenecks)

Find errors and exceptions

Mechanism: correlate monitoring, audit, configuration, identity, network and data signals on one timestamped incident timeline.

### 2 Understand dependencies in a…

Understand dependencies in a microservice architecture

Are we meeting time SLA?

### 3 Pinpoint service issues

Pinpoint service issues

Where I am throttled?

Failure mode: changing multiple layers before preserving evidence destroys causality and expands blast radius during an already uncertain event.

### 4 Review request behavior

Review request behavior

Identify users that are impacted

Evidence: observed facts, hypotheses, decisions, actions, verification results and communications remain separate but linked throughout the record.

**CURRENT OPERATIONS LENS**

Synthesize telemetry, configuration, identity, network and data evidence into one incident timeline and one justified recovery decision. Validate service limits and regional availability against current AWS documentation.


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# AWS Amplify - web and mobile applications

**1 A set of tools and**

A set of tools and services that helps you develop and deploy scalable full stack web and mobile applications

Amplify CLI

Amplify backend

**2 Authentication, Storage, API…**

Authentication, Storage, API (REST, GraphQL), CI/CD, PubSub, Analytics, AI/ML Predictions, Monitoring, …

Amazon S3 Amazon Cognito

connect frontend to backend using

**3 Connect your source code from**

Connect your source code from GitHub, AWS CodeCommit, Bitbucket, GitLab, or upload directly

Amazon SageMaker

Amplify Frontend Libraries

**4 configure backend using**

configure backend using

Amazon Lex Lambda DynamoDB

Amplify Console

Mechanism: correlate monitoring, audit, configuration, identity, network and data signals on one timestamped incident timeline.

**CURRENT OPERATIONS LENS**

Synthesize telemetry, configuration, identity, network and data evidence into one incident timeline and one justified recovery decision. Validate service limits and regional availability against current AWS documentation.


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# State of learning checkpoint

### 1 Let’s look how far we’ve

Let’s look how far we’ve gone on our learning journey

### 2 Mechanism: correlate monitoring…

Mechanism: correlate monitoring, audit, configuration, identity, network and data signals on one timestamped incident timeline.

### 3 Failure mode

Failure mode: changing multiple layers before preserving evidence destroys causality and expands blast radius during an already uncertain event.

### 4 Evidence: observed facts…

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TOPIC 10 · CONCEPT-LED LAB BRIEF

# Capstone — Resolve a Multi-Service CloudOps Incident

**LAB 12**

Apply A1–A7 in one connected incident by detecting impact, isolating the failed boundary, restoring service, validating data and documenting service-quality improvements.

Correlate synthetic evidence across monitoring, change, network, backup and security records; produce and execute a controlled recovery plan.

### YOU'LL BUILD

A timeline, evidence matrix, prioritised hypotheses, recovery runbook, validation record and post-incident action register.

### TOOLCHAIN

CloudWatch · CloudTrail · VPC Flow Logs · ELB · AWS Backup · Security Hub · Systems Manager

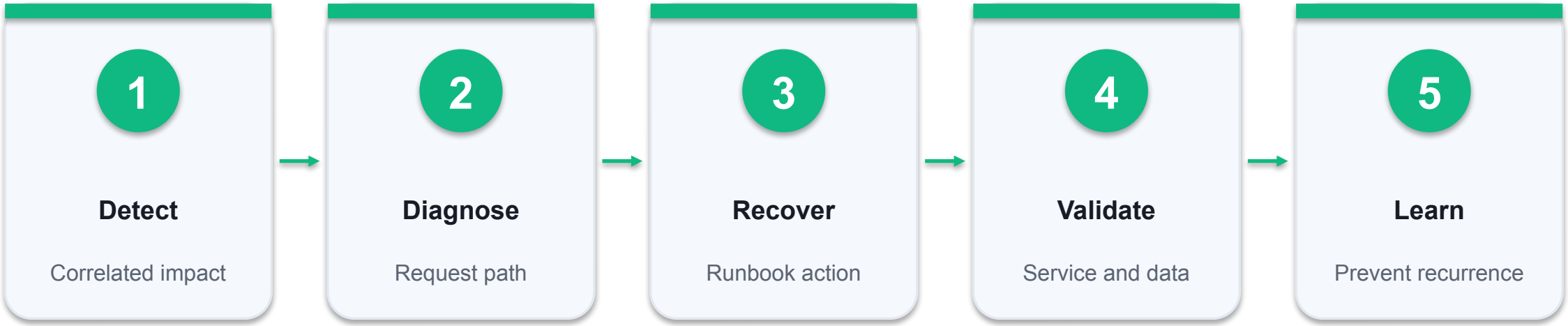
### DONE WHEN

All evidence IDs reconcile, the timeline precedes causal claims, recovery and rollback are explicit, validation covers service and data, A1–A7 are mapped, and validation emits LAB 12 PASS.


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LAB 12 · CONCEPTUAL WORKFLOW

# Lab 12 Evidence Model




**PROCEDURE BOUNDARY**

Detailed commands and numbered steps are in the Learner Guide and lab folder—not on these slides.


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LAB 12 · ACCEPTANCE EVIDENCE

# Capstone — Resolve a Multi-Service CloudOps Incident

## ✅ Expected result

All evidence IDs reconcile, the timeline precedes causal claims, recovery and rollback are explicit, validation covers service and data, A1–A7 are mapped, and validation emits LAB 12 PASS.

IF IT DOESN'T WORK

### Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

### Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

### Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.


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# Recap — Performance, Cost and Integrated Troubleshooting

1

**Explain**

Right-size and tune using utilisation, saturation, latency and business demand.

2

**Distinguish**

Unexpected spend can reveal leakage, misconfiguration, attack or failed cleanup.

3

**Apply**

A1–A7 in one connected incident by detecting impact, isolating the failed boundary, restoring service, validating data and documenting…

4

**Evidence**

All evidence IDs reconcile, the timeline precedes causal claims, recovery and rollback are explicit, validation covers service and data, A1–A7 are mapped, and…


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END

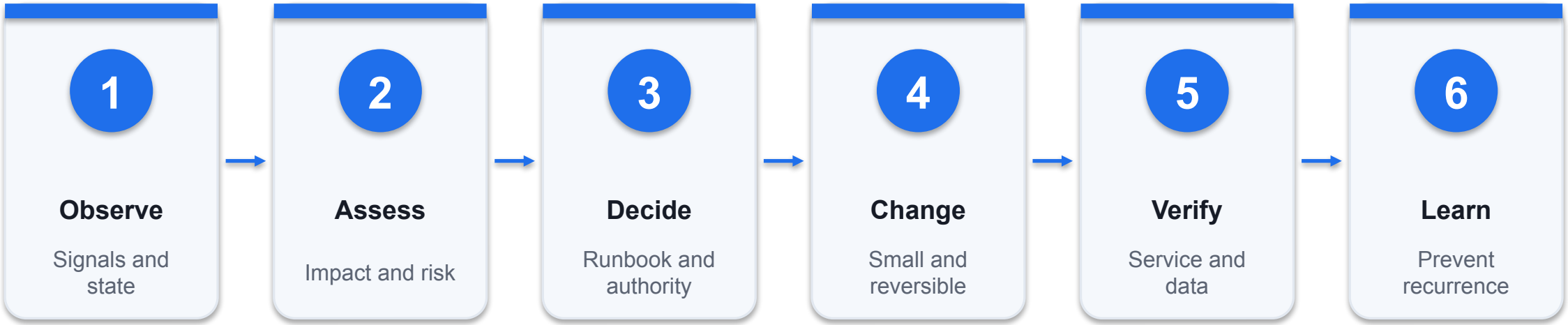
WRAP-UP

# Synthesis, Assessment & Next Steps


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LO1 → LO5

# The Connected CloudOps Control Loop




**CAPSTONE**

Lab 12 and the Case Study require one traceable operational narrative across the complete loop.


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LEARNING OUTCOMES

# What You Can Now Do

1

**LO1**

Maintain business continuity of AWS cloud operations to ensure stable service delivery.

2

**LO2**

Implement secure AWS cloud infrastructure setup such as data loading and performance monitoring.

3

**LO3**

Apply service quality standards to ensure security and compliance of AWS cloud operations.

4

**LO4**

Analyse and resolve processing issues to ensure continuity and disaster recovery in AWS cloud applications.

5

**LO5**

Troubleshoot networking performance issues at scale for optimal AWS infrastructure operation.


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FOUR LABS PER DAY

# Twelve-Lab Learning Journey

1

**Day 1**

Labs 01–04 · observe, triage, manage fleets and test high availability

2

**Day 2**

Labs 05–08 · automate change, prove recovery, enforce compliance and protect data

3

**Day 3**

Labs 09–12 · diagnose networks, fail over delivery, recover data and resolve a capstone incident


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CONTINUE YOUR LEARNING

# Recommended Courses

1. AWS Certified Solutions Architect Associate Training
2. AWS Certified Developer Associate Training
3. DevOps Engineering on AWS
4. AWS Certified Security Specialty Training
5. Kubernetes and Cloud Native Associate (KCNA) Training


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POST-COURSE SUPPORT

# Support

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2 **Telephone**
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3 **Website**
www.tertiarycourses.com.sg

4 **LMS-TMS**
lms-tms.tertiaryinfotech.com


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READY

# Assessment — Final Check

1

## WA-SAQ

Five open-ended questions · K1–K5 · 60 minutes.

2

## Case Study

Five connected tasks · A1–A7 · 60 minutes.

3

## Digital attendance

Complete assessment attendance and TRAQOM steps.

4

## Submission

Upload candidate papers through LMS-TMS; keys remain trainer-only.


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ON ASSESSMENT DAY

# Assessment Flow

1. **TRAQOM**
   Course feedback
2. **Attendance**
   Assessment QR
3. **WA-SAQ**
   5 questions · 60 min
4. **Case Study**
   5 tasks · 60 min
5. **Submit**
   LMS-TMS and sign-off

**APPEAL**

Use the documented reassessment and appeal process if you disagree with an assessment decision.


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# Digital Attendance (Mandatory)

1

## AM

Scan and submit at the start of the day.

2

## PM

Scan and submit after lunch.

3

## Assessment

Scan before starting the final assessment.

4

## Confirm

Do not close the page until submission succeeds.


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AWS CLOUD OPERATIONS

# Thank You!

Operate with evidence, change safely, verify recovery and leave the system better than you found it.


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